

Higher Categories I

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Introduction

About This Document

TODO

A Fairytale

At many places in homotopy theory interesting maps are uniquely determined up to homotopy but not on the nose. Let me give 4 examples:

- 1) The inverse to a homotopy equivalence
- 2) The classifying map $B \rightarrow BO(d)$ of a d -dimensional vector bundle
- 3) The lift in a HELP-diagram with one of the vertical maps a homotopy equivalence

$$\begin{array}{ccc}
 A & \longrightarrow & E \\
 \text{cofibration} \downarrow & \nearrow & \downarrow \text{fibration} \\
 X & \longrightarrow & B
 \end{array}$$

- 4) The n -fold multiplication

$$(\Omega X)^n \rightarrow \Omega X$$

Similar phenomena occur in algebra:

- 5) The map $F \rightarrow F'$ between free resolutions of $M \rightarrow N$.

Now, faced with such a situation it is often difficult to perform categorically flavoured operations in a meaningful way:

A simple example is the formation of fibres: f^{-1} is the pullback

$$\begin{array}{ccc}
 f^{-1}(y) & \longrightarrow & X \\
 \downarrow & \lrcorner & \downarrow f \\
 y & \xrightarrow{i} & Y
 \end{array}$$

but homotopic f and f' (or homotopic i, i') need not give homotopy equivalent pullbacks. To assign invariant meaning to them homotopical versions of such constructions were soon invented, in the case at hand, the *homotopy fibre*.

But this still not quite good enough:

- 1) While homotopy fibres of homotopic maps f and f' are homotopy equivalent, the (homotopy class of) homotopy equivalence depends on a choice of (homotopy class of) homotopy between f and f' , so the homotopy fibre is not functorial on the arrow category of the homotopy category.
- 2) The homotopy fibre is not a direct categorical construction anymore neither in the original category nor the homotopy category; it represents neither the functor of commuting diagrams nor of homotopy commuting diagrams. Rather a map

$$Z \rightarrow \text{hofib}_x(f)$$

represents a *homotopy coherent* diagram

$$\begin{array}{ccc}
 Z & \xrightarrow{\quad} & X \\
 \downarrow & \searrow & \downarrow f \\
 * & \xrightarrow{x} & Y
 \end{array}$$

(double arrows indicating homotopies)

The first problem is addressed by the observation that the maps in question are all determined up to contractible choice not just up to homotopy! Let me warn you that it is not true in examples 1), 2), 4) and 5) that the component of a given map in the mapping space is contractible, just that there is a contractible "space", in which the maps in question naturally live: for example in 2) the space of bundle maps

$$V \rightarrow EO(d) \times_{O(d)} \mathbb{R}^d$$

is contractible, in 4) the multiplications are given partitions of the interval into 4 parts i.e. the space

$$\{(t_1, t_2, t_3) \in I^3 \mid 0 < t_1 < t_2 < t_3 < 1\}$$

parametrizes the multiplications and is contractible. For a homotopy equivalence $g : Y \rightarrow X$ it is a space of pairs

$$\{(f, H) \in C(X, Y) \times C(X \times I, X) \mid H_0 = \text{id}_X, H_1 = g \circ f\} .$$

In particular, for any two choices of map as in 1)-5) not only does there exist a homotopy between them, but that homotopy is itself unique up to homotopy (so for example for a vector bundle $V \rightarrow B$ the homotopy fibre

$$\text{hofib}_\delta(B \rightarrow BSO(d))$$

is well-defined up to unique isomorphism in the homotopy category (once a point $\delta \in BSO(d)$ is fixed once and for all). But moreover any two homotopies witnessing this uniqueness are again unique up to homotopy, so further constructions of similar flavour are again possible without destroying functoriality. To exploit these observations a systematic category theory allowing for homotopy and contractibility instead of equality and uniqueness is required. The easiest way to achieve this is by considering topologically enriched categories. To accommodate non-topological examples it is in fact better to work combinatorially and

consider Kan-enriched categories. While the entirety of higher category theory can be developed for these objects, they carry a major defect: composition of morphisms is still strictly associative. While this is the case for most categories that serve as starting points ($\mathbf{Top}$, $\mathbf{sSet}$, $\mathbf{Ch}_R$, ...), it is not preserved by many categorical constructions, since (by design) these are only determined up to contractible choice. A related problem is that a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ of simplicial categories ought to be considered invertible if

- 1) it is essentially homotopically surjective, meaning for all $D \in \mathcal{D}$ there is a $C \in \mathcal{C}$ and a homotopy equivalence $F(C) \simeq D$ and
- 2) F induces a homotopy equivalence

$$\mathrm{Hom}(A, B) \rightarrow \mathrm{Hom}(FA, FB)$$

for all objects $A, B \in \mathcal{C}$.

Such functors are called *Dwyer-Kan equivalences*.

Although this notion restricts to usual equivalences of categories for discretely enriched $\mathcal{C}$ and $\mathcal{D}$ it is quite different from that of an enriched equivalence, which would require isomorphisms

$$\mathrm{Hom}(A, B) \rightarrow \mathrm{Hom}(FA, FB).$$

Here are two examples. Consider the full subcategory $\mathcal{C}$ of the simplicially enriched category $\mathbf{Top}$ spanned by contractible spaces. Then the functor $p: \mathcal{C} \rightarrow *$ to the one-point category (with trivial enrichment) is a Dwyer-Kan equivalence, since the space of continuous functions between contractible spaces is itself contractible. However, for any functor $s: * \rightarrow \mathcal{C}$ a natural transformation $\tau: s \circ p \Rightarrow \mathrm{id}_{\mathcal{C}}$ would require all diagrams of the form

$$\begin{array}{ccc} s(*) & \xrightarrow{\tau_X} & X \\ (s \circ p)(f) \downarrow & & \downarrow f \\ s(*) & \xrightarrow{\tau_Y} & Y \end{array}$$

to commute, which is clearly absurd since $s(*) \neq \emptyset$. Note, however, that a (unique) transformation $\mathrm{id}_{\mathcal{C}} \Rightarrow s \circ p$ exists just fine, showing that natural transformations, which are pointwise equivalences need not admit inverses, another desirable criterion that Kan-enriched categories fit. As a second example consider the category $\mathcal{N}[2]$ with three objects and six morphisms

$$\begin{array}{ccc} & \overset{\mathrm{id}}{\curvearrowright} & \\ & 1 & \\ f \nearrow & & \searrow g \\ \mathrm{id} \curvearrowleft 0 & \xrightarrow{gf} & 2 \curvearrowright \mathrm{id} \end{array}$$

We can give it the trivial enrichment. Consider also the simplicially enriched category $\mathcal{C}$, whose only difference is that we replace

$$\mathrm{Hom}_{\mathcal{N}[2]}(0, 2) = \mathrm{const}\{f, g\} \in \mathbf{sSet}$$

by

$$\mathrm{Hom}_{\mathcal{C}}(0, 2) = \mathrm{Sing} \, \mathbf{I}$$

with $g \circ f = (\Delta^0 \xrightarrow{0} \mathbf{I}) \in \mathrm{Sing}_0 \, \mathbf{I}$. Evidently, there are unique enriched functors

$$\mathcal{C} \rightarrow \mathcal{N}[2] \quad \text{and} \quad \mathcal{N}[2] \rightarrow \mathcal{C}$$

which are identity on objects and both are Dwyer-Kan equivalences. However, only one make the diagram

$$\begin{array}{c} \mathrm{id} \circlearrowleft \overset{0}{\bullet} \longrightarrow \overset{1}{\bullet} \circlearrowright \mathrm{id} = \mathcal{N}[1] \\ \begin{array}{ccc} & \begin{array}{c} \xrightarrow{0} \mathcal{N}[2] \\ \xrightarrow{1} \mathcal{N}[2] \\ \xrightarrow{2} \mathcal{N}[2] \end{array} & \\ \begin{array}{c} \xrightarrow{0} \mathcal{C} \\ \xrightarrow{1} \mathcal{C} \\ \xrightarrow{2} \mathcal{C} \end{array} & \begin{array}{c} \sim \\ \sim \end{array} & \end{array} \\ \Delta^0 \xrightarrow{0} \mathbf{I} \end{array}$$

commute. The worst offender, however, are Dwyer-Kan equivalences which do not admit inverses at all: for example, there is no reason to expect that the Dwyer-Kan equivalence

$$\left\{ \begin{array}{c} \text{finite cell complexes,} \\ \text{cellular}^1 \text{maps} \end{array} \right\} \subseteq \left\{ \begin{array}{c} \text{spaces homotopy equivalent to a finite cell complex,} \\ \text{continuous maps} \end{array} \right\}$$

admits any functor the other way that induces the inverse on homotopy categories. To remedy these deficits one considers a notion of higher category theory that is not equipped with a strict composition law. There are at least two good candidates: Joyal's *quasi-categories* and Rezk's *complete Segal spaces*. Given the recent development of the former at the hands of Lurie we will give preference to it and say little about the later notion. Quasi-categories were originally defined by Boardman and Vogt (who taught me topology in Osnabrück) during the development of E_∞ -spaces (the natural notion of commutative monoid that arises in higher categories): they attempted to define the morphisms between such objects as maps of the underlying spaces together with an E_∞ -structure on its cofibre extending the given ones. As cofibres do not commute with composition they did not quite obtain a category and instead formalized the resulting object (under the name "weak Kan complex"). This apparent defect was soon remedied by May's reformulation of their theory via his introduction of operads and thus stayed undeveloped for many years until Joyal demonstrated that much of ordinary category theory generalises to such weak Kan complexes (whence he renamed them). A quasi-category is (roughly speaking) a simplicial set $\mathcal{C}$, which admits compositions up to contractible choice; here one thinks of $\mathcal{C}_0$ as the objects and $\mathcal{C}_1$ as the morphisms of our higher category. The space of composites of two 1-simplices $f, g \in \mathcal{C}_1$ is then the pullback

$$\begin{array}{ccc} \mathrm{Comp}(f, g) & \longrightarrow & \mathrm{Hom}(\Delta^2, \mathcal{C}) \\ \downarrow & & \downarrow \\ * & \xrightarrow{(f, g)} & \mathrm{Hom}(\Lambda_1^2, \mathcal{C}) \end{array}$$

¹Suitably interpreted as a simplicial set the statement fails for the naive subspace of cellular maps inside all maps!

Warning: the actual definition requires the right hand vertical map to also be a Kan fibration!

We will develop the very basics of this notion. The main results we will hopefully show, say precisely that quasi-categories do not share the defects of Kan-enriched categories, i.e. that

- 1) functors are equivalences if they are fully faithful and essentially surjective (both in the homotopical sense) and
- 2) natural transformations are invertible if they consist of equivalences.

Furthermore, we will show that the classical homotopy theory of cell complexes is precisely the category theory of higher groupoids, a fact known as Grothendieck's *homotopy hypothesis*.

By way of compensation it is, however, quite a bit more difficult to produce interesting quasi-categories directly. We will therefore develop the *homotopy coherent nerve*, which produces a quasi-category from a Kan-enriched ordinary category. Both these last points require a bit a classical simplicial homotopy theory (which is the theory of Kan complexes) so we will treat that first.

Finally, let me just mention that the theories just discussed are only the start: they are all on models for $(\infty, 1)$ -category theory, meaning that all morphisms of degree ≥ 2 (starting with homotopies) are automatically invertible. For example, the (large) $(\infty, 1)$ -category Cat_∞ of (small) $(\infty, 1)$ -categories really wants to be an $(\infty, 2)$ -category (since not all natural transformations are equivalences).

We will have little to say on this. Other topics that will not make an appearance here:

- straightening / unstraightening
- Yoneda's Lemma
- operads / symmetric monoidal categories
- sheaves / hypersheaves
- spectra and stabilization
- systematic study of limits and colimits

Maybe if there is a part 2...

Chapter I.

Categories

I.1. Definition

A *category* $\mathcal{C}$ consists of two classes of O and M whose elements are called *objects* and *morphisms*, respectively, and four maps

$$i : O \rightarrow M, \quad s : M \rightarrow O, \quad t : M \rightarrow O$$

and

$$c : M \times_t M \rightarrow M$$

called *identity*, *source*, *target* and *composition* such that

- | | |
|--------------------------------------|--|
| 1) $s \circ i = \text{id}_O$ | 3) $c \circ (c \times_O \text{id}_M) = c \circ (\text{id}_M \times_{OC})$ |
| $t \circ i = \text{id}_O$ | |
| 2) $s \circ c = s \circ \text{pr}_2$ | 4) $c \circ ((i \circ t) \times \text{id}_M) \circ \Delta_M = \text{id}_M$ |
| $t \circ c = t \circ \text{pr}_1$ | $c \circ (\text{id}_M \times (i \circ s)) \circ \Delta_M = \text{id}_M$ |

For $x, y \in O$ we set

$$\text{Hom}_{\mathcal{C}}(x, y) = \{\varphi \in M \mid s(\varphi) = x, t(\varphi) = y\}$$

$\mathcal{C}$ is called *small* if both O and M are sets and *locally small* if $\text{Hom}_{\mathcal{C}}(x, y)$ is a set for every $x, y \in O$. More generally an object $x \in O$ is called *left* or *right locally small* if $\text{Hom}_{\mathcal{C}}(x, y)$ or $\text{Hom}_{\mathcal{C}}(y, x)$ are (in bijection with) sets for all $y \in O$.

For a category $\mathcal{C} = (O, M, \dots)$ one writes

$$x \in \mathcal{C}, \varphi : x \rightarrow y, \text{id}_x, \varphi \circ \psi$$

instead of

$$x \in O, \varphi \in \text{Hom}_{\mathcal{C}}(x, y), i(x), c(\varphi, \psi).$$

I.2. Remark

By 1) smallness is implied by M being a set, and for locally small categories also by the smallness of O .

I.3. Definition

A morphism $\varphi : x \rightarrow y$ is called a *split monomorphism*, or *split epimorphism* if it admits a *retraction* $\psi : y \rightarrow x$, that is $\psi \circ \varphi = \text{id}_x$, or a *section*, that is $\varphi \circ \psi = \text{id}_y$, respectively. It is called an *isomorphism* if it admits an *inverse*, that is $\psi : y \rightarrow x$ that is both a retraction and a section simultaneously.

I.4. Definition

A category $\mathcal{C}$ is called *essentially small* with *skeleton* S , a subset of O , if it is locally small and for each $x \in O$ there exists a $y \in S$ and an isomorphism $\varphi : x \rightarrow y$.

I.5. Definition

A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is a pair

$$F : O \rightarrow O' \quad , \quad F_* : M \rightarrow M'$$

that commutes with all structure maps, i.e. such that

$$\begin{aligned} 1) \quad F_* \circ i &= i \circ F & 2) \quad s' \circ F_* &= F \circ s \\ & & t' \circ F_* &= F \circ t \end{aligned}$$

$$3) \quad F_* \circ c = c \circ (F_* \times_O F_*)$$

We denote the class of functors $\mathcal{C} \rightarrow \mathcal{C}'$ by $\text{Fun}(\mathcal{C}, \mathcal{C}')$. F is called *full* or *faithful* if the map

$$(F_*, s, t) : M \rightarrow M' \times_{O' \times O'} O \times O^1$$

is surjective or injective, respectively. An *embedding* is a functor that is fully faithful.

I.6. Examples

i) Set, Top, Ring, R-Mod, Grp, Ab, Ord, Cat, ...

ii) For $\mathcal{M}$ a monoid there is the category $\mathbb{B}\mathcal{M}$ with

$$O = \{*\} \text{ and } M = \mathcal{M}$$

$i(*) = e$, and composition the multiplication in $\mathcal{M}$

iii) For every poset P there is the category NP with

$$O = P \text{ and } M = \{(i, j) \in P^2 \mid i \leq j\}$$

$i = \Delta_P, s = \text{pr}_1, t = \text{pr}_2$ and

$$c((p, q), (r, p)) = (r, q)$$

¹ $\{(x, y, \varphi) \in O^2 \times M' \mid \varphi : F(x) \rightarrow F(y)\}$

Chapter I. Categories

For $[n] = \{0, 1, 2, \dots, n\}$ we find $N[n]$ pictorially

$$\bullet \xrightarrow{(0,1)} \bullet \xrightarrow{(1,2)} \bullet \longrightarrow \dots \quad + \begin{array}{l} \text{composites } (i, j) \quad i < j + 1 \\ \text{identities } (i, i) \end{array}$$

We obtain an embedding $N : \text{Ord} \rightarrow \text{Cat}$.

iv) The various $\text{Hom}_{\mathcal{C}}(x, y)$ assemble into a functor $\text{Hom}_{\mathcal{C}} : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \text{Set}$.

I.7. Definition

A *natural transformation* $\tau : F \Rightarrow G$ of functors $\mathcal{C} \rightarrow \mathcal{C}'$ is a map

$$\tau : O \rightarrow M', \text{ s.t.}$$

$$\begin{array}{ll} 1) \quad s' \circ \tau = F & 2) \quad c' \circ (G \times (\tau \circ s)) \circ \Delta_M \\ & = c' \circ ((\tau \circ t) \times F) \circ \Delta_M \\ & t' \circ \tau = G \end{array}$$

From now on we will write $\text{obj}(\mathcal{C}) := O$ and $\text{mor}(\mathcal{C}) := M$.

I.8. Definition

Denote by $d_i : [n] \rightarrow [n+1]$ ($0 \leq i \leq n+1$) and $s_i : [n] \rightarrow [n-1]$ ($0 \leq i \leq n-1$) the unique order preserving injection and surjection, respectively, that misses i or hits i twice.

I.9. Definition

For categories $\mathcal{C}, \mathcal{C}'$ define the *category of functors* $\mathcal{C} \rightarrow \mathcal{C}'$ to have objects and morphisms

$$\text{Fun}(\mathcal{C} \times N[0], \mathcal{C}') \text{ and } \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}')$$

respectively; structure maps are

$$i = - \circ (\text{id}_{\mathcal{C}} \times N_* s_0), \quad s = - \circ (\text{id}_{\mathcal{C}} \times N_* d_1), \quad t = - \circ (\text{id}_{\mathcal{C}} \times N_* d_0)$$

and composition is given by

$$\begin{array}{ccc} \text{Fun}(\mathcal{C} \times N[2], \mathcal{C}') & \xrightarrow{- \circ N_* d_1} & \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \\ \downarrow \cong & & \\ \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \times_{\text{Fun}(\mathcal{C} \times N[0], \mathcal{C}')} & & \text{Fun}(\mathcal{C} \times N[1], \mathcal{C}') \end{array}$$

Warning. If $\mathcal{C}$ is small and $\mathcal{C}'$ locally small then $\text{Fun}(\mathcal{C}, \mathcal{C}')$ is locally small, but this can fail when $\mathcal{C}$ is only locally small!

I.10. Definition

A natural transformation that is an isomorphism in $\text{Fun}(\mathcal{C}, \mathcal{C}')$ is called a *natural isomorphism*.

I.11. Definition

A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is called an *equivalence* if there exists a functor $G : \mathcal{C}' \rightarrow \mathcal{C}$ and natural isomorphisms

$$F \circ G \Rightarrow \text{id}_{\mathcal{C}'} \quad \text{and} \quad G \circ F \Rightarrow \text{id}_{\mathcal{C}}.$$

I.12. Definition

The *essential image* $F(\mathcal{C})$ of a functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is the full² subcategory generated by all objects isomorphic to one in the image of $F : \mathcal{O} \rightarrow \mathcal{O}'$.

I.13. Warning

The actual image

$$(F(\mathcal{O}), F_*(M)) \subseteq (\mathcal{O}', M')$$

is not generally closed under composition in $\mathcal{C}'$, so does not form a subcategory. The notion of equivalence of categories is *not* that of isomorphism in Cat , which requires $F \circ G = \text{id}_{\mathcal{C}'}$ and $G \circ F = \text{id}_{\mathcal{C}}$!

I.14. Proposition

- i) A functor $F : \mathcal{C} \rightarrow \mathcal{C}'$ is an equivalence if and only if it is an embedding and essentially surjective, i.e. $F(\mathcal{C}) = \mathcal{C}'$.
- ii) A natural transformation is a natural isomorphism if and only if it takes values in isomorphisms.

Proof. MacLane ...

□

I.15. Corollary

Every embedding of categories is an equivalence onto its essential image³. For example, the inclusion of every full subcategory $\mathcal{C} \subseteq \mathcal{C}'$ containing an object from every isomorphism class of $\mathcal{C}'$ is an equivalence.

²i.e. with morphisms the entire Hom-sets.

³It need not be an isomorphism onto its image!

I.16. Definition

A category is called a *groupoid* if every morphism in it is an isomorphism. A typical example is the inclusion of the *groupoid core*

$$\text{Core}(\mathcal{C}) \subseteq \mathcal{C}$$

having the same objects as $\mathcal{C}$ but retaining only the isomorphisms of $\mathcal{C}$ as morphisms. Also the functor $\mathbb{B}$ factors as $\text{Grp} \xrightarrow{\mathbb{B}} \text{Grpd} \subseteq \text{Cat}$.

Every left/right locally small object $x \in \mathcal{C}$ determines the functors

$$\begin{array}{l|l} \text{Hom}_{\mathcal{C}}(x, -) = Y_x : \mathcal{C} \rightarrow \text{Set} & \text{Hom}_{\mathcal{C}}(-, x) = Y^x : \mathcal{C}^{\text{op}} \rightarrow \text{Set} \\ y \mapsto \text{Hom}_{\mathcal{C}}(x, y) & y \mapsto \text{Hom}_{\mathcal{C}}(y, x) \\ f \mapsto f \circ - & f \mapsto - \circ f \end{array}$$

known as the functors *(co)represented* by x .

The arguably most important statement of category theory is

I.17. Proposition (Yoneda's Lemma)

For all functors $F : \mathcal{C} \rightarrow \text{Set}$ and $G : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ and $x \in \mathcal{C}$ left or right locally small, the maps $\text{Nat}(Y_x, F) \rightarrow F(x)$ and $\text{Nat}(Y^x, G) \rightarrow G(x)$ given by

$$\tau \mapsto \tau_x(\text{id}_x)$$

are bijections, respectively.

Proof. MacLane ...

□

For locally small $\mathcal{C}$, the functors Y^x give rise to a single functor

$$Y^{\mathcal{C}} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set}) =: \mathcal{P}(\mathcal{C})^4$$

for example, as the image of $\text{Hom}_{\mathcal{C}} \in \text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \text{Set})$ under

$$\text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \text{Set}) \cong \text{Fun}(\mathcal{C}, \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set}))$$

I.18. Corollary/Definition

Let $\mathcal{C}$ be locally small. The *Yoneda embedding*

$$Y^{\mathcal{C}} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$$

is really an embedding. Its essential image is called the category of *representables*⁵ of $\mathcal{C}$.

Thus one can specify a functor $\mathcal{C} \rightarrow \mathcal{D}$ (with $\mathcal{D}$ locally small) up to unique isomorphism by giving a functor

$$\mathcal{C} \rightarrow \text{Fun}(\mathcal{D}^{\text{op}}, \text{Set})$$

⁴not usually locally small

⁵they are left locally small

and checking that its image is contained in the representables. Specialized to $\mathcal{C} = *$ this is a method of definition by *universal property*!

An example is:

I.19. Definition

An object $x \in \mathcal{C}^6$ is *left* or *right adjoint* to an object $y \in \mathcal{D}^6$ under a functor $F : \mathcal{C} \rightarrow \mathcal{D}$, if, respectively, there is a natural isomorphism

$$\tau^7 : Y_x \Longrightarrow Y_y \circ F \quad \text{or} \quad \tau^8 : Y^x \Longrightarrow Y^y \circ F^{\text{op}},$$

which is called a *witness* of the adjunction.

I.20. Examples

- i) The integers are left adjoint to the empty set under the forgetful functor $\text{Ring} \rightarrow \text{Set}$.
- ii) An R -module is free if and only if it is left adjoint to some set under the forgetful functor $R\text{-Mod} \rightarrow \text{Set}$ and a witness is precisely a choice of basis indexed by that set.
- iii) The groupoid core of $\mathcal{C}$ is right adjoint to $\mathcal{C}$ under the inclusion $\text{Grpd} \subseteq \text{Cat}$.
- iv) The set

$$\pi_0(\mathcal{C}) = \text{obj}(\mathcal{C}) / \sim,$$

where $\sim$ is the equivalence relation generated by setting

$$x \sim y \quad \text{if} \quad \text{Hom}_{\mathcal{C}}(x, y) \neq \emptyset,$$

is left adjoint to $\mathcal{C}$ under $\mathbb{D} : \text{Set} \rightarrow \text{Cat}$

If $\mathcal{D}_F \subseteq \mathcal{D}$ and $\mathcal{D}^F \subseteq \mathcal{D}$ denote the full subcategories spanned by the objects admitting a left or right adjoint, respectively, then the functors

$$\mathcal{D}_F^{\text{op}} \subseteq \mathcal{D}^{\text{op}} \xrightarrow{Y_{\mathcal{D}}} \text{Fun}(\mathcal{D}, \text{Set}) \xrightarrow{- \circ F} \text{Fun}(\mathcal{C}, \text{Set})$$

$$\mathcal{D}^F \subseteq \mathcal{D} \xrightarrow{Y^{\mathcal{D}}} \text{Fun}(\mathcal{D}^{\text{op}}, \text{Set}) \xrightarrow{- \circ F^{\text{op}}} \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$$

take values in the (co)representables. Thus they determine functors

$$L_F : \mathcal{D}_F \rightarrow \mathcal{C} \quad \text{and} \quad R_F : \mathcal{D}^F \rightarrow \mathcal{C}$$

together with natural isomorphisms

$$\text{Hom}_{\mathcal{C}} \circ (L_F^{\text{op}} \times \text{id}_{\mathcal{C}}) \Longrightarrow \text{Hom}_{\mathcal{D}} \circ (\text{id}_{\mathcal{D}_F^{\text{op}}} \times F)$$

⁶locally small, but see I.23

⁷i.e. $\text{Hom}_{\mathcal{C}}(x, -) \cong \text{Hom}_{\mathcal{D}}(y, F(-))$

⁸i.e. $\text{Hom}_{\mathcal{C}}(-, x) \cong \text{Hom}_{\mathcal{D}}(F(-), y)$

and

$$\mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R_F) \implies \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}^F})$$

of functors

$$\mathcal{D}_F^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathrm{Set} \quad \text{and} \quad \mathcal{C}^{\mathrm{op}} \times \mathcal{D}^F \rightarrow \mathrm{Set},$$

respectively. In case F happens to take values in $\mathcal{D}_F$ or $\mathcal{D}^F$ the functors L_F or R_F also determine F as R_{L_F} or L_{R_F} , respectively. We set:

I.21. Definition

An *adjunction* between functors $L : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ is a natural isomorphism

$$\tau : \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}}) \implies \mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R).$$

One often says that τ exhibits L as left adjoint to R or R as right adjoint to L in this situation.

I.22. Proposition/Definition

Any adjunction is both of the form

$$\mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}}) \xRightarrow{R_*^9} \mathrm{Hom}_{\mathcal{C}} \circ (R^{\mathrm{op}} L^{\mathrm{op}} \times R) \xRightarrow{u^*} \mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R)$$

and inverse to

$$\mathrm{Hom}_{\mathcal{C}} \circ (\mathrm{id}_{\mathcal{C}^{\mathrm{op}}} \times R) \xRightarrow{L_*^{10}} \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times LR) \xRightarrow{c_*} \mathrm{Hom}_{\mathcal{D}} \circ (L^{\mathrm{op}} \times \mathrm{id}_{\mathcal{D}})$$

for unique natural transformations

$$u : \mathrm{id}_{\mathcal{C}} \implies RL \quad \text{and} \quad c : LR \implies \mathrm{id}_{\mathcal{D}},$$

called the *unit* and *counit* of the adjunction. Such transformations c and u give rise to an adjunction if and only if the *triangle identities*

$$(L \xRightarrow{Lu} LRL \xRightarrow{cL} L) = \mathrm{id}_L$$

$$(R \xRightarrow{uR} RLR \xRightarrow{Rc} R) = \mathrm{id}_R$$

hold.

Proof. MacLane ...

□

I.23. Remark

This latter description does not require $\mathcal{C}$ and $\mathcal{D}$ to be locally small so extends the notion of adjunction to arbitrary categories. In particular, one can check that an adjunction $L : \mathcal{C} \rightleftarrows \mathcal{D} : R$ induces $- \circ R : \mathrm{Fun}(\mathcal{C}, \mathcal{E}) \rightleftarrows \mathrm{Fun}(\mathcal{D}, \mathcal{E}) : - \circ L$ and $L \circ - : \mathrm{Fun}(\mathcal{E}, \mathcal{C}) \rightleftarrows \mathrm{Fun}(\mathcal{E}, \mathcal{D}) : R \circ -$ by extending the unit/counit in the evident way.

⁹The tautological transformation $\mathrm{Hom}_{\mathcal{C}} \Rightarrow \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times F)$ for any $F : \mathcal{C} \rightarrow \mathcal{D}$.

¹⁰See footnote ⁹.

I.24. Proposition

A left adjoint functor L is an embedding if and only if the unit of adjunction $u : \text{id} \Rightarrow RL$ is a natural isomorphism, and dually a right adjoint is an embedding if and only if the counit $c : LR \Rightarrow \text{id}$ is an isomorphism.

Proof. MacLane ...

□

I.25. Definition

A functor admitting a fully faithful left or right adjoint is called a right or left *Bousfield localisation*.

I.26. Remark

In general a *localisation* of a category $\mathcal{C}$ at a class of morphisms $S \subseteq \text{mor}(\mathcal{C})$ is a functor $L : \mathcal{C} \rightarrow \mathcal{D}$ such that for any category $\mathcal{E}$ the induced functor

$$- \circ L : \text{Fun}(\mathcal{D}, \mathcal{E}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{E})$$

is an embedding with essential image

$$\{F : \mathcal{C} \rightarrow \mathcal{E} \mid F(s) \text{ is an isomorphism for all } s \in S\}.$$

For any S and $\mathcal{C}$ there exists a contractible 2-groupoid of such localisations, but note that the Yoneda Lemma does not apply here! Furthermore, even if $\mathcal{C}$ is locally small the localisation, usually denoted $\mathcal{C}[S^{-1}]$ need not be so. However, any Bousfield localisation is a localisation (at which class of morphisms?), where the latter problem does not occur!

Here is another general example of adjoints:

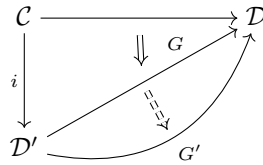
I.27. Definition

Given functors $i : \mathcal{C} \rightarrow \mathcal{C}'$ and $F : \mathcal{C} \rightarrow \mathcal{D}$ a *left or right Kan extension* of F along i is a functor $\mathcal{C}' \rightarrow \mathcal{D}$ that is left or right adjoint to F under

$$- \circ i : \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}).$$

Kan extensions are denoted $\text{Lan}_i(F)/\text{Ran}_i(F)$ or $i_!F/i_*F$.

Decoding, a witness for G being a left Kan extension is a natural transformation $\sigma : F \Rightarrow G \circ i$ such that any natural transformation $\sigma' : F \Rightarrow G' \circ i$ factors uniquely through σ ¹¹:



¹¹Succinctly: $\text{Nat}(\text{Lan}_i F, G') \cong \text{Nat}(F, G' \circ i)$.

I.28. Examples

- i) For a group homomorphism $\varphi: H \rightarrow G$ the equivalence

$$\text{Fun}(\mathbb{B}G, \mathbf{R}\text{-Mod}) \simeq \mathbf{R}[G]\text{-Mod}$$

translates left/right Kan extension along

$$\varphi^*: \mathbf{R}[G]\text{-Mod} \rightarrow \mathbf{R}[H]\text{-Mod}$$

into induction and coinduction

$$M \mapsto R[G] \otimes_{R[H]} M \quad \text{and} \quad M \mapsto \text{Hom}_{R[H]}(R[G], M)$$

- ii) Given an additive functor $F: \mathbf{R}\text{-Mod} \rightarrow \mathbf{S}\text{-Mod}$ (or any other abelian categories with enough projectives), e.g. $-\otimes_R M$ or $\text{Hom}_R(M, -)$ the left derived functors $\mathbb{L}_i F$ are given by the right Kan extension

$$\begin{array}{ccc} \text{CC}_R^{\geq 0} & \xrightarrow{\quad} & \text{CC}_S^{\geq 0} \\ \pi_R \downarrow & \nearrow & \downarrow \pi_S \\ \mathcal{D}^{\geq 0}(R) & \xrightarrow{\text{Ran}_{\pi_R}(\pi_S \circ F)} & \mathcal{D}^{\geq 0}(S) \end{array}$$

known as the total left derived (or sometimes hyperhomology) functor $\mathbb{L}F$, composed with the functor $-[0]: \mathbf{R}\text{-Mod} \rightarrow \mathcal{D}(R)$, that regards a module as a single degree chain complex, and the homology functor $H_i: \mathcal{D}(R) \rightarrow \mathbf{R}\text{-Mod}$.

- iii) A homotopy invariant functor $F: \mathbf{Top} \rightarrow \mathcal{C}$ is Kan extended from a homotopy invariant functor $\mathbf{CW} \rightarrow \mathcal{C}$ if and only if F sends weak homotopy equivalences to isomorphisms.
- iv) For $i: \mathcal{C} \rightarrow \mathcal{C}'$, $i': \mathcal{C}' \rightarrow \mathcal{C}''$ and $F: \mathcal{C} \rightarrow \mathcal{D}$ we find straight from the definitions that

$$\text{Lan}_{i'}(\text{Lan}_i(F)) \simeq \text{Lan}_{i' \circ i}(F).$$

On the other hand we set

I.29. Definition

A functor $G: \mathcal{D} \rightarrow \mathcal{E}$ preserves the left/right Kan extension of $F: \mathcal{C} \rightarrow \mathcal{D}$ along $i: \mathcal{C} \rightarrow \mathcal{C}'$ if the diagram

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} \mathcal{D} & \xrightarrow{G} \mathcal{E} \\ \downarrow i & \searrow G \circ F & \nearrow G \circ \text{Lan}_i F \\ \mathcal{C}' & & \end{array} \quad \begin{array}{ccc} \mathcal{C} & \xrightarrow{F} \mathcal{D} & \xrightarrow{G} \mathcal{E} \\ \downarrow i & \nwarrow G \circ F & \nearrow G \circ \text{Ran}_i F \\ \mathcal{C}' & & \end{array}$$

exhibits $G \circ \text{Lan}_i F / G \circ \text{Ran}_i F$ as the left/right Kan extension of GF along i .

I.30. Proposition

Left adjoints preserve left Kan extensions and right adjoints preserve right Kan extensions.

Proof. Let $L: \mathcal{D} \rightarrow \mathcal{E}$ be left adjoint to $R: \mathcal{E} \rightarrow \mathcal{D}$. Then for $G: \mathcal{C}' \rightarrow \mathcal{E}$

$$\begin{aligned} \text{Nat}(L \text{Lan}_i F, G) &\cong^{12} \text{Nat}(\text{Lan}_i F, RG) \\ &\cong \text{Nat}(F, RGi) \\ &\cong \text{Nat}(LF, Gi) \\ &\cong \text{Nat}(\text{Lan}_i LF, G) \end{aligned}$$

and tracing through the definitions shows that for $G = L \text{Lan}_i F$ we find

$$\text{id}_{L \text{Lan}_i F} \mapsto^{13} u \circ \text{Lan}_i F \mapsto^{14} u \circ y \mapsto^{15} c \circ L(u \circ y) =^{16} Ly$$

where $y: F \Rightarrow (\text{Lan}_i F) \circ i$ is the original witness of the Kan extension. $\square$

I.31. Definition

A Kan extension is called *absolute* if it is preserved by all functors. It is called *pointwise* if it is preserved by representables: For a left/right Kan extension this means preservation by

$$(Y^*)^{\text{op}}: \mathcal{D} \rightarrow \text{Set}^{\text{op}} \quad \text{or} \quad Y_*: \mathcal{D} \rightarrow \text{Set},$$

respectively.

I.32. Proposition

If $i: \mathcal{C} \rightarrow \mathcal{C}'$ admits a right or left adjoint j then any functor $F: \mathcal{C} \rightarrow \mathcal{D}$ admits a left or right Kan extension, respectively. It is given by $F \circ j$ and thus absolute and in particular pointwise.

Note the order reversal, succinctly

$$\text{Lan}_L(F) = F \circ R \quad \text{and} \quad \text{Ran}_R(F) = F \circ L.$$

Proof. By Remark I.23 the functor

$$- \circ j: \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}', \mathcal{D})$$

is left or right adjoint to $- \circ i$ if j is right or left adjoint to i . Thus $F \circ j$ satisfies the universal property of the Kan extension. $\square$

The adjunction witness determines isomorphisms

$$R \cong \text{Lan}_L(\text{id}_{\mathcal{C}}) \quad \text{and} \quad L \cong \text{Ran}_R(\text{id}_{\mathcal{D}})$$

¹²By I.23

¹³Definition of u .

¹⁴Definition of y .

¹⁵Definition of c .

¹⁶Triangle identity.

and these Kan extensions are absolute.

In fact, the converse holds as well: A functor R is right adjoint to L if and only if R is an absolute left Kan extension of $\text{id}_{\mathcal{C}}$ along L . The canonical map $\text{id}_{\mathcal{C}} \Rightarrow \text{Lan}_L(\text{id}_{\mathcal{C}}) \circ L$ is then exactly the counit of the adjunction.

As an important special case we also have:

I.33. Definition

Given a functor $F: \mathcal{J} \rightarrow \mathcal{D}$ the left and right Kan extensions of F along $r: \mathcal{J} \rightarrow *$ are called the *colimit* and *limit* of F , respectively, denoted $\text{colim } F$ ¹⁷ and $\text{lim } F$.

Let us unpack this again: Upon identifying $\text{Fun}(*, \mathcal{J}) \cong \mathcal{J}$, we find

$$\text{Hom}_{\mathcal{D}}(\text{colim } F, y) \cong \text{Nat}(F, \text{const}_{\mathcal{J}} y)$$

naturally in $y \in \mathcal{D}$, where $\text{const}_{\mathcal{J}} y$ is the constant functor with value y . The right hand side here consists of assignments $\tau: \text{obj}(\mathcal{J}) \rightarrow \text{mor}(\mathcal{D})$ with $\tau_x: F(x) \rightarrow y$ such that for all $f: x \rightarrow x'$ all triangles commute:

$$\begin{array}{ccc} F(x) & \xrightarrow{\tau_x} & y \\ f_* \downarrow & \nearrow \tau_{x'} & \\ F(x') & & \end{array}$$

The category $\mathcal{J}$ is often called the *shape* of the (co)limit.

I.34. Examples

- i) Colimits and limits of shape $\mathbb{D}S$ for $S \in \text{Set}$ are called *sums* (or *coproducts*) and *products*.
- ii) An element $x \in \mathcal{C}$ is called *initial* or *terminal* if it is a colimit or a limit, respectively, of the empty diagram $\mathbb{D}\emptyset \rightarrow \mathcal{C}$, i.e. if

$$\text{Hom}_{\mathcal{C}}(x, y) = \{*\} \quad \text{or} \quad \text{Hom}_{\mathcal{C}}(y, x) = \{*\}$$

for all $y \in \mathcal{C}$. Another formulation is that an object x is initial/terminal if and only if it is left/right adjoint to $r: \mathcal{C} \rightarrow *$, when regarded as a functor $* \rightarrow \mathcal{C}$.

- iii) The identification $\text{Fun}(\mathbb{B}G, \text{Set}) = \text{G-Set}$ translates limits and colimits into fixed points and orbits.
- iv) (Co)limits over the shapes NP for posets P can vary greatly: Colimits over $N\{(0,0), (0,1), (1,0)\}$ with the evident order are called *pushouts* and limits over $N\{(0,1), (1,0), (1,1)\}$ are called *pullbacks*.
- v) (Co)limits in functor categories.

From I.30 and I.32 we immediately obtain:

¹⁷Or informally $\text{colim}_{x \in \mathcal{C}} F(x)$

I.35. Corollary

- i) Left adjoint functors preserve colimits and right adjoints preserve limits.
- ii) If $\mathcal{J}$ admits an initial object $x \in \mathcal{J}$ then for every functor $F: \mathcal{J} \rightarrow \mathcal{C}$ we have $\lim F = F(x)$ and similarly $\operatorname{colim} F = F(y)$ if y is terminal in $\mathcal{J}$.

In particular, (co)limits as in ii) are absolute, which is fairly rare. We will see that the latter are precisely the colimits preserved by the Yoneda embedding.

We will need pointwise versions of I.23 and I.35 as well.

I.36. Observation

- i) Given $F: \mathcal{C} \rightarrow \mathcal{D}$ with partial left adjoint $L_F: \mathcal{D}_F \rightarrow \mathcal{C}$, and $G: \mathcal{E} \rightarrow \mathcal{D}$ with $\operatorname{im}(G) \subseteq \mathcal{D}_F$ then $L_F G \in \operatorname{Fun}(\mathcal{E}, \mathcal{C})$ is a left adjoint object to $G \in \operatorname{Fun}(\mathcal{E}, \mathcal{D})$ under

$$F \circ - : \operatorname{Fun}(\mathcal{E}, \mathcal{C}) \rightarrow \operatorname{Fun}(\mathcal{E}, \mathcal{D}).$$

- ii) Given $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{I} \rightarrow \mathcal{D}$ with $G(\mathcal{I}) \subseteq \mathcal{D}_F$ such that $\operatorname{colim}_{\mathcal{I}} G$ exists (in $\mathcal{D}$) with witness $y: G \Rightarrow \operatorname{const}_{\operatorname{colim} G}$. Then the following are equivalent:

- a) $\operatorname{colim}_{\mathcal{I}} G \in \mathcal{D}_F$
- b) $\operatorname{colim}_{\mathcal{I}} L_F G$ exists (in $\mathcal{C}$)

In this case the natural map

$$L_F G \xrightarrow{L_F y} L_F \operatorname{const}_{\operatorname{colim} G} = \operatorname{const}_{L_F \operatorname{colim} G}$$

exhibits $L_F \operatorname{colim}_{\mathcal{I}} G$ as a colimit of $L_F G$, i.e.

$$\operatorname{colim}_{\mathcal{I}} L_F G = L_F(\operatorname{colim}_{\mathcal{I}} G).$$

Similar statements hold for $\mathcal{D}^F, R^F$, right adjoints and limits.

We use this observation to establish another basic property of colimits:

I.37. Proposition

Given $G: \mathcal{I} \times \mathcal{J} \rightarrow \mathcal{C}$ denote by $G_0: \mathcal{I} \rightarrow \operatorname{Fun}(\mathcal{J}, \mathcal{C})$ and $G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ the curried functors. Then

- i) If $G_1(j): \mathcal{I} \rightarrow \mathcal{C}$ admits a colimit for all $j \in \mathcal{J}$, then G_0 admits a colimit and the canonical map

$$\operatorname{colim}_{\mathcal{I}} G_1(j) \rightarrow (\operatorname{colim}_{\mathcal{I}} G_0)(j)$$

is an isomorphism for all $j \in \mathcal{J}$; similarly for limits.

- ii) If $G_0(i): \mathcal{J} \rightarrow \mathcal{C}$ admits a colimit for each $i \in \mathcal{I}$ and $G_1(j): \mathcal{I} \rightarrow \mathcal{C}$ admits a colimit for each $j \in \mathcal{J}$, then

$$\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1) \text{ exists} \quad \text{iff} \quad \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} G_0) \text{ exists},$$

in which case they agree; similarly for limits.

One loosely says that "(co)limits in functor categories are computed pointwise", although this only applies to diagrams whose colimit *can* be computed pointwise: It can happen that $\operatorname{colim}_{\mathcal{I}} G_0$ exists "accidentally" without the colimits that could define its values existing. Similarly for part ii) one often says "colimits commute". The value is also given by $\operatorname{colim}_{\mathcal{I} \times \mathcal{J}} G$ but again this latter object may exist without either $\operatorname{colim}_{\mathcal{I}} G_0$ or $\operatorname{colim}_{\mathcal{J}} G_1$ existing.

Proof. Observe that by assumption

$$G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$$

takes values in $\operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*}$ with $r_{\mathcal{I}}^*: \mathcal{C} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ the "constant diagram" functor, as this latter category consists precisely of those functors admitting colimits over $\mathcal{I}$ (by definition of colimits!) Thus there is the partial adjoint

$$\operatorname{colim}_{\mathcal{I}}: \operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*} \rightarrow \mathcal{C}$$

and part i) of I.36 implies that $\operatorname{colim}_{\mathcal{I}} \circ G_1: \mathcal{J} \rightarrow \mathcal{C}$ is a left adjoint object to G_1 under

$$r_{\mathcal{I}}^* \circ - : \operatorname{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \operatorname{Fun}(\mathcal{J}, \operatorname{Fun}(\mathcal{I}, \mathcal{C})).$$

But using the equivalence

$$\operatorname{Fun}(\mathcal{J}, \operatorname{Fun}(\mathcal{I}, \mathcal{C})) \simeq \operatorname{Fun}(\mathcal{I} \times \mathcal{J}, \mathcal{C}) \simeq \operatorname{Fun}(\mathcal{I}, \operatorname{Fun}(\mathcal{J}, \mathcal{C}))$$

$r_{\mathcal{I}}^* \circ -$ corresponds to $- \circ r_{\mathcal{I}}$, and G_1 to G_0 , so $\operatorname{colim}_{\mathcal{I}} \circ G_1$ is a colimit of G_0 and clearly satisfies the desired formula. This gives i). For ii) we use part ii) of I.36 with $Fr_{\mathcal{I}}^*: \mathcal{C} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ and $L_F = \operatorname{colim}_{\mathcal{I}}$:

If $G_1: \mathcal{J} \rightarrow \operatorname{Fun}(\mathcal{I}, \mathcal{C})$ takes values in $\operatorname{Fun}(\mathcal{I}, \mathcal{C})_{r_{\mathcal{I}}^*}$, $\operatorname{colim}_{\mathcal{J}} G_1$ exists (which follows from the assumption by part i)) and $\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1)$ exists then also $\operatorname{colim}_{\mathcal{I}} \circ G_1$ exists and

$$\operatorname{colim}_{\mathcal{I}}(\operatorname{colim}_{\mathcal{J}} G_1) = {}^{18} \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} \circ G_1) = {}^{19} \operatorname{colim}_{\mathcal{J}}(\operatorname{colim}_{\mathcal{I}} G_0)$$

□

I.38. Definition

A category $\mathcal{C}$ is called *complete* or *cocomplete* if it admits all limits or colimits of small shape, respectively.

All the usual examples Grp, Set, Top, R-Mod, ... are both!

¹⁸By part ii) of I.35

¹⁹By i)

I.39. Corollary

If $\mathcal{D}$ is (co)complete then so is $\text{Fun}(\mathcal{C}, \mathcal{D})$ for every category $\mathcal{C}$, in particular $\mathbf{P}(\mathcal{C})$ is both complete and cocomplete for every $\mathcal{C}$. Furthermore, for any $F: \mathcal{C} \rightarrow \mathcal{C}'$ the functor

$$F^*: \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

preserves all limits and colimits.²⁰

In fact, we will see below that under a further smallness assumption F^* admits both adjoints $F_!$ and F_* (a.k.a. $\text{Lan}_F(-)$ and $\text{Ran}_F(-)$) from which the last statement also follows in that case.

Now we have:

I.40. Observation

The functors $Y^x: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ and $Y_x: \mathcal{C} \rightarrow \text{Set}$ preserve small limits for every (left locally small) $x \in \mathcal{C}$.²¹ For locally small $\mathcal{C}$ therefore the Yoneda embeddings

$$Y^{\mathcal{C}}: \mathcal{C} \rightarrow \mathbf{P}(\mathcal{C}) \quad \text{and} \quad Y_{\mathcal{C}}: \mathcal{C}^{\text{op}} \rightarrow \mathbf{P}(\mathcal{C}^{\text{op}})$$

preserve all limits.

Proof. We note that

$$\text{Hom}_{\mathcal{C}}(x, \lim_{\mathcal{I}} F) = \text{Nat}(\text{const}_{\mathcal{I}} x, F)$$

where the latter by inspection is $\lim_{\mathcal{I}} \text{Hom}_{\mathcal{C}}(x, F(-))$, with both consisting of assignments

$$\tau: \text{obj}(\mathcal{I}) \rightarrow \text{mor}(\mathcal{C})$$

with $\tau_i: x \rightarrow F(i)$, such that for all $f: i \rightarrow j \in \mathcal{I}$ the diagram

$$\begin{array}{ccc} x & \xrightarrow{\tau_i} & F(i) \\ \tau_j \downarrow & \swarrow F(f) & \\ & F(j) & \end{array}$$

commutes.

The first statement (for Y_x) follows, since

$$Y_x(\lim F) = \text{Hom}_{\mathcal{C}}(x, \lim F) = \lim_{\mathcal{C}} \text{Hom}(x, F(-)) = \lim_{\mathcal{C}} Y^x \circ F$$

The second one is similar:

$$Y^{\mathcal{C}}(\lim F) = \text{Hom}_{\mathcal{C}}(-, \lim F) = {}^{22} \lim \text{Hom}_{\mathcal{C}}(-, F) = \lim Y^{\mathcal{C}} \circ F$$

□

Colimits are very much *not* preserved by the Yoneda embedding! Before we analyse the situation, let's justify the smallness restriction in the definition of completeness.

²⁰Because they are pointwise!

²¹I.e. $\text{Hom}_{\mathcal{C}}(\text{colim}_{\mathcal{I}} F, y) = \lim_{\mathcal{I}^{\text{op}}} \text{Hom}(F(-), y) = \text{Hom}_{\mathcal{C}^{\text{op}}}(y, \lim_{\mathcal{I}^{\text{op}}} F^{\text{op}})$

²²Uses the fact that limits in $\mathbf{PC}$ are pointwise!

I.41. Proposition (Freyd)

If a category $\mathcal{C}$ admits coproducts for all diagrams of shape $\mathbb{D}(\text{mor}(\mathcal{C}))$, then $\mathcal{C}$ is equivalent to $N(P)$ for some poset P . In particular, small cocomplete categories are posets.

Proof. Let $x, y \in \mathcal{C}$ be two objects. Then form the sum $\sum_{\mathbb{D}(\text{mor}(\mathcal{C}))} x$, which exists by assumption. Then

$$\text{Hom}_{\mathcal{C}}\left(\sum_{\text{mor}(\mathcal{C})} x, y\right) = \prod_{\text{mor}(\mathcal{C})} \text{Hom}_{\mathcal{C}}(x, y)$$

where the former has cardinality at most that of $\text{mor}(\mathcal{C})$, whereas the latter has cardinality at least that of the power set of $\text{mor}(\mathcal{C})$, unless $|\text{Hom}_{\mathcal{C}}(x, y)| \leq 1$. $\square$

We are now ready to state the main theorem of this section:

I.42. Theorem

If $\mathcal{D}$ is cocomplete, and $\mathcal{C}$ small, then restriction along the Yoneda embedding induces an equivalence

$$(Y^{\mathcal{C}})^* : \text{Fun}^{\text{colim}}(\text{P}(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

where the superscript denotes the full subcategory on those functors that preserve all small colimits. The inverse functor is given by left Kan extension along $Y^{\mathcal{C}}$.

In other words, $\text{P}(\mathcal{C})$ is left 2-adjoint²³ to $\mathcal{C}$ under the inclusion of cocomplete categories with colimit preserving functors into all categories.

In particular,

$$\text{Fun}^{\text{colim}}(\text{Set}, \mathcal{C}) \simeq \mathcal{C}$$

means that Set is freely generated under colimits by any one-point set. Generally, $\text{P}(\mathcal{C})$ is the free cocompletion of $\mathcal{C}$.

The proof needs a few preparations:

I.43. Definition

Given functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{C}' \rightarrow \mathcal{D}$ define the *slice category* F/G by

$$\text{obj}(F/G) = \{(x, y, f) \in \text{obj}(\mathcal{C}) \times \text{obj}(\mathcal{C}') \times \text{mor}(\mathcal{D}) \mid f: F(x) \rightarrow G(y)\}$$

$$\text{Hom}_{F/G}((x, y, f), (x', y', f')) = \left\{ (g, g') \in \text{mor}(\mathcal{C}) \times \text{mor}(\mathcal{C}') \left| \begin{array}{l} \bullet g: x \rightarrow y \\ \bullet g': x' \rightarrow y' \\ \begin{array}{ccc} F(x) & \xrightarrow{f} & G(y) \\ F(g) \downarrow & & \downarrow G(g') \\ F(x') & \xrightarrow{f'} & G(y') \end{array} \end{array} \right. \right\}$$

If $F = \text{id}_{\mathcal{C}}$ one writes $\mathcal{C}/G$, and if $F: * \rightarrow \mathcal{C}$ one writes $F(*)/G$. As special cases we find

²³I.e. in the 2-categorical sense, because $(Y^{\mathcal{C}})^*$ is an equivalence, but not an iso!

- $\mathcal{D}/\mathcal{D} = \text{Fun}(\mathbb{N}[1], \mathcal{C})$, the *arrow category* of $\mathcal{D}$.
- $d/\mathcal{D}$ the *under category* of $d \in \mathcal{D}$:

$$\text{obj}(d/\mathcal{D}) = \{(x, f) \mid f: d \rightarrow \mathcal{D}\}$$

$$\text{Hom}_{d/\mathcal{D}}((x, f), (y, g)) = \left\{ \varphi: x \rightarrow y \mid \begin{array}{c} d \xrightarrow{f} x \\ g \downarrow \swarrow \varphi \\ y \end{array} \right\}$$

and similarly the *overcategory* $\mathcal{D}/d$.

There are always forgetful functors

$$\begin{array}{ccc} s: F/G \rightarrow \mathcal{C} & & t: F/G \rightarrow \mathcal{C}' \\ (x, y, f) \mapsto x & \text{and} & (x, y, f) \mapsto y \end{array}$$

and a tautological natural transformation

$$\begin{array}{c} \tau: Fs \Rightarrow Gt \\ \tau(x, y, f) = f \end{array}$$

Armed with this definition, we turn to the existence of Kan extensions:

I.44. Proposition

Given $i: \mathcal{C} \rightarrow \mathcal{C}'$ and $F: \mathcal{C} \rightarrow \mathcal{D}$ then the Kan extension $\text{Lan}_i F$ is pointwise if and only if the canonical map

$$\text{colim}(i/c' \xrightarrow{s} \mathcal{C} \xrightarrow{F} \mathcal{D}) \rightarrow (\text{Lan}_i F)(c')$$

is an isomorphism for all $c' \in \mathcal{C}'$. Dually, $\text{Ran}_i F$ is pointwise if and only if the natural map

$$(\text{Ran}_i F)(c') \rightarrow \text{lim}(c'/i \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D})$$

is an isomorphism.

If conversely the displayed (co)limits exist for all $c' \in \mathcal{C}'$, then they combine into Kan extensions of F along i (which are then automatically pointwise).

I.45. Corollary

If $\mathcal{C}$ is small, $\mathcal{C}'$ locally small²⁵ and $\mathcal{D}$ cocomplete or complete, respectively, then all functors $\mathcal{C} \rightarrow \mathcal{D}$ admit left or right Kan extensions along i , all of which are pointwise. In particular,

$$i^*: \text{Fun}(\mathcal{C}', \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

admits both adjoints in this case (namely Lan_i and Ran_i).

²⁴ $\text{obj}(i/c') = \{(c, f) \mid f: i(c) \rightarrow c'\}$

²⁵Makes i/c' and c'/i small for all $c' \in \mathcal{C}'$.

I.46. Proof of I.44

Assume that $\text{Ran}_i F$ is pointwise. recall that this means that

$$Y_d \circ \text{Ran}_i F = \text{Ran}_i(Y_d \circ F)$$

i.e. the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{C} & \xrightarrow{\quad} & \mathcal{D} & \xrightarrow{\text{Hom}(d, -)} & \text{Set} \\ & \searrow \text{Ran}_i F & & \nearrow & \\ i \downarrow & & & & \\ \mathcal{C}' & & & \nearrow \text{Ran}_i(\text{Hom}(d, F(-))) & \end{array}$$

Then we compute

$$\begin{aligned} \text{Hom}_{\mathcal{D}}(d, (\text{Ran}_i F)(i')) &= (Y_d \circ \text{Ran}_i F)(c') \\ &= \text{Nat}(Y_{c'}, Y_d \circ (\text{Ran}_i F)) \\ &= \text{Nat}(Y_{c'}, \text{Ran}_i(Y_d \circ F)) \\ &= \text{Nat}(Y_{c' \circ i}, Y_d \circ F) \\ &= \text{Nat}(\text{Hom}_{\mathcal{C}'}(c', i(-)), \text{Hom}_{\mathcal{D}}(d, F(-))). \end{aligned}$$

Now

$$\text{Nat}(\text{Hom}_{\mathcal{C}'}(c', i(-)), \text{Hom}_{\mathcal{D}}(d, F(-)))$$

consists of maps

$$\tau : \text{obj}(\mathcal{C}) \rightarrow \text{mor}(\text{Set})$$

$$\tau_c : \text{Hom}_{\mathcal{C}'}(c', i(c)) \rightarrow \text{Hom}_{\mathcal{D}}(d, Fc)$$

that are natural in c . Such a τ is exactly the same map as a map

$$\text{obj}(c'/i) \rightarrow \text{mor}(\text{Set})$$

that forms a natural transformation

$$\text{const}_{c'/i} d \Rightarrow F \circ t$$

Thus

$$\text{Hom}_{\mathcal{D}}(d, (\text{Ran}_i F)(c')) = \text{Nat}(\text{const}_{c'/i} d, F \circ t)$$

which exhibits $(\text{Ran}_i F)(c')$ as a limit of $F \circ t$, since $d \in \mathcal{D}$ was arbitrary.

Conversely, if all the functors $c'/i \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D}$ admit limits for all $c' \in \mathcal{C}'$, then they fit together into a functor $R: \mathcal{C}' \rightarrow \mathcal{D}$, with morphism component on $g: c' \rightarrow c''$ induced by

$$\begin{aligned} g^* : c''/i &\rightarrow c'/i \\ (x, f) &\mapsto (x, f \circ g) \end{aligned}$$

(note $F \circ t_{c'} \circ g^* = F \circ t_{c''}$) and the canonical map

$$\lim_{\mathcal{D}} A \rightarrow \lim_{\mathcal{C}} (A \circ B)$$

for functors $B: \mathcal{C} \rightarrow \mathcal{D}$, and $A: \mathcal{D} \rightarrow \mathcal{E}$.

R acquires a natural transformation

$$\lambda: R \circ i \Rightarrow F$$

by projecting $\lim^{i(c)/i} \mathcal{C} \xrightarrow{t} \mathcal{C} \xrightarrow{F} \mathcal{D}$ to the component of $(c, \text{id}_{i(c)}) \in {}^{i(c)}/i$. Given another functor and transformation $R' \circ i \xRightarrow{\tau} F$, then we obtain elements in

$$\text{Hom}_{\mathcal{D}}(R'(c'), R(c')) = \text{Nat}(\text{const}_{c'/i} R'(c'), F \circ t)$$

by

$$(x, f: c' \rightarrow i(x)) \mapsto (R'(c') \xrightarrow{f_*} R'(i(x)) \xrightarrow{\tau_x} F(x)).$$

These fit into a natural transformation $\sigma: R' \Rightarrow R$ that makes

$$\begin{array}{ccc} R'i & \xrightarrow{\sigma_i} & Ri \\ \tau \downarrow & \swarrow \lambda & \\ F & & \end{array}$$

commutative (by unwrapping the definitions). We leave it to the reader to check that σ is unique with this property, which shows R to be a right Kan extension of F along i .

That it is pointwise follows immediately from I.40. $\square$

Corollary I.45 yields the functor

$$\text{Lan}_{Y^{\mathcal{C}}} : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\text{P}(\mathcal{C}), \mathcal{D})^{26}$$

which we claim is inverse to the restriction in I.42. We first check that its essential image is indeed contained in $\text{Fun}^{\text{colim}}(\text{P}(\mathcal{C}), \mathcal{D})$: Since the Kan extension is pointwise and colimits in $\text{P}(\mathcal{C})$ are computed pointwise, this is immediate from part ii) of I.37 and I.44.

To see that $\text{Lan}_{Y^{\mathcal{C}}}$ really is an inverse we have:

I.47. Proposition

If $i: \mathcal{C} \rightarrow \mathcal{C}'$ is an embedding and $F: \mathcal{C} \rightarrow \mathcal{D}$ a functor with pointwise left/right Kan extension, then the structure maps

$$F \Rightarrow (\text{Lan}_i F) \circ i \quad \text{and} \quad (\text{Ran}_i F) \circ i \Rightarrow F$$

are natural equivalences.

In this case $\text{Lan}_i F$ and $\text{Ran}_i F$ really are extensions of F .

Proof. We saw (in I.46) that the structural map

$$\lambda_c : (\text{Ran}_i F)(i(c)) \rightarrow F(c)$$

is given by selecting the component of $(c, \text{id}_{i(c)}) \in {}^{i(c)}/i$ to give a map

$$(\text{Ran}_i F)(i(c)) = \lim^{i(c)/i} \mathcal{C} \xrightarrow{F} \mathcal{D} \rightarrow F(c).$$

²⁶ $\mathcal{D}$ is cocomplete.

But if i is an embedding then $(c, \text{id}_{i(c)})$ is initial in $i(c)/i$: A map

$$(c, \text{id}_{i(c)}) \rightarrow (x, f: i(c) \rightarrow i(x))$$

consists of a map $g: c \rightarrow x$ such that

$$\begin{array}{ccc} i(c) & \xrightarrow{f} & i(x) \\ \text{id}_{i(c)} \downarrow & \nearrow i(g) & \\ i(c) & & \end{array}$$

commutes. Thus $f = i(g)$ which determines g uniquely. Thus by part ii) of I.35 λ_c is an isomorphism and by part ii) of I.14 we are done. $\square$

In particular, this applies to Lan_{Y^C} and shows that the composite

$$\text{Fun}(\mathcal{C}, \mathcal{D}) \xrightarrow{\text{Lan}_{Y^C}} \text{Fun}(\text{P}(\mathcal{C}), \mathcal{D}) \xrightarrow{(Y^C)^*} \text{Fun}(\mathcal{C}, \mathcal{D})$$

is equivalent to the identity. To see the other direction we have:

I.48. Proposition

For every $F \in \text{P}(\mathcal{C})$ the tautological map

$$\text{colim}(Y^C/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C})) \Rightarrow F$$

induced by the tautological transformation

$$\begin{array}{ccc} Y^C/F & \xrightarrow{s} & \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C}) \\ & \Downarrow & \\ Y^C/F & \xrightarrow{t} & * \xrightarrow{F} \text{P}(\mathcal{C}) \end{array}$$

is an isomorphism.

Said more abstractly: The left Kan extension of Y^C along itself is the identity of $\text{P}(\mathcal{C})$. The category Y^C/F is often called the *category of objects* of F : By definition it consists of pairs (x, f) with $f: Y^x \Rightarrow F$. By Yoneda's Lemma this is equivalent to a pair (x, f) with $f \in F(X)$. Similarly, there is an identification

$$\text{Hom}((x, f), (y, g)) \cong \left\{ \varphi: x \rightarrow y \mid \varphi^* \hat{g} = \hat{f} \right\}.$$

The category Y^C/F is the simplest example of a Grothendieck construction, which Lurie's unstraightening construction generalises to ∞ -categories.

For us the important observation is that *every* $F \in \text{P}(\mathcal{C})$ is a colimit of representables. In particular, it illustrates the failure of Y^C to commute with colimits quite drastically.

Proof. We compute for $G: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$

$$\begin{aligned} \text{Nat}(\text{colim}(Y^C/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^C} \text{P}(\mathcal{C})), G) &= \\ &= \lim_{(x, f) \in (Y^C/F)^{\text{op}}} \text{Nat}(Y^x, G) \\ &= \lim_{(x, f) \in (Y^C/F)^{\text{op}}} G(x) \\ &= \lim((Y^C/F)^{\text{op}} \xrightarrow{s} \mathcal{C}^{\text{op}} \xrightarrow{G} \text{Set}) \end{aligned}$$

But this limit consists of assignments

$$\tau : \text{obj}(Y^{\mathcal{C}}/F) \rightarrow \text{obj}(\text{Set})$$

$$\tau(x, f) \in G(x)$$

such that for $\varphi: y \rightarrow x$ we have

$$\tau(y, \varphi^* f) = \varphi^* \tau(x, f).$$

This is the same data as that of a natural transformation $\tau: F \Rightarrow G$. Thus

$$\text{Nat}(\text{colim}(Y^{\mathcal{C}}/F \xrightarrow{s} \mathcal{C} \xrightarrow{Y^{\mathcal{C}}} P(\mathcal{C})), G) = \text{Nat}(F, G)$$

and the claim follows from Yoneda's Lemma after unwinding. $\square$

To prove I.42 we finally need the following lemma:

I.49. Lemma

Let $L: \mathcal{C} \rightleftarrows \mathcal{D}: R$ be an adjoint pair of functors. If

- i) L is an embedding, and
- ii) R detects isomorphisms, that is if Rf is an isomorphism in $\mathcal{D}$ then f is one in $\mathcal{C}$,

then L and R are inverse equivalences of categories, witnessed by the unit and counit of the adjunction.

Proof. As L is an embedding the unit $u: \text{id}_{\mathcal{C}} \Rightarrow RL$ is a natural isomorphism by I.24. To check that the counit

$$c: LP \Rightarrow \text{id}_{\mathcal{D}}$$

is an isomorphism as well, we may apply R (as natural isomorphisms are detected pointwise, and R detects these). But the triangle identity

$$(R \xrightarrow{uR} RLR \xrightarrow{Rc} R) = \text{id}_R$$

tells us that Rc is left-inverse to an isomorphism so one itself. $\square$

Finally we have:

I.50. Proof of I.42

We checked that

$$\text{Lan}_{Y^{\mathcal{C}}} : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(P(\mathcal{C}), \mathcal{D})$$

is an embedding as an application of I.46, and so the same holds for its restriction

$$\text{Lan}_{Y^{\mathcal{C}}} : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}^{\text{colim}}(P(\mathcal{C}), \mathcal{D})$$

as the right hand side was defined as a full subcategory. We are left to check that the restriction functor

$$Y_{\mathcal{C}}^* : \text{Fun}^{\text{colim}}(P(\mathcal{C}), \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

detects isomorphisms. This follows from I.47 as for a natural transformation $y: \mathcal{I} \Rightarrow \mathcal{J}$ of colimit-preserving functors $\mathbf{P}(\mathcal{C}) \rightarrow \mathcal{D}$, we have

$$\begin{array}{ccccc} \mathcal{I}(F) = \mathcal{I}(\operatorname{colim}_{(x,f) \in Y^{\mathcal{C}}/F} Y^x) & = & \operatorname{colim}_{(x,f) \in Y^{\mathcal{C}}/F} \mathcal{I}(Y^x) & & \\ Y_F \downarrow & & \downarrow & & \downarrow \operatorname{colim} y(Y^x) \\ \mathcal{J}(F) & = & \mathcal{J}(\operatorname{colim} Y^x) & = & \operatorname{colim} \mathcal{J}(Y^x) \end{array}$$

so if y is an isomorphism on representable it is an isomorphism. □

Chapter II.

Simplicial Sets

II.1. Definition

$\Delta :=$ full subcategory of the category of posets spanned by $\{[n] \mid n \geq 0\}$.

II.2. Definition

$\mathbf{sSet} := \mathbf{P}(\Delta) = \mathbf{Fun}(\Delta^{\text{op}}, \mathbf{Set})$.

II.3. Corollary

$\mathbf{sSet}$ is complete and cocomplete.

II.4. Definition

n -simplex, vertex, edge, face map, degeneracy map, degenerate simplex, non-degenerate simplex

II.5. Definition

standard n -simplex $\Delta^n := Y^\Delta([n])$

II.6. Remark

For $X \in \mathbf{sSet}$, we have

$$X_n \cong \mathbf{Hom}_{\mathbf{sSet}}(\Delta^n, X)$$

and

$$\text{colim}(Y^\Delta/X \xrightarrow{s} \Delta \xrightarrow{Y^\Delta} \mathbf{sSet}) \cong X.$$

The category Y^Δ/X is called the *category of simplices* of X .

II.7. Lemma

Every degenerate simplex is the degeneration of an unique non-degenerate simplex.

II.8. Example

For $\mathcal{C}$ a small category, the nerve $\mathcal{NC}$ is the simplicial set with

$$\mathcal{NC}_n := \text{obj}(\text{Fun}([n], \mathcal{C})).$$

for example, $\mathcal{N}[n] \cong \Delta^n$ and $\mathcal{NBM}$, M a monoid, is the bar construction on M .

II.9. Definition

Affine n -simplex

$$|\Delta^n| := \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid 0 \leq t_i \leq 1, \sum_{i=0}^n t_i = 1 \right\}$$

These form a functor $|\Delta^\bullet| : \Delta \rightarrow \text{Top}$.

II.10. Definition

The singular set functor $\text{Sing} : \text{sSet} \rightarrow \text{Top}$ is given by

$$\Delta^{\text{op}} \times \text{Top} \xrightarrow{|\Delta^\bullet|^{\text{op}} \times \text{id}} \text{Top}^{\text{op}} \times \text{Top} \xrightarrow{\text{Hom}_{\text{Top}}(-, -)} \text{Set}.$$

II.11. Definition

The geometric realisation functor $|-| : \text{sSet} \rightarrow \text{Top}$ is the unique colimit-preserving functor satisfying $|-| \circ Y^\Delta \cong |\Delta^\bullet|$.

II.12. Definition

The functors $|-|$ and Sing are adjoint.

II.13. Definition

For $k \geq 0$, let $i_k : \Delta_{\leq k} \rightarrow \Delta$ be the inclusion of the full subcategory on objects of cardinality at most $k + 1$.

The k -skeleton functor is given by $\text{sk}_k := (i_k)_! i_k^*$.

The k -coskeleton functor is given by $\text{cosk}_k := (i_k)_* i_k^*$.

II.14. Lemma

k -skeleton and k -coskeleton are adjoint functors.

The k -skeleton of X is the smallest simplicial subset which has the same simplices as X in degrees $\leq k$. All simplices in degrees $> k$ are degenerate.

II.15. Definition

$$\partial\Delta^n := \text{sk}_{n-1} \Delta^n$$

II.16. Proposition

For $X \in \mathbf{sSet}$ and $n \geq 0$ the square

$$\begin{array}{ccc} \coprod_{x \in X_n \text{ non-deg.}} \partial\Delta^n & \longrightarrow & \text{sk}_{n-1} X \\ \downarrow & & \downarrow \\ \coprod_{x \in X_n \text{ non-deg.}} \Delta^n & \longrightarrow & \text{sk}_n X \end{array}$$

is a pushout.

II.17. Corollary

The geometric realisation of a simplicial set X carries a canonical CW-structure given by

$$\emptyset \subseteq |\text{sk}_0 X| \subseteq |\text{sk}_1 X| \subseteq \dots \subseteq |X|.$$

In II.12 we've seen that $|-| : \mathbf{sSet} \rightleftarrows \mathbf{Top} : \text{Sing}$ are adjoint. More generally we have:

II.18. Proposition

If $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor, $\mathcal{C}$ essentially small and $\mathcal{D}$ cocomplete, then the colimit preserving extension $\widehat{F} : \mathbf{P}(\mathcal{C}) \rightarrow \mathcal{D}$ admits a right adjoint $R : \mathcal{D} \rightarrow \mathbf{P}(\mathcal{C})$ which satisfies

$$(R(d))(c) \cong \text{Hom}_{\mathcal{D}}(F(c), d)$$

naturally in $c \in \mathcal{C}$ and $d \in \mathcal{D}$. Furthermore, F itself admits an adjoint if and only if R takes values in the representables in which case the two adjoints correspond to one another under the Yoneda embedding.

The above case is $\Delta_{\mathbf{Top}} : \Delta \rightarrow \mathbf{Top}$, $[n] \mapsto \Delta_{\mathbf{Top}}^n (\subseteq \mathbb{R}^{n+1})$.

Here are two further examples.

II.19. Examples

- i) Let $*$: $\Delta \rightarrow \mathbf{Set}$ be the constant functor with value $*$. Then its adjoint is the functor

$$\text{const} : \mathbf{Set} \rightarrow \mathbf{sSet}.$$

The colimit preserving extension can be described in several ways. For example it has to be given by

$$X \mapsto \text{colim}_{\Delta^{\text{op}}} X$$

as this functor is also left adjoint to const . An explicit description is

$$X \mapsto \text{colim}(X_1 \xrightleftharpoons[d_1]{d_0} X_0);$$

or in other words $X_0/\sim$ where $\sim$ is generated by $d_0(y) \sim d_1(y)$ for every $y \in X_1$; the fact that this functor preserves colimits follows from

i) the functor

$$\text{Fun}(\Delta^{\text{op}}, \text{Set}) \xrightarrow{\text{rest.}} \text{Fun}(\{0\} \rightrightarrows \{0, 1\}^{\text{op}}, \text{Set})$$

preserves colimits since it admits right Kan extension as an adjoint, and

ii) colimits commute.

The fact that it restricts to the constant functor $\Delta \rightarrow \text{Set}$ follows since

$$\text{colim}(\Delta_1^n \xrightleftharpoons[d_1]{d_0} \Delta_0^n) = 0 :$$

for $i \leq j \in \Delta_0^n$ we have $d_0(i, j) = j$ and $d_1(i, j) = i$ so $i \sim j$.

II.20. Definition

For $X \in \text{sSet}$ we set $\pi_0 X = \text{colim}_{\Delta^{\text{op}}} X \in \text{Set}$

II.21. Proposition

The natural map $X_0 \rightarrow |X|$ induces a bijection

$$\pi_0 X \rightarrow \pi_0 |X|$$

for any simplicial set X .

Proof. This follows from cellular approximation / the theorem of Seifert-van Kampen: The inclusion $|X|^{(1)} \subseteq |X|$ induces an isomorphism on π_0 , and by II.16 and by $|X|^{(1)} = |\text{sk}_1 X|$ given by gluing an edge to X_0 for every non-degenerate 1-simplex $y \in X_1$. Thus $\pi_0 |X|$ is given by $X/\sim$ where $x \sim y$ if there is a series of edges in $|\text{sk}_1 X|$ connecting x and y . This is clearly the same relation as that defining $\pi_0 X$. $\square$

Continuing the examples in II.19, we have:

ii) the functor

$$\Delta \rightarrow \text{sSet}, [n] \mapsto X \times \Delta^n$$

has colimit preserving extension $Y \mapsto X \times Y$, since products (like all limits) are computed pointwise in sSet , and therefore commute with colimits as this is the case in Set , we obtain an adjoint

$$F(X, -) : \text{sSet} \rightarrow \text{sSet},$$

i.e. there are isomorphisms

$$\text{Hom}_{\text{sSet}}(Y \times X, Z) \cong \text{Hom}_{\text{sSet}}(Y, F(X, Z)),$$

and we find

$$F(X, Y)_n \cong \text{Hom}_{\text{sSet}}(X \times \Delta^n, Y) .$$

These properties formally imply that in fact

$$F(Y \times X, Z) \cong F(Y, F(X, Z))$$

as functors

$$(\text{sSet}^{\text{op}})^2 \times \text{sSet} \rightarrow \text{sSet} .$$

This provides us with a notion of homotopy of simplicial maps:

II.22. Definition

Two maps $f, g: X \rightarrow Y$ of simplicial sets are called *homotopic* if $[f] = [g] \in \pi_0 F(X, Y)$. A *simplicial homotopy* from f to g is an element

$$H \in F(X, Y)_1 = \text{Hom}_{\text{sSet}}(X \times \Delta^1, Y)$$

with $d_1 H = f$ and $d_0 H = g$.

Let me explicitly warn you: The existence of a simplicial homotopy is neither a symmetric nor a transitive relation in general.

To give an example, recall

$$\partial \Delta^n = \bigcup_{0 \leq i \leq n} \text{im}(\Delta^{n-1} \xrightarrow{d_i} \Delta^n) \subseteq \Delta^n$$

or equivalently

$$(\partial \Delta^n)_k = \{\varphi: [k] \rightarrow [n] \mid \varphi \text{ not surjective}\}$$

II.23. Definition

The j -th n -dimensional horn Λ_j^n ($0 \leq j \leq n$) is given by

$$\Lambda_j^n = \bigcup_{\substack{0 \leq i \leq n \\ i \neq j}} \text{im}(\Delta^{n-1} \xrightarrow{d_i} \Delta^n) \subseteq \partial \Delta^n$$

or equivalently

$$(\Lambda_j^n)_k = \{\varphi: [k] \rightarrow [n] \mid [n] \neq \text{im}(\varphi) \cup \{j\}\}$$

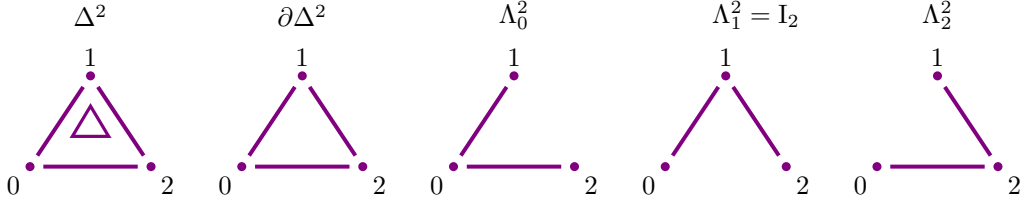
Similarly, we define the n -spine $I_n \subseteq \partial \Delta^n$ as

$$I_n = \bigcup_{0 \leq i \leq n-1} \text{im}(\Delta^1 \xrightarrow{(i, i+1)} \Delta^n)$$

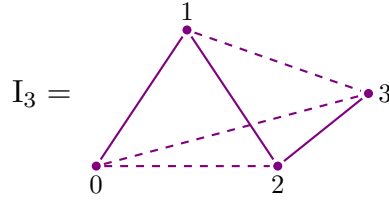
or equivalently

$$(I_n)_k = \{\varphi: [k] \rightarrow [n] \mid \text{im}(\varphi) \text{ consists of at most two consecutive numbers}\} .$$

In pictures:



Note that $\text{sk}_{n-2}(\Lambda_i^n) = \text{sk}_{n-2}(\Delta^n)$ whereas I_n is 1-skeletal. For $n \geq 3$ there is then an inclusion $I_n \subseteq \Lambda_i^n$ for all $0 \leq i \leq n$:



Now for the counterexamples consider

$$\begin{aligned} \text{Hom}_{\text{sSet}}(\Delta^\bullet, \Lambda_1^2) &\cong (\Lambda_1^2)_0 \cong \{0, 1, 2\} \\ \text{Hom}_{\text{sSet}}(\Delta^1, \Lambda_1^2) &\cong (\Lambda_1^2)_1 \cong \{(0, 1), (1, 2), (0, 0), (1, 1), (2, 2)\} \end{aligned}$$

meaning that there are simplicial paths

$$0 \rightsquigarrow 1 \quad \text{and} \quad 1 \rightsquigarrow 2$$

but not $1 \rightsquigarrow 0$ or $1 \rightsquigarrow 2$. We will isolate a better class of simplicial sets soon.

For now we give another example (continuing II.19):

iii) The functor

$$\Delta \cong \text{Ord} \xrightarrow{N} \text{Cat}$$

extends to a colimit preserving functor

$$h : \text{sSet} \rightarrow \text{Cat}$$

with right adjoint $\mathcal{N} : \text{Cat} \rightarrow \text{sSet}$

$$(\mathcal{N}\mathcal{C})_n = \text{Hom}_{\text{Cat}}(N[n], \mathcal{C})$$

as defined in II.8. As colimits in Cat are complicated, the functor h is a bit unpleasant to describe:

II.24. Construction/Proposition

For a simplicial set X let $hX \in \mathbf{Cat}$ have objects X_0 and morphisms are given as a certain quotient of

$$\sum_{n \geq 0} X_1 \times_{X_0} \dots \times_{X_0} X_1 \in \mathbf{Set}$$

i.e. of the set of strings $(f_1, \dots, f_n) \in X_1^n$ for some n with

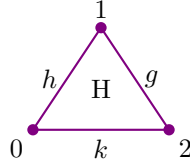
$$d_1 f_i = d_0 f_{i+1}$$

for all $0 \leq i \leq n-1$. The morphisms are given by the quotient with respect to the equivalence relation generated by

$$(f_1, \dots, f_i, g, h, f_{i+1}, \dots, f_n) \sim (f_1, \dots, f_i, k, f_{i+1}, \dots, f_n)$$

if there exists a 2-simplex $H \in X_2$ with

$$d_0 H = g, \quad d_1 H = k \quad \text{and} \quad d_2 H = h :$$



The source and target maps $\text{mor}(hX) \rightarrow \text{obj}(hX) = X_0$ are given by

$$s(f_1, \dots, f_n) = d_1(f_n) \quad \text{and} \quad t(f_1, \dots, f_n) = d_0 f_1,$$

respectively. Composition is then given by concatenation of strings and identities by

$$X_0 \xrightarrow{s_0} X_1 \rightarrow \text{mor}(hX).$$

Then hX really is a category and the resulting functor $h: \mathbf{sSet} \rightarrow \mathbf{Cat}$ is left adjoint to the nerve $\mathcal{N}: \mathbf{Cat} \rightarrow \mathbf{sSet}$.

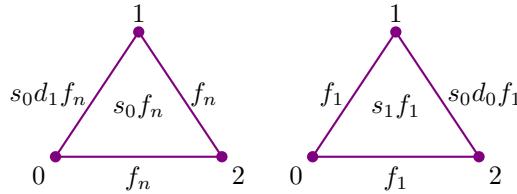
Proof. The only non-obvious step in checking that hX really is a well-defined category is probably the compatibility of composition and identities: However, we find

$$(f_1, \dots, f_n, s_0 d_1(f_n)) \sim (f_1, \dots, f_n)$$

through the 2-simplex $s_0(f_n)$, and

$$(s_0 d_0(f_1), f_1, \dots, f_n) \sim (f_1, \dots, f_n)$$

through $s_1(f_1)$:



Chapter II. Simplicial Sets

Let us then directly construct inverse map

$$\tau : \text{Hom}_{\text{Cat}}(hX, \mathcal{C}) \longleftrightarrow \text{Hom}_{\text{sSet}}(X, \mathcal{N}\mathcal{C}) : \sigma$$

as follows: Given $F : hX \rightarrow \mathcal{C}$ we define

$$\tau(F)_n : X_n \rightarrow \mathcal{N}_n \mathcal{C}$$

by

$$\begin{array}{ccc} X_n & \xrightarrow{\cong} & \text{Hom}_{\text{sSet}}(\Delta^n, X) \\ \downarrow & & \downarrow \text{restriction} \\ X_1 \times_{X_0} \dots \times_{X_0} X_1 & \xrightarrow{\cong} & \text{Hom}_{\text{sSet}}(\mathbf{I}_n, X) \\ \downarrow & & \\ \text{mor}(hX) \times_{\text{obj}(hX)} \dots \times_{\text{obj}(hX)} \text{mor}(hX) & \xrightarrow{F} & \text{mor}(\mathcal{C}) \times_{\text{obj}(\mathcal{C})} \dots \times_{\text{obj}(\mathcal{C})} \text{mor}(\mathcal{C}) \cong \mathcal{N}_n(\mathcal{C}) \end{array}$$

Conversely, given $g : X \rightarrow \mathcal{N}\mathcal{C}$, we define

$$\begin{aligned} \sigma(g) : \text{obj}(hX) &\rightarrow \text{obj}(\mathcal{C}) \\ X_0 &\mapsto \mathcal{N}_0 \mathcal{C} \end{aligned}$$

and $\sigma(g) \text{ mor}(hX) \rightarrow \text{mor}(\mathcal{C})$ on generators by

$$\sigma(g)(f_1, \dots, f_n) = g_1(f_1) \circ g_1(f_2) \circ \dots \circ g_1(f_n).$$

This is well-defined as $g(H) \in \mathcal{N}_2(\mathcal{C})$ witnesses $g(h) \circ g(i) = g(k)$ for

$$H = \begin{array}{ccc} & 1 & \\ i \swarrow & & \searrow h \\ 0 & \xrightarrow{k} & 2 \end{array}$$

It is easily checked that τ and σ are inverse and indeed well-defined. It is also clear that they are natural, so we are done. $\square$

The maps $\text{Seg}_i : X_n \rightarrow X_1$ induced by $[1] \rightarrow [n], 0 \mapsto i, 1 \mapsto i+1$ are called the *Segal maps* and always combine into the diagram

$$\begin{array}{ccc} X_n & \xrightarrow{(\text{Seg}_0, \dots, \text{Seg}_{n-1})} & X_1 \times_{X_0} \dots \times_{X_0} X_1 \\ \Downarrow & & \Downarrow \\ \text{Hom}_{\text{sSet}}(\Delta^n, X) & \longrightarrow & \text{Hom}_{\text{sSet}}(\mathbf{I}_n, X) \end{array}$$

occurring in the proof, but make sense for any simplicial object. A simplicial object for which they exhibit X_n as the pullback displayed on the right is called a *Segal object*.

The unit and counit

$$X \rightarrow \mathcal{N}hX \quad \text{and} \quad h\mathcal{N}\mathcal{C} \rightarrow \mathcal{C}$$

of the adjunction constructed in the proof are given by

$$\begin{aligned} X_n &\xrightarrow{\text{Seg}} X_1 \times_{X_0} \dots \times_{X_0} X_1 \rightarrow \text{mor}(hX) \times_{\text{obj}(hX)} \dots \times_{\text{obj}(hX)} \text{mor}(hX) \cong \mathcal{N}_n hX \\ \mathcal{N}_1 \mathcal{C} \times_{\mathcal{N}_0 \mathcal{C}} \dots \times_{\mathcal{N}_0 \mathcal{C}} \mathcal{N}_1 \mathcal{C} &= \text{mor } \mathcal{C} \times_{\text{obj } \mathcal{C}} \dots \times_{\text{obj } \mathcal{C}} \text{mor } \mathcal{C} \xrightarrow{\circ} \text{mor } \mathcal{C} \\ &\searrow \quad \quad \quad \nearrow \\ &\text{mor } h\mathcal{N}\mathcal{C} \end{aligned}$$

We next aim to prove:

II.25. Theorem

The functor $\mathcal{N}: \text{Cat} \rightarrow \text{sSet}$ is fully faithful and its essential image is given equivalently by all simplicial sets satisfying one of two equivalent conditions:

- a) the map $\text{Hom}(\Delta^n, X) \xrightarrow{\text{rest.}} \text{Hom}(\mathbf{I}_n, X)$ is bijective for all $n \geq 0$, in other words X is a Segal set, or
- b) the map $\text{Hom}(\Delta^n, X) \rightarrow \text{Hom}(\Lambda_i^n, X)$ is bijective for all $0 < i < n$.

One says that a simplicial set satisfying **b)** has "unique inner horn fillers"; said diagrammatically:

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & X \\ \downarrow & \nearrow \exists! & \\ \Delta^n & & \end{array}$$

for all $0 < i < n$. The proof will consist of several steps: We already observed that nerves satisfy **a)** and the first step will be to show that Segal sets satisfy **b)**. For simplicial sets X satisfying a weak version of **b)** there is a drastically simpler description of hX , which we will then use to show that the unit of a simplicial set satisfying **b)** and counit of an arbitrary category are isomorphisms. Before we dive into that let us, however, observe:

II.26. Lemma

For a topological space X the simplicial set $\text{Sing } X$ has (not-necessarily unique) horn fillers for all horns.

Proof. By adjunction the extension problems

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & \text{Sing } X \\ \downarrow \cap & \nearrow & \\ \Delta^n & & \end{array} \quad \text{and} \quad \begin{array}{ccc} |\Lambda_i^n| & \longrightarrow & X \\ \downarrow \cap & \nearrow & \\ |\Delta^n| & & \end{array}$$

are equivalent. As $|\Lambda_i^n| \subseteq |\Delta^n|$ is the inclusion of a (deformation) retract, the right hand one clearly admits a solution¹. $\square$

We can thus give the central definition of this course:

II.27. Definition

A simplicial set X is called a *Kan complex* if every horn $\Lambda_i^n \rightarrow X$ admits a filler, i.e.

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & X \\ \downarrow \cap & \nearrow \exists & \\ \Delta^n & & \end{array}$$

X is called a *quasi-category* if X admits fillers for all inner horns, i.e. those Λ_i^n with $0 < i < n$.

Thus a quasi-category is a hybrid of the nerve of a category and the singular complex of a space.

Before we look more closely at these definitions, let us prove II.25.

II.28. Lemma

Every Segal set X has unique inner horn fillers and in fact unique fillers for all horns Λ_i^n with $n \geq 4$.

Proof. Let $f: \Lambda_j^n \rightarrow X$ with $0 < j < n$ be given. In particular, $n \geq 2$ and since $n = 2$ forces $j = 1$ and $\Lambda_1^2 = I_2$ this case follows from an assumption. For $n \geq 3$ consider the inclusion $I_n \subseteq \Lambda_j^n$ and let $g: \Delta^n \rightarrow X$ be the unique extension of $f|_{I_n}: I_n \rightarrow X$. We have to show that $g|_{\Lambda_j^n} = f$. As

$$\Lambda_j^n = \bigcup_{\substack{0 \leq i \leq n \\ i \neq j}} \text{im}(\Delta^{n-1} \rightarrow \Delta^n)$$

it suffices to show that

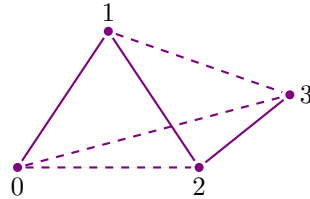
$$fd_i = gd_i: \Delta^{n-1} \rightarrow X \quad \text{for } i \neq j.$$

This is easy for $i = 0, n$ since in these cases

$$d_i(I_{n-1}) \subseteq I_n$$

so, since $f|_{I_n} = g|_{I_n}$ by definition both fd_i and gd_i are extensions of $fd_i|_{I_{n-1}} = gd_i|_{I_{n-1}}$. But by assumption there is only one such extension so $fd_i = gd_i$. We thus obtain that f equals g on all of

$$I_n \cup d_0 \Delta^{n-1} \cup d_n \Delta^{n-1} \subseteq \Lambda_j^n:$$



¹In fact, a contractible space of such solutions.

But now for $0 < i < n$ we still have

$$d_i I_{n-1} \subseteq I_n \cup d_0 \Delta^{n-1} \cup d_n \Delta^{n-1} :$$

a k -simplex on the left is a map $\varphi: [k] \rightarrow [n]$ that is monotone and whose image consists of at most two consecutive numbers in $\{0, \dots, i-1, i+1, \dots, n\}$. Thus $\varphi \in I_n$ unless

$$\text{im}(\varphi) = \{i-1, i+1\} .$$

But then $\varphi \in d_0 \Delta^n$ unless $i-1 = 0$, in which case $i+1 = 2 < n$, so $\varphi \in d_n \Delta^n$. Thus we can run the same argument again for all $0 < i < n$, $i \neq j$: The restriction of fd_i and gd_i to I_{n-1} agree by the above, so by uniqueness of extensions $fd_i = gd_i$. For the outer horns for $n \geq 4$ observe first that the previous argument shows that the restriction map

$$\text{Hom}(\partial \Delta^n, X) \rightarrow \text{Hom}(I_n, X)$$

is bijective for any Segal set (we only had to exclude $i = j$ from our reasoning previously, because f was not defined on $d_j \Delta^n$). But since

$$\begin{array}{ccc} \text{Hom}(\Delta^n, X) & \xrightarrow{\quad\quad\quad} & \text{Hom}(\partial \Delta^n, X) \\ & \searrow \cong & \swarrow \cong \\ & \text{Hom}(I_n, X) & \end{array}$$

commutes and has two isomorphisms it follows that every Segal set is 2-coskeletal. But then

$$\begin{aligned} \text{Hom}(\Lambda_j^n, X) &\cong \text{Hom}(\Lambda_j^n, \text{cosk}_2 X) \\ &\cong \text{Hom}(\text{sk}_2 \Lambda_j^n, X) \\ &\cong \text{Hom}(\text{sk}_2 \Delta^n, X) \\ &\cong \text{Hom}(\Delta^n, X) \end{aligned}$$

and tracing through shows the map given by restriction. $\square$

We then turn to the simpler description of hX when X admits (not necessarily unique!) inner horn fillers up to dimension 3.

II.29. Construction

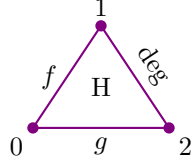
Assume X as described above, i.e. the maps

$$\begin{aligned} \text{Hom}_{\text{sSet}}(\Delta^2, X) &\rightarrow \text{Hom}_{\text{sSet}}(\Lambda_1^2, X) \\ \text{Hom}_{\text{sSet}}(\Delta^3, X) &\rightarrow \text{Hom}_{\text{sSet}}(\Lambda_1^3, X) \\ \text{Hom}_{\text{sSet}}(\Delta^3, X) &\rightarrow \text{Hom}_{\text{sSet}}(\Lambda_2^3, X) \end{aligned}$$

are surjective. Define a category πX , the *homotopy category* of X , as follows:

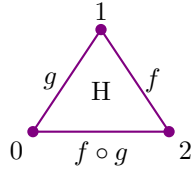
$$\text{obj}(\pi X) = X_0 \quad \text{mor}(\pi X) = X_1 / \sim$$

where $f \sim g$ if there exists $H \in X_2$ with $d_1(H) = g, d_2(H) = f$ and $d_0(H)$ degenerate:

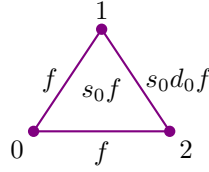


Let me immediately remark that the asymmetry of this definition is only apparent as we will show below. The source and target maps are given by $d_1: X_1/\sim \rightarrow X_0$ and $d_0: X_1/\sim \rightarrow X_0$, respectively, and identities by $s_0: X_0 \rightarrow X_1/\sim$. Composition is given by observing that a pair $(f, g) \in X_1^2$ with $d_0g = d_1f$ is precisely a map $\Lambda_1^2 \rightarrow X$, then choosing an extension $H: \Delta^2 \rightarrow X$ and setting

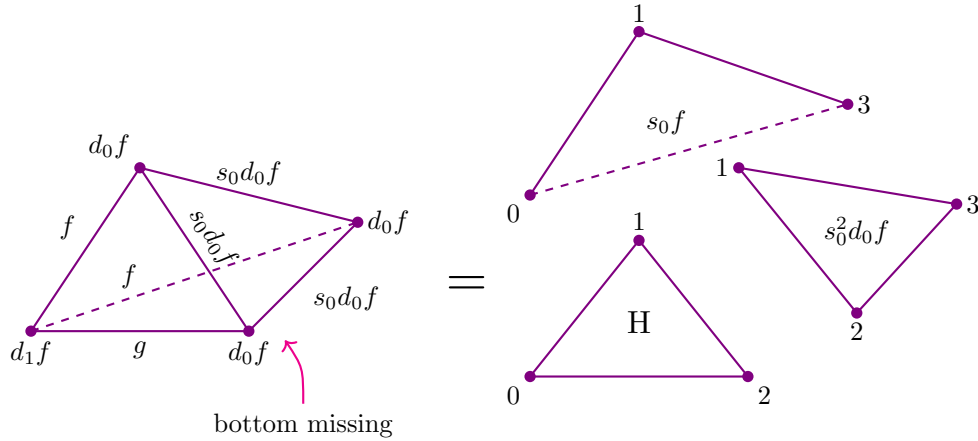
$$[f] \circ [g] = [d_1H] :$$



We now have to verify that all these pieces of structure are well-defined, starting with the act that $\sim$ really is an equivalence relation. Reflexivity is witnessed by $s_1: X_1 \rightarrow X_2$:

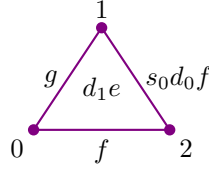


Symmetry is more involved: Let $H \in X_2$ be a witness for $f \sim g$. Then together with $s_0(f)$ and $s_0 s_0 d_0 f$, we can make a map $\Lambda_1^3 \rightarrow X$:

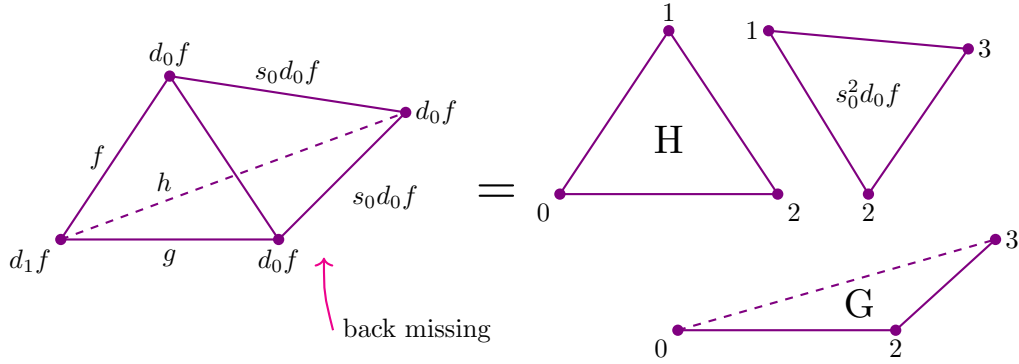


Chapter II. Simplicial Sets

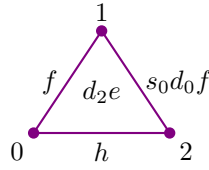
Now by assumption this map admits an extension $e: \Delta^3 \rightarrow X$, and its restriction $d_1(e): \Delta^2 \rightarrow X$ witnesses $g \sim f$:



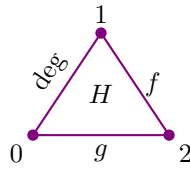
To see transitivity take H and $G \in X_2$ witnessing $f \sim g$ and $g \sim h$, respectively. Then together with $s_0^2 d_0 f$ we can make a map $\Lambda_2^3 \rightarrow X$ as follows:



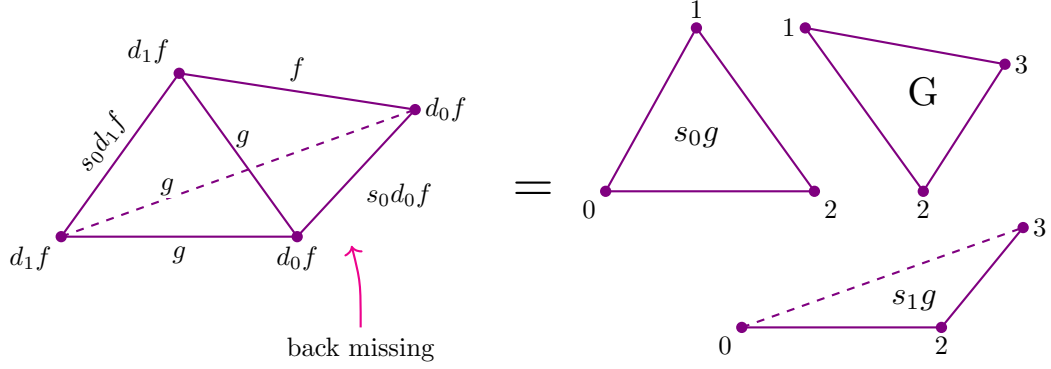
which we extend to $e: \Delta^3 \rightarrow X$ and $d_2 e: \Delta^2 \rightarrow X$ witnesses $f \sim h$:



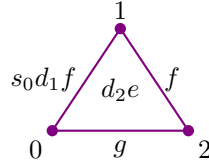
Let us next verify that $f \text{ im } g$ is also witnessed by $H \in X_2$:



(removing the asymmetry from the definition of $\sim$). Pick a witness G for $g \sim f$:

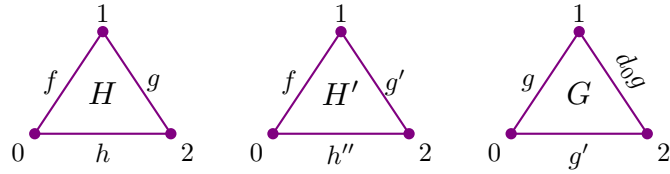


Then d_2e for an extension $e: \Delta^3 \rightarrow X$ does the job:



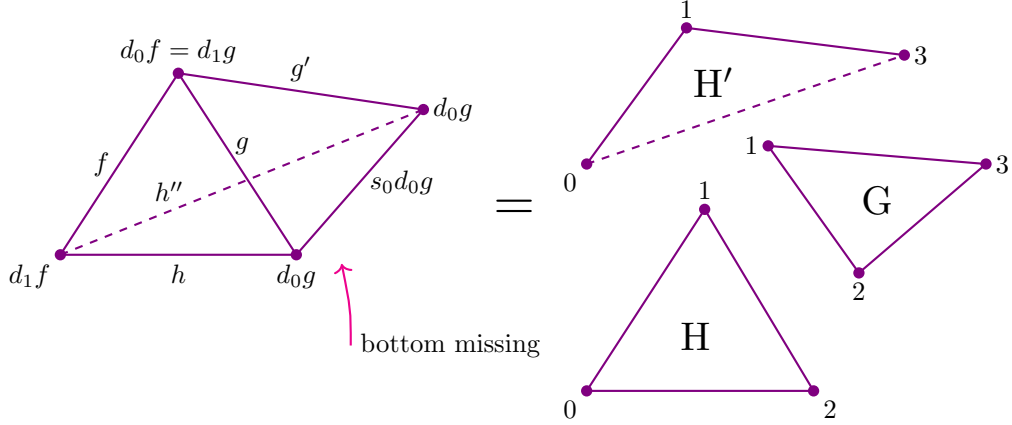
The converse holds by a similar argument, or by considering the opposite of X .² It is clear that source and target are well-defined and satisfy the requisite relations with the identities. We are left to show that composition is well-defined, associative and unital.

For the first item let $f \sim f'$ and $g \dim g'$ be composite 1-simplices. We have to show that an composite h of f and g is equivalent to any composite h' of f' and g' . To do so, it is best to show that both hands' are homotopic to a composite of f and g' . Let us do the former. So pick witnesses H for h being a composite of f and g , H' for h' being a composite of f and g' , and G for $g \sim g'$:

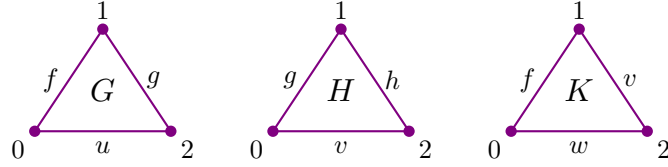


² $d_i^{\text{op}} = d_{n-i}: X_n \rightarrow X_{n-1}$

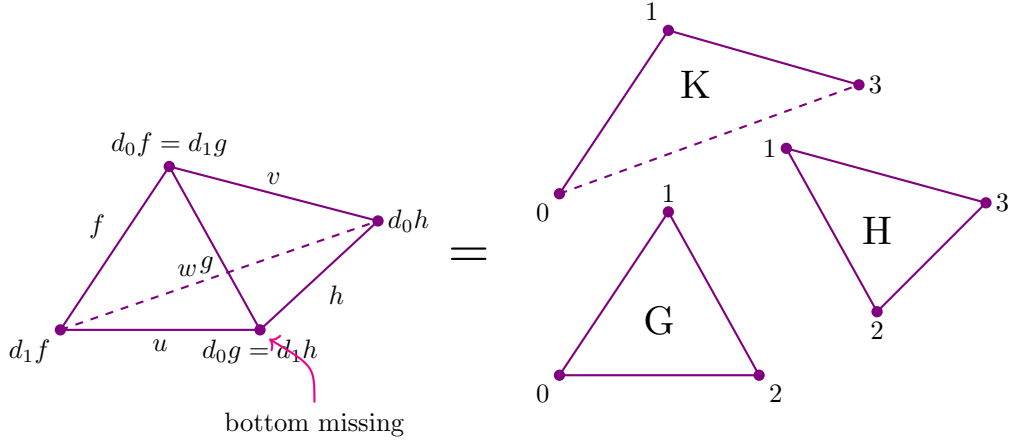
Then consider the map $\Lambda_1^3 \rightarrow X$:



Then the extension e , yada yada. The third item is now immediate from the two descriptions of equivalence. We are left with associativity. So let $f, g, h \in X_1$ be three composable morphisms and let u be a composition of f and g , v a composition of g and h and w a composition of f and v . We have to show that w is also a composition of u and h . So let G, H, K be witnesses as follows:



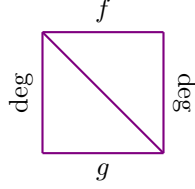
Then consider the map $\Lambda_1^3 \rightarrow X$:



and extend and done. Finally! □

II.30. Remark

A more systematic way of defining $f \sim g$ is the existence of a square $\Delta^1 \times \Delta^1 \rightarrow X$:



The equivalence to our definition is clear from transitivity of $\sim$.

II.31. Corollary

If X has inner horn fillers up to dimension 3 then there is a canonical isomorphism

$$\pi X \longleftrightarrow hX .$$

Proof. Both directions are the identity on objects.

To go from $hX \rightarrow \pi X$, take

$$\begin{array}{ccc} X_1 \times_{X_0} \dots \times_{X_0} X_1 & \longrightarrow & X_1/\sim \times_{X_0} \dots \times_{X_0} X_1/\sim \xrightarrow{\circ} X_1/\sim = \text{mor } \pi X \\ & \searrow & \nearrow \\ & \text{mor } hX & \end{array}$$

It is hopefully clear that this constitutes a functor.

Conversely,

$$\begin{array}{ccc} X_1 & \longrightarrow & \text{mor } hX \\ \downarrow & \nearrow & \\ X_1/\sim & = & \text{mor } \pi X \end{array}$$

Here, the compatibility with composition requires a (brief) moment's thought. Clearly

$$\pi X \rightarrow hX \rightarrow \pi X$$

is the identity (even before taking quotients), and

$$hX \rightarrow \pi X \rightarrow hX$$

sends $[f_1, \dots, f_n]$ to $[f_1] \circ \dots \circ [f_n]$, but each composition is witnessed by a relation in hX so $(f_1, \dots, f_n)$ is equivalent to any composite of the f_i in $\text{mor } hX$. $\square$

II.32. Proof of II.25

We have to verify that the unit $X \rightarrow \mathcal{N}hX$ for a simplicial set with unique inner horn fillers and the counit $h\mathcal{N}\mathcal{C} \rightarrow \mathcal{C}$ for an arbitrary category are isomorphisms. In both statements we are free to replace h by π because of II.28 and II.31. But this makes the second part tautological:

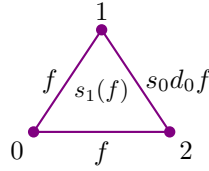
$$\pi\mathcal{N}\mathcal{C} \rightarrow \mathcal{C}$$

is essentially the identity, because the relation $\sim$ on $\mathcal{N}_1\mathcal{C}$ from the definition of π becomes equality.

A similar phenomenon occurs for the map

$$X \rightarrow \mathcal{N}\pi X.$$

As horn fillers of Λ_1^2 are unique in X the relation $\sim$ on X_1 is also equality: For $f \in X_1$ the simplex $s_1(f) \in X_2$ is the only 2-simplex σ with $d_1(\sigma) = f$ and $d_2(\sigma)$ degenerate:



Thus $X \rightarrow \mathcal{N}\pi X$ is the identity on both 0- and 1-simplices, in particular an isomorphism. As both X and $\mathcal{N}\pi X$ have unique inner horn fillers we can now use induction to deduce the result:

If $X_i \rightarrow \mathcal{N}_i\pi X$ is an isomorphism, then so is

$$\mathrm{Hom}_{\mathrm{sSet}}(Y, X) \rightarrow \mathrm{Hom}_{\mathrm{sSet}}(Y, \mathcal{N}\pi X)$$

for any i -dimensional Y :

$$\begin{aligned} \mathrm{Hom}_{\mathrm{sSet}}(Y, X) &\cong \mathrm{Hom}_{\mathrm{sSet}}(\mathrm{sk}_i Y, X) \\ &\cong \mathrm{Hom}_{\mathrm{sSet}}(Y, \mathrm{cosk}_i X) \\ &\cong \mathrm{Hom}_{\mathrm{sSet}}(Y, \mathrm{cosk}_i \mathcal{N}\pi X) \\ &\cong \mathrm{Hom}_{\mathrm{sSet}}(\mathrm{sk}_i Y, \mathcal{N}\pi X) \\ &\cong \mathrm{Hom}_{\mathrm{sSet}}(Y, \mathcal{N}\pi X) \end{aligned}$$

So (since we may assume $i \geq 1$) consider the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathrm{sSet}}(\Delta^{i+1}, X) & \longrightarrow & \mathrm{Hom}_{\mathrm{sSet}}(\Delta^{i+1}, \mathcal{N}\pi X) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\mathrm{sSet}}(\Lambda_1^{i+1}, X) & \longrightarrow & \mathrm{Hom}_{\mathrm{sSet}}(\Lambda_1^{i+1}, \mathcal{N}\pi X) \end{array}$$

then the vertical maps are isomorphisms by unique horn filling, and the lower horizontal one by induction and the fact that Λ_1^{i+1} is 1-dimensional. It follows that the top horizontal map is bijective as well, i.e.

$$X_{i+1} \rightarrow \mathcal{N}_{i+1}\pi X$$

is an isomorphism. □

Note that the proof does not even need all horns fillable. Unique fillability of the horns Λ_1^i and Λ_2^3 is enough to guarantee $X \cong \mathcal{N}\pi X$.

Chapter III.

Quasi-categories

We shall here develop just the language of quasi-categories just sufficiently to state the main results of the lecture course. The proofs are still very much out of reach.

We start off with some terminology: if $\mathcal{C}$ is a quasi-category than we shall refer to $\mathcal{C}_0$ as the objects of $\mathcal{C}$ and to $\mathcal{C}_1$ as its morphisms. The later will be denoted $f: x \rightarrow y$ if $d_0 f = y$ and $d_1 f = x$. The degenerate 1-simplex with boundaries x will be referred to as id_x .

In distinction with ordinary categories, however, quasi-categories have an entire simplicial set of morphisms between two objects:

III.1. Definition

Let $x, y \in \mathcal{C}$ be objects. We define $\text{Hom}_{\mathcal{C}}(x, y) \in \text{sSet}$ by

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(x, y) & \longrightarrow & F(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow (d_1^*, d_0^*) \\ \Delta^0 & \xrightarrow{(x, y)} & \mathcal{C} \times \mathcal{C} \end{array}$$

III.2. Lemma

For $x, y \in \mathcal{C}$ we have a canonical isomorphism

$$\pi_0 \text{Hom}_{\mathcal{C}}(x, y) \cong \text{Hom}_{\mathcal{C}}(x, y) .$$

Proof. On the one hand, $\pi_0 \text{Hom}_{\mathcal{C}}(x, y)$ is given by

$$\{f \in \mathcal{C}_1 \mid d_0 f = y, d_1 f = x\} / \sim$$

where $\sim$ is generated by $d_0 H \sim d_1 H$ for every $H \in \text{Hom}_{\mathcal{C}}(x, y)_1$. But decoding, this means $H: \Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$:

$$\begin{array}{ccc} & d_0 H & \\ \text{id}_x \swarrow & \square & \searrow \text{id}_y \\ & d_1 H & \end{array}$$

and we saw in II.30 that this is precisely the relation imposed on morphisms in $\pi\mathcal{C}$. Since II.31 $\pi\mathcal{C} \cong h\mathcal{C}$, the claim follows. $\square$

Furthermore, the conflict of notation is minimal:

III.3. Lemma

For $\mathcal{C}$ an ordinary small category and $x, y \in \mathcal{C}$ we have a canonical isomorphism

$$\mathrm{Hom}_{\mathcal{N}\mathcal{C}}(x, y) \cong \mathbb{D} \mathrm{Hom}_{\mathcal{C}}(x, y).$$

Proof. Exercise.

Here is the first result we do not prove yet:

III.4. Theorem A

For any $x, y \in \mathcal{C}$ the simplicial set $\mathrm{Hom}_{\mathcal{C}}(x, y)$ is Kan.

We will derive this from basic properties of left/right fibrations that we will develop soon. In any case, this is the sense in which a quasi-category has a space of morphisms between any two objects.

Next we turn to composition: For morphisms $g: x \rightarrow y$ and $f: y \rightarrow z$ in $\mathcal{C}$ we say that $h: x \rightarrow z$ is a composition of f and g if $[f] \circ [g] = [h]$ in $h\mathcal{C}$. Again this notion is underlain simplicial one:

III.5. Definition

Given $g: x \rightarrow y$ and $f: y \rightarrow z$ in $\mathcal{C}$ we define $\mathrm{comp}(f, g) \in \mathrm{sSet}$ by

$$\begin{array}{ccc} \mathrm{comp}(f, g) & \longrightarrow & F(\Delta^2, \mathcal{C}) \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{(f, g)} & F(\Lambda_1^2, \mathcal{C}) \end{array}$$

Warning: $\mathrm{comp}(f, g)_0 \not\subseteq \mathcal{C}_1$

It will be another exercise to show, that $\pi_0 \mathrm{comp}(f, g) = *$.

Here is another theorem to be proven later:

III.6. Theorem B

For any $f: y \rightarrow z$ and $g: x \rightarrow y$ in $\mathcal{C}$ we have that $\mathrm{comp}(f, g)$ is a contractible Kan set.

Again this follows from the theory of Kan fibrations and anodyne extensions, but is not easy to see directly. The condition that $\mathrm{comp}(f, g)$ be as above is almost equivalent to being a quasi-category; I had used that in the introduction to motivate the definition.

For an ordinary category $\mathcal{C}$ the right map in III.5 is an isomorphism (though this is not entirely trivial either) so $\mathrm{comp}(f, g) \simeq *$ in that case.

III.7. Definition

A morphism $f: x \rightarrow y$ in $\mathcal{C}$ is called an *equivalence* if there exists a morphism $g: y \rightarrow x$ such that id_x is a composition of f and g and id_y one of g and f .

III.8. Lemma

A morphism $f: x \rightarrow y$ in $\mathcal{C}$ is an equivalence if and only if $[f]: x \rightarrow y$ in $h\mathcal{C}$ is an isomorphism.

Proof. The 'if'-direction follows directly from exercise 3 of sheet 4 and $h\mathcal{C} \cong \pi\mathcal{C}$, while the 'only if'-direction is trivial. $\square$

III.9. Definition

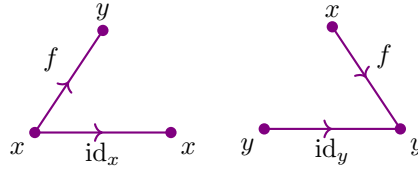
A quasi-category in which every morphism is an equivalence is called an *anima*, or quasi-groupoid or ∞ -groupoid.

So $\mathcal{C}$ is an anima if and only if $h\mathcal{C}$ is a groupoid. The new term anima reflects that these objects are fundamental to the theory of higher categories as sets are to ordinary categories, because of the following (together with [Theorem A](#)):

III.10. Proposition

Any Kan set is an anima.

Proof. Fill the horns



to obtain inverses in the homotopy category. $\square$

The converse is much more difficult:

III.11. Theorem C (Joyal)

Any anima is a Kan set.

And while we can define individual compositions, the construction of a composition map

$$\text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{C}}(x, z)$$

is beyond our current means. Its existence will follow from the result underlying [Theorem B](#). Here is the connection: The left hand side identifies with a simplicial-sub-set of $F(\Lambda_1^2)$,

which we will show equivalent to $F(\Delta^2, \mathcal{C})$ under restriction. Composing an inverse of that equivalence with

$$d_1^* : F(\Delta^2, \mathcal{C}) \rightarrow F(\Delta^1, \mathcal{C})$$

will then give the desired map, since $\text{Hom}_{\mathcal{C}}(x, z) \subseteq F(\Delta^1, \mathcal{C})$.

Next up, functors! A functor between quasi-categories is nothing but a map of simplicial sets $\mathcal{C} \rightarrow \mathcal{D}$ and a natural transformation is nothing but a homotopy of simplicial map. In fact:

III.12. Theorem D

The simplicial set $F(\mathcal{C}, \mathcal{D})$ is again a quasi-category and an anima if $\mathcal{D}$ is one.

In particular, we will denote $F(\mathcal{C}, \mathcal{D})$ by $\text{Fun}(\mathcal{C}, \mathcal{D})$, the *category of functors*, whenever $\mathcal{C}$ and $\mathcal{D}$ are quasi-categories. From the above we, in particular obtain notions of *natural equivalence*, i.e. an equivalence in $\text{Fun}(\mathcal{C}, \mathcal{D})$, and *equivalence of quasi-categories*, i.e. functors $\mathcal{C} \rightarrow \mathcal{D}$ and $\mathcal{D} \rightarrow \mathcal{C}$ whose composites are naturally equivalent to the identities of $\mathcal{C}$ and $\mathcal{D}$, respectively. One of the last things we (hopefully) will show is:

III.13. Theorem E

A natural transformation $\eta: F \Rightarrow G$ is a natural equivalence of functors $\mathcal{C} \rightarrow \mathcal{D}$, if and only if for every $x \in \mathcal{C}$, the morphism

$$\eta_x : F(x) \rightarrow G(x)$$

in $\mathcal{D}$ is an equivalence.

III.14. Theorem F

A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is an equivalence of quasi-categories, if and only if it induces an equivalence of categories $h\mathcal{C} \rightarrow h\mathcal{D}$ and for every $x, y \in \mathcal{C}$ a homotopy equivalence

$$\text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y)).$$

Clearly, there should also be (non-small) quasi-categories of (small) quasi-categories and (small) anima, to be called Cat_{∞} and An . Denoting the full subcategories of sSet spanned by the anima and quasi-categories by $\mathcal{N}\text{Kan} \subseteq \text{qCat} \subseteq \text{sSet}$, there should be inclusions

$$\mathcal{N}\text{Kan} \subseteq \text{An} \quad \text{and} \quad \mathcal{N}\text{qCat} \subseteq \text{Cat}_{\infty}$$

which are equalities on 0- and 1-simplices, but the right hand side will be much bigger. The difference is best illustrated by 2-simplices in $\mathcal{N}\text{Kan}$ and An . For the former we obtain a commuting triangle, while for the latter it should consist of three maps f, g, h and a homotopy $H: f \circ g \Rightarrow h$:

$$(\mathcal{N}\text{Kan})_1 \ni \begin{array}{c} \bullet \\ f \swarrow \quad \searrow g \\ \bullet \quad \circlearrowright \quad \bullet \\ h \end{array} \quad (\text{An})_1 \ni \begin{array}{c} \bullet \\ f \swarrow \quad \searrow g \\ \bullet \quad \Downarrow \quad \bullet \\ h \end{array}$$

Our ultimate goal will be to analyse the quasi-category $\mathbf{An}$ enough to prove *Grothendieck's homotopy hypothesis*:

III.15. Theorem G (Cordier-Joyal)

The quasi-category $\mathbf{An}$ is equivalent to the quasi-category of cell complexes. In particular, $\mathrm{Hom}_{\mathbf{An}}(X, Y)$ is equivalent to the simplicial set (in fact, Kan set) of maps $|X| \rightarrow |Y|$, for all anima X, Y .

Here is a sample of a theorem we will not get close to showing this semester:

Theorem (Lurie) For every quasi-category $\mathcal{C}$, there exists a functor

$$\mathrm{Hom}_{\mathcal{C}} : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathbf{An}$$

extending the anima $\mathrm{Hom}_{\mathcal{C}}$ defined earlier.

Note that such a functor is required to even share a version of Yoneda's Lemma, the necessity of which I hope to have convinced you of.

Chapter IV.

Simplicially Enriched Categories

The goal of this section is to introduce the homotopy coherent nerve of Cordier and Porter which will be our main source of examples of quasi-categories. First we will have to define the input objects, and this will take us back to ordinary category theory for a while.

IV.1. Definition

A *monoidal structure* on a category $\mathcal{C}$ consists of

- i) a functor $\mu: \mathcal{C}^2 \rightarrow \mathcal{C}$, usually written as

$$(X, Y) \mapsto X \boxtimes Y,$$

- ii) an object $\mathbb{1} \in \mathcal{C}$, regarded as a functor

$$\mathbb{1} : * \rightarrow \mathcal{C}$$

- iii) a natural isomorphism

$$\alpha : \mu \circ (\mu \times \text{id}_{\mathcal{C}}) \Rightarrow \mu \circ (\text{id}_{\mathcal{C}} \times \mu)$$

of functors $\mathcal{C}^3 \rightarrow \mathcal{C}$, and

- iv) natural isomorphisms

$$\eta_L : \text{id}_{\mathcal{C}} \Rightarrow \mu \circ (\mathbb{1} \times \text{id}_{\mathcal{C}})$$

$$\eta_R : \text{id}_{\mathcal{C}} \Rightarrow \mu \circ (\text{id}_{\mathcal{C}} \times \mathbb{1})$$

such that the following two diagrams commute:

$$\begin{array}{ccc} \mu \circ (\text{id} \times \mu) \circ (\mu \times \text{id} \times \text{id}) & \xlongequal{\quad} & \mu \circ (\mu \times \text{id}) \circ (\text{id} \times \text{id} \times \mu) \\ \uparrow \alpha & & \downarrow \alpha \\ \mu \circ (\mu \times \text{id}) \circ (\mu \times \text{id} \times \text{id}) & & \mu \circ (\text{id} \times \mu) \circ (\text{id} \times \text{id} \times \mu) \\ \downarrow \alpha & & \uparrow \alpha \\ \mu \circ (\mu \times \text{id}) \circ (\text{id} \times \mu \times \text{id}) & \xrightarrow[\alpha]{} & \mu \circ (\text{id} \times \mu) \circ (\text{id} \times \mu \times \text{id}) \end{array}$$

$$\begin{array}{ccc}
 & \eta_R \nearrow & \mu \circ (\mu \times \text{id}) \circ (\text{id} \times \mathbb{1} \times \text{id}) \\
 \mu = \mu \circ (\text{id} \times \text{id}) & & \downarrow \alpha \\
 & \eta_L \searrow & \mu \circ (\text{id} \times \mu) \circ (\text{id} \times \mathbb{1} \times \text{id})
 \end{array}$$

These diagrams encode that

$$\begin{array}{ccc}
 & ((X \boxtimes Y) \boxtimes W) \boxtimes Z & \\
 \nearrow & & \searrow \\
 (X \boxtimes Y) \boxtimes (Y \boxtimes Z) & & (X \boxtimes (Y \boxtimes W)) \boxtimes Z \\
 \searrow & & \nearrow \\
 X \boxtimes (Y \boxtimes (W \boxtimes Z)) \doteq X \boxtimes ((Y \boxtimes W) \boxtimes Z)
 \end{array}$$

and

$$\begin{array}{ccc}
 & (X \boxtimes \mathbb{1}) \boxtimes Y & \\
 \nearrow & & \downarrow \\
 X \boxtimes Y & & X \boxtimes (\mathbb{1} \boxtimes Y) \\
 \searrow & &
 \end{array}$$

commute. In Mac Lane's book there is a third axiom, namely $\eta_L = \eta_R: \mathbb{1} \rightarrow \mathbb{1} \boxtimes \mathbb{1}$, which we will need later, so let me prove it.

IV.2. Lemma (Kelly)

The diagrams

$$\begin{array}{ccc}
 (X \boxtimes Y) & \xrightarrow{\eta_R} & (X \boxtimes Y) \boxtimes \mathbb{1} \\
 & \searrow \text{id}_X \boxtimes \eta_R & \downarrow \alpha \\
 & & X \boxtimes (Y \boxtimes \mathbb{1})
 \end{array}
 \quad
 \begin{array}{ccc}
 (X \boxtimes Y) & \xrightarrow{\eta_L} & \mathbb{1} \boxtimes (X \boxtimes Y) \\
 & \searrow \eta_L \boxtimes \text{id}_Y & \downarrow \alpha \\
 & & (\mathbb{1} \boxtimes X) \boxtimes Y
 \end{array}$$

commute for all $X, Y \in \mathcal{C}$, and consequently

$$\eta_R = \eta_L: \mathbb{1} \rightarrow \mathbb{1} \boxtimes \mathbb{1}.$$

Proof. Consider the diagram

$$\begin{array}{ccccc}
 ((1 \boxtimes 1) \boxtimes Y) & \xrightarrow{\alpha} & (1 \boxtimes (1 \boxtimes X)) \boxtimes Y & & \\
 \uparrow (\eta_R \boxtimes \text{id}_X) \boxtimes \text{id}_Y & \nearrow (\text{id}_1 \boxtimes \eta_L) \boxtimes \text{id}_Y & \searrow \alpha & & \\
 (1 \boxtimes (1 \boxtimes X)) \boxtimes Y & & & & 1 \boxtimes ((1 \boxtimes X) \boxtimes Y) \\
 \searrow \alpha & & \nearrow \text{id}_1 \boxtimes (\eta_L \boxtimes \text{id}_Y) & & \downarrow \alpha \\
 & 1 \boxtimes (X \boxtimes Y) & \xrightarrow{\text{id}_1 \boxtimes \eta_L} & 1 \boxtimes (1 \boxtimes (X \boxtimes Y)) & \\
 & & & ? &
 \end{array}$$

Then commutativity of the right triangle in the claim is then equivalent to that of the triangle labeled with a question mark, as the latter is $1 \boxtimes -$ applied to the former (and $1 \boxtimes - \stackrel{\eta_L}{\cong} \text{id}_C$ is an equivalence of categories). But because of the left and middle diagrams commuting and consisting of isomorphisms, this equivalent to the commutativity of the outer diagram

$$\begin{array}{ccccc}
 ((1 \boxtimes 1) \boxtimes Y) & \xrightarrow{\alpha} & (1 \boxtimes (1 \boxtimes X)) \boxtimes Y & & \\
 \uparrow (\eta_R \boxtimes \text{id}_X) \boxtimes \text{id}_Y & \searrow \alpha & & \searrow \alpha & \\
 (1 \boxtimes (1 \boxtimes X)) \boxtimes Y & & (1 \boxtimes 1) \boxtimes (X \boxtimes Y) & & 1 \boxtimes ((1 \boxtimes X) \boxtimes Y) \\
 \searrow \alpha & & \uparrow \eta_R \boxtimes (\text{id}_X \boxtimes \text{id}_Y) & & \downarrow \alpha \\
 & 1 \boxtimes (X \boxtimes Y) & \xrightarrow{\text{id}_1 \boxtimes \eta_L} & 1 \boxtimes (1 \boxtimes (X \boxtimes Y)) &
 \end{array}$$

But this follows because of the pink stuff (left parallelogram - naturality of α , upper right pentagon - pentagon axiom, lower triangle - unit axiom). The claim for η_R follows by the dual argument. To obtain $\eta_L = \eta_R: 1 \rightarrow 1 \boxtimes 1$ observe that it again suffices to prove $\eta_L \boxtimes 1 = \eta_R \boxtimes 1: 1 \boxtimes 1 \rightarrow (1 \boxtimes 1) \boxtimes 1$. But from naturality of η_L we find the commutative diagram

$$\begin{array}{ccc}
 1 & \xrightarrow{\eta_L} & 1 \boxtimes 1 \\
 \eta_L \downarrow & & \downarrow 1 \boxtimes \eta_L \\
 1 \boxtimes 1 & \xrightarrow{\eta_L} & 1 \boxtimes (1 \boxtimes 1)
 \end{array}$$

which shows

$$1 \boxtimes \eta_L = \eta_L: 1 \boxtimes 1 \rightarrow 1 \boxtimes (1 \boxtimes 1).$$

But the first part of the proposition then gives $1 \boxtimes \eta_L = (\eta_L \boxtimes 1) \circ \alpha$. But from the unit axiom we also find $1 \boxtimes \eta_L = (\eta_R \boxtimes 1) \circ \alpha$, so

$$(\eta_R \boxtimes 1) \circ \alpha = (\eta_L \boxtimes 1) \circ \alpha$$

which gives the claim since α is an isomorphism. $\square$

Now Mac Lane proved that given the commutativity of these diagrams, *every* sensible diagram in $\mathcal{C}$ built from objects of $\mathcal{C}$ by tensoring and units commutes; more formally, we showed that there is a canonical isomorphism between any two meaningful ways of positioning brackets and units in the expression

$$X_1 \boxtimes X_2 \boxtimes \dots \boxtimes X_n,$$

e.g.

$$((X_1 \boxtimes 1) \boxtimes X_2) \boxtimes X_3 \quad \text{and} \quad (1 \boxtimes (X_1 \boxtimes X_2)) \boxtimes (1 \boxtimes X_3)$$

but not

$$(X_1 \boxtimes 1 \boxtimes X_2) \boxtimes X_3,$$

for any $X_1, \dots, X_n \in \mathcal{C}$ that is any two “formal” ways of moving between such expressions give the same result. Of course such an isomorphism is generally not unique: On the one hand for the stupid reason that there can for example be isomorphisms $(X \boxtimes Y) \boxtimes Z \rightarrow X \boxtimes (Y \boxtimes Z)$ that have nothing to do with the associator, but also for the more subtle reason that it may happen that two differently bracketed expressions are “accidentally” equal in $\mathcal{C}$, which allows one to build non-commuting diagrams made solely from associators, by skipping some rebracketings. See Mac Lane’s book for a precise statement of this *coherence theorem*.

IV.3. Remark

Of course the conclusion of the coherence theorem should really be the *definition* of a monoidal category, and the theorem should be that it suffices to check the pentagon and unit axioms to establish a monoidal structure! The same already goes for associativity in algebra: its definition should be that any bracketing of $a_1 \cdot \dots \cdot a_n$ gives the same result and it is a proposition that $(a_1 a_2) a_3 = a_1 (a_2 a_3)$ already implies this. This becomes visible when associativity is replaced by weaker conditions such as *alternativity* (meaning that bracketing of $a_1 \cdot \dots \cdot a_n$ only need to agree if $|\{a_1, \dots, a_n\}| \leq 2$) or *power-associativity* (rebracketing requirement only for $|\{a_1, \dots, a_n\}| = 1$, i.e. $a_1 = a_2 = \dots = a_n$). That rebracketing for expressions of length 3 suffices is a non-trivial theorem of Artin in the former case, and simply false in the latter!

In the case of quasi-categories there can also not be a coherence theorem, albeit for a different reason: The commutativity of a rebracketing diagram is then a *datum* and not a condition and larger and larger diagrams require higher and higher simplices as witnesses for their commutativity. The definition of a monoidal quasi-category is therefore necessarily quite different from IV.1; in fact, it is much more elegant as it avoids the choices we shall have to make in the examples below.

IV.4. Examples

- i) Using your favorite model of the product of two sets and your favorite one element set $*$ ($\{\emptyset\}$, no?) we obtain a functor

$$- \times - : \text{Set} \rightarrow \text{Set}$$

and as part of the package come bijections

$$(X \times Y) \times Z \longleftrightarrow X \times (Y \times Z)$$

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and

$$* \times Z \longleftrightarrow Z \longleftrightarrow Z \times *$$

which give the associator and unit isomorphisms of a monoidal structure on $\mathbf{Set}$.

- ii) Let $\mathcal{C}$ be a category which admits finite products. Then pick a functor

$$\mu : \mathcal{C}^2 \rightarrow \mathcal{C},$$

which upon postcomposition with Yoneda's embedding $\mathcal{C} \rightarrow \mathbf{P}(\mathcal{C})$ becomes naturally isomorphic to

$$\mathcal{C}^2 \xrightarrow{Y_{\mathcal{C}^2}} \mathbf{P}(\mathcal{C}^2) \xrightarrow{\Delta^*} \mathbf{P}(\mathcal{C})$$

say via an isomorphism φ and let furthermore $*$ $\in \mathcal{C}$ be a terminal object; the assumptions exactly guarantee the existence of such objects!

Unwinding definitions the functor μ sends (X, Y) to a choice of product $X \times Y$. Now φ induces isomorphisms of both composites of

$$\mu \circ (\mathrm{id}_{\mathcal{C}} \circ \mu) \quad \text{and} \quad \mu \circ (\mu \circ \mathrm{id}_{\mathcal{C}})$$

and the Yoneda embedding of $\mathcal{C}$ with

$$\mathcal{C}^3 \xrightarrow{Y_{\mathcal{C}^3}} \mathbf{P}(\mathcal{C}^3) \xrightarrow{\Delta} \mathbf{P}(\mathcal{C})$$

and the resulting isomorphism between the two composites is a candidate associator. Similarly,

$$\mu \circ (\mathrm{id}_{\mathcal{C}} \times *) \quad \text{and} \quad \mu \circ (* \times \mathrm{id}_{\mathcal{C}})$$

composed with $Y_{\mathcal{C}}$ both represent the Yoneda embedding of $\mathcal{C}$ itself, so are canonically equivalent to the identity. Checking the axioms is tedious but easy.

what we have just constructed is a *cartesian* monoidal structure on $\mathcal{C}$; by definition this is a monoidal structure with terminal unit such that the resulting maps

$$\begin{aligned} X \boxtimes Y &\longrightarrow X \boxtimes 1 \xrightarrow{Y_R^{-1}} X \\ X \boxtimes Y &\longrightarrow 1 \boxtimes Y \xrightarrow{Y_L^{-1}} Y \end{aligned}$$

exhibit $X \boxtimes Y$ as a product of X and Y . Such a structure is not unique but once we have introduced monoidal functors, it is easy to see that they form the objects of a contractible groupoid, so are essentially unique, i.e. the choices occurring in the construction above do not really matter.

A dual discussion applies to *cocartesian* monoidal structures on categories with finite coproducts.

- iii) The most prominent example of a monoidal structure that is neither cartesian nor cocartesian is the tensor product $\otimes_R$ on the category $\mathbf{R}\text{-Mod}$, for R a commutative ring, the unit is the ring R itself. Similarly, for a not necessarily commutative ring A the category of A - A -bimodules, which is just $A \otimes A^{\mathrm{op}}\text{-Mod}$, also carries a monoidal structure given by $\otimes_A$ with unit A .

- iv) For a category $\mathcal{C}$ the category $\text{Fun}(\mathcal{C}, \mathcal{C})$ carries a monoidal structure given by composition with unit $\text{id}_{\mathcal{C}}$. This is an example of a *strict* monoidal category, that is one where the associator and unitors α, η_L, η_R are in fact identities.
- v) For M a monoid the category $\mathbb{D}M$ is also strictly monoidal. Similarly a commutative monoid N gives rise to a monoidal structure on $\mathbb{B}N$.
- vi) A monoidal structure on $\mathcal{C}$ induces one on $\mathcal{C}^{\text{op}}$, by sending $(\mu, \mathbb{1}, \alpha, \eta_L, \eta_R)$ to $(\mu^{\text{op}}, \mathbb{1}, \alpha^{-1}, \eta_L^{-1}, \eta_R^{-1})$, and also a reverse monoidal structure on $\mathcal{C}$ via $(\mu \circ \tau, \mathbb{1}, \alpha^{-1} \circ \sigma, \eta_R, \eta_L)$, where $\sigma: \mathcal{C}^3 \rightarrow \mathcal{C}^3$, $\tau: \mathcal{C}^2 \rightarrow \mathcal{C}^2$ are the order reversals. For example,

$$\mathbb{D}(M^{\text{op}}) \simeq (\mathbb{D}M)^{\text{rev}}, \quad \text{Fun}(\mathcal{C}^{\text{op}}, \mathcal{C}^{\text{op}}) \simeq \text{Fun}(\mathcal{C}, \mathcal{C})^{\text{op}}$$

as monoidal categories.

IV.5. Definition

A *lax monoidal functor* between monoidal categories $\mathcal{C}$ and $\mathcal{D}$ consists of

- i) a functor $\mathcal{C} \rightarrow \mathcal{D}$
- ii) a natural transformation

$$\lambda: \mu_{\mathcal{C}} \circ (F \times F) \Rightarrow F \circ \mu_{\mathcal{D}}$$

of functors $\mathcal{C}^2 \rightarrow \mathcal{D}$

- iii) a map

$$u: \mathbb{1}_{\mathcal{D}} \rightarrow F(\mathbb{1}_{\mathcal{C}})$$

such that

$$\begin{array}{ccc} (FX \boxtimes_{\mathcal{D}} FY) \boxtimes_{\mathcal{D}} FZ & \xrightarrow{\lambda} & F(X \boxtimes_{\mathcal{C}} Y) \boxtimes_{\mathcal{D}} FZ \\ \alpha_{\mathcal{D}} \downarrow & & \downarrow \lambda \\ FX \boxtimes_{\mathcal{D}} (FY \boxtimes_{\mathcal{D}} FZ) & & F((X \boxtimes_{\mathcal{C}} Y) \boxtimes_{\mathcal{C}} Z) \\ \lambda \downarrow & & \downarrow \alpha_{\mathcal{C}} \\ FX \boxtimes_{\mathcal{D}} F(Y \boxtimes_{\mathcal{C}} Z) & \xrightarrow{\lambda} & F(X \boxtimes_{\mathcal{C}} (Y \boxtimes_{\mathcal{C}} Z)) \end{array}$$

and

$$\begin{array}{ccc} F(X) \xrightarrow{F\eta_R^{\mathcal{C}}} F(X \boxtimes \mathbb{1}_{\mathcal{C}}) & & F(X) \xrightarrow{F\eta_L^{\mathcal{C}}} F(\mathbb{1}_{\mathcal{C}} \boxtimes X) \\ \eta_R^{\mathcal{D}} \downarrow & \uparrow \lambda & \eta_L^{\mathcal{D}} \downarrow & \uparrow \lambda \\ F(X) \boxtimes_{\mathcal{D}} \mathbb{1}_{\mathcal{D}} \xrightarrow{u} F(X) \boxtimes F(\mathbb{1}_{\mathcal{C}}) & & \mathbb{1}_{\mathcal{D}} \boxtimes_{\mathcal{D}} F(X) \xrightarrow{u} F(\mathbb{1}_{\mathcal{C}}) \boxtimes_{\mathcal{D}} F(X) \end{array}$$

commute. The triple (F, λ, u) is called (strong) *monoidal* if both λ and u are isomorphisms. Dually, an *oplax monoidal functor* has as data

- i) a functor $F: \mathcal{C} \rightarrow \mathcal{D}$

ii) a natural transformation

$$\omega : F \circ \mu_{\mathcal{D}} \Rightarrow \mu_{\mathcal{C}} \circ (F \times F)$$

iii) a map

$$c : F(1_{\mathcal{C}}) \rightarrow 1_{\mathcal{D}}$$

satisfying dual axioms.

Note that a monoidal functor in particular also gives rise to an oplax monoidal one (by inverting λ and u), so we find categories

$$\text{MonCat}^{\text{lax}} \supseteq \text{MonCat} \subseteq \text{MonCat}^{\text{oplax}}.$$

IV.6. Examples

i) If $\mathcal{C}$ and $\mathcal{D}$ carry cartesian monoidal structures then any functor $F : \mathcal{C} \rightarrow \mathcal{D}$ admits a unique oplax monoidal structure: We have to provide maps

$$F(X \times Y) \rightarrow FX \times FY \quad \text{and} \quad F(*_{\mathcal{C}}) \rightarrow *_{\mathcal{D}}.$$

But for the latter there is a unique choice since $*_{\mathcal{D}}$ is terminal, and for the former we have to take the map induced by

$$\begin{aligned} F(X \times Y) &\rightarrow F(X \times *_{\mathcal{C}}) \xrightarrow{F \eta_R^{-1}} F(X) \\ F(X \times Y) &\rightarrow F(*_{\mathcal{C}} \times Y) \xrightarrow{F \eta_L^{-1}} F(Y) \end{aligned}$$

Similarly, if $\mathcal{C}$ and $\mathcal{D}$ carry cocartesian structures then all functors $\mathcal{C} \rightarrow \mathcal{D}$ have unique lax monoidal structures.

These structures are strong monoidal if and only if F preserves finite products or coproducts, respectively.

ii) For an adjoint pair $L : \mathcal{C} \rightleftarrows \mathcal{D} : R$ between monoidal structures, there is a bijection between

$$\{\text{oplax monoidal structures on } L\} \quad \text{and} \quad \{\text{lax monoidal structures on } R\}$$

iii) Here is a concrete example: Consider the adjoint functors

$$\text{free} : \text{Set} \rightleftarrows \text{Ab} : \text{forget}$$

Then we have 4 monoidal structures to play with: $(\text{Set}, +)$, $(\text{Set}, \times)$, $(\text{Ab}, \oplus)$ and $\text{Ab}, \otimes$. For the cocartesian structures we find the lax structures given by the evident maps

$$\begin{aligned} \mathbb{Z}[S] \oplus \mathbb{Z}[T] &\rightarrow \mathbb{Z}[S + T], & 0 \rightarrow 0 &= \mathbb{Z}[\emptyset] && \in \text{Ab} \\ A + B &\rightarrow A \oplus B, & \emptyset \rightarrow 0 &&& \in \text{Set} \end{aligned}$$

of which the former is clearly strong monoidal (this proves the linear algebra fact, that bases of products are unions of bases). As there are no maps $0 \rightarrow \emptyset$ the forgetful functor does not carry an oplax structure.

For the cartesian structures we find the oplax structures given by

$$\begin{aligned} \mathbb{Z}[S \times T] &\rightarrow \mathbb{Z}[S] \oplus \mathbb{Z}[T], & \mathbb{Z}[*] &= \mathbb{Z} \rightarrow 0 & \in \text{Ab} \\ A \oplus B &\xrightarrow{\text{id}} A \times B, & 0 &\rightarrow + & \in \text{Set} \end{aligned}$$

where clearly the latter functor is strong monoidal as part [i](#)) implies. Conversely, the free functor this time also carries a lax structure: The zero map

$$\mathbb{Z}[S] \oplus \mathbb{Z}[T] \xrightarrow{\circ}, \quad 0 \rightarrow \mathbb{Z} = \mathbb{Z}[*] \in \text{Ab}$$

Finally, the free functor is also strong monoidal

$$\text{free} : (\text{Set}, \times) \rightarrow (\text{Ab}, \otimes)$$

since

$$\mathbb{Z}[S \times T] \cong \mathbb{Z}[S] \otimes \mathbb{Z}[T], \quad \mathbb{Z}[*] \cong \mathbb{Z}.$$

This also admits a categorical proof, but let us just observe that it gives a lax monoidal structure to the forgetful functor, and once you're done the exercise proving [ii](#)) it will unwind to give

$$A \times B \rightarrow A \otimes B, \quad * \xrightarrow{1} \mathbb{Z}$$

where the former map is the canonical bilinear form, which can conversely be used to construct the map

$$\mathbb{Z}[S \times T] \rightarrow \mathbb{Z}[S] \otimes \mathbb{Z}[T]$$

via [ii](#)). I don't think there are any refinements in the final case $(\text{Set}, +)$ and $(\text{Ab}, \otimes)$.

iv) the functors

$$\text{Hom}_{\mathcal{C}}(\mathbb{1}, -) : \mathcal{C} \rightarrow \text{Set}, \quad \text{Hom}_{\mathcal{C}}(-, \mathbb{1}) : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$$

are both lax monoidal for the cartesian structure on Set : We require a map

$$* \rightarrow \text{Hom}_{\mathcal{C}}(\mathbb{1}, \mathbb{1})$$

in both cases, for which we take $\text{id}_{\mathbb{1}}$, and then maps

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(\mathbb{1}, X) \times \text{Hom}_{\mathcal{C}}(\mathbb{1}, Y) &\rightarrow \text{Hom}_{\mathcal{C}}(\mathbb{1}, X \boxtimes Y) \\ \text{Hom}_{\mathcal{C}}(X, \mathbb{1}) \times \text{Hom}_{\mathcal{C}}(Y, \mathbb{1}) &\rightarrow \text{Hom}_{\mathcal{C}}(X \boxtimes Y, \mathbb{1}) \end{aligned}$$

for which we take

$$(f : \mathbb{1} \rightarrow X, g : \mathbb{1} \rightarrow Y) \mapsto (\mathbb{1} \xrightarrow{\eta} \mathbb{1} \boxtimes \mathbb{1} \xrightarrow{f \otimes g} X \boxtimes Y)$$

and the dual construction. Checking the axioms is a good exercise.

IV.7. Definition

Let $F, G: \mathcal{C} \rightarrow \mathcal{D}$ be lax monoidal functors. Then a transformation $\tau: F \rightarrow G$ is called *monoidal* if the diagrams

$$\begin{array}{ccc} FX \boxtimes_{\mathcal{D}} FY & \xrightarrow{\lambda_F} & F(X \boxtimes_{\mathcal{C}} Y) \\ \downarrow \tau & & \downarrow \tau \\ GX \boxtimes_{\mathcal{D}} GY & \xrightarrow{\lambda_G} & G(X \boxtimes_{\mathcal{C}} Y) \end{array} \quad \begin{array}{ccc} & & F(1_{\mathcal{C}}) \\ & u_F \nearrow & \downarrow \tau \\ 1_{\mathcal{D}} & & \\ & u_G \searrow & \\ & & G(1_{\mathcal{C}}) \end{array}$$

commute for all $X, Y \in \mathcal{C}$.

The analogous definition applies to transformations of oplax monoidal functors.

Note that for a strong monoidal functor the two arising notions of monoidal transformation agree. We obtain full inclusions

$$\text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D}) \supseteq \text{Fun}^{\text{mon}}(\mathcal{C}, \mathcal{D}) \subseteq \text{Fun}^{\text{oplax}}(\mathcal{C}, \mathcal{D}).$$

Note furthermore, that these definitions provide us with the notions monoidal natural isomorphism of functor (which can again be tested pointwise) and monoidal equivalence of categories. Here a priori, there are three notions which might be called lax, oplax and strong monoidal equivalence. These turn out to agree: Two lax monoidal functors

$$F: \mathcal{C} \rightleftarrows \mathcal{D}: G$$

with $F \circ G$ and $G \circ F$ monoidally isomorphic to the respective identities, are automatically strong monoidal (as one easily checks by writing out the definitions) Let me immediately warn you that the same does not apply if $F: \mathcal{C} \rightarrow \mathcal{D}$ is lax monoidal and an equivalence of categories, see the example [ii](#)) below.

A word on terminology: Many authors call such transformation lax monoidal and oplax monoidal. There is, however, nothing lax about the transformations or their monoidality, but rather only about the functors they run between. Clearly, that's no good!

IV.8. Examples

- i) If $\mathcal{C}$ and $\mathcal{D}$ are cartesian monoidal then any natural transformation between functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$ is monoidal for the canonical oplax structure of F and G . In particular, the forgetful functor

$$\text{Fun}^{\text{oplax}}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

is an isomorphism of categories in this case, and similarly

$$\text{Fun}^{\text{lax}}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

is an isomorphism for cocartesian $\mathcal{C}$ and $\mathcal{D}$.

- ii) If $\mathcal{C}$ admits a *zero-object*, i.e. one that is simultaneously initial and terminal, and both finite products and coproducts, then the identity functor $\text{id}_{\mathcal{C}}$ admits a canonical oplax refinement

$$(\mathcal{C}, +) \rightarrow (\mathcal{C}, \times)$$

or equivalently, lax refinement

$$(\mathcal{C}, \times) \rightarrow (\mathcal{C}, +)$$

given by the unique map on units, and

$$X + Y \rightarrow X \times Y$$

induced by

$$X + Y \longrightarrow X + 0 \xrightarrow{\eta_R^{-1}} X$$

$$X + Y \longrightarrow 0 + X \xrightarrow{\eta_L^{-1}} Y$$

or equivalently

$$X \xrightarrow{\eta_R} X \times 0 \longrightarrow X \times Y$$

$$X \xrightarrow{\eta_L} 0 \times X \longrightarrow X \times Y;$$

concisely, by the matrix

$$\begin{pmatrix} \text{id}_X & 0 \\ 0 & \text{id}_Y \end{pmatrix} : X + Y \rightarrow X \times Y,$$

where 0 denotes the unique map of any object factoring through the zero-object. This refinement is not strong for example for $*/_{\text{Set}}$.

iii) If $R \rightarrow S \rightarrow T$ are homomorphisms of commutative rings then the functors

$$\text{R-Mod} \rightarrow \text{S-Mod}, \quad M \mapsto S \otimes_R M$$

$$\text{R-Mod} \rightarrow \text{T-Mod} \xrightarrow{\text{forget}} \text{S-Mod}, \quad M \mapsto T \otimes_R M$$

are lax monoidal for the respective tensor product and the upper even strongly so. The map $S \rightarrow T$ then induces a monoidal transformation

$$S \otimes_R - \implies T \otimes_R -.$$

We are ready to give the main definition of this section:

IV.9. Definition

Let $\mathcal{V}$ be a monoidal category. Then a $\mathcal{V}$ -enriched category $\mathcal{C}$ consists of

- i) a class O
- ii) a map $F: O \times O \rightarrow \text{obj}(\mathcal{V})$
- iii) a map $i: O \rightarrow \text{mor}(\mathcal{V})$
- iv) a map

$$\text{comp}: O \times O \times O \rightarrow \text{mor}(\mathcal{V})$$

such that

a)

$$s(\text{comp}(x, y, z)) = F(y, z) \boxtimes F(x, y)$$

and

$$t(\text{comp}(x, y, z)) = F(x, z)$$

b)

$$s(i(x)) = \mathbb{1}, \quad t(i(x)) = F(x, x)$$

c) the diagrams

$$\begin{array}{ccc} (F(y, z) \boxtimes F(x, y)) \boxtimes F(w, x) & \xrightarrow{\text{comp}} & F(x, z) \boxtimes F(w, x) \\ \downarrow \alpha & & \downarrow \text{comp} \\ F(y, z) \boxtimes (F(x, y) \boxtimes F(w, x)) & & \\ \downarrow \text{comp} & & \\ F(y, z) \boxtimes F(w, y) & \xrightarrow{\text{comp}} & F(w, z) \end{array}$$

and

$$\begin{array}{ccc} F(x, y) & \xrightarrow{\text{id}} & F(x, y) \\ \downarrow \eta_R & & \uparrow \text{comp} \\ F(x, y) \boxtimes \mathbb{1} & \xrightarrow{i} & F(x, y) \boxtimes F(x, x) \end{array} \quad \begin{array}{ccc} F(x, y) & \xrightarrow{\text{id}} & F(x, y) \\ \downarrow \eta_L & & \uparrow \text{comp} \\ \mathbb{1} \boxtimes F(x, y) & \xrightarrow{i} & F(y, y) \boxtimes F(x, y) \end{array}$$

commute.

IV.10. Examples

- i) A $(\text{Set}, \times)$ -enriched category is just an ordinary category.
- ii) A $(\text{Top}, \times)$ -enriched category is an ordinary category $\mathcal{C}$ together with a topology on each of the sets $\text{Hom}_{\mathcal{C}}(x, y)$ such that composition

$$\text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(w, x) \rightarrow \text{Hom}_{\mathcal{C}}(w, y)$$

is continuous. Let me warn you that the compact-open topology does not unfortunately provide such an enrichment on Top itself. Some restrictions need to be imposed, making topologically enriched categories somewhat annoying to work with.

- iii) An $(\text{Ab}, \otimes)$ -enriched category translates to an ordinary category together with an abelian group structure on each $\text{Hom}_{\mathcal{C}}(x, y)$ such that composition

$$\text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(w, x) \rightarrow \text{Hom}_{\mathcal{C}}(w, y)$$

is biadditive. Since the question arose in the lecture let me give a brief discussion of the existence and uniqueness of such enrichments. First of all, we did not need inverses for any of the discussions above, and the same decoding works for $(\text{CMon}, \otimes)$ -enriched

categories. Now a category $\mathcal{C}$ which admits both finite products and finite coproducts, which agree (in the sense that the initial element is terminal, and the arising lax transformation

$$(\mathcal{C}, +) \rightarrow (\mathcal{C}, \times)$$

is actually strong), admits an *unique* $(\mathbf{CMon}, \otimes)$ -enrichment.

Conversely, if a $(\mathbf{CMon}, \otimes)$ -enriched category admits one of finite products or finite coproducts, then it necessarily also admits the other, and they agree. Without the existence of products or coproducts, however, even $(\mathbf{Ab}, \otimes)$ -enrichments need not be unique! For example an $(\mathbf{Ab}, \otimes)$ -enriched category with a single object is the same thing as a ring, and the underlying category precisely encodes the multiplicative monoid.

$$\mathbb{Z}_4 \quad \text{and} \quad \mathbb{Z}_2[t]/t^2$$

are two rings with isomorphic multiplicative monoids, but different additive groups.

IV.11. Remark

Definition IV.9 is not as symmetric in the objects and morphisms as our original definition of a category (it looks more like the usual form!) An analogue of the former style of object exists as well, a *category object initial to $\mathcal{V}$* , but outside the case of $\mathcal{V} = (\mathbf{Set}, \times)$ this notion is different from that of a $\mathcal{V}$ -enriched category: In particular, the objects of a $\mathcal{V}$ -internal category are also an object of $\mathcal{V}$! In fact the definition of $\mathcal{V}$ -internal categories requires more structure than just a monoidal one on $\mathcal{V}$, one needs to be able to form relative $\boxtimes$ -products in $\mathcal{V}$. In the case of cartesian monoidal $\mathcal{V}$, these correspond to pullbacks (which featured in our definition of composition in an ordinary category).

IV.12. Definition

A $(\mathbf{Cat}, \times)$ -enriched category is called a (strict) *2-category*.

Decoding, a 2-category thus consists of objects (namely the object class), morphisms (the objects of the morphism categories) and 2-morphisms (the morphisms of the morphism categories). The objects and morphisms form an usual category, whereas the 2-morphisms can be composed in two ways: Once via the composition in the category of morphisms $F(x, y)$ and once via the composition functors

$$F(x, y) \times F(w, x) \rightarrow F(w, x).$$

IV.13. Examples

- i) the 2-category $\mathbf{Cat}_2$ consisting of
 - objects: (small) categories
 - morphisms: functors
 - 2-morphisms: natural transformations

ii) the 2-categories

$$\text{MonCat}_2^{\text{lax}} \supseteq \text{MonCat}_2 \subseteq \text{MonCat}_2^{\text{oplax}}$$

consisting of

objects: (small) monoidal categories

morphisms: lax / strong / oplax monoidal functors

2-morphisms: monoidal transformations

iii) the 2-category Grp_2 consisting of

objects: groups

morphisms: group homomorphisms

2-morphisms: elements, in the following sense

$$\text{Hom}_{F(G,H)}(\varphi, \psi) = \{h \in H \mid \varphi \cdot h = h \cdot \psi\}$$

I leave it to you to figure the two ways of composing group elements.

iv) the not-usually-useful 2-category Top_2 consisting of

objects: topological spaces

morphisms: continuous maps

2-morphisms: homotopy classes of homotopies.

The last example admits many analogues such as $\text{sSet}_2, (\text{CC}_R)_2, \dots$. We will discuss those in the (better!) context of simplicially enriched categories:

IV.14. Definition

A *simplicially enriched category* is a category enriched in $(\text{sSet}, \times)$.

As a first example let us mention simplicial sets themselves via the function complexes

$$F(x, y) \in \text{sSet}$$

constructed in [ii](#)). In fact, it is an exercise to show that any cartesian closed category is canonically enriched over itself. We will soon see that there are many more simplicial enrichments than one might first guess. For now let us briefly discuss the relation of simplicially enriched categories with simplicial objects in categories:

IV.15. Construction/Lemma

Let $\mathcal{C}$ be simplicially enriched. Then for $i \in \mathbb{N}$ consider the category $\mathcal{C}_i \in \text{Cat}$ with

$$\begin{aligned} \text{obj}(\mathcal{C}_i) &= \text{obj}(\mathcal{C}) \\ \text{mor}(\mathcal{C}_i) &= \sum_{x, y \in \mathcal{C}} F(x, y); \end{aligned}$$

i.e. $\text{Hom}_{\mathcal{C}_i}(x, y)$ consists of the i -simplices of $F(x, y)$. It is (hopefully) clear that $\mathcal{C}_i$ really is a category and furthermore one readily checks that the $\mathcal{C}_i$ assemble into a simplicial category

$$S(\mathcal{C}) : \Delta^{\text{op}} \rightarrow \text{Cat}, \quad [i] \mapsto \mathcal{C}_i$$

with induced functors the identity on objects and the structure maps in $F(x, y)$ on isomorphisms. Conversely, given a simplicial category $T : \Delta^{\text{op}} \rightarrow \text{Cat}$ such that the functor

$$\Delta^{\text{op}} \xrightarrow{T} \text{Cat} \xrightarrow{\text{obj}} \text{Set}$$

is constant with value O , then T yields a simplicially enriched category $\mathcal{C}$ with objects O and morphisms $F(x, y) \in \text{sSet}$ given by

$$F_{\mathcal{C}}(x, y)_i = \text{Hom}_{T(i)}(x, y)$$

with the evident simplicial structure induced from T . These processes are evidently inverse to one another.

IV.16. Remark

Unwinding definitions a simplicial object in categories is in fact the same as a category object internal to $(\text{sSet}, \times)$.

IV.17. Definition

An *enriched functor* G between $\mathcal{V}$ -enriched categories $\mathcal{C} \rightarrow \mathcal{D}$ consists of

- i) a map $G_o : \text{obj}(\mathcal{C}) \rightarrow \text{obj}(\mathcal{D})$
- ii) a map

$$G_m : \text{obj}(\mathcal{C})^2 \rightarrow \text{mor}(\mathcal{V})$$

such that

$$s \circ G_m(x, y) = F_{\mathcal{C}}(x, y), \quad t \circ G_m(x, y) = F_{\mathcal{D}}(G_o x, G_o y)$$

and the diagrams

$$\begin{array}{ccc} F_{\mathcal{C}}(y, z) \boxtimes F_{\mathcal{C}}(x, y) & \xrightarrow{G_m \boxtimes G_m} & F_{\mathcal{D}}(G_o y, G_o z) \boxtimes F_{\mathcal{D}}(G_o x, G_o y) \\ \downarrow & & \downarrow \\ F_{\mathcal{C}}(x, z) & \xrightarrow{G_m} & F_{\mathcal{D}}(G_o x, G_o z) \end{array}$$

$$\begin{array}{ccc} & F_{\mathcal{C}}(x, x) & \\ & \downarrow G_m & \\ \mathbb{1} & \begin{array}{c} \nearrow \\ \searrow \end{array} & F_{\mathcal{D}}(G_o x, G_o x) \end{array}$$

commute. An *enriched natural transformation* τ from G to G' consists of a map

$$\tau : \text{obj}(\mathcal{C}) \rightarrow \text{mor}(\mathcal{V})$$

with

$$s \circ \tau(x) = \mathbb{1}, \quad t \circ \tau(x) = F_{\mathcal{D}}(Gx, G'x)$$

such that

$$\begin{array}{ccccc} F_{\mathcal{C}}(x, y) & \xrightarrow{\eta_R} & \mathbb{1} \boxtimes F_{\mathcal{C}}(x, y) & \longrightarrow & F_{\mathcal{D}}(Gy, G'y) \boxtimes F_{\mathcal{D}}(Gx, Gy) \\ \downarrow \eta_R & & & & \downarrow \circ \\ F_{\mathcal{C}}(x, y) \boxtimes \mathbb{1} & \xrightarrow{G \boxtimes \tau} & F_{\mathcal{D}}(G'x, G'y) \boxtimes F_{\mathcal{D}}(Gx, G'x) & \xrightarrow{\circ} & F_{\mathcal{D}}(Gx, G'y) \end{array}$$

commutes.

We obtain a category $\text{Cat}^{\mathcal{V}}$ and a 2-category $\text{Cat}_2^{\mathcal{V}}$ consisting of $\mathcal{V}$ -enriched categories, functors and natural transformations in the latter case.

We next extract an underlying ordinary category from an enriched one (hence the name). To this end we generally observe:

IV.18. Construction/Lemma

A lax monoidal functor $\Phi : \mathcal{V} \rightarrow \mathcal{W}$ induces a 2-functor

$$\Phi_* : \text{Cat}_2^{\mathcal{V}} \rightarrow \text{Cat}_2^{\mathcal{W}}$$

and this is part of a 2-functor

$$\begin{array}{c} \text{MonCat}^{\text{lax}} \rightarrow 2\text{Cat}^1 \\ \mathcal{V} \mapsto \text{Cat}_2^{\mathcal{V}} \end{array}$$

Proof. To define Φ_* on objects

$$\begin{aligned} \text{obj}(\Phi_*\mathcal{C}) &:= \text{obj}(\mathcal{C}), & F_{\Phi_*\mathcal{C}}(x, y) &:= \Phi F_{\mathcal{C}}(x, y) \\ i_{\Phi_*\mathcal{C}}(x) &:= \Phi(i_{\mathcal{C}}x) \circ u : \mathbb{1}_{\mathcal{W}} \rightarrow \Phi(\mathbb{1}_{\mathcal{V}}) \rightarrow \Phi F_{\mathcal{C}}(x, x) \end{aligned}$$

with composition

$$\Phi F_{\mathcal{C}}(y, z) \boxtimes_{\mathcal{W}} \Phi F_{\mathcal{C}}(x, y) \xrightarrow{\lambda} \Phi(F_{\mathcal{C}}(y, z) \boxtimes_{\mathcal{V}} F_{\mathcal{C}}(x, y)) \xrightarrow{\Phi(- \circ -)} \Phi F_{\mathcal{C}}(x, z).$$

To define Φ_* on morphisms send a $\mathcal{V}$ -enriched functor $G : \mathcal{C} \rightarrow \mathcal{D}$ to

$$\begin{aligned} (\Phi_*G)_o &:= G : \text{obj}(\Phi_*\mathcal{C}) = \text{obj}(\mathcal{C}) \rightarrow \text{obj}(\mathcal{D}) = \text{obj}(\Phi_*\mathcal{D}) \\ (\Phi_*G)_m &= \Phi(G_m) : F_{\Phi_*\mathcal{C}}(x, y) = \Phi F_{\mathcal{C}}(x, y) \rightarrow \Phi F_{\mathcal{D}}(Gx, Gy) = F_{\Phi_*\mathcal{D}}(Gx, Gy). \end{aligned}$$

¹ $\text{Cat}_2^{\text{Cat}}$, 2-category of 2-categories, 2-functors and 2-natural transformations

Finally, to define Φ_* on 2-morphisms send $\tau: G \Rightarrow G'$, a $\mathcal{V}$ -enriched transformation, to

$$(\Phi_*\tau)(x) := \Phi(\tau(x)) \circ u : \mathbb{1}_{\mathcal{W}} \rightarrow \Phi(\mathbb{1}_{\mathcal{V}}) \rightarrow \Phi F_{\mathcal{C}}(Gx, G'x) = F_{\Phi_*\mathcal{C}}((\Phi_*G)x, (\Phi_*G')x).$$

One readily checks that these data really give a 2-functor $\Phi_*: \text{Cat}_2^{\mathcal{V}} \rightarrow \text{Cat}_2^{\mathcal{W}}$ as desired. This construction gives the morphism part of a 2-functor

$$\text{MonCat}_2^{\text{lax}} \rightarrow 2\text{Cat}.$$

To obtain it on 2-morphisms, let $\tau: \Phi \rightarrow \Psi$ be a monoidal transformation. Then we have to give for each monoidal category $\mathcal{C}$ a morphism

$$* \rightarrow \text{Fun}^{\mathcal{W}}(\Phi_*\mathcal{C}, \Psi_*\mathcal{C})^2$$

in Cat , i.e. a $\mathcal{W}$ -enriched functors $\Phi_*\mathcal{C} \rightarrow \Psi_*\mathcal{C}$. On objects we take

$$\text{id}_{\text{obj}(\mathcal{C})} : \text{obj}(\Phi_*\mathcal{C}) \rightarrow \text{obj}(\Psi_*\mathcal{C})$$

and on morphisms

$$F_{\Phi_*\mathcal{C}}(x, y) = \Phi F_{\mathcal{C}}(x, y) \xrightarrow{\tau} \Psi F_{\mathcal{C}}(x, y) = F_{\Psi_*\mathcal{C}}(x, y).$$

Again there are a lot of compatibilities to be checked; but I shan't do that. $\square$

IV.19. Definition

For $\mathcal{C}$ a $\mathcal{V}$ -enriched category, let $u\mathcal{C} \in \text{Cat}$, the *underlying category* of $\mathcal{C}$, denote the image of $\mathcal{C}$ under

$$(\text{Hom}_{\mathcal{V}}(\mathbb{1}, -))_* : \text{Cat}_2^{\mathcal{V}} \rightarrow \text{Cat}_2^{\text{Set}} = \text{Cat}_2.$$

For example, we have

$$\text{Hom}_{\text{Top}}(*, -) : \text{Top} \rightarrow \text{Set} \quad \text{and} \quad \text{Hom}_{\text{Ab}}(\mathbb{Z}, -) : \text{Ab} \rightarrow \text{Set}$$

are the forgetful functors, whereas

$$\text{Hom}_{\text{Cat}}(*, -) = \text{obj} \quad \text{and} \quad \text{Hom}_{\text{sSet}}(\Delta^0, -) = \text{ev}_0$$

so the underlying categories are given by forgetting topologies, the group structure, the 2-morphisms and higher simplices. Here are other examples of changes in enrichment:

IV.20. Examples

- i) the functors $\text{ev}_i : \text{sSet} \rightarrow \text{Set}$ all preserve products since they are given by $\text{Hom}_{\text{sSet}}(\Delta^i, -)$, so give rise to functors

$$(\text{ev}_i)_* : \text{Cat}_2^{\text{sSet}} \rightarrow \text{Cat}_2$$

²Category of $\mathcal{W}$ -enriched functors

that we used in IV.15. Having defined enriched functors that Lemma immediately upgrades to give that the resulting functor

$$\mathrm{Cat}^{\mathrm{sSet}} \rightarrow \mathrm{sCat}$$

is an embedding with essential image the simplicial categories whose image under

$$\mathrm{obj} : \mathrm{sCat} \rightarrow \mathrm{sSet}$$

is constant.

In particular, since sCat is bicomplete by I.39 and $\mathrm{obj} : \mathrm{Cat} \rightarrow \mathrm{Set}$ preserves both limits and colimits it follows that $\mathrm{Cat}^{\mathrm{sSet}}$ is bicomplete as well (this is a cheap trick avoiding further analysis of colimits in $\mathrm{Cat}^{\mathrm{sSet}}$).

- ii) the functor $\mathrm{Sing} : \mathrm{Top} \rightarrow \mathrm{sSet}$ preserves products so every $(\mathrm{Top}, \times)$ -enriched category gives a $(\mathrm{sSet}, \times)$ -enriched category. The converse is almost true: The functor $|\cdot| : \mathrm{sSet} \rightarrow \mathrm{Top}$ carries its tautological oplax monoidal structure

$$|X \times Y| \rightarrow |X| \times |Y|$$

and for finite simplicial sets this a homeomorphisms (this can be checked explicitly for simplices and then follows, since the product functor $\times Z$ commutes with colimits as $\mathrm{Ing} \ a \ Z$ is locally compact Hausdorff).

In general, however, it is not a homeomorphism and one needs to slightly alter the category of topological spaces to allow for maps that are only continuous on compact subsets (or better after precomposition with all maps from compact Hausdorff spaces). The latter category is called kTop and turns out to be bicomplete with the inclusion $\mathrm{Top} \rightarrow \mathrm{kTop}$ preserving limits, but not generally colimits. It does, however, preserve colimits of compactly generated spaces, such as $|X|$ for any $X \in \mathrm{sSet}$, so the realisation/singular set adjunction remains unchanged

$$|\cdot| : \mathrm{sSet} \rightleftarrows \mathrm{kTop}$$

and any $(\mathrm{sSet}, \times)$ -enriched category gives a $(\mathrm{kTop}, \times)$ -enriched one and conversely.

- iii) the functors $\pi_0 : (k) \mathrm{Top} \rightarrow \mathrm{Set}$ and $\pi_0 : \mathrm{sSet} \rightarrow \mathrm{Set}$ both preserve finite products. This is obvious in the former case (where π_0 in fact preserves all limits), but slightly more subtle for sSet . Let me sketch a counterexample for π_0 commuting with arbitrary products here: Let X be the infinite spine with $X_0 = \mathbb{Z}$, and non-degenerate 1-simplices also given by $\mathbb{Z}$, with boundary $d_i(z) = z + i$, and no higher non-degenerate simplices, or in other words

$$X = \mathrm{colim}(\mathrm{I}_0 \xrightarrow{d_1} \mathrm{I}_1 \xrightarrow{d_2} \mathrm{I}_2 \xrightarrow{d_3} \mathrm{I}_3 \rightarrow \dots).$$

Then $\pi_0 X = 0$ since $|X| \cong \mathbb{R}_{\geq 0}$, and so also $\prod_{\mathbb{N}} \pi_0 X = 0$. But $\pi_0 \prod_{i \in \mathbb{N}} X \neq 0$, as the 0-simplices $(0, \dots, 0, \dots)$ and $(0, 1, 2, 3, \dots)$ of $\prod_{i \in \mathbb{N}} X$ cannot be connected by a finite sequence of 1-simplices in $\prod_{i \in \mathbb{N}} X$. In fact, $\pi_0 \prod_{i \in \mathbb{N}} X$ consists of all sequences in $\mathbb{N}$ modulo the relation of bounded difference. Anyway, we obtain

$$(\pi_0)_* : \mathrm{Cat}^{(k) \mathrm{Top}} \rightarrow \mathrm{Cat} \quad \text{and} \quad (\pi_0)_* : \mathrm{Cat}^{\mathrm{sSet}} \rightarrow \mathrm{Cat}$$

sending $\mathcal{C}$ to its *homotopy category*.

- iv) various flavors of discrete functors, e.g. $\mathbb{D}: \mathbf{Set} \rightarrow \mathbf{Cat}$, $\text{const}_{\Delta^{\text{op}}}: \mathbf{Set} \rightarrow \mathbf{Set}$ and $\mathbf{Set} \rightarrow \mathbf{Top}$ preserve products and these give rise to functors (even embeddings)

$$\mathbf{Cat} \rightarrow 2\mathbf{Cat}, \quad \mathbf{Cat} \rightarrow \mathbf{Cat}^{\mathbf{sSet}}, \quad \mathbf{Cat} \rightarrow \mathbf{Cat}^{\mathbf{Top}}$$

- v) Lemma IV.18 in particular implies that adjoint functors

$$L: \mathcal{V} \rightleftarrows \mathcal{W}: R$$

between monoidal categories, such that both L and R are lax monoidal and the unit and counit are monoidal (a lax monoidal adjunction³) give rise to adjoint functors

$$L_*: \mathbf{Cat}^{\mathcal{V}} \rightleftarrows \mathbf{Cat}^{\mathcal{W}}: R_*.$$

Essentially all functors above come in such adjoint pairs.

- vi) The adjunction

$$h: \mathbf{sSet} \rightleftarrows \mathbf{Cat}: \mathcal{N}$$

is also monoidal for the cartesian structures (by IV.8 i) and exercise 3, sheet 5). We obtain functors

$$h_*: \mathbf{Cat}^{\mathbf{sSet}} \rightleftarrows 2\mathbf{Cat}: \mathcal{N}_*$$

embedding $2\mathbf{Cat}$ into $\mathbf{Cat}^{\mathbf{sSet}}$ and conversely sending a simplicially enriched category to its *homotopy 2-category*. Let me immediately warn you that a quasi-category does *not* have a strict homotopy 2-category. Rather there is a notion of a weak 2-category, traditionally known as a *bicategory* introduced by Benabou, and every quasi-category has a homotopy bicategory (though we shall not even define bicategories rigorously in this course). A typical example of a bicategory is

$$\mathbf{BiMod} = \begin{cases} \text{objects} & \text{rings} \\ \text{morphisms} & \text{bimodules} \\ \text{2-morphisms} & \text{bimodule homomorphisms} \end{cases}$$

with the composition of two morphisms

$$M \in R \otimes S^{\text{op}}\text{-Mod} \quad \text{and} \quad N \in S \otimes T^{\text{op}}\text{-Mod}$$

being $M \otimes_S N \in R \otimes T^{\text{op}}\text{-Mod}$, which is not a strictly associative operation.

Exercise 2, sheet 5 together with III.12 in particular imply that the adjunction between h_* and $\mathcal{N}_*$ restricts to

$$h_*: \mathbf{Cat}^{\mathbf{Kan}} \rightleftarrows \mathbf{Cat}^{\mathbf{Grpd}}: \mathcal{N}_*$$

Our next goal is to produce examples of Kan-enriched simplicial categories, but before going there let us briefly open Pandora's box:

IV.21. Definition

A *strict n -category* is a category enriched in the category of strict $(n-1)$ -categories⁴. Thus a n -category has objects and i -morphisms for $1 \leq i \leq n$.

³In fact, there's a notion of adjointness in any 2-category; so the same works for $\mathbf{Cat}_2^{\mathcal{V}} \rightleftarrows \mathbf{Cat}_2^{\mathcal{W}}$.

⁴Under cartesian product

IV.22. Definition

A strict (n, k) -category is a strict n -category for which all i -morphisms with $i > k$ are invertible.

By convention a 0-category is just a set, and then a 1-category is an ordinary category. A (n, n) -category is just a n -category, and a $(n, 0)$ -category is a n -groupoid. A (n, k) -category is enriched in $(n-1, k-1)$ -categories and forgetting the top layer of morphisms gives a $(n-1, k)$ -category; this process is also defined inductively starting with the monoidal functor

$$\text{obj} : 1\text{Cat} = \text{Cat} \rightarrow \text{Set} = 0\text{Cat}$$

and using the monoidal functor

$$\text{obj}_n : n\text{Cat} \rightarrow (n-1)\text{Cat}$$

to induce

$$\text{obj}_{n+1} = (\text{obj}_n)_* : (n+1)\text{Cat} \rightarrow n\text{Cat}.$$

Similarly, a n -category has a homotopy $(n-1)$ -category by starting the same process with π_0 instead of obj .

IV.23. Definition

A strict ∞ -category $\mathcal{C}$ consists of a sequence $(C_i, \mathcal{C}_i)$ where $\mathcal{C}_i \in i\text{Cat}$ and $\varphi_i : \pi \mathcal{C}_{i+1} \rightarrow C_i$ is an isomorphism of i -categories. A strict ∞ -functor is a sequence of i -functors⁵ $\mathcal{C}_i \rightarrow \mathcal{C}'_i$ strictly commuting with the φ_i . $\mathcal{C}$ is called a (∞, k) -category if C_i is a (i, k) -category for all $i \geq k$.

These objects form categories ωCat and $(\omega, k)\text{Cat}$ for $0 \leq k$, and indeed there are equivalences

$$\text{Cat}^{\omega\text{Cat}} \simeq \omega\text{Cat}, \quad \text{Cat}^{(\omega, k)\text{Cat}} \simeq (\omega, k+1)\text{Cat}$$

by shifting. The only examples for $n \geq 3$ that I am immediately aware of arise by a similar iteration: The category of (small) n -categories actually forms a (large) $(n+1)$ -category $n\text{Cat}_{n+1}$; for example between two 2-natural transformations of 2-functors $\sigma, \tau : G \Rightarrow G', G, G' : \mathcal{C} \rightarrow \mathcal{D}$ there is a notion of *modification* μ , consisting of an assignment of a natural transformation

$$\mu(x) : \sigma(x) \rightarrow \tau(x),$$

where $\sigma(x), \tau(x) \in F_{\mathcal{D}}(G(x), G'(x))$, the category of morphisms $Gx \rightarrow G'x$, making the diagram

$$\begin{array}{ccc} \sigma(x') \circ Gf & \xrightarrow{\mu} & \tau(x') \circ Gf \\ \parallel & & \parallel \\ G'f \circ \sigma(x) & \xrightarrow{\mu} & G'f \circ \tau(x) \end{array}$$

in $F_{\mathcal{D}}(Gx, G'x')$ commutes.

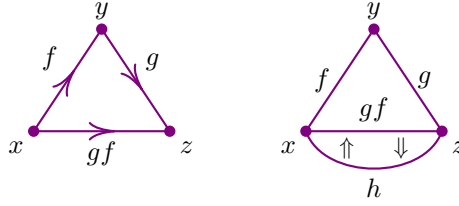
In particular, the category of all small ω -categories organizes into an ω -category $\omega\text{Cat}_{\omega}$, and (ω, k) -categories into an $(\omega, k+1)$ -category $(\omega, k)\text{Cat}_{(\omega, k+1)}$.

⁵I.e. $i-1$ Cat-enriched functors

Now quasi-categories want to be a weak form of $(\omega, 1)$ -categories and indeed Duskin and Street constructed nerves

$$\mathbf{Bicat} \rightarrow \mathbf{sSet}, \quad \omega\mathbf{Cat} \rightarrow \mathbf{sSet}$$

which take values in quasi-categories, when restricted to $(\mathbf{Bi}, 1)\mathbf{Cat}$ and $(\omega, 1)\mathbf{Cat}$. The former is well-documented in Lurie's *Kerodon*, and quite useful, whereas details of the latter were worked out in Verity's unpublished thesis. The constructions again rely on II.19 and proceed by building certain cosimplicial objects in $\mathbf{Bicat}$ and $\omega\mathbf{Cat}$. In the latter case these are known as Street's *orientals*, obtained from $n \rightarrow \mathcal{N}[n]$ by freely adding a new morphism $n_1 \rightarrow n_k$ for every string $n_1 \rightarrow n_2 \rightarrow \dots \rightarrow n_k$ and then higher morphisms to relate it to the actual composite $n_1 \rightarrow n_k$, and their composites, and their composites... Let me just mention that Street's nerve is *not* fully faithful: For example consider the 2-categories



with $F(x, z) \cong$  in the second example. Then a strict 2-functor from left to right that is the identity on objects is automatically the inclusion, whereas a weak 2-functor can send f to f and g to g , but gf to h !

Now finally for the examples of Kan enriched categories: The general construction mechanism is that of the simplicial function sets

$$F(X, Y)_n = \mathbf{Hom}_{\mathbf{sSet}}(X \times \Delta^n, Y).$$

IV.24. Construction

Let $\mathcal{C}$ be a monoidal category and $\mathcal{D}: \Delta \rightarrow \mathcal{C}$ a cosimplicial object in $\mathcal{C}$, equipped with

- i) a *coproduct* $d: \mathcal{D} \Rightarrow \mathcal{D} \boxtimes \mathcal{D}$, where the target is

$$\Delta \xrightarrow{\Delta} \Delta \times \Delta \xrightarrow{\mathcal{D} \times \mathcal{D}} \mathcal{C} \times \mathcal{C} \xrightarrow{- \boxtimes -} \mathcal{C}$$

- ii) a *counit* map

$$c: \mathcal{D} \Rightarrow \text{const}_{\Delta} \mathbb{1}_{\mathcal{C}}, \quad \text{i.e. a map } \mathcal{D}[0] \Rightarrow \mathbb{1}_{\mathcal{C}}$$

such that

$$\begin{array}{ccccc} \mathcal{D} & \xRightarrow{d} & \mathcal{D} \boxtimes \mathcal{D} & \xRightarrow{\text{id} \boxtimes d} & \mathcal{D} \boxtimes (\mathcal{D} \boxtimes \mathcal{D}) \\ & \searrow d & & & \uparrow \alpha_{\mathcal{C}} \\ & & \mathcal{D} \boxtimes \mathcal{D} & \xRightarrow{d \boxtimes \text{id}} & (\mathcal{D} \boxtimes \mathcal{D}) \boxtimes \mathcal{D} \end{array}$$

and

$$\begin{array}{ccc} \mathcal{D} \boxtimes \mathcal{D} & \xrightarrow{\text{id} \boxtimes c} & \mathcal{D} \boxtimes \text{const}_{\Delta} 1_{\mathcal{C}} \\ \swarrow d & & \nearrow \eta_R \\ & \mathcal{D} & \end{array} \quad \begin{array}{ccc} \mathcal{D} \boxtimes \mathcal{D} & \xrightarrow{c \boxtimes \text{id}} & \text{const}_{\Delta} 1_{\mathcal{C}} \boxtimes \mathcal{D} \\ \swarrow d & & \nearrow \eta_R \\ & \mathcal{D} & \end{array}$$

commute.

Then we can define a simplicially enriched category $\mathcal{C}_{\mathcal{D}} \in \text{Cat}^{\text{sSet}}$ with

$$\begin{aligned} \text{obj } \mathcal{C}_{\mathcal{D}} &= \text{obj } \mathcal{C} \\ F_{\mathcal{C}_{\mathcal{D}}}(X, Y)_n &= \text{Hom}_{\mathcal{C}}(X \boxtimes \mathcal{D}[n], Y) \\ \text{id}_{\mathcal{C}_{\mathcal{D}}} : X \boxtimes \mathcal{D}[0] &\xrightarrow{\text{id} \times c} X \boxtimes 1_{\mathcal{C}} \xrightarrow{\eta_R^{-1}} X \end{aligned}$$

and composition of $f : X \boxtimes \mathcal{D}[n] \rightarrow Y$ and $g : Y \boxtimes \mathcal{D}[n] \rightarrow Z$ given by

$$X \boxtimes \mathcal{D}[n] \xrightarrow{\text{id} \boxtimes d} X \boxtimes (\mathcal{D}[n] \boxtimes \mathcal{D}[n]) \xrightarrow{\alpha_{\mathcal{C}}^{-1}} (X \boxtimes \mathcal{D}[n]) \boxtimes \mathcal{D}[n] \xrightarrow{f \boxtimes \text{id}} Y \boxtimes \mathcal{D}[n] \xrightarrow{g} Z$$

which is readily checked to be unital and associative. The counit also provides a canonical functor

$$c_* : \mathcal{C} \rightarrow u\mathcal{C}_{\mathcal{D}}$$

which is the identity on objects and send $f : X \rightarrow Y$ to

$$X \boxtimes \mathcal{D}[0] \xrightarrow{\text{id} \boxtimes c} X \boxtimes 1_{\mathcal{C}} \xrightarrow{\eta_R^{-1}} X \xrightarrow{f} Y.$$

The functor c_* is an isomorphism if c itself is one.

Let's postpone the discussion of the functoriality of this construction for a moment and look at examples:

IV.25. Examples

- i) Applied to $\mathcal{C} = \text{sSet}$ with the cartesian structure and $\mathcal{D} = \Delta^0$ the cosimplicial object given by the standard simplices with its unique (!) comonoid structure given by

$$\Delta : \Delta^n \rightarrow \Delta^n \times \Delta^n \quad \text{and} \quad \text{id} : \Delta^0 \rightarrow \Delta^0$$

we recover the cartesian enrichment

$$F(X, Y)_n = \text{Hom}_{\text{sSet}}(X \times \Delta^n, Y).$$

The composition can here also be described by

$$f \circ g = X \times \Delta^n \xrightarrow{(g, \text{pr}_2)} Y \times \Delta^n \xrightarrow{f} Z^6$$

This enrichment is of course not remotely Kan. However, Theorems C and D imply that its restriction to $\text{Kan} \subseteq \text{sSet}$ gives a Kan enrichment.

⁶The same is true for any cartesian $\mathcal{C}$.

Similarly, investing Theorem C and D fully we can also produce a Kan enrichment on $\mathbf{qCat} \subseteq \mathbf{sSet}$ by setting

$$F_{\mathbf{qCat}}(X, Y) = \text{Core } F(X, Y).$$

For now, we note that we obtain simplicial enrichments on both

$$\mathbf{Kan} \quad \text{and} \quad \mathbf{qCat}.$$

- ii) Taking $\mathcal{C} = (\mathbf{k})\mathbf{Top}$ and $\mathcal{D} = |\Delta^0| = \Delta_{\text{top}}^0$ gives the simplicial enrichment

$$F(X, Y_n) = C(X \times |\Delta^n|, Y)$$

on $(\mathbf{k})\mathbf{Top}$. It is Kan by the same argument we used to show that $\text{Sing } X$ is always Kan:

A horn

$$\begin{array}{ccc} X \times |\Lambda_i^n| & \longrightarrow & Y \\ & \lrcorner \cap & \nearrow \exists \\ & X \times |\Delta^n| & \end{array}$$

can be filled since $|\Lambda_i^n|$ is a retract of $|\Delta^n|$.

- iii) Taking $\mathcal{C} = \mathbf{CC}_R$. monoidal under $\otimes_R$ and

$$\mathcal{D}_n = C^{\text{sup}}(\Delta^n)^7$$

with diagonal given by the Alexander-Whitney map

$$C^{\text{sup}}(\Delta^n) \rightarrow C^{\text{sup}}(\Delta^n \times \Delta^n) \rightarrow C^{\text{sup}}(\Delta^n) \otimes C^{\text{sup}}(\Delta^n)$$

and counit the identity map

$$C^{\text{sup}}(\Delta^0) \rightarrow \mathbb{Z}[0]$$

gives a simplicial enrichment; even one in simplicial R -modules. As simplicial abelian groups always have underlying simplicial sets that are Kan, this (and any such “additive” enrichment) are Kan.

- iv) One such additive enriched is on the category $\mathbf{sAb}$ with the degree wise tensor product and

$$\mathcal{D}_n = \mathbb{Z}[\Delta^n]$$

which inherits its diagonal from Δ^n by the strong monoidality of $\mathbb{Z}[-]: \mathbf{Set} \rightarrow \mathbf{Ab}$. Note that there is always a canonical injection

$$F_{\mathbf{sAb}}(X, Y) \hookrightarrow F_{\mathbf{sSet}}(X, y).$$

- v) Finally, here is a non-example: The category $\mathbf{CW}$ of cell complexes and cellular maps admits the monoidal structure $-\times_K-$, the Kelley-product, which has as underlying set the usual product, but is topologised with the cellular topology

$$\text{colim}_{\substack{A \subseteq X, B \subseteq Y \\ \text{finite subcomplexes}}} A \times B := X \times_k Y.$$

⁷Either ordered simplicial complex $([n], \mathbf{P}[n])$ or normalised chains on $\Delta^n \in \mathbf{sSet}$

However, $X \times_k Y$ is *not* the categorical product in CW: The maps $\text{id}: \Delta^1 \rightarrow \Delta^1$ and $\text{id}: \Delta^1 \rightarrow \Delta^1$ do *not* combine into a cellular map

$$\Delta: \Delta^1 \rightarrow \Delta^1 \times \Delta^1$$

if we give $\Delta^1 \times \Delta^1$ the product cell structure!

Thus the cosimplicial object formed by the simplices Δ_{top}^n with their evident cell structure does not admit a diagonal. We proceed differently:

Define

$$F_{\text{CW}}^b(X, Y)_n = \left\{ \varphi: X \times \Delta_{\text{top}}^n \rightarrow Y \times \Delta_{\text{top}}^n \left| \begin{array}{l} \varphi \text{ is cellular} \\ \varphi(X \times \alpha \Delta^i) \subseteq Y \times \alpha \Delta^i \\ \alpha: [i] \hookrightarrow [n] \end{array} \right. \right\}$$

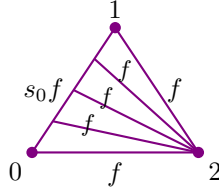
the simplicial set of *block-cellular maps*. Its boundary maps are given by restricting the maps φ in the evident fashion, and degeneracies are most easily described by a recipe: For $f: X \times \Delta^n \rightarrow Y \times \Delta^n$ define $s_i f: X \times \Delta^{n+1} \rightarrow Y \times \Delta^{n+1}$ component wise by

$$X \times \Delta^{n+1} \xrightarrow{\text{id} \times s_i} X \times \Delta^n \xrightarrow{f} Y \times \Delta^n \xrightarrow{\text{pr}_1} Y$$

to the simplex by mapping $(x, t_0, \dots, t_{n+1})$ to

$$\frac{t_i}{t_i + t_{i+1}} d_i \text{pr}_2 f(x, t_0, \dots, \overbrace{t_i + t_{i+1}, \dots, t_{n+1}}^{\in \Delta^n}) + \frac{t_{i+1}}{t_i + t_{i+1}} d_{i+1} \text{pr}_2 f(x, t_0, \dots, t_i + t_{i+1}, \dots, t_{n+1})$$

which is hopefully illustrated by the picture below.



It is readily checked that this really gives a simplicial set, and it is an exercise on sheet 6 to show that it is Kan, giving $\text{CW}_{\text{Cell}}^b$.

These spaces of block cellular maps admit forgetful maps

$$F_{\text{CW}}^b(X, Y)_n \rightarrow \{\varphi: X \times \Delta^n \rightarrow Y \mid \varphi \text{ is cellular}\}$$

by projecting to the Y factor, and we will show that this map is a homotopy equivalence of Kan sets.

There is a similar block version of the enrichment Top^b of Top :

$$F_{\text{Top}}^b(X, Y)_n = \left\{ \varphi: X \times \Delta^n \rightarrow Y \times \Delta^n \left| \begin{array}{l} \varphi(X \times \alpha \Delta^i) \subseteq Y \times \alpha \Delta^i \\ \alpha: [i] \hookrightarrow [n] \end{array} \right. \right\}$$

and the projection $F_{\text{Top}}^b \rightarrow F_{\text{Top}}$ admits a section

$$\begin{aligned} F_{\text{Top}}(X, Y) &\rightarrow F_{\text{Top}}^b(X, Y) \\ (\varphi: X \times \Delta^n \rightarrow Y) &\mapsto ((\varphi, \text{pr}_2): X \times \Delta^n \rightarrow Y \times \Delta^n) \end{aligned}$$

which is easily shown a homotopy equivalence of Kan complexes. This will be another exercise.

There result simplicially enriched functors

$$CW_{\text{Cell}}^b \rightarrow CW^b \subseteq k\text{Top}^b \leftarrow \text{Top}^b \rightleftharpoons \text{Top}$$

the former and latter three of which are examples of

IV.26. Definition

An enriched functor $G: \mathcal{C} \rightarrow \mathcal{D}$ of Kan-enriched categories is called a *Dwyer-Kan equivalence* if

i) G induces an equivalence

$$\pi\mathcal{C} \rightarrow \pi\mathcal{D}$$

on homotopy categories

ii) the map

$$G: F_{\mathcal{C}}(x, y) \rightarrow F_{\mathcal{D}}(Gx, Gy)$$

is a homotopy equivalence of Kan complexes for all $x, y \in \mathcal{C}$.

Similarly, the Eilenberg-Zilber theorem shows that the Dold-Kan correspondence gives Dwyer-Kan equivalences between

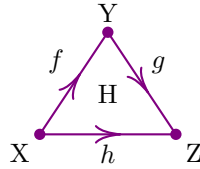
$$\text{sAb} \quad \text{and} \quad \text{CC}_R^{\geq 0}$$

in the enrichment discussed above. We will maybe review this later in the course, when we have established Theorem F. The fact that Dwyer-Kan equivalences are not invertible is the worst defect of simplicially enriched categories.

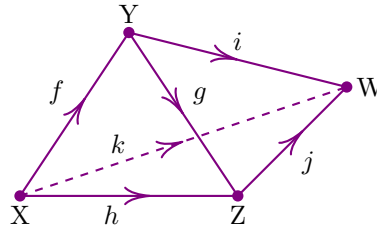
To this end recall, what the coherent nerve $\mathcal{N}^c$ is supposed to do:

$$(\mathcal{N}^c\mathcal{C})_0 = \text{obj}(\mathcal{C}) \quad \text{and} \quad (\mathcal{N}^c\mathcal{C})_1 = \text{mor}(u\mathcal{C})$$

whereas $(\mathcal{N}^c\mathcal{C})_2$ is supposed to consist of diagrams that do not necessarily commute but rather come equipped with an element $H \in F_{\mathcal{C}}(X, Z)_1$ with $d_0H = h$ and $d_1H = gf$.



Similarly, an element of $(\mathcal{N}^c\mathcal{C})_3$ is to consist of 6 maps as indicated below,



one homotopy (i.e. 1-simplex) for each face; this data produces two parallel homotopies namely

$$jgf \Rightarrow jh \Rightarrow k \quad \text{and} \quad jgf \Rightarrow if \Rightarrow k$$

and as the final piece of data we want a homotopy between these homotopies, i.e. a map $\Delta^1 \times \Delta^1 \rightarrow F_C(X, W)$ restricting appropriately. This is accomplished by enumerating such maps, homotopies and so on by paths through the simplex as follows:

IV.27. Construction

For a totally ordered set $\mathcal{J}$ and $i \leq j \in \mathcal{J}$ set

$$\mathcal{J}_{i,j} = \{I \subseteq [i, j] \mid i, j \in I\}$$

the poset of paths from i to j , ordered by refinement, i.e. inclusion.

For $i \leq j \leq k$ there is a map

$$\begin{aligned} \mathcal{J}_{j,h} \times \mathcal{J}_{i,j} &\rightarrow \mathcal{J}_{i,k} \\ (S, T) &\mapsto S \cup T \end{aligned}$$

which is clearly order preserving. We obtain a poset enriched category $\mathfrak{P}\mathcal{J}$ with

$$\text{obj } \mathfrak{P}\mathcal{J} = \mathcal{J} \quad F_{\mathfrak{P}\mathcal{J}}(i, j) = \mathcal{J}_{i,j}$$

with composition given by the union map above and identities the singletons.

IV.28. Definition

Define the simplicially enriched categories $\mathcal{L}\mathcal{J}$ as the image of $\mathfrak{P}\mathcal{J}$ under

$$\text{Cat}^{\text{Ord}} \xrightarrow{\mathbb{N}_*} \text{Cat}^{\text{Cat}} \xrightarrow{\mathcal{N}_*} \text{Cat}^{\text{sSet}}$$

IV.29. Corollary/Definition

We obtain a cosimplicial object

$$\mathcal{L} : \mathbb{A} \rightarrow \text{Cat}^{\text{sSet}}$$

sending $[n]$ to $\mathcal{L}[n]$. Its associated adjunction

$$\mathcal{L} : \text{sSet} \rightleftarrows \text{Cat}^{\text{sSet}} : \mathcal{N}^c$$

defines the *coherent nerve*, explicitly given by

$$(\mathcal{N}^c \mathcal{C})_n = \text{Hom}_{\text{Cat}^{\text{sSet}}}(\mathcal{L}([n]), \mathcal{C}).$$

To analyse this construction, observe that

$$[n]_{i,j} \cong \prod_{i < k < j} \{0, 1\}$$

via

$$S \mapsto \left(k \mapsto \begin{cases} 0 & k \notin S \\ 1 & k \in S \end{cases} \right).$$

IV.30. Observation

This isomorphism induces an isomorphism

$$F_{\mathcal{L}_n}(i, j) \cong \prod_{i < k < j} \Delta^1.$$

In particular, we find

$$\mathcal{L}(0) \cong \text{const}_* N[0] \quad \text{and} \quad \mathcal{L}(1) \cong \text{const}_* N[1]$$

discrete and $\pi\mathcal{L}(n) \cong N[n]$ for all n .

In $\mathcal{L}(2)$ we find

$$F_{\mathcal{L}(2)}(i, j) \cong \Delta^0$$

except that

$$F_{\mathcal{L}(2)}(0, 2) \cong \Delta^1$$

as desired. A similar analysis shows that maps $\mathcal{L}(3) \rightarrow \mathcal{C}$ correspond exactly to diagrams in $\mathcal{C}$ as previously discussed.

IV.31. Theorem (Cordier-Porter)

If $\mathcal{C}$ is Kan-enriched, then $\mathcal{N}^c\mathcal{C} \in \text{sSet}$ is a quasi-category.

IV.32. Theorem H

If $\mathcal{C}$ is Kan-enriched, then there are canonical equivalences

$$F_{\mathcal{C}}(x, y) \simeq \text{Hom}_{\mathcal{N}^c\mathcal{C}}(x, y)$$

of Kan complexes.

For the proof of [IV.31](#) we used to understand $\mathcal{L}(\Lambda_i^n)$ for $0 < i < n$.

IV.33. Lemma

The inclusion $\Lambda_i^n \subseteq \Delta^n$ for $0 < i < n$ induces an inclusion⁸ of simplicial categories

$$\mathcal{L}(\Lambda_i^n) \rightarrow \mathcal{L}(\Delta^n)$$

with

$$\begin{aligned} \text{obj } \mathcal{L}(\Lambda_i^n) &= \text{obj } \mathcal{L}(\Delta^n) = \{0, \dots, n\} \\ F_{\mathcal{L}(\Lambda_i^n)}(j, k) &= F_{\mathcal{L}(n)}(j, k) = \mathcal{N}([n]_{j,k}) \end{aligned}$$

⁸But not an embedding!

except for

$$F_{\mathcal{L}\Lambda_i^n}(0, n) = \bigcup_{0 < j < n} \mathcal{N}(\{I \in [n]_{0,n} \mid j \in I\}) \cup \bigcup_{\substack{0 < j < n \\ i \neq j}} \mathcal{N}(\{I \in [n]_{0,n} \mid j \notin I\}) \subseteq F_{\mathcal{L}(n)}(0, n)$$

The proof is an exercise on sheet 6. Note that under the isomorphism of IV.30 the nerves on the right correspond precisely to the faces of the cube $\prod_{0 < j < n} \Delta^1$: Let for $\epsilon \in \{0, 1\}$

$$e_i^\epsilon : \prod_{\substack{0 < j < n \\ j \neq i}} \Delta^1 \rightarrow \prod_{0 < j < n} \Delta^1$$

denote the embedding induced by

$$d_\epsilon : \Delta^0 \rightarrow \Delta^1$$

in the i -th coordinate, i.e. e_i^0 is the inclusion of the i -th bottom face of the cube and e_i^1 the inclusion of the i -th top face we find

IV.34. Corollary

Under the isomorphism of IV.30 the inclusion

$$F_{\mathcal{L}(\Lambda_i^n)}(0, n) \subseteq F_{\mathcal{L}(n)}(0, n)$$

becomes

$$\bigcup_{\substack{0 < j < n \\ i \neq j}} \text{im}(e_j^0) \cup \bigcup_{0 < j < n} \text{im}(e_j^1) \subseteq \prod_{0 < j < n} \Delta^1.$$

In particular, this inclusion becomes a homotopy equivalence after realisation.

$$\begin{array}{ccccccc} \bullet_1 & \subseteq & \begin{array}{c} \bullet_1 \\ | \\ \bullet_0 \end{array} & \begin{array}{cc} 10 & 11 \\ | & | \\ 00 & 01 \end{array} & \subseteq & \square & \begin{array}{cc} 10 & 11 \\ | & | \\ 00 & 01 \end{array} & \subseteq & \square \\ |F_{\Lambda_1^2}(0, 2)| \subseteq |F_{\Delta^2}(0, 2)| & & |F_{\Lambda_1^3}(0, 3)| \subseteq |F_{\Delta^3}(0, 3)| & & |F_{\Lambda_2^3}(0, 3)| \subseteq |F_{\Delta^3}(0, 3)| \end{array}$$

A degree wise injection of simplicial sets, that becomes a homotopy equivalence after realisation is known as an *anodyne extension*⁹. Horn and spine inclusions are examples as well.

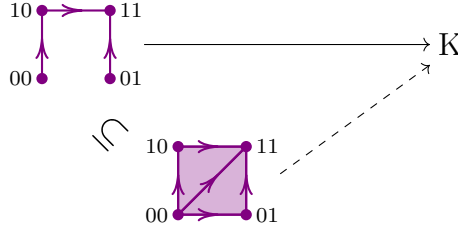
One of the main results of the next section will be that Kan complexes in fact admit fillers for *any* anodyne extension, i.e. if $f : X \rightarrow Y$ is anodyne and K Kan then

$$f^* : \text{Hom}_{\text{sSet}}(Y, K) \rightarrow \text{Hom}_{\text{sSet}}(X, K)$$

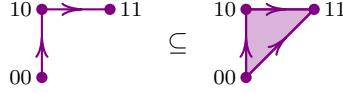
is surjective. In the case of the inclusions occurring as $F_{\mathcal{L}(\Lambda_i^n)}(0, n) \subseteq F_{\mathcal{L}(\Delta^n)}(0, n)$ this statement can also be seen by hand: For $n = 0$ this inclusion simply is a horn, for $n = 1$ a

⁹Though this will not be the definition!

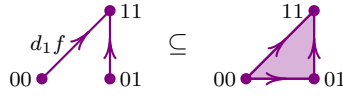
filler for



can clearly be constructed by first filling the horn



and then using the result $f: \Delta^2 \rightarrow K$ to fill



A similar argument works for larger n , but as we will show the general statement soon, let us assume anodyne fillers for a moment and dive in to

IV.35. Proof of IV.31

We have to solve all extension problems

$$\begin{array}{ccc} \Lambda_i^n & \xrightarrow{f} & \mathcal{N}^c \mathcal{C} \\ \downarrow \cap & \nearrow & \\ \Delta^n & & \end{array}$$

but by adjunction they are equivalent to

$$\begin{array}{ccc} \mathcal{L}(\Lambda_i^n) & \xrightarrow{G} & \mathcal{C} \\ \downarrow \cap & \nearrow & \\ \mathcal{L}(\Delta^n) & & \end{array}$$

From Lemma IV.33 it now follows that an extension of g only has to be defined on $F_{\mathcal{L}(\Delta^n)}(0, n)$. Furthermore, an arbitrary extension of

$$\begin{array}{ccc} F_{\mathcal{L}(\Lambda_i^n)}(0, n) & \xrightarrow{G_*} & F_{\mathcal{C}}(G(0), G(n)) \\ \downarrow \cap & \nearrow & \\ F_{\mathcal{L}(\Delta^n)}(0, n) & & \end{array}$$

Chapter IV. Simplicially Enriched Categories

will give a functor as desired since the only object mapping to 0 is 0 (and $F_{\mathcal{L}(\Delta^n)}(0, 0) = *$) and receiving a map from n is n (and similarly $F_{\mathcal{L}(\Delta^n)}(n, n) = *$), so there are no compatibilities with composition to be checked. But since $F_{\mathcal{C}}(G(0), G(n))$ is assumed Kan a filler exists as the inclusion on the left is anodyne. $\square$

Chapter V.

Simplicial Homotopy Theory

V.1. Motivation and basic definitions

We proved in *Topology II* that the canonical map

$$|\mathrm{Sing}(X)| \rightarrow X$$

is a weak equivalence for every topological space X , so we “know” that simplicial sets are sufficient to model (weak) homotopy types. We would like to try to expand on this idea by showing that it is possible to “do homotopy theory” in the category of simplicial sets just as one usually does in topological spaces. One of the key outcomes of this discussion will be the observation that there is essentially no difference between simplicial sets and topological spaces as far as homotopy theory is concerned. Of course, the translation between these two categories will be performed through the familiar adjunction between geometric realisation and the singular set functor. However, our goal will be to develop a homotopy theory for simplicial sets independently of topological spaces and only make the comparison at the very end.

To get things off the ground, let us attempt to find simplicial analogues of the key notions that allow us to do homotopy theory in topological spaces: these are fibrations, cofibrations and weak equivalences.

Let us first find an appropriate notion of fibration. The following terminology is not only useful for this purpose, but will reappear throughout the entire lecture course.

V.1.1. Definition

Let $\mathcal{C}$ be a category, let $f: X \rightarrow Y$ be a morphism in $\mathcal{C}$ and let Σ be a class of morphisms in $\mathcal{C}$.

- i) The map f has the *right lifting property* (or RLP for short), with respect to Σ if for every commutative square

$$\begin{array}{ccc} S & \xrightarrow{x} & X \\ \downarrow s & & \downarrow f \\ T & \xrightarrow{y} & Y \end{array}$$

with $s \in \Sigma$ there exists a commutative diagram of the following form:

$$\begin{array}{ccc} S & \xrightarrow{x} & X \\ \downarrow s & \nearrow \bar{y} & \downarrow f \\ T & \xrightarrow{y} & Y \end{array}$$

- ii) The map f has the *left lifting property* (or LLP) with respect to Σ if for every commutative square

$$\begin{array}{ccc} X & \xrightarrow{x} & S \\ \downarrow f & & \downarrow s \\ Y & \xrightarrow{y} & T \end{array}$$

with $s \in \Sigma$ there exists a commutative diagram of the following form:

$$\begin{array}{ccc} X & \xrightarrow{x} & S \\ \downarrow f & \nearrow \bar{y} & \downarrow s \\ Y & \xrightarrow{y} & T \end{array}$$

V.1.2. Notation

Let $\mathcal{C}$ be a category with finite products and pushouts. For $i: A \rightarrow B$ and $f: X \rightarrow Y$ morphisms in $\mathcal{C}$, let

$$i \boxtimes f : (A \times Y) \cup_{A \times X} (B \times X) \rightarrow B \times Y$$

be the canonical arrow induced by the universal property of the pushout.

Recall that a map of topological spaces $p: E \rightarrow B$ is a (Serre) fibration if it has the RLP for the set of inclusions

$$\{(S^{n-1} \hookrightarrow D^n) \boxtimes (0 \hookrightarrow [0, 1]) \mid n \geq 1\}.$$

There is an evident analogue of these maps in simplicial sets, namely

$$(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (0 \hookrightarrow \Delta^1).$$

However, note that there is a homeomorphism

$$(S^{n-1} \hookrightarrow D^n) \boxtimes (0 \hookrightarrow [0, 1]) \cong (S^{n-1} \hookrightarrow D^n) \boxtimes (1 \hookrightarrow [0, 1])$$

which has no simplicial analogue. This is the reason why it will also be important to consider the maps

$$(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (1 \hookrightarrow \Delta^1).$$

V.1.3. Definition

- i) A map of simplicial sets is a *left fibration* if it has the RLP with respect to the set

$$LC := \{(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (0 \hookrightarrow \Delta^1) \mid n \geq 1\}.$$

- ii) A map of simplicial sets is a *right fibration* if it has the RLP with respect to the set

$$RC := \{(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (1 \hookrightarrow \Delta^1) \mid n \geq 1\}.$$

- iii) A map of simplicial sets is a Kan fibration if it is both a left and a right fibration, i.e. if it has the RLP with respect to the set

$$C := LC \cup RC.$$

V.1.4. Remark

With this definition, it is not clear that a simplicial set X is Kan if and only if $X \rightarrow \Delta^0$ is a Kan fibration. This will be shown in V.2.15.

Similarly, cofibrations in topological spaces were defined in terms of the homotopy extension property. We could try to mimic this definition directly but this turns out to be a bad idea. For example, we would certainly expect the inclusion $(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (\{0\} \hookrightarrow \Delta^1)$ to be a cofibration. However, if we took “cofibration” to mean “has the homotopy extension property with respect to all simplicial sets”, then this map would have a left inverse. It is easy to check in the case $n = 1$ that no such retraction exists (no does it for any other n).

On the other hand, we can think of every (left) homotopy extension problem as a commutative diagram

$$\begin{array}{ccc} A \times \Delta^1 \cup_{A \times \{0\}} B \times \{0\} & \longrightarrow & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ B \times \Delta^1 & \longrightarrow & \Delta^0 \end{array}$$

In particular, for $A = \partial\Delta^n$ and $B = \Delta^n$ we have already *defined* this problem to have a solution if $X \rightarrow \Delta^0$ is a left fibration.

V.1.5. Definition

A morphism $i: A \rightarrow B$ is a *left cofibration* if $i \boxtimes (\{0\} \hookrightarrow \Delta^1)$ has the LLP with respect to all Kan fibrations $X \rightarrow \Delta^0$.

V.1.6. Remark

- i) It would actually be more natural to require $X \rightarrow \Delta^0$ to be a left Kan fibration in the definition of a left cofibration. We will see later that every left fibration over Δ^0 is a Kan fibration.
- ii) One could equally well consider *right cofibrations* and *fibrations*, and develop a parallel theory. We will show soon that this is not a parallel theory, but the *same* theory.

V.1.7. Examples

- i) By definition, the inclusion $\partial\Delta^n \rightarrow \Delta^n$ is a left cofibration for all $n \geq 0$ if we set $\partial\Delta^0 := \emptyset$.
- ii) The map $\emptyset \rightarrow B$ is a left cofibration for every simplicial set B .

Finally, recall that the following statements are equivalent for a map $f: X \rightarrow Y$ of CW-complexes:

- i) f is a homotopy equivalence;
- ii) f induces a bijection on π_0 and an isomorphism on all homotopy groups with respect to every choice of base point;

iii) for every space Z , the map f induces a weak homotopy equivalence

$$f^* : \text{Hom}_{\text{Top}}(Y, Z) \rightarrow \text{Hom}_{\text{Top}}(X, Z);$$

iv) for every space Z , the map f induces a bijection

$$f^* : \pi_0 \text{Hom}_{\text{Top}}(Y, Z) \rightarrow \pi_0 \text{Hom}_{\text{Top}}(X, Z).$$

When defining cofibrations, we already observed that homotopies in the category of simplicial sets are more rigid than homotopies in topological spaces, so it is unlikely that copying condition i) will give a sensible notion. For example, our geometric intuition tells us that $(\partial\Delta^n \hookrightarrow \Delta^n) \boxtimes (0 \hookrightarrow \Delta^1)$ should be an equivalence, but it is certainly not a homotopy equivalence. Conditions ii) and iii) would require us to first develop a sensible notion of homotopy groups for simplicial sets. So we will try to adopt iv) as our definition. Note that $\pi_0 \text{Map}(Y, Z)$ will correspond to homotopy classes of maps $Y \rightarrow Z$. If we were to require this condition for every simplicial set Z , we would run into the same problems we faced when defining cofibrations. Hence, we only require condition iv) to hold when Z is a Kan complex (see again V.1.6).

V.1.8. Definition

A map of simplicial sets $f: X \rightarrow Y$ is a *weak equivalence* if it induces a bijection

$$f^* : \pi_0 F(Y, K) \rightarrow \pi_0 F(X, K)$$

for every left Kan complex K .

V.1.9. Definition

A morphism $f: X \rightarrow Y$ is a *retract* of a morphism $f': X' \rightarrow Y'$ if it is a retract in $\text{Fun}([1], \mathcal{C})$, i.e. there exists a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & X' & \xrightarrow{p} & X \\ \downarrow f & & \downarrow f' & & \downarrow f \\ Y & \xrightarrow{j} & Y' & \xrightarrow{q} & Y \end{array}$$

such that $pi = \text{id}_X$ and $qj = \text{id}_Y$.

V.1.10. Definition

Let $\mathcal{C}$ be a category and let Σ be a class of morphisms in $\mathcal{C}$.

- i) The class Σ satisfies the *two-out-of-six property* if the following holds: Given three composable morphisms

$$X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} X_2 \xrightarrow{f_3} X_3$$

such that $f_2 f_1$ and $f_3 f_2$ belong to Σ , then f_1 , f_2 , f_3 and $f_3 f_2 f_1$ also belong to Σ .

- ii) The class Σ is *closed under retracts*, i.e. every morphism which is a retract of a member of Σ also belongs to Σ .

V.1.11. Proposition

The class of weak equivalences satisfies the two-out-of-six property and is closed under retracts. $\square$

Proof. This follows directly since bijections satisfy two-out-of-six and are closed under retracts.

V.2. Fibrations via Horn-Filling

As already promised, there is an easier criterion to check whether a map of simplicial sets is a left or right fibration:

V.2.1. Theorem

Let $f: X \rightarrow Y$ be a map of simplicial sets.

- i) f is a left fibration if and only if it has the RLP with respect to the set of left horn inclusions

$$LH := \{\Lambda_i^n \rightarrow \Delta^n \mid n \geq 1, 0 \leq i < n\}.$$

- ii) f is a right fibration if and only if it has the RLP with respect to the set of right horn inclusions

$$RH := \{\Lambda_i^n \rightarrow \Delta^n \mid n \geq 1, 0 < i \leq n\}.$$

- iii) f is a Kan fibration if and only if it has the RLP with respect to the set of horn inclusions $H := LH \cup RH$.¹

The proof of the theorem relies on the observation that we may enlarge the class of morphisms against which we test the RLP without changing the class of morphisms which possess the RLP.

V.2.2. Lemma

Let $\mathcal{C}$ be a category with countable colimits and let $f: X \rightarrow Y$ be a morphism in $\mathcal{C}$.

- i) Suppose that f has the RLP with respect to $s: S \rightarrow T$ and let $g: S \rightarrow Z$ be arbitrary. Then f has the RLP with respect to the induced map $Z \rightarrow Z \cup_S T$.
- ii) Suppose that f has the RLP with respect to s' and that s is a retract of s' . Then f has the RLP with respect to s .
- iii) If $\{s_i: S_i \rightarrow T_i\}_{i \in I}$ is an arbitrary family of morphisms and f has the RLP with respect to every s_i , then f has the RLP with respect to the coproduct $\coprod_{i \in I} s_i: \coprod_{i \in I} S_i \rightarrow \coprod_{i \in I} T_i$.
- iv) Let $S_0 \xrightarrow{s_1} S_1 \xrightarrow{s_2} \dots$ be a sequence of composable arrows. If f has the RLP with respect to every s_n , then f has the RLP with respect to the canonical map $S_0 \rightarrow \text{colim}_n S_n$.

Proof. Draw the appropriate diagrams, see Marcus Land's "Introduction to infinity-categories", Lemma 3.8. $\square$

¹These are the standard definitions one will find in the literature.

V.2.3. Remark

Note that f has the RLP with respect to s if and only if s has the LLP with respect to f . Lemma V.2.2 can therefore also be read as a statement about preservation of the LLP.

V.2.4. Definition

Let $\mathcal{C}$ be a category and let Σ be a class of morphisms in $\mathcal{C}$. Then Σ is *saturated* if it is closed under the four constructions appearing in V.2.2:

- i) S is *closed under pushouts*: If $s: S \rightarrow T$ is in Σ and $g: S \rightarrow Z$ is an arbitrary morphism, then the induced morphism $Z \rightarrow Z \cup_S T$ is also in Σ .
- ii) S is *closed under retracts*: If s' is a retract of a morphism $s \in \Sigma$, then $s' \in \Sigma$.
- iii) S is *closed under coproducts*: If $\{s_i: S_i \rightarrow T_i\}_{i \in I}$ is a family of morphisms in Σ , then $\coprod_{i \in I} s_i: \coprod_{i \in I} S_i \rightarrow \coprod_{i \in I} T_i$ is in Σ .
- iv) S is *closed under countable compositions*: If $S_0 \xrightarrow{s_1} S_1 \xrightarrow{s_2} \dots$ is a sequence of composable arrows in S , then $S_0 \rightarrow \text{colim}_n S_n$ is in Σ .

Every class of morphisms Σ in a category $\mathcal{C}$ is contained in a smallest saturated class of morphisms $\text{sat}(\Sigma)$, which we call the *saturation* of Σ . Since any intersection of saturated classes is saturated,

$$\text{sat}(\Sigma) = \bigcap_{\substack{S \subseteq \text{mor}(\mathcal{C}) \\ \Sigma \subseteq S \text{ saturated}}} S.$$

Note that the indexing set of the intersection is non-empty since $\text{mor}(\mathcal{C})$ certainly is saturated.

With this terminology, lemma V.2.2 can be rephrased as saying that a morphism has the RLP with respect to a class Σ if and only if it has the RLP with respect to $\text{sat}(\Sigma)$.

V.2.5. Remark

Note that the saturation of a class of injective maps in $\mathbf{sSet}$ contains only injections.

There are numerous classes of morphisms we have considered already and whose saturation we might be interested in.

V.2.6. Definition

- i) The class B of *boundary inclusions* is given by

$$B := \{\partial \Delta^n \hookrightarrow \Delta^n \mid n \geq 1\}.$$

- ii) The class I of injections is given by

$$I := \{i: X \rightarrow Y \mid i \text{ injective}\}.$$

Observe that I is a saturated class. As a little warm-up, we consider the classes B and I .

V.2.7. Proposition

The following classes of morphisms coincide:

- i) $\text{sat}(B)$;
- ii) I .

Proof. Exercise. □

V.2.8. Corollary

If a map of simplicial sets has the RLP with respect to B , then it is a Kan fibration and a homotopy equivalence.

Proof. Let $f: X \rightarrow Y$ have the RLP with respect to B . By V.2.7, it has the RLP with respect to all inclusions, so it is a Kan fibration. Since $\emptyset \rightarrow Y$ is an injection, there exists a section $s: Y \rightarrow X$. Since $fsf = f$, we have a commutative diagram

$$\begin{array}{ccc} X \times \{0, 1\} & \xrightarrow{sf \sqcup \text{id}_X} & X \\ (\emptyset \rightarrow X) \boxtimes (\{0, 1\} \rightarrow \Delta^1) \downarrow & & \downarrow f \\ X \times \Delta^1 & \longrightarrow & Y \end{array}$$

where the map $X \times \Delta^1 \rightarrow Y$ is the constant homotopy of f . Since f has the RLP with respect to all injections, there exists a homotopy $X \times \Delta^1 \rightarrow X$ witnessing $sf \simeq \text{id}_X$. □

Obviously, $LC \subseteq C$, $RC \subseteq C$, $LH \subseteq H$ and $RH \subseteq H$. The analogous inclusions hold for the saturations of these classes. □

V.2.9. Proposition

Let $\widetilde{LC}$ denote the class of morphisms

$$\{(X \rightarrow Y) \boxtimes (\{0\} \hookrightarrow \Delta^1) \mid X \rightarrow Y \text{ injective}\}.$$

Then the following holds:

- i) $\widetilde{LC} \subseteq \text{sat}(LC)$;
- ii) $LC \subseteq \text{sat}(LH)$;
- iii) $LH \subseteq \text{sat}(LC)$;

Proof. For i), we use again that for every injection $j: X \rightarrow Y$ there is a filtration

$$X = Y^{(-1)} \subseteq Y^{(0)} \subseteq Y^{(1)} \subseteq Y^{(2)} \subseteq \dots \subseteq Y$$

such that

$$Y^{(n)} \cong Y^{(n-1)} \cup_{\coprod_{i \in I_n} \partial \Delta^n} \coprod_{i \in I_n} \Delta^n,$$

$\operatorname{colim}_n Y^{(n)} \cong Y$ and j becomes identified with the canonical map $X \rightarrow \operatorname{colim}_n Y^{(n)}$. Since pushouts commute with each other and $- \times \Delta^1$ commutes with pushouts, there exists for every $n \geq 0$ a pushout

$$\begin{array}{ccc} \coprod_{i \in I_n} ((\Delta^n \times \{0\})_{\partial \Delta^n \times \{0\}} (\partial \Delta^n \times \Delta^1)) & \longrightarrow & (Y \times \{0\})_{Y^{(n-1)} \times \{0\}} (Y^{(n-1)} \times \Delta^1) \\ \downarrow & & \downarrow \\ \coprod_{i \in I_n} (\Delta^n \times \Delta^1) & \longrightarrow & (Y \times \{0\})_{Y^{(n)} \times \{0\}} (Y^{(n)} \times \Delta^1) \end{array}$$

Consequently, the sequence of maps

$$\dots \rightarrow (Y \times \{0\})_{Y^{(n-1)} \times \{0\}} (Y^{(n-1)} \times \Delta^1) \rightarrow (Y \times \{0\})_{Y^{(n)} \times \{0\}} (Y^{(n)} \times \Delta^1) \rightarrow \dots$$

lies in $\operatorname{sat}(LC)$. It follows that

$$(Y \times \{0\})_{X \times \{0\}} (X \times \Delta^1) \rightarrow \operatorname{colim}_n ((Y \times \{0\})_{Y^{(n)} \times \{0\}} (Y^{(n)} \times \Delta^1)) \cong Y \times \Delta^1$$

also lies in $\operatorname{sat}(LC)$.

For **ii)**, it suffices to show that $(\partial \Delta^n \rightarrow \Delta^n) \boxtimes (\{0\} \rightarrow \Delta^1)$ can be obtained by filling left horns in the domain. Note that $\Delta^n \times \Delta^1 \cong N([n] \times [1])$, so simplices correspond to order-preserving maps $\alpha: [k] \rightarrow [n] \times [1]$. Such a simplex is non-degenerate if and only if α is injective.

For $k < n$, α factors as $[k] \rightarrow [n-1] \times [1] \xrightarrow{d_i \times \operatorname{id}} [n] \times [1]$ for some i . So every k -simplex is contained in $\partial \Delta^n \times \Delta^1$ for $k < n$.

Let $p: [n] \times [1] \rightarrow [n]$ and $q: [n] \times [1] \rightarrow [1]$ be the two projection maps. A n -simplex α is not contained in $(\Delta^n \times \{0\}) \cup_{\partial \Delta^n \times \{0\}} (\partial \Delta^n \times \Delta^1)$ if and only if $p\alpha = \operatorname{id}_{[n]}$ and $\operatorname{im}(q\alpha) \neq \{0\}$. Moreover, $(\Delta^n \times \{0\}) \cup_{\partial \Delta^n \times \{0\}} (\partial \Delta^n \times \Delta^1)$ does not contain any non-degenerate $n+1$ -simplices.

A $(n+1)$ -simplex α is uniquely determined by the number $j(\alpha) \in [n]$ satisfying $q\alpha(j(\alpha)) = 0$ and $q\alpha(j(\alpha) + 1) = 1$. Hence, there are precisely $n+1$ non-degenerate $(n+1)$ -simplices. These come with a canonical ordering $\alpha_0, \dots, \alpha_n$ by requiring that $j(\alpha_i) = i$.

Consider α_n . Then there is exactly one face which does not lie in $(\Delta^n \times \{0\}) \cup_{\partial \Delta^n \times \{0\}} (\partial \Delta^n \times \Delta^1)$, namely $d_n \alpha_n$. The remaining faces of α_n define a map

$$\Lambda_n^{n+1} \rightarrow (\Delta^n \times \{0\}) \cup_{\partial \Delta^n \times \{0\}} (\partial \Delta^n \times \Delta^1),$$

and filling this inner horn adds precisely α_n and $d_n \alpha_n$ to give a simplicial subset $X_n \subseteq \Delta^n \times \Delta^1$.

For α_{n-1} , observe that $d_{n+1} \alpha_{n-1}$ lies in $\partial \Delta^n \times \Delta^1$, $d_n \alpha_{n-1} = d_n \alpha_n$ and $d_k \alpha_{n-1}$ also lies in $\partial \Delta^n \times \Delta^1$ for $k < n-1$. So all faces of α_{n-1} except $d_{n-1} \alpha_{n-1}$ define a map $\Lambda_{n-1}^{n+1} \rightarrow X_n$. Filling this horn adds α_{n-1} and $d_{n-1} \alpha_{n-1}$ to X_n to obtain X_{n-1} .

Proceeding like this inductively, one ends up with α_0 . All faces of α_0 except $d_0 \alpha_0$ have been added in previous steps. Filling the corresponding (left!) horn $\Lambda_0^{n+1} \rightarrow X_1$ adds the last two missing simplices α_0 and $d_0 \alpha_0$ to X_1 to give $X_0 = \Delta^n \times \Delta^1$. There are closed formulas for all the face relations which are not difficult to figure out. If you want to look them up, see Goerss and Jardine's "Simplicial homotopy theory", proposition 4.2. This shows **ii)**.

Finally, consider the inclusion of a left horn $\Lambda_k^n \rightarrow \Delta^n$, i.e. $k < n$. Then there is a retract diagram

$$\begin{array}{ccccc} \Lambda_k^n & \xrightarrow{j|_{\Lambda_k^n}} & (\Delta^n \times \{0\}) \cup_{\Lambda_k^n \times \{0\}} (\Lambda_k^n \times \Delta^1) & \xrightarrow{r_k|_{\dots}} & \Lambda_k^n \\ \downarrow & & \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{j} & \Delta^n \times \Delta^1 & \xrightarrow{r_k} & \Delta^n \end{array}$$

in which j is given by the inclusion $\Delta^n \cong \Delta^n \times \{1\} \rightarrow \Delta^n \times \Delta^1$ and r_k is induced by the map of posets

$$r_k : [n] \times [1] \rightarrow [n], \quad (l, i) \mapsto \begin{cases} l & i = 1 \text{ or } l < k, \\ k & \text{else} \end{cases}.$$

Note that $r_k j = \text{id}$, and r_k does not restrict as indicated in the diagram: we only need to check that the inclusion $[n] \setminus \{k\} \subseteq \text{im}(r_k \alpha)$ is impossible. If $\alpha : [p] \rightarrow [n] \times \{0\}$ is a simplex, then $r_k \alpha$ is a map $[p] \rightarrow [k]$. Since $k < n$, $\text{im}(r_k \alpha)$ cannot contain n . Suppose that $\alpha : [p] \rightarrow [n] \times [1]$ is a simplex in $\Lambda_k^n \times \Delta^1$. Then $\text{im}(r_k \alpha) \subseteq \text{im}(q \alpha) \cup \{k\}$, where $q : [n] \times [1] \rightarrow [n]$ is the projection. Since $[n] \setminus \{k\}$ is not a subset of $\text{im}(q \alpha)$, it cannot be a subset of $\text{im}(r_k \alpha)$ either. This proves [iii](#). $\square$

V.2.10. Corollary

$$\text{sat}(LH) = \text{sat}(LC) = \text{sat}(\widetilde{LC})$$

Proof. note that $LC \subseteq \widetilde{LC}$. By [V.2.9](#), we have

$$\text{sat}(LH) \subseteq \text{sat}(LC) \subseteq \text{sat}(\widetilde{LC}) \subseteq \text{sat}(LC) \subseteq \text{sat}(LH).$$

$\square$ Analogous statements hold for the right cylinders and right horn inclusions. Instead of saying that all proofs work identically, we employ a little trick.

V.2.11. Definition

Define an automorphism $\iota : \Delta \rightarrow \Delta$ by setting $\iota([n]) := [n]$ and $\iota(\alpha : [m] \rightarrow [n]) := \iota_n^{\text{op}} \alpha^{\text{op}} \iota_m$, where

$$\iota_n : [n] \rightarrow [n]^{\text{op}}, \quad i \mapsto n - i.$$

Denote the induced isomorphism of categories $- \circ \iota^{\text{op}} : \text{sSet} \rightarrow \text{sSet}$ by $(-)^{\text{op}}$ and call the image X^{op} of a simplicial set under this functor the *opposite* of X .

V.2.12. Example

$$\iota(d_i) = d_{n-i}.$$

V.2.13. Remark

1. For a category $\mathcal{C}$, there is a natural isomorphism $N(\mathcal{C})^{\text{op}} \cong N(\mathcal{C}^{\text{op}})$.
2. The functor ι preserves inner horn inclusions, so X^{op} is a quasi-category if X is a quasi-category.

V.2.14. Corollary

The following classes of morphisms coincide:

1. $\text{sat}(RC)$;
2. $\text{sat}(RH)$;
3. the saturation of

$$\widetilde{RC} := \{(X \rightarrow Y) \boxtimes (\{1\} \hookrightarrow \Delta^1) \mid Y \rightarrow X \text{ injective}\}.$$

Proof. Note that $(-)^{\text{op}}$ induces bijections $LC \sim RC$, $L \sim R$ and $\widetilde{LC} \sim \widetilde{RC}$. As an isomorphism of categories, $(-)^{\text{op}}$ also preserves saturations, so the corollary follows from V.2.10. $\square$

Combining V.2.10 and V.2.14, we immediately obtain the following:

V.2.15. Corollary

The following classes of morphisms coincide:

1. $\text{sat}(C)$;
2. $\text{sat}(H)$;
3. the saturation of

$$\widetilde{C} := \{X \times \{i\} \cup Y \times \Delta^1 \hookrightarrow X \times \Delta^1 \mid Y \rightarrow X \text{ injective}, i = 0, 1\}.$$

Proof. By V.2.9 (and its “opposite”), we have the following inclusions:

1. $LH \cup RH \subseteq \text{sat}(LC) \cup \text{sat}(RC) \subseteq \text{sat}(LC \cup RC)$;
2. $\widetilde{LC} \cup \widetilde{RC} \subseteq \text{sat}(LH) \cup \text{sat}(RH) \subseteq \text{sat}(LH \cup RH)$;
3. $LC \cup RC \subseteq \widetilde{LC} \cup \widetilde{RC}$.

Hence, we have

$$\text{sat}(C) \subseteq \text{sat}(\widetilde{LC} \cup \widetilde{RC}) \subseteq \text{sat}(H) \subseteq \text{sat}(C).$$

$\square$

V.2.16. Definition

The class of morphisms described in V.2.15 is called the class of *anodyne extensions*.² Similarly, V.2.10 and V.2.14 describe the classes of *left anodyne extensions* and *right anodyne extensions*, respectively.

Proof of theorem V.2.1. Immediate from V.2.2 and V.2.15.

²The terminology is due to Gabriel-Zisman. “Anodyne” is Greek and translates “painless”.

V.2.17. Remark

In particular, this shows that X is a Kan complex if and only if $X \rightarrow \Delta^0$ is a Kan fibration. Since all inclusions $F_{\mathcal{L}(\Lambda_k^n)}(0, n) \subseteq F_{\mathcal{L}(\Delta^n)}(0, n)$ lie in $\tilde{C}$ by IV.34, the proof of theorem IV.31 is now complete.

V.2.18. Corollary

A map of simplicial sets is

1. a left fibration if and only if it has the RLP with respect to all left anodyne extensions;
2. a right fibration if and only if it has the RLP with respect to all right anodyne extensions;
3. a Kan fibration if and only if it has the RLP with respect to all anodyne extensions.

V.2.19. Corollary

For every simplicial set X there exists an anodyne extension $X \rightarrow \overline{X}$, where $\overline{X}$ is a Kan complex.

Proof. For any simplicial set X , define X' by pushout

$$\begin{array}{ccc} \coprod_f: \Lambda_i^n \rightarrow X & \longrightarrow & X \\ \downarrow & & \downarrow \\ \coprod_f: \Lambda_i^n \rightarrow X & \longrightarrow & X' \end{array}$$

where the coproduct runs over all maps from all possible horns to X . Note that $X \rightarrow X'$ is an anodyne extension.

Given a fixed simplicial set X , let $X_0 := X$ and define $X_{n+1} := \overline{X_n}$. Then the Yoneda lemma implies that every horn in $\overline{X} := \text{colim}_n X_n$ can be filled. Moreover, the canonical map $X \rightarrow \overline{X}$ is a left anodyne extension.

Using V.2.3, V.2.2 is enough to show that cofibrations admit a much easier description than our original definition suggests.

V.2.20. Theorem

A map of simplicial sets is a left cofibration if and only if it is injective.³

Proof. Exercise: Fix an injection j . Let Σ be a saturated class of morphisms. Then the class of morphisms

$$S := \{i \mid i \boxtimes j \in \Sigma\}$$

is saturated.

Consequently, the class of left cofibrations is saturated. Since $\partial\Delta^n \rightarrow \Delta^n$ is a left cofibration

³As in the case of fibrations, this is usually the definition of a cofibration.

for all $n \geq 0$, [V.2.10](#) implies that every injection is a left cofibration.

Let $i: A \rightarrow B$ be a cofibration. By [V.2.19](#), there is an anodyne extension

$$B \times \{0\} \cup_{A \times \{0\}} A \times \Delta^1 \rightarrow K$$

for some Kan complex K . Since i is a left cofibration, this injection extends to a map $B \times \Delta^1 \rightarrow K$. Consequently, $B \times \{0\} \cup_{A \times \{0\}} A \times \Delta^1 \rightarrow B \times \Delta^1$ is injective. Since i can be recovered from this map via the inclusion of $A \times \{1\}$, it follows that i is injective. $\square$

V.2.21. Corollary

The classes of left cofibrations, right cofibrations and injections coincide.

From now on, we will only (and interchangeably) use the terms “injection” and “cofibration”.

V.2.22. Proposition

Let $i: A \rightarrow B$ be a left anodyne extension and let $j: X \rightarrow Y$ be a cofibration. Then

$$(A \times Y) \cup_{A \times X} (B \times X) \rightarrow B \times Y$$

is left anodyne.

Proof. For fixed j , the class of morphisms

$$S := \{k: A \rightarrow B \mid k \boxtimes j \text{ left anodyne}\}$$

is saturated by the exercise set in the proof of [V.2.20](#).

By [V.2.10](#), it suffices to check that $l \boxtimes (\{0\} \rightarrow \Delta^1)$ lies in S for all cofibrations $l: A \rightarrow B$. Since $- \times X$ and $- \times Y$ commute with pushouts, the morphism

$$\left((B \times \{0\}) \cup_{A \times \{0\}} (A \times \Delta^1) \right) \times Y \cup_{((B \times \{0\}) \cup_{A \times \{0\}} (A \times \Delta^1)) \times X} (B \times \Delta^1) \times X \rightarrow (B \times \Delta^1) \times Y$$

can be identified with the morphism

$$\left((B \times Y \times \{0\}) \cup_{A \times Y \times \{0\}} (A \times Y \times \Delta^1) \right) \cup_{(B \times X \times \{0\}) \cup_{A \times X \times \{0\}} (A \times X \times \Delta^1)} (B \times X \times \Delta^1) \rightarrow (B \times Y \times \Delta^1)$$

which is left anodyne by [V.2.10](#).

V.2.23. Corollary

If $i: A \rightarrow B$ is a cofibration and $p: X \rightarrow Y$ is a left/right/Kan fibration, then the induced map

$$F(B, X) \rightarrow F(A, X) \times_{F(A, Y)} F(B, Y)$$

is a left/right/Kan fibration. In particular, the following holds:

i) If B is any simplicial set and $p: X \rightarrow Y$ is a left/right/Kan fibration, then

$$p^*: F(B, X) \rightarrow F(B, Y)$$

is a left/right/Kan fibration.

ii) If $i: A \rightarrow B$ is a cofibration and X is a Kan complex, then

$$i^*: F(B, X) \rightarrow F(A, X)$$

is a Kan fibration of Kan complexes.

Proof. Every lifting problem

$$\begin{array}{ccc} \Lambda_k^n & \longrightarrow & F(B, X) \\ \downarrow & & \downarrow \\ \Delta^n & \longrightarrow & F(A, X) \times_{F(A, Y)} F(B, Y) \end{array}$$

corresponds to a commutative diagram

$$\begin{array}{ccc} (B \times \Lambda_k^n) \cup_{A \times \Lambda_k^n} (A \times \Delta^n) & \longrightarrow & X \\ \downarrow & & \downarrow p \\ B \times \Delta^n & \longrightarrow & Y \end{array}$$

The left vertical map is an anodyne extension by [V.2.22](#), and hence has the LLP with respect to p . The resulting lift provides the required lift of the original lifting problem. $\square$

V.2.24. Corollary

Let K be a Kan complex and let X be a simplicial set. Then $F(X, K)$ is a Kan complex.

Proof. This is [V.2.23 i\)](#) for $p: K \rightarrow \Delta^0$. $\square$

[V.2.24](#) shows that the simplicially enriched category $\mathbf{Kan}$ of Kan complexes is enriched in Kan complexes. So the following definition is justified.

V.2.25. Definition

Define the quasi-category of *anima* to be

$$\mathbf{An} := \mathcal{N}^c(\mathbf{Kan}).$$

Note that the name $\mathbf{An}$ is abusive at this point in time. we have defined “anima” is such a way that it is a theorem that “anima” is synonymous to “Kan complex”. Our notation anticipates that we will prove this eventually.

Also, let us emphasise that at the time of writing the name “anima” is very much non-standard. Most people prefer the term “spaces” and write $\mathcal{S}$ or $\mathbf{Spc}$ instead of $\mathbf{An}$.

V.2.26. Corollary

Let $i: A \rightarrow B$ be a left/right/ $\emptyset$ anodyne extension and let $p: X \rightarrow Y$ be a left/right/Kan fibration. Then the induced map

$$F(B, X) \rightarrow F(A, X) \times_{F(A, Y)} F(B, Y)$$

has the RLP with respect to I . In particular, it is a Kan fibration and a homotopy equivalence (see V.2.8).

Proof. Since $\text{sat}(B) = I$, it is enough to consider B . Any lifting problem

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & F(B, X) \\ \downarrow & \nearrow & \downarrow \\ \Delta^n & \longrightarrow & F(A, X) \times_{F(A, Y)} F(B, Y) \end{array}$$

corresponds to a lifting problem

$$\begin{array}{ccc} (A \times \Delta^n) \cup_{A \times \partial\Delta^n} (B \times \partial\Delta^n) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow p \\ B \times \Delta^n & \longrightarrow & Y \end{array}$$

The left vertical map is left/right/ $\emptyset$ anodyne by V.2.22, so there exists a solution. $\square$

V.2.27. Corollary

Every anodyne extension is a cofibration and a weak equivalence.

Proof. We only need to show that every anodyne extension is a weak equivalence. This follows from V.2.26 for $Y = \Delta^0$ by taking π_0 .

V.2.28. Definition

- i) A category $\mathcal{I}$ is *filtered* if for every diagram $F: \mathcal{J} \rightarrow \mathcal{I}$ where $\mathcal{J}$ has only finitely many objects and morphisms there exists an object $i \in \mathcal{I}$ and a natural transformation $F \Rightarrow \text{const}_i$.
- ii) A *filtered colimit* is a colimit of a diagram indexed by a filtered category.
- iii) Let $\mathcal{C}$ be a category with all filtered colimits. Then $c \in \mathcal{C}$ is *compact* if $\text{Hom}_{\mathcal{C}}(c, -)$ commutes with all filtered colimits.
- iv) Let $\mathcal{C}$ be a cocomplete category. Then $c \in \mathcal{C}$ is *tiny* if $\text{Hom}_{\mathcal{C}}(c, -)$ commutes with all colimits.

V.2.29. Examples

- i) Every directed poset is a filtered category, e.g. $\mathbb{N}$.
- ii) A set X is compact in $\mathbf{Set}$ if and only if X is finite.
- iii) A simplicial set X is compact in $\mathbf{sSet}$ if and only if X is finite.
- iv) A module M over some ring R is compact in $\mathbf{R-Mod}$ if and only if M is finitely generated.

The following statement is often referred to as the “small object argument”:

V.2.30. Proposition

Let $\mathcal{C}$ be a cocomplete category. Let S be a set of morphisms such that the domain of each morphism in S is compact. Then every morphism $f: X \rightarrow Y$ of simplicial sets can be factored as $f = p \circ i$, where i lies in $\text{sat}(S)$ and p has the RLP with respect to S .

Proof. For an arbitrary map $g: V \rightarrow W$, let

$$\text{LP}(S, g) := \{(a: A \rightarrow V, b: B \rightarrow W, i: A \rightarrow B) \mid i \in S, g \circ a = b \circ i\}.$$

Define V' as the pushout

$$\begin{array}{ccc} \coprod_{(a,b,i) \in \text{LP}(S,g)} A & \xrightarrow{\Sigma a} & V \\ \downarrow \coprod i & & \downarrow g \\ \coprod_{(a,b,i) \in \text{LP}(S,g)} B & \xrightarrow{\Sigma b} & V' \end{array}$$

Then there is a canonically induced map $V' \rightarrow W$, and the map $V \rightarrow V'$ lies in the saturation of S .

Now define $X^0 := X$ and $X^{n+1} := (X^n)'$. Set $E := \text{colim}_n X^n$. By definition, the induced map $i: X \rightarrow E$ is in $\text{sat}(S)$. Moreover, the induced map $p: E \rightarrow Y$ satisfies $p \circ i = f$.

Consider a lifting problem

$$\begin{array}{ccc} A & \xrightarrow{a} & V \\ \downarrow j & \nearrow i & \downarrow f \\ B & \xrightarrow{b} & W \end{array}$$

with $j \in S$. Since A is compact by assumption, a factors as

$$a: A \rightarrow X^n \rightarrow \text{colim}_n X^n = E$$

for some $n \in \mathbb{N}$. By construction, the induced lifting problem

$$\begin{array}{ccc} A & \xrightarrow{a'} & X^{n+1} \\ \downarrow j & \nearrow i & \downarrow f \\ B & \xrightarrow{b} & Y \end{array}$$

has a solution, and this also provides a solution to the original lifting problem. $\square$

Note that this contains [V.2.19](#) as a special case (take $S = H$ and f to be the projection map $X \rightarrow \Delta^0$).

We can now prove a converse to [V.2.18](#).

V.2.31. Corollary

Let $\mathcal{C}$ be a cocomplete category and S a set of maps such that the domain of every morphism in S is compact. Then

$$\text{sat}(S) = \left\{ f: X \rightarrow Y \mid \begin{array}{l} f \text{ has the LLP with respect to the class of morphisms} \\ \text{which have the RLP with respect to } S \end{array} \right\} =: \bar{S}$$

In particular, a morphism is a left/right/ $\emptyset$ anodyne extension if and only if it has the LLP with respect to every left/right/Kan fibration.

Proof. Since $\bar{S}$ is a saturated class by V.2.2 and contains S , we have $\text{sat}(S) \subseteq \bar{S}$. Let $f: X \rightarrow Y$ be in $\bar{S}$. By V.2.30, factor f into a map $i: X \rightarrow E$ in $\text{sat}(S)$ followed by a map $p: E \rightarrow Y$ which has the RLP with respect to S . The lifting problem

$$\begin{array}{ccc} X & \xrightarrow{i} & E \\ \downarrow f & \nearrow s & \downarrow p \\ Y & \xrightarrow{\text{id}} & Y \end{array}$$

has a solution since f has the LLP with respect to p . These maps define a retract diagram

$$\begin{array}{ccccc} X & \xrightarrow{\text{id}} & X & \xrightarrow{\text{id}} & X \\ \downarrow f & & \downarrow i & & \downarrow f \\ Y & \xrightarrow{s} & E & \xrightarrow{p} & Y \end{array}$$

So f is in $\text{sat}(S)$. □

V.3. Simplicial homotopy groups

V.3.1. Proposition

Let K be a Kan complex. Then πX is a groupoid. In particular, homotopy is an equivalence relation on $\text{Hom}_{\text{sSet}}(\Delta^0, K)$.

Proof. Let $x, y \in X$ and let e be an edge such that $d_0(e) = y$ and $d_1(e) = x$. Consider e and $s_0(x)$ as a map $\Lambda_0^2 \rightarrow X$. For any filler F , the edge $d_0(F)$ provides a morphism $y \rightarrow x$ in πX such that $d_0 F \circ e = \text{id}_x$. Similarly, e and $s_0(x)$ also define a map $\Lambda_2^2 \rightarrow X$, and d_2 of any filler provides a right inverse to e . Since isomorphisms satisfy two-out-of-six, e is an isomorphism in πX .

V.3.2. Corollary

Let K be a Kan complex, let X be a simplicial set, and let A be a simplicial subset of X . Then homotopy relative A is an equivalence relation on $\text{Hom}_{\text{sSet}}(X, K)$.

Proof. Let $f: A \rightarrow K$ be given and consider the pullback

$$\begin{array}{ccc} F(X, K)_f & \longrightarrow & F(X, K) \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{f} & F(A, K) \end{array}$$

Then $F(X, K)_f$ is a Kan complex by [V.2.23](#). Homotopy of 0-simplices in $F(X, K)_f$ corresponds precisely to homotopy relative A of maps $X \rightarrow K$ which restrict to the given map f on A . So the corollary follows from [V.3.1](#). $\square$

V.3.3. Corollary

Every weak equivalence between Kan complexes is a homotopy equivalence.

Proof. Let $f: K \rightarrow L$ be a weak equivalence between Kan complexes. Since $f^*: \pi_0(L, K) \rightarrow \pi_0(K, K)$ is a bijection, there exists by [V.3.2](#) a map $g: L \rightarrow K$ such that $gf \simeq \text{id}_K$. Since $f g f \simeq f$, the injectivity of f^* implies that $f g \simeq \text{id}_L$ as well. $\square$

V.3.4. Definition

Let $A \subseteq X$ and $B \subseteq Y$ be inclusions of simplicial subsets. Define $F((X, A), (Y, B))$ as the pullback

$$\begin{array}{ccc} F((X, A), (Y, B)) & \longrightarrow & F(X, Y) \\ \downarrow & & \downarrow \\ F(A, B) & \longrightarrow & F(A, Y) \end{array}$$

Note that $F((X, A), (Y, B))$ is a Kan complex if B and Y are Kan complexes (by [V.2.23](#) and [V.2.24](#)).

V.3.5. Definition

For $n \geq 0$, let

$$\square^n := (\Delta^1)^n \cong N([1]^n)$$

be the n -cube. The simplicial subset

$$\partial \square^n := \{\alpha: [k] \rightarrow [1]^n \mid \exists 1 \leq i \leq n: \text{pr}_i \alpha \text{ constant}\}$$

is called the *boundary* of the n -cube. We agree that $\partial \square^0 = \emptyset$.

V.3.6. Definition

Let X be a Kan complex and $x \in X_0$. Define

$$\pi_n(X, x) := \pi_0(F((\square^n, \partial \square^n), (X, x)))$$

Let Y be a Kan complex and $y \in Y$. Define

$$\pi_1(Y, y) := \text{End}_{\pi_Y}(y)$$

Unwinding definitions and using [V.2.30](#), there is a natural isomorphism

$$\pi_n(X, x) \cong \pi_1(F((\square^{n-1}, \partial \square^{n-1}), (X, x)), c_x)$$

for all $n \geq 1$, where c_x denotes the constant map with value x . So the composition law in the homotopy category equips $\pi_x(X, x)$ with the structure of a monoid.

Conjugating the identification

$$\pi_n(X, x) \cong \pi_1(F((\square^{n-1}, \partial \square^{n-1}), (X, x)), c_x)$$

by the transposition $\tau_{1,i}$ acting on the factors of $(\Delta^1)^n$, we in fact obtain n monoid structures $\{\cdot_i\}_{i \leq n}$ on $\pi_n(X, x)$ which have the same neutral element (namely the class represented by the constant map).

V.3.7. Proposition

Let X be a Kan complex and $x \in X$. Then $\pi_n(X, x)$ is a group for all $n \geq 1$. For $n \geq 2$, the compositions $\cdot_i$ all agree and are commutative.

Proof. Since $F((\square^{n-1}, \partial \square^{n-1}), (X, x))$ is a Kan complex for all $n \geq 1$, the first assertion follows from the fact that πY is a groupoid for every Kan complex Y .

Let $\alpha, \beta, \gamma, \delta: (\square^n, \partial \square^n) \rightarrow (X, x)$ represent elements in $\pi_n(X, x)$ for $n \geq 2$. We claim that

$$([\alpha] \cdot_1 [\beta]) \cdot_2 ([\gamma] \cdot_1 [\delta]) = ([\alpha] \cdot_2 [\gamma]) \cdot_1 ([\beta] \cdot_2 [\delta]).$$

If this holds, the Eckmann-Hilton argument shows that $\cdot_1 = \cdot_2$ and that this composition is abelian.

The four given elements induce a map $\Lambda_1^2 \times \Lambda_1^2 \times \square^{n-2} \rightarrow X$. Decoding the definition of the composition laws, the left hand composition amounts to extending this map along the anodyne extensions

$$\Lambda_1^2 \times \Lambda_1^2 \times \square^{n-2} \rightarrow \Delta^2 \times \Lambda_1^2 \times \square^{n-2} \rightarrow \Delta^2 \times \Delta^2 \times \square^{n-2},$$

while the right hand side corresponds to extending along the anodyne extensions

$$\Lambda_1^2 \times \Lambda_1^2 \times \square^{n-2} \rightarrow \Lambda_1^2 \times \Delta^2 \times \square^{n-2} \rightarrow \Delta^2 \times \Delta^2 \times \square^{n-2}.$$

Note that we are ignoring various boundary conditions for legibility. Both extensions provide an extension along the anodyne extension

$$\Lambda_1^2 \times \Lambda_1^2 \times \square^{n-2} \rightarrow \Delta^2 \times \Delta^2 \times \square^{n-2}.$$

Since such an extension is unique up to homotopy (V.2.26), this proves the claim.

For arbitrary $\cdot_i$ and $\cdot_j$, the same proof works after appropriate permutation of the factors. $\square$

Of course, one can construct the group structure on $\pi_n(X, x)$ directly and observe that there are n different compositions on $\pi_n(X, x)$. As soon as there are at least two, the Eckmann-Hilton argument applies.

Each π_n defines a functor

$$\pi_n : \text{Kan}_* \rightarrow \text{Grp}$$

which is invariant under pointed simplicial homotopies.

V.3.8. Theorem

Let $p: X \rightarrow Y$ be a Kan fibration of Kan complexes and $x \in X_0$. Let F denote the fibre of p over $p(x)$. Then there exists a natural long exact sequence

$$\dots \rightarrow \pi_{n+1}(X, x) \xrightarrow{p_*} \pi_{n+1}(Y, p(x)) \xrightarrow{\partial} \pi_n(F, x) \xrightarrow{i_*} \pi_n(X, x) \xrightarrow{p_*} \pi_n(Y, p(x)) \rightarrow \dots$$

In low degrees, exactness means the following:

- i) There is an action $\pi_1(Y, p(x)) \times \pi_0(F, x) \rightarrow \pi_0(F, x)$ such that
- the image of $\pi_*: \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is precisely the stabilizer of $[x] \in \pi_0(F)$;
 - two elements in $\pi_0(F, x)$ map to the same element in $\pi_0(X, x)$ if and only if they lie in the same orbit of this action.
- ii) An element in $\pi_0(X, x)$ maps to the basepoint in $\pi_0(Y, p(x))$ if and only if they lie in the image of $\pi_0(F, x) \rightarrow \pi_0(X, x)$.

Proof. Since we are using cubes to model homotopy groups, the proof is more or less identical to the proof for topological spaces. Set $y := p(x)$. We begin by defining the boundary map. Let $\alpha: (\square^{n+1}, \partial\square^{n+1}) \rightarrow (Y, y)$ represent any element in $\pi_{n+1}(Y, y)$. Since $(\partial\square^n \rightarrow \square^n) \boxtimes (\{0\} \rightarrow \Delta^1)$ is anodyne, the lifting problem

$$\begin{array}{ccc} (\Delta^1 \times \partial\square^n) \cup_{\{0\} \times \partial\square^n} (\{0\} \times \square^n) & \xrightarrow{\text{const}_x} & X \\ \downarrow & \nearrow \bar{\alpha} & \downarrow p \\ \Delta^1 \times \square^n & \xrightarrow{\alpha} & Y \end{array}$$

has a solution. Set

$$\partial[\alpha] := [\bar{\alpha}]_{\{1\} \times \square^n}.$$

This is well-defined: let $h: \square^{n+1} \times \Delta^1 \cong \Delta^1 \times \square^n \times \Delta^1 \rightarrow Y$ be a homotopy relative $\partial\square^{n+1}$ from α to α' . Any two lifts $\bar{\alpha}$ and $\bar{\alpha}'$ as above define a map $(\Delta^1 \times \square^n \times \partial\Delta^1) \rightarrow X$. This map glues with the constant map on

$$((\Delta^1 \times \partial\square^n) \cup_{\{0\} \times \partial\square^n} (\{0\} \times \square^n)) \times \Delta^1$$

to define a partial lift of h . Since $(\{0\} \rightarrow \Delta^1) \boxtimes (\partial\square^n \rightarrow \square^n)$ is anodyne, so is

$$((\{0\} \rightarrow \Delta^1) \boxtimes (\partial\square^n \rightarrow \square^n)) \boxtimes (\partial\Delta^1 \rightarrow \Delta^1).$$

Any lift $\bar{h}: \Delta^1 \times \square^n \times \Delta^1 \rightarrow X$ witnesses that α and α' are homotopic relative $\partial\square^n$. So ∂ is well-defined. If $n \geq 1$, we claim that ∂ is a group homomorphism. Two elements $[\alpha], [\beta]$ define a map

$$\Delta^1 \times \Lambda_1^2 \times \square^{n-1} \rightarrow Y.$$

Similarly, the lifts $\bar{\alpha}$ and $\bar{\beta}$ glue to a map on $\Delta^1 \times \Lambda_1^2 \times \square^{n-1}$ which has an extension to a map $\gamma: \Delta^1 \times \Delta^2 \times \square^{n-1} \rightarrow X$. Then $\gamma|_{\{1\} \times d_1\Delta^2 \times \square^{n-1}}$ defines $\partial[\alpha] \cdot \partial[\beta]$. On the other hand, $p \circ \gamma$ is an extension of the original map, and $(p \circ \gamma)|_{\{1\} \times d_1\Delta^2 \times \square^{n-1}}$ defines $[\alpha] \cdot_2 [\beta]$, so

$$\partial([\alpha] \cdot_2 [\beta]) = \partial[\alpha] \cdot \partial[\beta].$$

Since $\cdot_2 = \cdot$, this proves that ∂ is a homomorphism. Let $e \in F_0$ and $[\alpha] \in \pi_1(Y, y)$. Solve the lifting problem

$$\begin{array}{ccc} \Delta^0 & \xrightarrow{e} & X \\ \downarrow & \nearrow \bar{\alpha} & \downarrow p \\ \Delta^1 & \xrightarrow{\alpha} & Y \end{array}$$

to define $[\alpha][e] := [\bar{\alpha}(1)]$. It is easy to check that this is well-defined and defines a group action. We recover the definition of the connecting map by letting $\pi_1(Y, y)$ act on $[x]$. We begin by showing exactness in low degrees.

Exactness at $\pi_0(X)$: The composition $\pi_0(F) \rightarrow \pi_0(X) \rightarrow \pi_0(Y)$ obviously hits the basepoint. Suppose that $x' \in X$ such that $[p(x')] = [y] \in \pi_0(Y)$. Then the lifting problem

$$\begin{array}{ccc} \Delta^0 & \xrightarrow{x'} & X \\ \downarrow & \nearrow h & \downarrow p \\ \Delta^1 & \xrightarrow{p(x') \simeq y} & Y \end{array}$$

has a solution, and $h(1)$ is a point in F such that $[h(1)] = [x'] \in \pi_0(X)$.

Exactness at $\pi_0(F)$: Suppose that $[e], [e'] \in p_0(F)$ satisfy $[\alpha][e] = [e']$ for some $[\alpha] \in \pi_1(Y, y)$. Then, by definition, there exists $\bar{\alpha}: \Delta^1 \rightarrow X$ such that $\bar{\alpha}(0) = e$ and $\bar{\alpha}(1)$ is homotopic to $[e']$ in F . This implies that e and e' are homotopic in X . Conversely, suppose that $[e] = [e'] \in \pi_0(X)$. Then there exists a homotopy $h: \Delta^1 \rightarrow X$ from e to e' . The map ph defines an element in $\pi_1(Y, y)$ since $p(e) = y = p(e')$. By definition, we have $[ph][e] = [e']$. So the fibres of $\pi_0(F) \rightarrow \pi_0(X)$ correspond precisely to the orbits of the $\pi_1(Y, y)$ -action on $\pi_0(F)$.

Exactness at $\pi_1(Y, y)$: Let $[\alpha] \in \pi_1(X, x)$. Then α itself is a solution to the lifting problem

$$\begin{array}{ccc} \Delta^0 & \xrightarrow{x} & X \\ \downarrow & \nearrow & \downarrow p \\ \Delta^1 & \xrightarrow{p\alpha} & Y \end{array}$$

so $[p\alpha][x] = [x]$. Conversely, suppose that $[\alpha] \in \pi_1(Y, y)$ satisfies $[\alpha][x] = [x]$. Then α lifts by definition of the action to a loop in X . So the image of $p_*: \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is precisely the stabilizer of $[x] \in \pi_0(F)$.

Exactness at $\pi_n(X, x)$ for $n \geq 1$: The composition p_*i_* is obviously trivial. Suppose that $p_*[\alpha] = 1 \in \pi_n(Y, y)$. Then there exists a homotopy $h: \square^n \times \Delta^1 \rightarrow Y$ relative $\partial\square^n$ from $p\alpha$ to the constant map with value y . Since p is a Kan fibration, the lifting problem

$$\begin{array}{ccc} (\square^n \times \{0\}) \cup_{\partial\square^n \times \{0\}} (\partial\square^n \times \Delta^1) & \xrightarrow{\alpha \cup \text{const}_x} & X \\ \downarrow & & \downarrow p \\ \square^n \times \Delta^1 & \xrightarrow{h} & Y \end{array}$$

has a solution. The endpoint of this solution provides a preimage of $[\alpha]$ under i_* .

Exactness at $\pi_n(F, x)$ for $n \geq 1$: Let $[\alpha] \in \pi_{n+1}(Y, y)$. Then $\partial[\alpha]$ is by construction represented by a map which is homotopic to const_x relative $\partial\square^n$ in X . Consequently, $i_*\partial$ is trivial. Conversely, suppose that $[i\alpha]$ is trivial in $\pi_n(X, x)$ for some $[\alpha] \in \pi_n(F, x)$. Then there exists a homotopy (note we are free to choose the direction of the homotopy) $h: \Delta^1 \times \square^n \rightarrow X$ relative $\partial\square^n$ from const_x to $i\alpha$. Then, by definition, h represents an element in $\pi_{n+1}(Y, y)$ with $\partial[h] = [\alpha]$.

Exactness at $\pi_{n+1}(Y, y)$ for $n \geq 1$: Let $[\alpha] \in \pi_{n+1}(X, x)$. Then α solves the lifting problem used to define $\partial[p\alpha]$. Since α is constant on $\partial\square^n$, this proves that ∂p_* is trivial. Conversely, suppose that $\partial[\alpha] = [\bar{\alpha}|_{\{1\} \times \square^n}] = 1$ for some $[\alpha] \in \pi_{n+1}(Y, y)$. Then there exists a homotopy

$h: \Delta^1 \times \square^n \rightarrow F$ relative $\partial \square^n$ from $\bar{\alpha}|_{\{1\} \times \square^n}$ to const_x . Since p is a Kan fibration, the lifting problem

$$\begin{array}{ccc} (\Lambda_1^2 \times \square^n) \cup_{\Lambda_1^2 \times \partial \square^n} (\Delta^2 \times \partial \square^n) & \xrightarrow{(\alpha \cup h) \cup \text{const}_x} & X \\ \downarrow & & \downarrow p \\ \Delta^2 \times \square^n & \xrightarrow{\alpha(s_1 \times \text{id})} & Y \end{array}$$

has a solution $\bar{h}$, where s_1 denotes the appropriate degeneration map. Then $\bar{h}|_{d_1 \Delta^2 \times \square^n}$ defined an element in $\pi_{n+1}(X, x)$ such that $[p\bar{h}|_{d_1 \Delta^2 \times \square^n}] = [\alpha]$. $\square$

V.3.9. Lemma

Let $f: (X, x) \rightarrow (Y, y)$ be a pointed map of pointed Kan complexes. Then f is a simplicial homotopy equivalence if and only if it is a pointed simplicial homotopy equivalence.

Proof. The basepoint inclusions induce Kan fibrations

$$\text{ev}_x: F(X, Z) \rightarrow Z, \quad \text{ev}_y: F(Y, Z) \rightarrow Z$$

for every Kan complex Z . For $z \in Z_0$, the fibres of ev_x and ev_y over z are precisely $F((X, x), (Z, z))$ and $F((Y, y), (Z, z))$, respectively. Consider the induced map on long exact sequences in low degrees:

$$\begin{array}{ccccccc} \pi_1(Z, z) & \xrightarrow{\partial_Y} & \pi_0 F((Y, y), (Z, z)) & \longrightarrow & \pi_0 F(Y, Z) & \longrightarrow & \pi_0(Z) \\ \downarrow \text{id} & & \downarrow f_p^* & & \sim \downarrow f^* & & \downarrow \text{id} \\ \pi_1(Z, z) & \xrightarrow{\partial_X} & \pi_0 F((X, x), (Z, z)) & \longrightarrow & \pi_0 F(X, Z) & \longrightarrow & \pi_0(Z) \end{array}$$

By a diagram chase, f_p^* is surjective. Inserting $(Z, z) = (X, x)$, there exists a pointed map $g: (Y, y) \rightarrow (X, x)$ such that gf is pointed homotopic to id_X . In particular, g is an unpointed homotopy equivalence. Repeating the argument with g in place of f , we obtain a pointed map $f': (X, x) \rightarrow (Y, y)$ such that $f'g$ is pointed homotopic to id_Y . By two-out-of-six, it follows that f is a pointed homotopy equivalence. $\square$

V.3.10. Corollary

Let $f: (X, x) \rightarrow (Y, y)$ be a map of pointed Kan complexes such that $f: X \rightarrow Y$ is a homotopy equivalence. Then f induces isomorphisms

$$f_*: \pi_n(X, x) \xrightarrow{\cong} \pi_n(Y, y)$$

for all n .

Our next goal is to prove a converse to V.3.10.

V.3.11. Theorem

A map $f: X \rightarrow Y$ between Kan complexes is a homotopy equivalence if and only if f induces a bijection on π_0 and an isomorphism $\pi_n(X, x) \rightarrow \pi_n(Y, f(x))$ for all $x \in X_0$ and $n \geq 1$.

We begin by characterizing contractible Kan complexes by their homotopy groups. The proof relies on a description of the homotopy groups in terms of simplices.

V.3.12. Definition

For X a Kan complex, $x \in X_0$ and $n \geq 1$, define

$$\pi_n^\Delta(X, x) := \pi_0(F((\Delta^n, \partial\Delta^n), (X, x))).$$

The comparison of π_n^Δ with π_n rests on the following lemma.

V.3.13. Lemma

Let $A_1 \subseteq A_2 \subseteq X$ be simplicial sets and suppose that $A_1 \subseteq A_2$ is an anodyne extension. Then the restriction map

$$F((X, A_2), (Y, B)) \rightarrow F((X, A_1), (Y, B))$$

induces a bijection on π_0 .

Proof. The map under consideration comes from a map of pullback squares

$$\begin{array}{ccccc} & & F((X, A_2), (Y, B)) & \xrightarrow{\quad} & F(X, Y) \\ & \swarrow & \downarrow & & \swarrow \text{id} \\ F((X, A_1), (Y, B)) & \xrightarrow{\quad} & F(X, Y) & & \\ \downarrow & & \downarrow & & \downarrow \\ & & F(A_2, B) & \xrightarrow{\quad} & F(A_2, Y) \\ \swarrow & & \downarrow & & \swarrow \\ F(A_1, B) & \xrightarrow{\quad} & F(A_1, Y) & & \end{array}$$

The restriction maps $F(A_2, B) \rightarrow F(A_1, B)$ and $F(A_2, Y) \rightarrow F(A_1, Y)$ have the RLP with respect to I by V.2.26.

Let $a: A_1 \rightarrow A_2$, $i_1: A_1 \rightarrow X$, $i_2: A_2 \rightarrow X$ and $j: B \rightarrow Y$ denote the various inclusion maps.

We first prove surjectivity on π_0 . Let $f: (X, A_1) \rightarrow (Y, B)$ be a vertex in $F((X, A_1), (Y, B))$. Let $f|_{A_1}: A_1 \rightarrow B$ be the induced map. Then $f|_{A_1}$ has a lift to map $f_2: A_2 \rightarrow B$. By assumption, $j f_2 a = j f|_{A_1} = j i_1: A_1 \rightarrow Y$. Since $F(A_2, Y) \rightarrow F(A_1, Y)$ has the RLP with respect to I , there exists a homotopy $f i_2 \simeq j f_2$ relative A_1 .

Since $f i_2$ extends to X and $F(X, Y) \rightarrow F(A_2, Y)$ is a Kan fibration, we obtain a homotopy $X \times \Delta \rightarrow Y$ relative A_1 from f to a map f' satisfying $f' i_2 = j f_2$. By construction, this homotopy gives a 1-simplex in $F((X, A_1), (Y, B))$ from f to f' , and f' is a map $(X, A_2) \rightarrow (Y, B)$.

For injectivity, let $f_0, f_1: (X, A_2) \rightarrow (Y, B)$ be two maps such that there exists a homotopy $h: X \times \Delta^1 \rightarrow Y$ from f_0 to f_1 which restricts to a homotopy $h|_{A_1}: A_1 \times \Delta^1 \rightarrow B$ from $f_0|_{A_1}$ to $f_1|_{A_1}$. Since $F(A_2, B) \rightarrow F(A_1, B)$ has the RLP with respect to I , the homotopy $h|_{A_1}$ extends to a homotopy $k: A_2 \times \Delta^1 \rightarrow B$ from $f_0|_{A_2} \simeq f_1|_{A_2}$.

Consider the map

$$A_2 \times (\Delta^1 \times \{0\}) \cup_{\partial\Delta^1 \times \{0\}} \partial\Delta^1 \times \Delta^1 \rightarrow B$$

which is given by $h|_{A_2}$ on $A_2 \times \Delta^1 \times \{0\}$ and by the constant homotopies on f_e on $A_2 \times \{e\} \times \Delta^1$. This map glues with k to give a map

$$A_2 \times (\Delta^1 \times \partial\Delta^1 \cup_{\partial\Delta^1 \times \{0\}} \partial\Delta^1 \times \Delta^1) \rightarrow B,$$

which extends to a map $A_2 \times \Delta^1 \times \Delta^1$. Since $h|_{A_2}$ extends to a homotopy on X and $F(X, Y) \rightarrow F(A_2, Y)$ is a Kan fibration, there exists a map

$$\bar{h} : X \times \Delta^1 \times \Delta^1 \rightarrow Y$$

such that $\bar{h}|_{X \times \Delta^1 \times \{0\}} = h$ and $\bar{h}|_{X \times \{e\} \times \Delta^1}$ is the constant homotopy on f_e . The endpoint $\bar{h}|_{X \times \Delta^1 \times \{1\}}$ is a homotopy $f_0 \simeq f_1$ which restricts to a homotopy $f_0|_{A_2} \simeq f_1|_{A_2}$.

V.3.14. Proposition

There is a natural bijection

$$\pi_n^\Delta(X, x) \sim \pi_n(X, x).$$

Proof. By definition,

$$\pi_n(X, x) = \pi_0(F((\square^n, \partial\square^n), (X, x))).$$

Let $C^n \subseteq \square^n$ denote the simplicial subset obtained by removing the simplex

$$\sigma^n : [n] \rightarrow [1]^n, \quad i \mapsto \underbrace{1 \dots 1}_{i \text{ times}} \underbrace{0 \dots 0}_{n-i \text{ times}}.$$

Claim: *The inclusion $\partial\square^n \rightarrow C^n$ is anodyne.* For someone who is good at finite combinatorics, this can probably be proven directly by successively filling horns. We will derive it later from results that are independent of the present discussion, but would lead us on a tangent right now. Consequently, the restriction map induces by V.3.13 a bijection

$$\pi_0(F((\square^n, C^n), (X, x))) \xrightarrow{\sim} \pi_0(F((\square^n, \partial\square^n), (X, x))).$$

By inspection, we have for every pair of simplicial sets $A \subseteq B$ a natural isomorphism

$$F((B, A), (X, x)) \cong F((B/A, *), (X, x)).$$

Consequently, we have natural bijections

$$\begin{aligned} \pi_0(F((\square^n, C^n), (X, x))) &\sim \pi_0(F((\square^n/C^n, *), (X, x))) \\ &\sim \pi_0(F((\Delta^n/\partial\Delta^n, *), (X, x))) \\ &\sim \pi_0(F((\Delta^n, \partial\Delta^n), (X, x))). \end{aligned}$$

□

V.3.15. Remark

The bijection $\pi_n^\Delta(X, x) \rightarrow \pi_n(X, x)$ from V.3.14 sends a map $\alpha : (\Delta, \partial\Delta^n) \rightarrow (X, x)$ to the map $(\square^n, \partial\square^n) \rightarrow (X, x)$ which is given by α on the simplex σ^n appearing in the proof and is constant on all other simplices.

V.3.16. Lemma

Let X, x be a pointed Kan complex such that $\pi_n(X, x)$ is trivial for all $n \geq 0$. Let $f: B \rightarrow X$ be a map, let $A \subseteq B$ and let $h: A \times \Delta^1 \rightarrow X$ be a homotopy from $f|_A$ to the constant map with value x .

Then there exists an extension of h to a homotopy $B \times \Delta^1 \rightarrow X$ from f to the constant map with value x .

Proof. The extension may be constructed simplex by simplex, so it suffices to consider $B = \Delta^n$ and $A = \partial\Delta^n$. Since $(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\{0\} \rightarrow \Delta^1)$ is anodyne, there exists a homotopy $h': \Delta^n \times \Delta^1 \rightarrow X$ such that $h'|_{\Delta^n \times \{0\}} = f$ and $h'|_{\partial\Delta^n \times \{1\}} = \text{const}_x$. Therefore, $f' := h'|_{\Delta^n \times \{1\}}$ represents an element in $\pi_n^\Delta(X, x)$. Since $\pi_x(X, x)$ is trivial, [V.3.14](#) guarantees the existence of a homotopy h'' relative $\partial\Delta^n$ from f' to the constant map with value x . The two homotopies h' and h'' glue with the map

$$\partial\Delta^n \times \Delta^2 \xrightarrow{\partial\Delta^n \times s_1} \partial\Delta^n \times \Delta^1 \xrightarrow{h} X$$

to give a map

$$\tilde{h}: (\Delta^n \times \Lambda_1^2) \cup_{\partial\Delta^n \times \Lambda_1^2} (\partial\Delta^n \times \Delta^2) \rightarrow X.$$

Since $(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2)$ is anodyne, there exists an extension of $\tilde{h}$ to $\Delta^n \times \Delta^2$. restricting to $\Delta^n \times d_1\Delta^2$ yields the desired extension of h . $\square$

V.3.17. Corollary

Let X be a Kan complex. The following are equivalent:

- i) The map $X \rightarrow \Delta^0$ has the RLP with respect to I .
- ii) The map $X \rightarrow \Delta^0$ is a homotopy equivalence.
- iii) X is non-empty and $\pi_n(X, x)$ is trivial for all $x \in X_0$ and for every $n \geq 0$.
- iv) $\pi_0(X)$ is a singleton and there exists some $x \in X_0$ such that $\pi_n(X, x)$ is trivial for all $n \geq 1$.

Proof. i) implies ii) by [V.2.8](#). ii) implies iii) by [V.3.10](#). iii) obviously implies iv). Suppose iv) holds. Let

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ \Delta^n & \longrightarrow & \Delta^0 \end{array}$$

be a lifting problem. By [V.3.16](#), there exists a homotopy from f to the constant map with value x . Since $(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\{1\} \rightarrow \Delta^1)$ is anodyne, we obtain a homotopy $\Delta^n \times \Delta^1 \rightarrow X$ whose restriction to $\Delta^n \times \{0\}$ is the desired lift. $\square$

V.3.18. Corollary

Let $p: X \rightarrow Y$ be a Kan fibration. The following are equivalent:

- i) p has the RLP with respect to I .
- ii) For every $y \in Y$, the fibre $p^{-1}(y)$ is contractible.
- iii) For every $y \in Y$, the set $\pi_0(p^{-1}(y))$ is a singleton and there exists some $x \in p^{-1}(y)$ such that $\pi_n(p^{-1}(y), x)$ is trivial for all $n \geq 1$.

Proof. Assume i). Then, being a pullback, $p^{-1}(y) \rightarrow \Delta^0$ has the RLP with respect to I . Then ii) iii) follow by V.3.17. Note that ii) and iii) are equivalent, also by V.3.17. Suppose ii) holds and that we are given a lifting problem

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{\partial\bar{\sigma}} & X \\ \downarrow & & \downarrow p \\ \Delta^n & \xrightarrow{\sigma} & Y \end{array}$$

Let $h: \Delta^n \times \Delta^1 \rightarrow \Delta^n$ be a simplicial homotopy from id to the constant map with value n (this exists e.g. by Exercise 2 on Sheet 4). Then $\sigma \circ h$ is a homotopy from σ to the constant map with value $\sigma(n)$.

Then $(\sigma \circ h)|_{\partial\Delta^n \times \Delta^1}$ admits a lift to a homotopy k in X whose endpoint is a map $\partial\tau: \partial\Delta^n \rightarrow p^{-1}(\sigma(n))$. Since $p^{-1}(\sigma(n))$ is assumed to be contractible, V.3.17 implies that $\partial\tau$ extends to a map $\tau: \Delta^n \rightarrow p^{-1}(\sigma(n))$. Since $(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\{1\} \rightarrow \Delta^1)$ is anodyne, there exists a homotopy $\bar{k}$ extending k such that $\bar{k}|_{\Delta^n \times \{1\}} = \tau$ and $\bar{k}|_{\partial\Delta^n \times \{0\}} = \partial\bar{\sigma}$. So $\bar{\sigma} := \bar{k}|_{\Delta^n \times \{0\}}$ is the desired lift. $\square$

Proof of Theorem V.3.11. One direction is provided by V.3.10.

For the converse, assume that f induces a bijection on π_0 and an isomorphism $\pi_n(X, x) \rightarrow \pi_n(Y, f(x))$ for all $x \in X_0$ and $n \geq 1$. By V.2.30, factor f into an anodyne extension $i: X \rightarrow \bar{X}$ followed by a Kan fibration $p: \bar{X} \rightarrow Y$. Since i is a weak equivalence of Kan complexes by V.2.27, it is a homotopy equivalence. So we only need to show that p is an equivalence.

We show that all fibres of p are contractible.

Exercise. Show that the fibres over any two vertices in the same path component are homotopy equivalent.

Since f is surjective on π_0 , we may therefore only consider $y \in Y_0$ such that there exists $x \in X_0$ with $f(x) = y$. By V.3.17 and the long exact sequence of a fibration (V.3.8), it suffices to show that $p_*: \pi_n(\bar{X}, i(x)) \rightarrow \pi_n(Y, y)$ is a bijection for all $n \geq 0$. Since

$$\begin{array}{ccc} \pi_n(X, x) & \xrightarrow{i_*} & \pi_n(\bar{X}, i(x)) \\ & \searrow f_* & \swarrow p_* \\ & \pi_n(Y, y) & \end{array}$$

commutes, f_* is assumed to be a bijection and i_* is a bijection by V.3.10, the theorem follows. $\square$

V.3.19. Remark

It should in principle be possible to prove Theorem V.3.11 without factoring the morphism, working with relative homotopy groups instead. However, this would force us to consider homotopy classes of maps of triples, which would certainly not reduce the amount of bookkeeping we would have to do.

V.4. Ex^∞

The discussion in V.3 provides us with a good understanding of Kan fibrations which are homotopy equivalences. However, it remains unclear how these relate to Kan fibrations which are weak equivalences. For example, if $p: X \rightarrow Y$ is any Kan fibration, we could apply the small object argument (V.2.30) twice to replace Y by a Kan complex $j: Y \rightarrow \bar{Y}$ and then the map jp into an anodyne map $i: X \rightarrow \bar{X}$ followed by a Kan fibration $\bar{p}: \bar{X} \rightarrow \bar{Y}$:

$$\begin{array}{ccc} X & \xrightarrow{i} & \bar{X} \\ p \downarrow & & \downarrow \bar{p} \\ Y & \xrightarrow{j} & \bar{Y} \end{array}$$

However, we are unable to control the effect of this construction on the fibres. This is remedied by a more sophisticated construction called Ex^∞ .

V.4.1. Theorem

There exists a functor $\text{Ex}^\infty: \text{sSet} \rightarrow \text{sSet}$ and a natural transformation $\rho^\infty: \text{id}_{\text{sSet}} \rightarrow \text{Ex}^\infty$ such that the following holds:

- i) $\text{Ex}^\infty \Delta^0 \cong \Delta^0$;
- ii) $\text{Ex}^\infty X$ is Kan complex for every simplicial set X ;
- iii) $\rho^\infty: X \rightarrow \text{Ex}^\infty X$ is a weak equivalence for every simplicial set X ;
- iv) Ex^∞ preserves finite limits;
- v) Ex^∞ preserves Kan fibrations.

Before we start with the proof of this theorem, we establish some important consequences.

V.4.2. Corollary

Ex^∞ preserves and detects weak equivalences.

Proof. If $f: X \rightarrow Y$ is a map, it follows from the commutative square

$$\begin{array}{ccc} X & \xrightarrow{\rho^\infty} & \text{Ex}^\infty X \\ f \downarrow & & \downarrow \text{Ex}^\infty f \\ Y & \xrightarrow{\rho^\infty} & \text{Ex}^\infty Y \end{array}$$

and V.4.1 that f is a weak equivalence if and only if $\text{Ex}^\infty f$ is a weak equivalence. $\square$

V.4.3. Corollary

Let $p: X \rightarrow Y$ be a Kan fibration and a weak equivalence. Then p has the RLP with respect to I .

Proof. By V.4.1, we have a commutative square

$$\begin{array}{ccc} X & \xrightarrow{\rho^\infty} & \mathrm{Ex}^\infty X \\ p \downarrow & & \downarrow \mathrm{Ex}^\infty p \\ Y & \xrightarrow{\rho^\infty} & \mathrm{Ex}^\infty Y \end{array}$$

Since p is a weak equivalence, $\mathrm{Ex}^\infty p$ is a homotopy equivalence. Consequently, all fibres of $\mathrm{Ex}^\infty p$ are contractible. For any $y \in Y_0$, we have a weak equivalence $\rho^\infty: p^{-1}(y) \rightarrow (\mathrm{Ex}^\infty p)^{-1}(\rho^\infty(y))$. Since Ex^∞ preserves finite limits, the target of this map is the fibre of $\mathrm{Ex}^\infty p$ over $\rho^\infty(y)$. Consequently, $p^{-1}(y)$ is contractible. It follows that p has the RLP with respect to I . $\square$

V.4.4. Corollary

If $f: X \rightarrow Y$ is a cofibration and a weak equivalence, then it is anodyne.

Proof. Factor f into an anodyne map $i: X \rightarrow \bar{X}$ followed by a Kan fibration $p: \bar{X} \rightarrow Y$. Any lifting problem against a Kan fibration $q: V \rightarrow W$ gives rise to a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\quad} & V \\ i \downarrow & & \downarrow q \\ \bar{X} & \xrightarrow{p} & Y \longrightarrow W \end{array}$$

There exists a lift $l: \bar{X} \rightarrow V$. Since f is a weak equivalence, so is p . By V.4.3, p has the RLP with respect to I . In particular, there exists by V.2.8 a section s . Then $l \circ s$ is the desired lift.

V.4.5. Corollary (HELP)

Let $i: A \rightarrow B$ be a cofibration and let $p: X \rightarrow Y$ be a Kan fibration. If i or p is a weak equivalence, then the canonical map

$$F(B, X) \rightarrow F(A, X) \times_{F(A, Y)} F(B, Y)$$

has the RLP with respect to I .

Proof. If i is a weak equivalence, it is an anodyne extension by V.4.4. Then the corollary follows from V.2.23.

As usual, any right lifting problem with respect to $\partial\Delta^n \rightarrow \Delta^n$ corresponds to a lifting problem

$$\begin{array}{ccc} \Delta^n \times A \cup_{\partial\Delta^n \times A} \partial\Delta^n \times B & \longrightarrow & X \\ (\partial\Delta^n \rightarrow \Delta^n) \boxtimes i \downarrow & & \downarrow p \\ \Delta^n \times B & \longrightarrow & Y \end{array}$$

If p is a weak equivalence, it has the RLP with respect to I by V.4.3, so this lifting problem has a solution. $\square$

V.4.6. Remark

Informally, V.4.5 tells us that “the space of solutions to any lifting problem

$$\begin{array}{ccc} A & \longrightarrow & X \\ i \downarrow & & \downarrow p \\ B & \longrightarrow & Y \end{array}$$

is contractible”.

The proof of V.4.1 occupies the remainder of this section. We begin by defining all the relevant constructions.

V.4.7. Definition

Let P be a poset. Define $\text{chains}(P)$ as the poset whose elements are finite, linearly ordered subsets of P such that a chain c is smaller than a chain c' if c is a suborder of c' .

This defines an endofunctor on the category of posets.

V.4.8. Definition

Define $\text{sd } \Delta^\bullet : \Delta \rightarrow \text{sSet}$ as the functor sending $[n]$ to $N(\text{chains}[n])$. The (*barycentric*) *subdivision* $\text{sd} : \text{sSet} \rightarrow \text{sSet}$ is the unique colimit-preserving extension of $\text{sd } \Delta$. By Exercise 2 on Sheet 3. sd has a right adjoint $\text{Ex} : \text{sSet} \rightarrow \text{sSet}$.

V.4.9. Definition

The *last vertex map* is the natural transformation

$$\lambda : \text{sd} \rightarrow \text{id}_{\text{sSet}}$$

induced by the natural map

$$\text{chains}[n] \rightarrow [n], \quad c \mapsto \max c.$$

Denote by ρ the adjoint transformation

$$\rho : \text{id}_{\text{sSet}} \rightarrow \text{Ex}.$$

V.4.10. Definition

Define $\text{Ex}^\infty : \text{sSet} \rightarrow \text{sSet}$ as the colimit of

$$\text{id}_{\text{sSet}} \xrightarrow{\rho} \text{Ex} \xrightarrow{\rho \text{Ex}} \text{Ex}^2 \xrightarrow{\rho \text{Ex}^2} \text{Ex}^3 \rightarrow \dots$$

Let $\rho^\infty : \text{id}_{\text{sSet}} \rightarrow \text{Ex}^\infty$ be the induced natural transformation.

V.4.11. Corollary

Ex^∞ commutes with finite limits.

Proof. Since Ex is a right adjoint, it commutes with arbitrary limits. Filtered colimits commute with finite limits in Set (by inspection), so the same statement holds in sSet . This implies the corollary. $\square$

Since Ex arises as the right adjoint of sd , everything we want to know about Ex will be deduced from appropriate statements about the subdivision functor.

V.4.12. Proposition

Let X be a simplicial set. Then every horn in $\text{Ex } X$ has a filler in $\text{Ex}^2 X$.

Proof. For every map $f: \Lambda_k^n \rightarrow \text{Ex } X$, we have to show that $\Lambda_k^n \xrightarrow{f} \text{Ex } X \xrightarrow{\rho_{\text{Ex } X}} \text{Ex}^2 X$ admits a filler. Let $g: \text{sd } \Lambda_k^n \rightarrow X$ be the map determined by f . The map adjoint to $\rho_{\text{Ex } X} \circ f$ is given by $\lambda_{\text{Ex } X} \circ \text{sd } f$, which is the same as $f \circ \lambda_{\Lambda_k^n}$ by naturality of λ . The adjoint of this map is

$$\text{sd}^2 \Lambda_k^n \xrightarrow{\text{sd } \lambda_{\Lambda_k^n}} \text{sd } \Lambda_k^n \xrightarrow{g} X.$$

Consequently, we have to show that the dotted arrow in the diagram

$$\begin{array}{ccc} \text{sd}^2 \Lambda_k^n & \xrightarrow{\text{sd } \lambda} & \text{sd } \Lambda_k^n \\ \downarrow & & \downarrow g \\ \text{sd}^2 \Delta^n & \cdots \cdots \cdots & X \end{array}$$

exists and makes the resulting square commute. Note that it suffices to find the dotted arrow in the case $g = \text{id}$.

Observe that $\text{sd } \Lambda_k^n$ is the nerve of the poset $P := \text{chains}[n] \setminus \{[n], [n] \setminus \{k\}\}$, so we have to specify a map of posets

$$\text{chains chains}[n] \rightarrow P, \quad (c_0 \subseteq \dots \subseteq c_r) \mapsto \{\widetilde{\text{max}}(c_0), \dots, \widetilde{\text{max}}(c_r)\},$$

where $\widetilde{\text{max}}$ is defined by

$$\widetilde{\text{max}}(c) := \begin{cases} k & c = [n] \text{ or } c = [n] \setminus \{k\}, \\ \max c & \text{else.} \end{cases}$$

It is straightforward to check that this is well-defined, and the map evidently extends $\text{sd } \lambda$. $\square$

V.4.13. Corollary

$\text{Ex}^\infty X$ is a Kan complex for every simplicial set X .

Proof. Since Λ_k^n is compact, every map $\Lambda_k^n \rightarrow \text{Ex}^\infty$ factors through some $\text{Ex}^n X$. Assume $n \geq 1$. Then $\Lambda_k^n \rightarrow \text{Ex}^n X \xrightarrow{\rho_{\text{Ex}^n X}} \text{Ex}^{n+1} X$ extends to a map $\Delta^n \rightarrow \text{Ex}^{n+1} X$ by V.4.12, and this provides also a filler in $\text{Ex}^\infty X$. $\square$

We consider the question whether Ex preserves Kan fibrations next. By the adjunction between sd and Ex , this reduces to the question whether sd preserves anodyne extensions.

V.4.14. Lemma

Let J be a non-empty, finite set. The inclusion $L_J := \mathcal{P}(J) \setminus \{J\} \rightarrow \mathcal{P}(J)$ induces a left anodyne map on nerves.

Proof. Pick some element $j \in J$. Then the inclusion map $N(L_J) \rightarrow N(\mathcal{P}(J))$ is isomorphic to the map

$$(N(L_{J \setminus \{j\}}) \times \Delta^1) \bigcup_{N(L_{J \setminus \{j\}}) \times \{0\}} (N(\mathcal{P}(J \setminus \{j\})) \times \{0\}) \rightarrow N(\mathcal{P}(J \setminus \{j\})) \times \Delta^1.$$

This map is left anodyne by V.2.22. $\square$

V.4.15. Proposition

The subdivision functor preserves anodyne extensions.

Proof. Since sd is a colimit-preserving functor, the class of morphisms

$$\Sigma := \{i \mid \text{sd } i \text{ is anodyne}\}$$

is saturated. Hence it suffices to show that $H \subseteq \Sigma$. Recall that $\text{sd } \Delta^n$ is the nerve of the poset $\text{chains}[n]$, and that Λ_k^n is the nerve of the subposet $P := \text{chains}[n] \setminus \{[n], [n] \setminus \{k\}\}$. In this proof, we identify $\text{chains}[n]$ with the set $\mathcal{P}_*([n])$ of non-empty subsets of $[n]$. Set $[n]_k := [n] \setminus \{k\}$. The map

$$\mathcal{P}([n]_k) \rightarrow \mathcal{P}_*([n]), \quad S \mapsto S \cup \{k\}$$

is injective. Let $X \subseteq \text{sd } \Delta^n$ denote the image of the induced map on nerves. Then $X \cap \text{sd } \Lambda_k^n \cong N(L_{[n]_k})$, and there is a pushout square

$$\begin{array}{ccc} N(L_{[n]_k}) \cong X \cap \text{sd } \Lambda_k^n & \longrightarrow & \text{sd } \Lambda_k^n \\ \downarrow & & \downarrow \\ N(\mathcal{P}([n]_k)) \cong X & \longrightarrow & Y \end{array}$$

Consequently, $\text{sd } \Lambda_k^n \rightarrow X$ is left anodyne by V.4.14.

Consider now the injective map

$$(\mathcal{P}([n]_k) \setminus \{\emptyset\}) \times [1] \rightarrow \mathcal{P}_*([n]), \quad (S, r) \mapsto \begin{cases} S & r = 0, \\ c(S) = S \cup \{k\} & r = 1, \end{cases}$$

and denote its image by Z . Then

$$Z \cap Y \cong N((\mathcal{P}([n]_k) \setminus \{\emptyset\}) \times \{1\}) \bigcup_{N((\mathcal{P}([n]_k) \setminus \{\emptyset, [n]_k\}) \times \{1\})} N((\mathcal{P}([n]_k) \setminus \{\emptyset, [n]_k\}) \times \{1\})$$

and there is a pushout square

$$\begin{array}{ccc} Z \cap Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & \text{sd } \Delta^n \end{array}$$

The left vertical map is right anodyne by V.2.22, so $X \rightarrow \text{sd } \Delta^n$ is also right anodyne. Consequently, the composition $\text{sd } \Lambda_k^n \rightarrow X \rightarrow \text{sd } \Delta^n$ is anodyne. $\square$

V.4.16. Corollary

Ex^n preserves Kan fibrations for every $n \in \mathbb{N} \cup \{\infty\}$.

Proof. Suppose by induction that Ex^n preserves Kan fibrations. Any lifting problem for $\text{Ex}^{n+1} p$, where $p: X \rightarrow Y$ is a Kan fibration, corresponds to a lifting problem

$$\begin{array}{ccc} \text{sd } \Lambda_k^n & \longrightarrow & \text{Ex}^n X \\ \downarrow & & \downarrow \text{Ex}^n p \\ \text{sd } \Delta^n & \longrightarrow & \text{Ex}^n Y \end{array}$$

Since $\text{sd } \Lambda_k^n \rightarrow \text{sd } \Delta^n$ is anodyne by V.4.15, a solution to the lifting problem exists.

The statement for $n = \infty$ follows since Λ_k^n and Δ^n are compact objects in sSet . $\square$

Since $\text{Ex}^\infty \Delta^0$ is obviously isomorphic to Δ^0 , V.4.11, V.4.13 and V.4.16 take care of V.4.1 with the exception of iii).

V.4.17. Proposition

Let $X, Y: \mathbb{N} \rightarrow \text{sSet}$ be diagrams and let $f: X \rightarrow Y$ be a transformation. Suppose that $x(n): X(n) \rightarrow X(n+1)$ and $y(n): Y(n) \rightarrow Y(n+1)$ are cofibrations for all n , and that $f(n): X(n) \rightarrow Y(n)$ is a weak equivalence for every n .

Then the induced map $\text{colim}_{\mathbb{N}} X \rightarrow \text{colim}_{\mathbb{N}} Y$ is a weak equivalence.

Proof. Let $g: V \rightarrow W$ be an arbitrary transformation of diagrams $\mathbb{N} \rightarrow \text{sSet}$. We claim that g can be factored as $g = pi$, where $i(n)$ is anodyne for every n and $p(n)$ is a Kan fibration for every n .

This factorisation can be built by induction. In degree 0, apply V.2.20 to factor $g(0)$ into an anodyne $i(0): V(0) \rightarrow \bar{V}(0)$ and a Kan fibration $p(0): \bar{V}(0) \rightarrow W(0)$. Then the induced map $j(1): V(1) \rightarrow V(1) \cup_{V(0)} \bar{V}(0)$ is anodyne and the map $\bar{V}(0) \rightarrow V(1) \cup_{V(0)} \bar{V}(0)$ is a cofibration.

Moreover, $g(1)$ induces a map $V(1) \cup_{V(0)} \bar{V}(0) \rightarrow W(0)$. Factor this map into an anodyne map $j'(1): V(1) \cup_{V(0)} \bar{V}(0) \rightarrow \bar{V}(1)$ and a Kan fibration $p(1): \bar{V}(1) \rightarrow W(1)$. Set $i(1) := j'(1) \circ j(1)$.

Processing like this inductively produces the desired factorisation.

Apply the claim to the unique transformation $Y \rightarrow \Delta^0$ to obtain a transformation $j: Y \rightarrow \bar{Y}$ such that $j(n)$ is anodyne for all n and $\bar{Y}(n)$ is a Kan complex for every n . Apply the claim a second time to $jf: X \rightarrow \bar{Y}$, producing a pointwise anodyne transformation $i: X \rightarrow \bar{X}$ and a pointwise Kan fibration $p: \bar{X} \rightarrow \bar{Y}$. Since $Y(n)$ is a Kan complex and $p(n)$ is a Kan fibration, $\bar{X}(n)$ is a Kan complex for every n . Moreover, $p(n)$ is a weak equivalence, so it has the RLP with respect to I by V.3.18, V.3.8 and V.3.11.

Since $\partial \Delta^n$ and Δ^n are compact, it follows directly that $\text{colim}_{\mathbb{N}} p$ has the RLP with respect to I . In particular, $\text{colim}_{\mathbb{N}} p$ is a weak equivalence.

Therefore, it suffices to show the proposition in the case when all components of the transformation are anodyne: then both $\text{colim}_{\mathbb{N}} j$ and $\text{colim}_{\mathbb{N}} i$ are weak equivalences, which readily implies that $\text{colim}_{\mathbb{N}} f$ is a weak equivalence.

Hence, assume that $f(n)$ is anodyne for all n . Let K be a Kan complex. We have to show that the induced map

$$(\text{colim}_{\mathbb{N}} f)^*: \pi_0 F(\text{colim}_{\mathbb{N}} Y, K) \rightarrow \pi_0 F(\text{colim}_{\mathbb{N}} X, K)$$

is a bijection. This map fits into a commutative square

$$\begin{array}{ccc} \pi_0 F(\operatorname{colim}_{\mathbb{N}} Y, K) & \xrightarrow{(\operatorname{colim}_{\mathbb{N}} f)^*} & \pi_0 F(\operatorname{colim}_{\mathbb{N}} X, K) \\ \cong \downarrow & & \downarrow \cong \\ \pi_0(\lim_{\mathbb{N}^{\text{op}}} F(Y, K)) & \xrightarrow{(\lim_{\mathbb{N}^{\text{op}}} f)^*} & \pi_0(\lim_{\mathbb{N}^{\text{op}}} F(X, K)) \end{array}$$

Let us show directly that the lower horizontal map is a bijection.

We claim in slightly greater generality that if we have a transformation between towers of Kan fibrations of Kan complexes

$$\begin{array}{ccccccc} \dots & \longrightarrow & U(2) & \xrightarrow{u(2)} & U(1) & \xrightarrow{u(1)} & U(0) \\ & & \downarrow q(2) & & \downarrow q(1) & & \downarrow q(0) \\ \dots & \longrightarrow & V(2) & \xrightarrow{v(2)} & V(1) & \xrightarrow{v(1)} & V(0) \end{array}$$

such that $q(n)$ has the RLP with respect to I for all n , then the induced map $\pi_0(\lim_{\mathbb{N}^{\text{op}}} U) \rightarrow \pi_0(\lim_{\mathbb{N}^{\text{op}}} V)$ is a bijection.

Given $(b_n)_n \in \lim_{\mathbb{N}^{\text{op}}} V$, there exists a sequence $(a'_n)_n \prod_{\mathbb{N}} U(n)$ such that $q(n)(a'_n) = b_n$.

Set $a_0 := a'_0$. Since $q(0)(u(1)(a'_1)) = b_0$, the constant homotopy on b_0 lifts to a homotopy $u(1)(a'_1) \simeq a_0$. Since $u(1)$ is a Kan fibration, this homotopy lifts to a homotopy $h_1: \Delta^1 \rightarrow U(1)$ from a'_1 to some vertex a_1 . Note that $u(1)(a_1) = a_0$. Moreover, $q(1) \circ h_1$ provides a homotopy $b_1 \simeq q(1)(a_1)$ such that $v(1) \circ q(1) \circ h_1$ is the constant homotopy on b_0 . Continuing inductively, we find a vertex $(a_n)_n \in \lim_{\mathbb{N}^{\text{op}}} U$ and a homotopy $\Delta^1 \rightarrow \lim_{\mathbb{N}^{\text{op}}} V$ from $(b_n)_n$ to $(q(n)(a_n))_n$. So the comparison map is surjective.

Suppose that $(a_n)_n, (a'_n)_n$ are two vertices such that there exists a homotopy $\Delta^1 \rightarrow \lim_{\mathbb{N}^{\text{op}}} V$ from $(q(n)(a_n))_n$ to $(q(n)(a'_n))_n$, i.e. a sequence of homotopies $(h_n: \Delta^1 \rightarrow V(n))_n$ such that h_n is a homotopy $q(n)(a_n) \simeq q(n)(a'_n)$ and $v(n+1) \circ h_{n+1} = h_n$. Since each $q(n)$ has the RLP with respect to I , there exists a sequence of homotopies $(k'_n: \Delta^1 \rightarrow U(n))_n$ such that k'_n is a homotopy $a_n \simeq a'_n$. Set $k_0 := k'_0$.

Consider the map $H_0: \partial \square^2 \rightarrow U(0)$ which is given by $u(1)k'_1$ and k_0 on two opposite faces, respectively. Then $q(0) \circ H_0$ extends to the map $\square^2 \xrightarrow{\text{pr}} \Delta^1 \xrightarrow{h_0} V(0)$. As $q(0)$ has the RLP with respect to I , the map H_0 extends to a map $K_0: \square^2 \rightarrow U(0)$ lifting the given map. Since $u(1)$ is a Kan fibration, K_0 lifts further to a homotopy $k'_1 \simeq k_1$ (relative endpoints). Again, one continues this procedure inductively to obtain a homotopy $\Delta^1 \rightarrow \lim_{\mathbb{N}^{\text{op}}} U$ from $(a_n)_n$ to $(a'_n)_n$. This proves injectivity, and thus finishes the proof of the proposition. $\square$

V.4.18. Proposition

Let

$$\begin{array}{ccc} X & \xrightarrow{f} & X' \\ p \downarrow & & \downarrow p' \\ Y & \xrightarrow{g} & Y' \end{array}$$

be a commutative diagram such that p and p' are Kan fibrations of Kan complexes and g is a weak equivalence. Then the following are equivalent:

- i) f is a weak equivalence.
- ii) The induced map $p^{-1}(y) \rightarrow (p')^{-1}(g(y))$ is a homotopy equivalence for every $y \in Y_0$.

Proof. Exercise.

V.4.19. Proposition

The last vertex map $\lambda: \text{sd } X \rightarrow X$ is a weak equivalence for every simplicial set X .

Proof. Since sd preserves colimits and cofibrations, it follows from V.4.17 that it suffices to show that $\lambda: \text{sd } \text{sk}_n(X) \rightarrow \text{sk}_n(X)$ is a weak equivalence for every n . We do this by induction on n . For $n = 0$, one sees directly that $\lambda_{\text{sk}_0(X)}$ is an isomorphism. Let K be a Kan complex. The vertical maps in the commutative square

$$\begin{array}{ccc} F(\text{sk}_{n+1}(X), K) & \xrightarrow{\lambda_{\text{sk}_{n+1}(X)}^*} & F(\text{sd } \text{sk}_{n+1}(X), K) \\ \downarrow & & \downarrow \\ F(\text{sk}_n(X), K) & \xrightarrow{\lambda_{\text{sk}_n(X)}^*} & f(\text{sd } \text{sk}_n(X), K) \end{array}$$

are Kan fibrations of Kan complexes, and $\lambda_{\text{sk}_n(X)}^*$ is a weak equivalence by induction hypothesis.

By V.4.18, it suffices to show that the induced maps on vertical fibres are weak equivalences. Consider the pullback square

$$\begin{array}{ccc} F(\text{sk}_{n+1} X, K) & \longrightarrow & F(\coprod_{I_{n+1}} \partial \Delta^{n+1}, K) \\ \downarrow & & \downarrow \\ F(\text{sk}_n X, K) & \longrightarrow & F(\coprod_{I_{n+1}} \Delta^{n+1}, K) \end{array}$$

arising from the pushout describing the attachment of $(n+1)$ -simplices. Since sd preserves colimits, there is an analogous square in which all sources of function complexes have bin replaced by their subdivisions. Consequently, for $f \in F(\text{sk}_n(X), K)$ the induced map on fibres can be identified with

$$\{f'\}_{F(\coprod_{I_{n+1}} \partial \Delta^{n+1}, K)} \times_{F(\coprod_{I_{n+1}} \Delta^{n+1}, K)} \rightarrow \{f'\}_{F(\coprod_{I_{n+1}} \text{sd } \partial \Delta^{n+1}, K)} \times_{F(\coprod_{I_{n+1}} \text{sd } \Delta^{n+1}, K)},$$

where $f': \coprod_{I_{n+1}} \partial \Delta^{n+1} \rightarrow K$ is the map induced by f .

Again by V.4.18, it suffices to show that the two horizontal maps in the diagram

$$\begin{array}{ccc} F(\coprod_{I_{n+1}} \Delta^{n+1}, K) & \xrightarrow{\lambda^*} & F(\coprod_{I_{n+1}} \text{sd } \Delta^{n+1}, K) \\ \downarrow & & \downarrow \\ F(\coprod_{I_{n+1}} \partial \Delta^{n+1}, K) & \xrightarrow{\lambda^*} & F(\coprod_{I_{n+1}} \text{sd } \partial \Delta^{n+1}, K) \end{array}$$

are homotopy equivalences. The lower horizontal map is again a homotopy equivalence by induction hypothesis.

A product of homotopy equivalences is a homotopy equivalence, so it suffices to see that $\lambda_{\Delta^{n+1}}: \text{sd } \Delta^{n+1} \rightarrow \Delta^{n+1}$ is a homotopy equivalence. Since both $\text{sd } \Delta^{n+1}$ and Δ^{n+1} are nerves of posets with a largest element, this is immediate from Exercise 2 on Sheet 4. $\square$

V.4.20. Lemma

Let X be a simplicial set and let K be a Kan complex. Then the bijection $\text{Hom}_{\text{sSet}}(\text{sd } X, K) \cong \text{Hom}_{\text{sSet}}(X, \text{Ex } K)$ induces a bijection

$$\pi_0 F(\text{sd } X, K) \cong \pi_0 F(X, \text{Ex } K).$$

Proof. Suppose that $h: \text{sd } X \times \Delta^1 \rightarrow K$ is a homotopy. Then the adjoint morphism to

$$\text{sd}(X \times \Delta^1) \rightarrow \text{sd } X \times \text{sd } \Delta^1 \xrightarrow{\text{sd } X \times \lambda_{\Delta^1}} \text{sd } X \times \Delta^1 \xrightarrow{h} K$$

provides a homotopy between the morphisms adjoint to h_0 and h_1 .

Suppose that $X \times \Delta^1 \rightarrow \text{Ex } K$ is a homotopy. Let $h: \text{sd}(X \times \Delta^1) \rightarrow K$ be the adjoint morphism. Let k denote the composition $\text{sd}(X \times \Delta^1) \rightarrow \text{sd } X \xrightarrow{h_0} K$. Since $X \times \{0\} \rightarrow X \times \Delta^1$ is anodyne, the restriction map

$$F(\text{sd}(X \times \Delta^1), K) \rightarrow F(\text{sd}(X \times \{0\}), K)$$

has the RLP with respect to I by V.4.15 and V.2.26. Since h and k restrict to the same map $\text{sd}(X \times \{0\}) \rightarrow K$, there exists a homotopy $\text{sd}(X \times \Delta^1) \times \Delta^1 \rightarrow K$ between h and k . The restriction of this homotopy to $\text{sd}(X \times \{1\}) \times \Delta^1$ provides the required homotopy between h_0 and h_1 . $\square$

V.4.21. Corollary

The transformation $\rho: K \rightarrow \text{Ex } K$ induces a bijection

$$\pi_0 F(X, K) \xrightarrow{\sim} \pi_0 F(X, \text{Ex } K)$$

for every Kan complex K and every simplicial set X .

Proof. By V.4.20, there is a commutative triangle

$$\begin{array}{ccc} & \pi_0(X, \text{Ex } K) & \\ \rho_* \nearrow & \downarrow \cong & \\ \pi_0 F(X, K) & & \pi_0 F(\text{sd } X, K) \\ \lambda_* \searrow & & \end{array}$$

The lower diagonal arrow is a bijection by V.4.19. $\square$

V.4.22. Lemma

Let X be a simplicial set. Then there is a simplicial homotopy $\rho_{\text{Ex } X} \simeq \text{Ex } \rho_X$.

Proof. Define the map of posets

$$h : \text{chains}(\text{chains}[n]) \times [q] \rightarrow \text{chains}[n]$$

$$(c_0 \subseteq \dots \subseteq c_k, i) \mapsto \begin{cases} \max c_0 \leq \dots \leq \max c_k & i = 0, \\ c_k & i = 1 \end{cases}$$

Note that $\max c_0 \leq \dots \leq \max c_k$ is a suborder of c_k , so this is well-defined. Taking nerves, we obtain a homotopy from $\text{sd } \lambda_{\Delta^n}$ to $\lambda_{\text{sd } \Delta^n}$. This is natural in n , so we obtain a natural transformation $\text{sd}^2(-) \times \Delta^1 \Rightarrow \text{sd}$ (using that $- \times \Delta^1$ commutes with arbitrary colimits). Since there is also a natural transformation $\text{sd}^2(- \times \Delta^1) \Rightarrow \text{sd}^2(-) \times \Delta^1$, we obtain a natural transformation $\text{sd}^2(- \times \Delta^1) \Rightarrow \text{sd}$. Using the solution to Exercise 1 on Sheet 4, we obtain a natural transformation $\text{Ex} \Rightarrow F(\Delta^1, \text{Ex}^2(-))$ which provides (by “currying”) the required homotopy $\rho \text{Ex} \simeq \text{Ex } \rho$. $\square$

V.4.23. Proposition

The transformation $\rho : X \rightarrow \text{Ex } X$ is a weak equivalence for every simplicial set X .

Proof. Let K be a Kan complex. Consider the diagram

$$\begin{array}{ccc} \pi_0 F(\text{Ex } X, K) & \xrightarrow{\rho_X^*} & \pi_0 F(X, K) \\ (\rho_K)_* \downarrow & \swarrow \text{dashed} & \downarrow (\rho_K)_* \\ \pi_0 F(\text{Ex } X, \text{Ex } K) & \xrightarrow{\rho_X^*} & \pi_0 F(X, \text{Ex } K) \end{array}$$

The dashed diagonal map is induced by the functoriality of Ex : note that Ex preserves simplicial homotopy since a homotopy $X \times \Delta^1 \rightarrow K$ gives rise to a homotopy $\text{Ex } X \times \Delta^1 \rightarrow \text{Ex } X \times \text{Ex } \Delta^1 \cong \text{Ex}(X \times \Delta^1) \rightarrow K$. Both vertical maps are bijections by V.4.21. Then the lower triangle commutes by the naturality of ρ .

For $f : \text{Ex } X \rightarrow K$, we have by V.4.22

$$\text{Ex}(f \circ \rho_X) = \text{Ex } f \circ \text{Ex } \rho_X \simeq \text{Ex } f \circ \rho_{\text{Ex } X} = \rho_K \circ f.$$

So the upper triangle commutes as well.

It follows that all maps in the diagram are bijections. $\square$

V.4.24. Remark

Note that ρ is injective. Since λ is evidently surjective, this follows from the commutativity of

$$\begin{array}{ccc} & \text{Hom}_{\text{sSet}}(Y, \text{Ex } X) & \\ (\rho_X)_* \nearrow & \downarrow \cong & \\ \text{Hom}_{\text{sSet}}(Y, X) & & \text{Hom}_{\text{sSet}}(\text{sd } Y, X) \\ \lambda_Y^* \searrow & & \end{array}$$

Proof of V.4.1. i) is obvious. ii), iv) and v) hold by V.4.11, V.4.13 and V.4.16, respectively. So we are left with showing that $\rho^\infty: X \rightarrow \text{Ex}^\infty X$ is a weak equivalence for every simplicial set X . By V.4.23, each composite $X \rightarrow \text{Ex}^n X$ is a weak equivalence for every n . By V.4.17, it follows that ρ^∞ is also a weak equivalence. $\square$

V.5. The comparison with the homotopy theory of CW-complexes

To finish our discussion of simplicial homotopy theory, we outline the relation to the homotopy theory of CW-complexes. We will assume some facts about the geometric realisation functor. This is summarized in the following theorem:

V.5.1. Theorem

Let Kan denote the simplicially enriched category of Kan complexes and let CW denote the simplicially enriched category of CW-complexes (by taking Sing of the spaces of all continuous maps). Then both

$$|-| : \text{Kan} \rightarrow \text{CW}$$

and

$$\text{Sing} : \text{CW} \rightarrow \text{Kan}$$

are Dwyer-Kan equivalences, i.e.

- they induced essentially surjective functors on homotopy categories and
- they induce weak equivalences on all mapping spaces.

The remainder of this section is occupied by the proof of V.5.1.

V.5.2. Proposition

Geometric realisation $|-| : \text{sSet} \rightarrow \text{kTop}$ commutes with finite products.

Proof. Gabriel-Zisman ...

V.5.3. Corollary

Both $\text{Sing}: \text{kTop} \rightarrow \text{sSet}$ and $|-| : \text{sSet} \rightarrow \text{kTop}$ define enriched functors which are adjoint. More precisely, the adjunction $(|-|, \text{Sing})$ induces natural bijections

$$F_{\text{kTop}}(|-|, -) \cong F_{\text{sSet}}(-, \text{Sing } -) : \text{sSet}^{\text{op}} \times \text{kTop} \rightarrow \text{sSet}.$$

Proof. Let $f: X \times \Delta^n \rightarrow Y$. By V.5.2, the map $|f|$ may be regarded as a map

$$|\Delta^n| \times |X| \cong |\Delta^n \times X| \rightarrow |Y|,$$

and therefore defines an n -simplex in $F_{\mathbf{kTop}}(|X|, |Y|)$.

An n -simplex in $F_{\mathbf{kTop}}(T, U)$ is given by a map $f: |\Delta^n| \times T \rightarrow U$ which induces a map

$$\Delta^n \times \text{Sing } T \xrightarrow{u_{\Delta^n} \times \text{Sing } T} \text{Sing } |\Delta^n| \times \text{Sing } T \cong \text{Sing}(|\Delta^n| \times T) \xrightarrow{\text{Sing } f} \text{Sing } U.$$

It is straightforward to check that these rules do define enriched functors.

For the second part, observe that an n -simplex $\Delta^n \rightarrow F_{\mathbf{kTop}}(|X|, T)$ induces by V.5.2 an unique map

$$|X \times \Delta^n| \cong |X| \times |\Delta^n| \rightarrow T,$$

which by adjunction is the same as a map $\Delta^n \rightarrow F(X, \text{Sing } T)$. $\square$

V.5.4. Lemma

There are natural isomorphisms

$$\pi_0(-) \xrightarrow{\sim} \pi_0(\text{Sing}(-)) : \mathbf{kTop} \rightarrow \mathbf{Set}$$

and

$$\pi_n(-) \xrightarrow{\sim} \pi_n(\text{Sing}(-)) : \mathbf{kTop}_* \rightarrow \mathbf{Grp}$$

for every $n \geq 0$.

Proof. Exercise. $\square$

V.5.5. Corollary

Both $|-|$ and Sing preserve weak equivalences.

Proof. Exercise. $\square$

V.5.6. Theorem

The following are equivalent:

- i) Simplicial approximation holds for maps from finite simplicial sets: Let $A \subseteq X$ be finite simplicial sets and let Y be a simplicial set. Suppose that $f_0: A \rightarrow Y$ and $g: |X| \rightarrow |Y|$ are maps such that $g|_{|A|} = |f_0|$.

Then there exists for some n a map $f: \text{sd}^n X \rightarrow Y$ such that

$$\begin{array}{ccccc} \text{sd}^n A & \xrightarrow{\lambda^n} & A & \xrightarrow{f_0} & Y \\ \downarrow & & \nearrow f & & \\ \text{sd}^n X & & & & \end{array}$$

commutes and a homotopy $|f| \simeq g|_{\lambda^n}$ relative $|\text{sd}^n A|$.

- ii) Simplicial approximation holds for maps into Kan complexes: Let $A \subseteq X$ be simplicial sets and let K be a Kan complex. Suppose that $f_0: A \rightarrow K$ and $g: |X| \rightarrow |K|$ are maps such that $g|_{|A|} = |f_0|$.

Then there exists a map $f: X \rightarrow K$ and a homotopy $|f| \simeq g$ relative $|A|$.

- iii) The unit morphism $X \rightarrow \text{Sing } |X|$ is a homotopy equivalence for every Kan complex.

- iv) The unit morphism $X \rightarrow \text{Sing } |X|$ is a weak equivalence for every simplicial set.
- v) Geometric realisation induces an equivalence $F(X, K) \xrightarrow{\cong} F(|X|, |K|)$ for every simplicial set X and every Kan complex K .

Proof. We only sketch why [i](#)) is equivalent [ii](#)). If [i](#)) holds, one constructs an approximation simplex by simplex. This reduces the problem to the case $A = \partial\Delta^n$ and $X = \Delta^n$. In this case, the simplicial approximation whose existence is guaranteed by [i](#)) yields a commutative square

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{f_0} & K \\ \downarrow & & \downarrow \rho^n \\ \Delta^n & \xrightarrow{f} & \text{Ex}^n K \end{array}$$

Since ρ^n is an anodyne map between Kan complexes, it has a retraction. This yields the desired extension.

Conversely, we may replace Y by $\text{Ex}^\infty Y$ and apply [ii](#)). Since X is assumed to be finite, the resulting map $X \rightarrow \text{Ex}^\infty Y$ factors through some $\text{Ex}^n Y$ and gives thus rise to a map $\text{sd}^n X \rightarrow Y$ with the desired properties.

We now show [ii](#)) $\Rightarrow$ [iii](#)). By one of the triangle identities, the composition

$$|X| \xrightarrow{|u_X|} |\text{Sing } |X|| \xrightarrow{c_{|X|}} |X|$$

is the identity. By [ii](#)), there exists a map $k: \text{Sing } |X| \rightarrow X$ extending id_X such that $|k|$ is homotopic to $c_{|X|}$ relative $|X|$. Let $h: |\text{Sing } |X|| \times [0, 1] \rightarrow |X|$ be such a homotopy. Since $|\Delta^1| \cong [0, 1]$ and $|-|$ commutes with finite products, this corresponds to a map

$$|\text{Sing } |X| \times \Delta^1| \rightarrow |X|$$

whose adjoint provides a deformation retraction of $\text{Sing } |X|$ onto $|X|$.

For [iii](#)) $\Rightarrow$ [iv](#)), consider the commutative square

$$\begin{array}{ccc} X & \xrightarrow{u_X} & \text{Sing } |X| \\ \rho^\infty \downarrow & & \downarrow \text{Sing } |\rho^\infty| \\ \text{Ex}^\infty & \xrightarrow{u_{\text{Ex}^\infty X}} & \text{Sing } |\text{Ex}^\infty X| \end{array}$$

The vertical maps are weak equivalences by [V.4.1](#) and [V.5.5](#), and the lower horizontal map is a weak equivalence by assumption. So u_X is also a weak equivalence.

For [iv](#)) $\Rightarrow$ [v](#)), consider the commutative diagram

$$\begin{array}{ccc} & & F_{\text{CW}}(|X|, |K|) \\ & \nearrow | - | & \downarrow \cong \\ F_{\text{sSet}}(X, K) & & \\ & \searrow (u_K)_* & \downarrow \\ & & F_{\text{sSet}}(X, \text{Sing } |K|) \end{array}$$

Since u_K is a homotopy equivalence by assumption, **v**) follows.

Finally, we show that **v**) $\Rightarrow$ **ii**). There is a map of Kan fibrations of Kan complexes

$$\begin{array}{ccc} F_{\text{Set}}(X, K) & \xrightarrow{|\cdot|} & F_{\text{CW}}(|X|, |K|) \\ \downarrow & & \downarrow \\ F_{\text{Set}}(A, K) & \xrightarrow{|\cdot|} & F_{\text{CW}}(|A|, |K|) \end{array}$$

and by assumption both horizontal maps are weak equivalences. It follows from [V.4.18](#) that the induced map on fibres over f_0 and $|f_0|$ is a homotopy equivalence. Picking a preimage of g yields the desired simplicial approximation. $\square$

Standing assumption: Let us assume for a moment that one of the simplicial approximation statements holds.

V.5.7. Corollary

The counit $|\text{Sing } T| \rightarrow T$ is a weak equivalence for every $T \in \text{Top}$.

Proof. By [V.5.4](#), it suffices to check that $\text{Sing } c_T: \text{Sing } |\text{Sing } T| \rightarrow \text{Sing } T$ induces bijections on all π_n . By one of the triangle identities, the identity on $\text{Sing } T$ factors as

$$\text{Sing } T \xrightarrow{u_{\text{Sing } T}} \text{Sing } |\text{Sing } T| \xrightarrow{\text{Sing } c_T} \text{Sing } T.$$

Since $u_{\text{Sing } T}$ is a homotopy equivalence, $\text{Sing } c_T$ is also a homotopy equivalence. So the corollary follows from [V.3.10](#). $\square$

V.5.8. Corollary

The geometric realisation detects weak equivalences, i.e. if $f: X \rightarrow Y$ is a map of simplicial sets such that $|f|$ is a weak equivalence, then f is a weak equivalence.

Proof. By [V.5.5](#), $\text{Sing } |f|$ is a weak equivalence. Then it follows from the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ u_X \downarrow & & \downarrow u_Y \\ \text{Sing } |X| & \xrightarrow{\text{Sing } |f|} & \text{Sing } |Y| \end{array}$$

and [V.5.6](#) that f is a weak equivalence. $\square$

Proof of [V.5.1](#). [V.5.6](#) deals almost with the entire statement since it already tells us that Sing is essentially surjective on homotopy categories and that $|\cdot|$ induces weak equivalences on mapping spaces. By [V.5.7](#), the functor $|\cdot|$ is also essentially surjective on homotopy categories.

Using V.5.7 a second time, the commutative diagram

$$\begin{array}{ccc}
 & F_{\text{sSet}}(\text{Sing } T, \text{Sing } U) & \\
 \text{Sing} \nearrow & & \downarrow \cong \\
 F_{\text{CW}}(T, U) & & F_{\text{CW}}(|\text{Sing } T|, U) \\
 c_T^* \searrow & & \\
 & &
 \end{array}$$

shows that Sing also induces weak equivalences on mapping spaces. $\square$

In the remainder of this section, we outline why simplicial approximation holds for simplicial sets. Our discussion follows Jardine, but many of the properties in which we are interested (plus many other, more subtle properties) are also studied in WJR13, Section 2. We assume the classical simplicial approximation theorem for simplicial complexes. To formulate it, we need to recall the barycentric subdivision of a simplicial complex: Given an (abstract) simplicial complex (V, S) , we define $\text{sd}(V, S)$ as the abstract simplicial complex whose set of vertices is given by S , and whose n -simplices are given by strictly ascending chains $s_0 \subseteq s_1 \subseteq \dots \subseteq s_n$ of length n .

V.5.9. Remark

Recall that any locally ordered simplicial complex gives rise to a simplicial set by formally adjoining degeneracies (see Exercise 3 on Sheet 3). Denote this functor by

$$(-)^+ : \text{SC}^{\text{lo}} \rightarrow \text{sSet}.$$

Canonical examples of locally ordered simplicial complexes are those arising from posets: For a poset P , let $V(P)$ be the simplicial complex with vertex set P and n -simplices strictly ascending chains $p_0 < \dots < p_n$. The simplicial complex $V(P)$ is often called the nerve of the poset P , but we will avoid that terminology because the nerve of a poset is a simplicial set for us. However, note that there is a natural isomorphism $NP \cong V(P)^+$, so the two notions are not unrelated.

Moreover, it is straightforward to check that $(\text{sd } V(P))^+ \cong \text{sd } V(P)^+$ for any poset P .

V.5.10. Proposition

Let $U \subseteq V$ and W be simplicial complexes. Let $f_0 : U \rightarrow W$ be a simplicial map and let $g : |V| \rightarrow |W|$ be a map such that $g|_{|U|} = |f_0|$. Suppose that V is finite.

Then there exist some n and a map $f : \text{sd}^n V \rightarrow W$ such that

$$\begin{array}{ccccc}
 \text{sd}^n U & \xrightarrow{\lambda^n} & U & \xrightarrow{f_0} & W \\
 \downarrow & & \nearrow f & & \\
 \text{sd}^n V & & & &
 \end{array}$$

commutes and a homotopy $|f| \simeq g|\lambda^n|$ relative $|\text{sd}^n U|$.

Proof. Exercise, possibly. If not, I will insert a textbook reference here later. $\square$

Our line of approach is to deduce a simplicial approximation theorem for simplicial sets by turning the simplicial sets in question into simplicial complexes. Here is a particularly brutal way of building a simplicial complex from simplicial set:

V.5.11. Definition

Let X be a simplicial set. Let $\text{nd } X$ denote the set of non-degenerate simplices of X . Denoting by $\langle x \rangle$ the smallest simplicial subset generated by a non-degenerate simplex x , equip $\text{nd } X$ with the partial order given by

$$x \leq y \quad :\Leftrightarrow \quad \langle x \rangle \subseteq \langle y \rangle .$$

The *Barratt nerve* $B X$ of X is defined as

$$B X := N(\text{nd } X) .$$

Any map $f: X \rightarrow Y$ of simplicial sets induces a map of posets $\text{nd } f: \text{nd } X \rightarrow \text{nd } Y$ (since every simplex is the unique, possibly trivial, degeneration of a unique non-degenerate simplex), and thus induces a map $B f: B X \rightarrow B Y$. Therefore, we obtain a functor

$$B : \mathbf{sSet} \rightarrow \mathbf{sSet} .$$

V.5.12. Remark

In general, the Barratt nerve of a simplicial set has little to do with the original simplicial set. For example, $B(\Delta^1 / \partial \Delta^1)$ is the nerve of the poset $[1]$ and therefore simplicially contractible. On the other hand, $B(V, S)^+ \cong \text{sd}(V, S)^+$ for any locally ordered simplicial complex (V, S) .

V.5.13. Lemma

For every simplicial set X , the canonical maps

$$\text{colim}_{x \in \text{nd } X} \text{sd} \langle x \rangle \rightarrow \text{sd } X \quad \text{and} \quad \text{colim}_{x \in \text{nd } X} B \langle x \rangle \rightarrow B X$$

are isomorphisms.

Proof. Since $\text{colim}_{x \in \text{nd } X} \text{sd} \langle x \rangle \cong X$, the first part follows because sd is colimit-preserving.

Let $\xi = x_0 \leq \dots \leq x_n$ be an n -simplex in $B X$. Then ξ is also an n -simplex in $\langle x_n \rangle$. So $\text{colim}_{x \in \text{nd } X} B \langle x \rangle \rightarrow B X$ is surjective.

Suppose that $\xi = x_0 \leq \dots \leq x_k \in B \langle x \rangle$ and $\eta = y_0 \leq \dots \leq y_l \in B \langle y \rangle$ map to the same simplex in $B X$. Then ξ and η define the same sequence in $\text{nd } X$. In particular, $y_l = x_k$. Call this simplex z . Then ξ and η both lie in $B \langle z \rangle$ and define the same element there. So $\xi = \eta$ in $\text{colim}_{x \in \text{nd } X} B \langle x \rangle$. $\square$

By V.5.9, the restrictions of sd and B along Yoneda embedding $\Delta \rightarrow \mathbf{sSet}$ agree. By the universal property of the left Kan extension, there is an induced natural transformation

$$\pi : \text{sd} \rightarrow B .$$

V.5.14. Lemma

Let X be a simplicial set. the map $\pi: \text{sd } X \rightarrow \text{B } X$ is

- i) surjective in all degrees;
- ii) bijective on vertices.

In particular, two n -simplices in $\text{sd } X$ map to the same simplex in $\text{B } X$ if and only if they have the same set of vertices.

Proof. By definition, an n -simplex in $\text{B } X$ is a sequence $x_0 \leq \dots \leq x_n$ in $\text{nd } X$. Choose injective maps δ_i such that $x_i = \delta_i^*(x_n)$. Then each δ_i can be considered a simplex in $\Delta^{|x_n|}$, and thus $\delta_0 \leq \dots \leq \delta^n$ defines an n -simplex in $\text{sd } \Delta^{|x_n|}$. the pair $(\delta_0 \leq \dots \leq \delta^n, x_n)$ defines an n -simplex in $\text{sd } X$ which maps to the original n -simplex in $\text{B } X$ under π .

For any two vertices in $\text{sd } X$, choose k, l minimal such that they can be represented as

$$\Delta^0 \xrightarrow{v} \text{sd } \Delta^k \xrightarrow{\text{sd } x} \text{sd } X \quad \text{and} \quad \Delta^0 \xrightarrow{w} \text{sd } \Delta^l \xrightarrow{\text{sd } y} \text{sd } X$$

for non-degenerate simplices x and y of X . By the minimality of k and l , $v = [k]$ and $w = [l]$. If $\pi_X \circ \text{sd } x \circ v = \pi_X = \text{sd } y \circ w$, then use $(\text{B } X)_0 = \text{nd } X$ and $\text{B } \Delta^k \cong \text{sd } \Delta^k$ to see that

$$x = \text{B } x(v) = \pi_X(\text{sd } x(v)) = \pi_X(\text{sd } y(w)) = \text{B } y(w) = y.$$

□

V.5.15. Definition

A non-degenerate simplex $x \in X_n$ is *regular* if

$$\begin{array}{ccc} \Delta^{n-1} & \xrightarrow{d_n(x)} & \langle d_n(x) \rangle \\ d_n \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{x} & \langle x \rangle \end{array}$$

is a pushout square.

A simplicial set is *regular* if all its non-degenerate simplices are regular.

V.5.16. Example

If the characteristic map of every non-degenerate simplex in X is injective (i.e. X is *non-singular*), then X is regular. In particular, the nerve of every poset is regular.

V.5.17. Proposition

The subdivision $\text{sd } X$ is regular for every simplicial set X .

Proof. Let $c: \Delta^n \rightarrow \text{sd } X$ be a non-degenerate simplex. Choose k minimal such that c factors as

$$c: \Delta^n \xrightarrow{\kappa} \text{sd } \Delta^k \xrightarrow{\text{sd } x} \text{sd } \text{sk}_k(X)$$

for some non-degenerate k -simplex x of X and some non-degenerate n -simplex κ of $\text{sd } \delta^k$ (if κ was degenerate, c would also be degenerate; if x was degenerate, k would not have been

minimal). We may assume $k > 0$. Since k is minimal, $\kappa(n) = [k]$. In particular, $d_n(\kappa)$ is a simplex in $\text{sd } \partial \Delta^k$. Therefore, there is an induced commutative diagram

$$\begin{array}{ccccc} \Delta^{n-1} & \xrightarrow{d_n(\kappa)} & \text{sd } \partial \Delta^k & \longrightarrow & \text{sd } \partial \langle x \rangle \\ d_n \downarrow & & \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{\kappa} & \text{sd } \Delta^k & \longrightarrow & \text{sd } \langle x \rangle \end{array}$$

where $\partial \langle x \rangle$ denotes the simplicial set $\langle x \rangle$ with the unique top-dimensional simplex x removed. The right hand square is a pushout. Consider the outer square

$$\begin{array}{ccc} \Delta^{n-1} & \xrightarrow{d_n(c)} & \text{sd } \partial \langle x \rangle \\ d_n \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{c} & \text{sd } \langle x \rangle \end{array}$$

If y is a simplex in Δ^n which does not lie in $d_n \Delta^n$, its image contains $[n]$. Hence $c(y) \notin \text{sd } \partial \langle x \rangle$. Consequently, this square is a pullback. Since this square fits into the commutative diagram

$$\begin{array}{ccccc} \Delta^{n-1} & \xrightarrow{d_n(\kappa)} & \langle d_n(c) \rangle & \hookrightarrow & \text{sd } \partial \langle x \rangle \\ d_n \downarrow & & \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{\kappa} & \langle x \rangle & \hookrightarrow & \text{sd } \langle x \rangle \end{array}$$

and all arrows in the right hand square are monomorphisms as indicated, the square

$$\begin{array}{ccc} \Delta^{n-1} & \xrightarrow{d_n(c)} & \langle d_n(c) \rangle \\ d_n \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{c} & \langle c \rangle \end{array}$$

is also a pullback. Therefore, it suffices to show that the induced map

$$\Delta^n \setminus d_n \Delta^n \rightarrow \langle c \rangle \setminus \langle d_n(c) \rangle$$

is a degree-wise bijection. This map is surjective since $c: \Delta^n \rightarrow \langle c \rangle$ is surjective.

Suppose that σ, τ are in $\Delta^n \setminus d_n \Delta^n$ such that $c(\sigma) = c(\tau) \in \langle c \rangle \subseteq \text{sd } \langle x \rangle$. Then $c(\sigma)$ and $c(\tau)$ factor through $\text{sd } \Delta^k$, so $\kappa(\sigma) = \kappa(\tau)$. Since $\text{sd } \Delta^k$ is non-singular and κ is non-degenerate, $\sigma = \tau$. This proves the proposition. $\square$

V.5.18. Definition

A simplicial set X is *op-regular* if X^{op} is regular, i.e. for every n -simplex x the diagram

$$\begin{array}{ccc} \Delta^{n-1} & \xrightarrow{d_0(c)} & \langle d_0(x) \rangle \\ d_0 \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{x} & \langle x \rangle \end{array}$$

is a pushout. Let $\text{sSet}^{\text{opreg}}$ denote the full subcategory of sSet spanned by the op-regular simplicial sets.

V.5.19. Proposition

There exists a natural transformation

$$\gamma : B|_{\mathbf{sSet}^{\text{opreg}}} \Rightarrow \text{id}$$

such that $\lambda|_{\mathbf{sSet}^{\text{opreg}}} = \gamma \circ \pi|_{\mathbf{sSet}^{\text{opreg}}}$.

Proof. If such a factorisation γ_X exists for every op-regular simplicial set X in some full subcategory $\mathcal{C}$ of $\mathbf{sSet}^{\text{opreg}}$, then the collection $(\gamma_X)_{X \in \mathcal{C}}$ is automatically a natural transformation on $\mathcal{C}$: for every $f: X \rightarrow Y$ we have

$$\pi_X \circ B f \circ \gamma_Y = \gamma_Y \circ \pi_Y \circ \text{sd } f = \lambda_Y \circ \text{sd } f = f \circ \lambda_X = f \circ \lambda_X \circ \pi_X.$$

Since π is an epimorphism by V.5.14, naturality of γ follows. So we only need to show that γ_X exists for every $X \in \mathbf{sSet}^{\text{opreg}}$.

By V.5.13 and the preliminary observation, the existence of γ_X follows if we can show that for every non-degenerate simplex $x \in X$ there exists a commutative triangle

$$\begin{array}{ccc} \text{sd}\langle x \rangle & \xrightarrow{\pi_{\langle x \rangle}} & B\langle x \rangle \\ & \searrow \lambda_{\langle x \rangle} & \swarrow \gamma_x \\ & \langle x \rangle & \end{array}$$

We proceed by induction on the dimension of x . If x is a 0-simplex, $\text{sd}_{\text{op}}\langle x \rangle$, $B_{\text{op}}\langle x \rangle$ and $\langle x \rangle$ are all isomorphic to Δ^0 , and the existence of γ_x follows trivially.

Suppose that the dimension of x is $n > 0$. By induction hypothesis, there exists a commutative triangle

$$\begin{array}{ccc} \text{sd}\langle d_n(x) \rangle & \xrightarrow{\pi_{\langle d_0(x) \rangle}} & B\langle d_n(x) \rangle \\ & \searrow \lambda_{\langle d_0(x) \rangle} & \swarrow \gamma_{d_0(x)} \\ & \langle d_n(x) \rangle & \end{array}$$

Note the slight abuse of notation: $d_0(x)$ is not necessarily non-degenerate. However, it is the unique degeneration $d_0(x) = s(y)$ of a non-degenerate simplex y , and evidently $\langle y \rangle = \langle d_0(x) \rangle$. Since X is op-regular, $\lambda_{\langle x \rangle}$ fits into a map of pushout squares (the front and back face of the cube):

$$\begin{array}{ccccc} & \text{sd } \Delta^{n-1} & \xrightarrow{\quad} & \text{sd}\langle d_0(x) \rangle & \\ & \swarrow \lambda & \downarrow & \swarrow \lambda & \\ \Delta^{n-1} & \xrightarrow{\quad} & \langle d_n(x) \rangle & & \\ \downarrow d_0 & \downarrow & \downarrow & \downarrow & \\ & \text{sd } \Delta^n & \xrightarrow{\quad} & \text{sd}\langle x \rangle & \\ & \swarrow \lambda & \downarrow & \swarrow \lambda & \\ \Delta^n & \xrightarrow{x} & \langle x \rangle & & \end{array}$$

Consequently, we only need to factor the composite map

$$\text{sd } \Delta^n \xrightarrow{\text{sd } x} \text{sd}\langle x \rangle \xrightarrow{\lambda_{\langle x \rangle}} \langle x \rangle$$

in a way which is compatible with the given factorisation on $\text{sd } \Delta^{n-1}$.

Therefore, suppose that $\sigma, \tau: \Delta^k \rightarrow \text{sd } \Delta^n$ are k -simplices such that $\pi_{\langle x \rangle} \circ \text{sd}(x) \circ \sigma = \pi_{\langle x \rangle} \circ \text{sd}(x) \circ \tau$. Write $\sigma = \sigma_0 \leq \dots \leq \sigma_k$ and $\tau = \tau_0 \leq \dots \leq \tau_k$. By V.5.14 this means that

$$\{x \circ \sigma_0, \dots, x \circ \sigma_k\} = \{x \circ \tau_0, \dots, x \circ \tau_k\}.$$

We have to show that

$$\lambda_{\langle x \rangle} \circ \text{sd}(x) \circ \sigma = \lambda_{\langle x \rangle} \circ \text{sd}(x) \circ \tau.$$

Let i be the smallest index such that 0 lies in the image of σ_i and τ_i . Then $\sigma_j = \tau_j$ for all $j \geq i$ since x is injective on that part of Δ^n . Set $i := k + 1$ if no such index exists.

If $i = k + 1$, then σ and τ both map to $\text{sd } d_0 \Delta^n$. This case is covered by the induction hypothesis. So suppose $i < k + 1$.

If $i = 0$, then $\sigma = \tau$, so obviously $\lambda \circ \text{sd}(x) \circ \sigma = \lambda_{\langle x \rangle} \circ \text{sd}(x) \circ \tau$.

Suppose that $0 < i < k + 1$. Then $\lambda \circ \sigma_j = \lambda \circ d_0(\sigma_j)$ and $\lambda \circ \tau_j = \lambda d_0(\tau_j)$ for all $j \geq i$. Consequently, the simplices

$$\sigma' := \sigma_0 \leq \dots \leq \sigma_{i-1} \leq d_0(\sigma_i) \leq \dots \leq d_0(\sigma_k)$$

and

$$\tau' := \tau_0 \leq \dots \leq \tau_{i-1} \leq d_0(\tau_i) \leq \dots \leq d_0(\tau_k)$$

both lie in $\text{sd } d_0 \Delta^n$ and map to the same simplex as σ and τ under $\lambda \circ \text{sd}(X)$, respectively. Since $x \circ d_0(\sigma_j) = x \circ \sigma_j \circ d_0 = x \circ \tau_j \circ d_0 = x \circ d_0(\tau_j)$, the simplices σ' and τ' map to the same simplex under π , and therefore satisfy $\lambda \circ \text{sd}(x) \circ \sigma' = \lambda \circ \text{sd}(x) \circ \tau'$ by induction. This finishes the proof. $\square$

For the following discussion, abbreviate the composite functor

$$(-)^{\text{op}} \circ \text{sd} : \text{sSet} \rightarrow \text{sSet}$$

by sd_{op} . By V.5.17, sd_{op} takes values in the full subcategory of op-regular simplicial sets. It comes equipped with a natural transformation

$$\varphi : \text{sd}_{\text{op}} \rightarrow \text{id}_{\text{sSet}},$$

the *first vertex map*, induced by the transformation $(\text{chains}[n])^{\text{op}} \rightarrow [n]$, $c \mapsto \min c$.

V.5.20. Lemma

There is a natural isomorphism $\Phi: \text{sd} \cong \text{sd} \circ (-)^{\text{op}}$.

Let $\Psi: \text{sd}^2 \cong \text{sd} \text{sd}_{\text{op}}$ denote the induced isomorphism. Then there exists a simplicial homotopy

$$\text{sd}^2 X \times \Delta^1 \rightarrow X$$

from $\varphi \lambda \Psi$ to λ^2 .

Proof. It suffices to show that the lemma holds on the full subcategory of standard simplices.

For any poset P , Φ is given by

$$\Phi_P : \text{chains } P \rightarrow \text{chains}(P^{\text{op}}), \quad c \mapsto c \text{ with reversed order.}$$

The transformation λ^2 is induced by

$$\text{chains}(\text{chains}[n]) \rightarrow [n], \quad (c_0 \subseteq \dots \subseteq c_r) \mapsto \max c_r,$$

and $\varphi\lambda\Psi$ is induced by

$$\text{chains}(\text{chains}[n]) \rightarrow [n], \quad (c_0 \subseteq \dots \subseteq c_r) \mapsto \min c_0.$$

Since $\min c_0 \leq \max c_r$, these two maps combined define a natural map

$$\text{chains}(\text{chains}[n]) \times [1] \rightarrow [n]$$

which induces the required homotopy. $\square$

V.5.21. Theorem (Simplicial approximation for simplicial sets)

Let $A \subseteq X$ be finite simplicial sets and let Y be a simplicial set. Suppose that $f_0: A \rightarrow Y$ and $g: |X| \rightarrow |Y|$ are maps such that $g|_A = |f_0|$.

Then there exists for some k a map $f: \text{sd}^k \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} X \rightarrow Y$ such that

$$\begin{array}{ccc} \text{sd}^k \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} A & \xrightarrow{\varphi\lambda\varphi\lambda^k} & A \xrightarrow{f_0} Y \\ \downarrow & \nearrow f & \\ \text{sd}^k \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} X & & \end{array}$$

commutes and a homotopy $|f| \simeq g|_{\varphi\lambda\varphi\lambda^k}|$ relative $|\text{sd}^k \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} A|$.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} |\text{sd} \text{sd}_{\text{op}} A| & \xrightarrow{|\text{sd} \text{sd}_{\text{op}} f_0|} & |\text{sd} \text{sd}_{\text{op}} Y| \\ \downarrow |\text{sd} \text{sd}_{\text{op}} i| & & \downarrow |\varphi\lambda| \\ |\text{sd} \text{sd}_{\text{op}} X| & \xrightarrow{g|\varphi\lambda|} & |Y| \end{array}$$

Since $\text{sd} \text{sd}_{\text{op}} i$ is a cofibration (even the inclusion of a sub-CW complex) and $|\varphi\lambda|$ is a homotopy equivalence by V.5.20 and V.4.19, there exists a map

$$\bar{g}: |\text{sd} \text{sd}_{\text{op}} X| \rightarrow |\text{sd} \text{sd}_{\text{op}} Y|$$

such that $\bar{g} \circ |\text{sd} \text{sd}_{\text{op}} i| = |\text{sd} \text{sd}_{\text{op}} f_0|$ and $|\varphi\lambda| \circ \bar{g} \simeq g|\varphi\lambda|$ relative $|\text{sd} \text{sd}_{\text{op}} A|$.

Using V.5.17 and V.5.19, we can consider the maps

$$\begin{array}{ccc} \text{B} \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} A & \xrightarrow{\text{B} \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} f_0} & \text{B} \text{sd}_{\text{op}} Y \\ \downarrow \text{B} \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} i & & \\ \text{B} \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} X & & \end{array}$$

and

$$\tilde{g}: |\text{B} \text{sd}_{\text{op}} \text{sd} \text{sd}_{\text{op}} X| \xrightarrow{|\varphi\gamma|} |\text{sd} \text{sd}_{\text{op}} X| \xrightarrow{\bar{g}} |\text{sd} \text{sd}_{\text{op}} Y| \xrightarrow{\pi} |\text{B} \text{sd}_{\text{op}} Y|.$$

V.5.10 applies to these to yield a commutative diagram

$$\begin{array}{ccc}
 \mathrm{sd}^n B \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A & \xrightarrow{\lambda^n} & B \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A \xrightarrow{B \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} f_0} B \mathrm{sd}_{\mathrm{op}} Y \\
 \downarrow & & \nearrow f \\
 \mathrm{sd}^n B \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} X & &
 \end{array}$$

such that $|f|$ is homotopic to $\tilde{g}|\lambda^n|$ relative $|\mathrm{sd}^n B \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A|$. These maps can be composed with the appropriate instances of π and φ to obtain a commutative diagram

$$\begin{array}{ccc}
 \mathrm{sd}^{n+1} \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A & \xrightarrow{\varphi \lambda \varphi \lambda^{n+1}} A \xrightarrow{f_0} Y \\
 \downarrow & & \nearrow f \circ \mathrm{sd}^n \pi \\
 \mathrm{sd}^{n+1} \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} X & &
 \end{array}$$

such that

$$|f \circ \mathrm{sd}^n \pi| \simeq |\varphi \gamma| \tilde{g} |\lambda^n \mathrm{sd}^n \pi| \simeq g |\varphi \lambda \varphi \lambda^{n+1}|$$

relative $|\mathrm{sd}^{n+1} \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A|$. □

V.5.22. Corollary

All statements listed in V.5.6 hold.

Proof sketch. We prove i). In this situation, V.5.21 gives a simplicial approximation

$$\begin{array}{ccc}
 \mathrm{sd}^k \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} A & \xrightarrow{\varphi \lambda \varphi \lambda^k} A \xrightarrow{f_0} Y \\
 \downarrow & & \nearrow f' \\
 \mathrm{sd}^k \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} X & &
 \end{array}$$

Compose this approximation with the canonical map $\rho^\infty: K \rightarrow \mathrm{Ex}^\infty Y$. Since $A \rightarrow X$ is a cofibration and $\mathrm{Ex}^\infty Y$ is Kan, there exists by V.5.20 a map

$$f: \mathrm{sd}^{k+3} X \rightarrow \mathrm{Ex}^\infty Y$$

such that f is simplicially homotopic to $\mathrm{sd}^{k+3} X \cong \mathrm{sd}^k \mathrm{sd}_{\mathrm{op}} \mathrm{sd} \mathrm{sd}_{\mathrm{op}} X \xrightarrow{\rho^\infty \circ f} \mathrm{Ex}^\infty Y$ and extends $\rho^\infty f_0 \lambda^{k+3}$.

As X is assumed finite, $\mathrm{sd}^{k+3} X$ is also finite, so the approximation factors through some $\mathrm{Ex}^n Y$. Adjoining to a map $\mathrm{sd}^{k+n+3} X \rightarrow Y$ yields the desired approximation. □

Chapter VI.

Categorical Homotopy Theory

VI.1. Basic definitions

Our discussion of simplicial homotopy theory in the previous chapter was essentially concerned with studying the interplay between the notions of *fibration*, *cofibration* and *weak equivalence* on simplicial sets. In fact, we had three notions of fibration, but let us concentrate on Kan fibrations.

The tools we used to get a grip on things were essentially the following:

- i) the notion of a *saturated set* (V.2.4);
- ii) the *small object argument* (V.2.30);
- iii) the long exact sequence of a fibration and the analogue of Whitehead's theorem (V.3.8 and V.3.11);
- iv) the extension functor Ex^∞ (V.4.1).

Here is an attempt at illustrating the main relations we established between these notions:

TODO

Note in addition that the implications labeled with Ex^∞ can be shown to hold for maps between Kan complexes without using Ex^∞ .

We have not shown yet that weak equivalences induce equivalences on mapping spaces if the target is a Kan complex:

VI.1.1. Lemma

Let $f: X \rightarrow Y$ be a weak equivalence of simplicial sets. Then for every Kan complex K , the natural map

$$f^*: F(Y, K) \rightarrow F(X, K)$$

is a homotopy equivalence.

Proof. By V.3.11, it suffices to check that f^* induces a bijection on π_0 and isomorphisms on all π_n for $n \geq 1$. The map is a bijection on π_0 by assumption. By the small object argument (V.2.30), there exists a Kan fibration $p: P \rightarrow K$ such that P is contractible. Let G be the fibre of p over any vertex of K . Then the long exact sequence of p (V.3.8) implies that $\pi_{n+1}(F(X, K)) \cong \pi_n(F(X, G))$, so f^* induces isomorphisms on π_1 . Iterating this argument yields the analogous statement for all $n \geq 1$. $\square$

For quasicategories, we will try to develop a similar picture as far as it is possible to develop a parallel theory. Note that the mere possibility of defining analogous, but different concepts is very specific to the setting of simplicial sets. We originally decided, drawing on our intuition from topological spaces, to base our “homotopy theory” on lifting properties with respect to various cylinder inclusions. Then we observed that these lifting properties can equivalently be described in terms of horn inclusions. The theory of quasicategories is entirely based on horn inclusions; there is no description of quasicategories in terms of filling conditions for cylinders.

One of the key points in simplicial homotopy theory was the fact that all notions behave very reasonably on simplicial sets for which the projection to Δ^0 is a Kan fibration. Since quasicategories are defined by requiring the existence of inner horn fillers, there is a very natural notion of fibration to consider:

VI.1.2. Definition

Let IH denote the set of *inner horn inclusions*:

$$IH := \{\Lambda_k^n \rightarrow \Delta^n \mid 0 < k < n\}$$

VI.1.3. Definition

An *inner fibration* is a map of simplicial sets which has the RLP with respect to IH .

VI.1.4. Definition

A map is *inner anodyne* if it lies in $\text{sat}(IH)$.

VI.1.5. Corollary

- i) A map is an inner fibration if and only if it has the RLP with respect to all inner anodyne maps.
- ii) A map is inner anodyne if and only if it has the LLP with respect to all inner fibrations.

Proof. i) is clear from V.2.2. Every inner anodyne map has the LLP with respect to inner fibrations by V.2.2, and the converse of ii) is provided by V.2.31. $\square$

Despite our best efforts to come up with a notion of cofibration which is analogous to the three different notions of fibration we had before (left, right and Kan), we ended up with the same class of morphisms, namely the injections. Since we do not have any better idea what we should do now (and since homotopy extension properties do not make any sense in the absence of cylinders), we will stick to this notion of cofibration.

Defining a notion of weak equivalence was already a somewhat subtle point in the previous chapter. For maps between quasicategories, we discussed a notion of equivalence in Chapter IV; at the moment, there is no reasonable guess what a notion of “categorical equivalence” might be for arbitrary simplicial sets.

VI.1.6. Definition

A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is a *Joyal equivalence* if there exist a functor $G: \mathcal{D} \rightarrow \mathcal{C}$ and natural equivalences $GF \sim \text{id}_{\mathcal{C}}$ and $FG \sim \text{id}_{\mathcal{D}}$.

VI.1.7. Remark

Note that, the way we have defined natural equivalences, there is no ambiguity about the directionality of the natural equivalences $GF \sim \text{id}_{\mathcal{C}}$ and $FG \sim \text{id}_{\mathcal{D}}$ since natural equivalences can always be inverted.

VI.2. Inner anodyne maps

As with anodyne maps, we need a better understanding of the class of inner anodynes to get things going. In the following, we use the shorthand notation $S \boxtimes T$ for two classes of morphisms S and T to mean

$$S \boxtimes T := \{s \boxtimes t \mid s \in S, t \in T\}.$$

VI.2.1. Lemma

Every inner horn inclusion $\Lambda_k^n \rightarrow \Delta^n$ (i.e. $0 < k < n$) is a retract of $(\Lambda_k^n \rightarrow \Delta^n) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2)$.

Proof. Define maps of posets

$$s : [n] \rightarrow [n] \times [2]$$

$$i \mapsto \begin{cases} (i, 0) & i < k \\ (i, 1) & i = k \\ (i, 2) & i > k \end{cases}$$

and

$$r : [n] \times [2] \rightarrow [n]$$

$$(i, j) \mapsto \begin{cases} i & i < k, j = 0 \\ i & i > k, j = 2 \\ k & \text{else} \end{cases}$$

Evidently, $rs = \text{id}_{[n]}$. We need to check that s and r restrict appropriately on nerves. Observe that a simplex $\alpha : [p] \rightarrow [n]$ lies in Λ_k^n if and only if $[n] \setminus \{k\} \not\subseteq \text{im } \alpha$.

Note that the composition of s with the projection $\text{pr} : [n] \times [2] \rightarrow [n]$ onto the first factor is also the identity. Hence, $\text{pr } s\alpha$ lies in Λ_k^n , so $s\alpha$ is in $\Lambda_k^n \times \Delta^2$.

Let $\alpha : [p] \rightarrow [n] \times [2]$ represent a simplex in $\Lambda_k^n \times \Delta^2$, i.e. $[n] \setminus \{k\} \not\subseteq \text{im}(\text{pr } \alpha)$. Then there exists $i \neq k$ such that $i \notin \text{im}(\text{pr } \alpha)$. Since $i \neq k$, the preimage of i under r consists of precisely one point, namely either $(i, 0)$ for $i < k$ or $(i, 2)$ for $i > k$. Since $i \notin \text{im}(\text{pr } \alpha)$, we have $\text{im}(\alpha) \cap r^{-1}(i) = \emptyset$ and thus $i \notin \text{im}(r\alpha)$. Hence $r\alpha$ defines a simplex in Λ_k^n .

If $\alpha : [p] \rightarrow [n] \times [2]$ represents a simplex in $\Delta^n \times \Lambda_1^2$, then there exists $j \in \{0, 2\}$ such that $j \notin \text{im}(\text{pr}' \alpha)$, where $\text{pr}' : [n] \times [2] \rightarrow [2]$ is the projection onto the second factor. Then it follows from the definition of r that $\text{im}(r\alpha)$ contains either no $i < k$ (if $j = 0$) or no $i > k$ (if $j = 2$). Since $0 < k < n$, both the sets $\{i \mid 0 \leq i < k\}$ and $\{i \mid k < i \leq n\}$ are non-empty, so there exists some $i \neq k$ with $i \notin \text{im}(r\alpha)$. $\square$

VI.2.2. Proposition

The following classes of morphisms generate the same saturated class:

- i) IH ;
- ii) $B \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}$;
- iii) $I \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}$;
- iv) $I \boxtimes IH$.

Proof. Obviously,

$$\text{sat}(B \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}) \subseteq \text{sat}(I \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}) \subseteq \text{sat}(I \boxtimes IH).$$

The inclusion

$$I \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\} \subseteq \text{sat}(B \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\})$$

follows as in V.2.9 by considering the relative skeletal filtration of an injection.

Let i be an arbitrary injection. By VI.2.1, the map $i \boxtimes (\Lambda_k^n \rightarrow \Delta^n)$ is a retract of

$$i \boxtimes ((\Lambda_k^n \rightarrow \Delta^n) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2)) \cong (i \boxtimes (\Lambda_k^n \rightarrow \Delta^n)) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2).$$

Since $i \boxtimes (\Lambda_k^n \rightarrow \Delta^n)$ is an injection, this shows

$$I \boxtimes IH \subseteq \text{sat}(I \boxtimes (\Lambda_1^2 \rightarrow \Delta^2)).$$

Also by VI.2.1, IH is contained in the saturation of $I \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}$.

So the only thing left to show is that $(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2)$ is contained in the saturation of IH . We do this by explicit horn-filling.

For $n = 0$, note that

$$(\partial\Delta^n \rightarrow \Delta^n) \boxtimes (\Lambda_1^2 \rightarrow \Delta^2) \cong (\Lambda_1^2 \rightarrow \Delta^2),$$

so we may assume $n > 0$.

The simplicial set $\Delta^n \times \Delta^2$ is the nerve of the poset $[n] \times [2]$, so non-degenerate simplices are given by injective maps $[p] \rightarrow [n] \times [2]$. For $p < n$, the composition of any map $\alpha: [n] \times [2]$ with the projection $\text{pr}: [n] \times [2] \rightarrow [n]$ cannot be surjective, so such a simplex is contained in $\partial\Delta^n \times \Delta^2$. In particular, $(\partial\Delta^n \times \Delta^2) \cup_{\partial\Delta^n \times \Lambda_1^2} (\Delta^n \times \Lambda_1^2)$ is only missing n -simplices,

$(n+1)$ -simplices and $(n+2)$ -simplices. Moreover, all $(n+2)$ -simplices are missing.

Any injective map $[n+2] \rightarrow [n] \times [2]$ is uniquely determined by the two indices at which its value jumps in the second coordinate. Hence we can enumerate all non-degenerate $(n+2)$ -simplices in $\Delta^n \times \Delta^2$ as $(\tau_{i,j})_{0 \leq i \leq j \leq n}$, where we set

$$\tau_{ij}(k) := \begin{cases} (k, 0) & 0 \leq k \leq i \\ (k-1, 1) & i+1 \leq k \leq j+1 \\ (k-2, 2) & j+2 \leq k \leq n+2 \end{cases}$$

To glue these simplices in, we need to understand their face structure:

- 1) For $l \notin \{i, i+1, j+1, j+2\}$, $\text{pr} \circ d_l \tau_{ij}$ is not surjective, So $d_l \tau_{ij} \in \partial\Delta^n \times \Delta^2$.

2) The simplex $d_{j+2}\tau_{ij}$ is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k \leq i \\ (k-1, 1) & i+1 \leq k \leq j+1 \\ (k-1, 2) & j+2 \leq k \leq n+1 \end{cases}$$

Call this simplex σ_{ij} . It is defined for $0 \leq i \leq j < n$, and it is a non-degenerate $(n+1)$ -simplex in $\Delta^n \times \Delta^2$. Since both $\text{pr} \circ \sigma_{ij}$ and $\text{pr}' \circ \sigma_{ij}$ are surjective, where $\text{pr}' : [n] \times [2] \rightarrow [2]$ is the projection onto the second coordinate, none of these simplices is contained in $\partial\Delta^n \times \Delta^2 \cup_{\partial\Delta^n \times \Lambda_1^2} (\Delta^n \times \Lambda_1^2)$.

3) For $i < j$, the simplex $d_{j+1}\tau_{ij}$ is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k \leq i \\ (k-1, 1) & i+1 \leq k \leq j \\ (k-1, 2) & j+1 \leq k \leq n+1 \end{cases}$$

So $d_{j+1}\tau_{ij} = \sigma_{i,j-1}$.

4) If $i = 0$, then $0 \notin \text{pr}' \circ d_i\tau_{ij}$, so $d_i\tau_{ij} \in \Delta^n \times \Lambda_1^2$.

5) If $i > 0$, then $d_i\tau_{ij}$ is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k < i \\ (k, 1) & i \leq k \leq j \\ (k-1, 2) & j+1 \leq k \leq n+1 \end{cases}$$

so $d_i\tau_{ij} = d_i\tau_{i-1,j}$.

The remaining face $d_{i+1}\tau_{ij}$ is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k \leq i \\ (k, 1) & i+1 \leq k \leq j \\ (k-1, 2) & j+1 \leq k \leq n+1 \end{cases}$$

This entails:

- 6) For $k < i$, $d_{i+1}\tau_{ij}$ is not a face of τ_{kj} for any $k < i$: this holds since $(i, 0) \in \text{im}(d_{i+1}\tau_{ij})$, but $(i, 0) \notin \text{im}(\tau_{kj})$.
- 7) For $l < j$, $d_{i+1}\tau_{ij}$ is not a face of τ_{kl} : this holds since $(j, 1) \in \text{im}(d_{i+1}\tau_{ij})$, but $(j, 1) \notin \text{im}(\tau_{kl})$.

The collection of $(n+1)$ -simplices $(\sigma_{ij})_{0 \leq i \leq j < n}$ naturally appeared as faces of the τ_{ij} , so let us understand their face structure next:

- 8) For $l \notin \{i, i+1\}$, $\text{pr} \circ d_l\sigma_{ij}$ is not surjective, so $d_l\sigma_{ij} \in \partial\Delta^n \times \Delta^2$.
- 9) If $i = 0$, then $0 \notin \text{im}(\text{pr} \circ d_i\sigma_{ij})$, so $d_0\sigma_{0j} \in \partial\Delta^n \times \Delta^2$.

10) If $i > 0$, the face $d_i\sigma_{ij}$ is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k < i \\ (k, 1) & i \leq k \leq j \\ (k, 2) & j+1 \leq k \leq n \end{cases}$$

so $d_i\sigma_{ij} = d_i\sigma_{i-1,j}$.

Once more, the interesting remaining face is $d_{i+1}\sigma_{ij}$. This face is given by

$$k \mapsto \begin{cases} (k, 0) & 0 \leq k \leq i \\ (k, 1) & i+1 \leq k \leq j \\ (k, 2) & j+1 \leq k \leq n \end{cases}$$

Observe the following:

- 11) $d_{i+1}\sigma_{ij} \notin (\partial\Delta^n \times \Delta^2) \cup_{\partial\Delta^n \times \Lambda_1^2} (\Delta^n \times \Lambda_1^2)$: this holds since $\{0, 2\} \subseteq \text{im}(\text{pr}' \circ d_{i+1}\sigma_{ij})$ and $\text{pr} \circ d_{i+1}\sigma_{ij}$ is surjective.
- 12) For $k < i$ and $l \leq j$, $d_{i+1}\sigma_{ij}$ is not a face of σ_{kl} : this holds since $(i, 0) \in \text{im}(d_{i+1}\sigma_{ij})$ and $(i, 0) \notin \text{im}(\sigma_{kl})$.
- 13) For $l < j$, $d_{i+1}\sigma_{ij}$ is not a face of σ_{kl} : if $i < j$, then $(j, 1) \in \text{im}(d_{i+1}\sigma_{ij})$, but $(j, 1) \notin \text{im}(\sigma_{kl})$. If $i = j$, then $k \leq l < j = i$, so 12) applies.

Now we only need to assemble the information we have gathered. Define $X(0, 0) := (\partial\Delta^n \times \Delta^2) \cup_{\partial\Delta^n} (\Delta^n \times \Lambda_1^2)$. Proceeding by induction, we build simplicial subsets of $\Delta^n \times \Delta^2$ by letting $X(i+1, j)$ be given by the pushout

$$\begin{array}{ccc} \Lambda_{i+1}^{n+1} & \xrightarrow{\sigma_{ij}|_{\dots}} & X(i, j) \\ \downarrow & & \downarrow \\ \Delta^{n+1} & \xrightarrow{\sigma_{ij}} & X(i+1, j) \end{array}$$

for $0 \leq i \leq j < n$ and setting $X(0, j+1) := X(j+1, j)$. We have to check that the pushout squares above actually exist as claimed:

- For the start of induction, 8) and 9) imply that σ_{00} induces a map $\Lambda_1^{n+1} \rightarrow X(0, 0)$. 11) shows that σ_{00} induces an injective map $X(1, 0) \rightarrow \Delta^n \times \Delta^2$.
- For $i = 0$ and $j \geq 1$, 8) and 9) imply that σ_{0j} induces a map $\Lambda_1^{n+1} \rightarrow X(0, j)$. 11) and 13) show that σ_{0j} induces an injective map $X(1, j) \rightarrow \Delta^n \times \Delta^2$.
- For $i \geq 1$ and $j \geq 1$, 8) and 10) imply that σ_{ij} induces a map $\Lambda_1^{n+1} \rightarrow X(i, j)$. 11), 12) and 13) show that σ_{ij} induces an injective map $X(i+1, j) \rightarrow \Delta^n \times \Delta^2$.

Observe that $X(i, j) \rightarrow X(i+1, j)$ is inner anodyne. Thus, so is $X(0, j) \rightarrow X(0, j+1)$ and therefore $X(0, 0) \rightarrow X(0, n)$. Set $Y(0, 0) := X(0, n)$.

Then we do a second induction to define for $i \leq j \leq n$ the simplicial subset $Y(i+1, j)$ by a pushout

$$\begin{array}{ccc} \Lambda_{i+1}^{n+2} & \xrightarrow{\tau_{ij}|_{\dots}} & Y(i, j) \\ \downarrow & & \downarrow \\ \Delta^{n+2} & \xrightarrow{\tau_{ij}} & Y(i+1, j) \end{array}$$

and by defining $Y(0, j+1) := Y(j+1, j)$. Again, we have to check that the pushout squares above actually exist as claimed:

- For the start of the induction, [1\)](#), [2\)](#) and [4\)](#) show that τ_{00} induces a map $\Lambda_{i+1}^{n+1} \rightarrow Y(0, 0)$. By the formula for $d_1\tau_{00}$, one observes that $\text{im}(d_1\tau_{00}) = \{0, 2\}$, and none of the $(n+1)$ -simplices in $Y(0, 0)$ has this property. So the pushout $Y(1, 0)$ is in fact a simplicial subset of $\Delta^n \times \Delta^2$.
- For $i = 0$ and $j \geq 1$, [1\)](#), [2\)](#), [3\)](#) and [4\)](#) imply that τ_{ij} induces a map $\Lambda_{i+1}^{n+1} \rightarrow Y(0, j)$. The remaining face $d_1\tau_{0j}$ is not of the form σ_{kl} , and by [6\)](#) and [7\)](#) it is also not a face if any of the preceding τ_{kl} . So $Y(1, j)$ is a simplicial subset of $\Delta^n \times \Delta^2$.
- For $i, j \geq 1$, [1\)](#), [2\)](#), [3\)](#) and [5\)](#) imply that τ_{ij} induces a map $\Lambda_{i+1}^{n+1} \rightarrow Y(i, j)$. The remaining face $d_{i+1}\tau_{ij}$ is not of the form σ_{kl} , and by [6\)](#) and [7\)](#) it is also not a face of any of the preceding τ_{kl} . So $Y(i+1, j)$ is a simplicial subset of $\Delta^n \times \Delta^2$.

All the inclusions $Y(i, j) \rightarrow Y(i+1, j)$ are inner anodyne by definition, and therefore so is $Y(0, 0) \rightarrow Y(0, n+1)$. Since $Y(0, n+1)$ contains all non-degenerate $(n+2)$ -simplices of $\Delta^n \times \Delta^2$, we have $Y(0, n+1) = \Delta^n \times \Delta^2$. Consequently, the inclusion $(\partial\Delta^n \times \Delta^2) \cup_{\partial\Delta^n \times \Lambda_1^2} (\Delta^n \times \Lambda_1^2) = X(0, 0) \rightarrow X(0, n) = Y(0, 0) \rightarrow Y(0, n+1) = \Delta^n \times \Delta^2$ is inner anodyne as claimed. $\square$

VI.2.3. Corollary

Let i be an inner anodyne morphism and j a cofibration. Then $i \boxtimes j$ is inner anodyne.

Proof. The class

$$\Sigma := \{i \mid i \boxtimes j \text{ inner anodyne}\}$$

is saturated by Exercise 3 on Sheet 7. Hence, it suffices to check that $IH \subseteq \Sigma$. This is true by [VI.2.2](#). $\square$

VI.2.4. Corollary

Let $i: A \rightarrow B$ be a cofibration and let $p: X \rightarrow Y$ be an inner fibration. Then induced map

$$F(B, X) \rightarrow F(A, X) \times_{F(A, Y)} F(B, Y)$$

is an inner fibration. If i is inner anodyne, this map is a trivial Kan fibration.

Proof. As usual, every lifting problem

$$\begin{array}{ccc} U & \longrightarrow & F(B, X) \\ j \downarrow & & \downarrow \\ V & \longrightarrow & F(A, X) \times_{F(A, Y)} F(B, Y) \end{array}$$

translates to a lifting problem

$$\begin{array}{ccc} (U \times B) \cup_{U \times A} (V \times A) & \longrightarrow & X \\ i \boxtimes j \downarrow & & \downarrow \\ V \times B & \longrightarrow & Y \end{array}$$

If i or j is inner anodyne, then $i \boxtimes j$ is inner anodyne by VI.2.3. This implies the corollary. $\square$

VI.2.5. Corollary

A simplicial set $\mathcal{C}$ is a quasicategory if and only if the restriction map $F(\Delta^2, \mathcal{C}) \rightarrow F(\Lambda_1^2, \mathcal{C})$ is a trivial Kan fibration.

Proof. Suppose that $\mathcal{C}$ is a quasicategory, i.e. $\mathcal{C} \rightarrow \Delta^0$ is an inner fibration. Since $\Lambda_1^2 \rightarrow \Delta^2$ is inner anodyne, it follows from VI.2.4 that $F(\Delta^2, \mathcal{C}) \rightarrow F(\Lambda_1^2, \mathcal{C})$ is a trivial Kan fibration. Conversely, VI.2.2 implies that it suffices to show that $\mathcal{C} \rightarrow \Delta^0$ has the RLP with respect to $I \boxtimes \{\Lambda_1^2 \rightarrow \Delta^2\}$. Let $i: A \rightarrow B$ be a cofibration. Since every lifting problem

$$\begin{array}{ccc} (A \times \Delta^2) \cup_{A \times \Lambda_1^2} (B \times \Lambda_1^2) & \longrightarrow & \mathcal{C} \\ i \boxtimes (\Lambda_1^2 \rightarrow \Delta^2) \downarrow & & \downarrow \\ B \times \Delta^2 & \longrightarrow & \Delta^0 \end{array}$$

corresponds to a lifting problem

$$\begin{array}{ccc} A & \longrightarrow & F(\Delta^2, \mathcal{C}) \\ i \downarrow & & \downarrow \\ B & \longrightarrow & F(\Lambda_1^2, \mathcal{C}) \end{array}$$

a solution exists by assumption. $\square$

VI.2.6. Corollary (Theorem B)

Let $f: y \rightarrow z$ and $g: x \rightarrow y$ be edges in a quasicategory $\mathcal{C}$. Then $\text{comp}(f, g) := \Delta^0 \times_{F(\Lambda_1^2, \mathcal{C})} F(\Delta^2, \mathcal{C})$ is a contractible Kan complex.

Proof. This is immediate from VI.2.4 and V.3.18. $\square$

VI.2.7. Remark

VI.2.4 also implies that the “mapping space”

$$\mathrm{Hom}_{\mathcal{C}}(x, y) = \Delta^0 \times_{\mathcal{C} \times \mathcal{C}} F(\Delta^1, \mathcal{C})$$

between any two objects x, y in a quasicategory $\mathcal{C}$ is itself a quasicategory (because $\partial\Delta^1 \rightarrow \Delta^1$ is a cofibration). Of course, this is still quite a bit away from the statement of [Theorem A](#) (claiming that it is a Kan complex).

VI.2.8. Corollary [Theorem D, first part]

If $\mathcal{C}$ is a quasicategory and X is a simplicial set, then $F(X, \mathcal{C})$ is also a quasicategory.

Proof. Apply [VI.2.4](#) with $i = \emptyset \rightarrow X$ and $p: \mathcal{C} \rightarrow \Delta^0$. □

As a consequence of [VI.2.8](#), we will prefer to write $\mathrm{Fun}(X, \mathcal{C})$ instead of $F(X, \mathcal{C})$ if $\mathcal{C}$ is a quasicategory.

[VI.2.8](#) puts us in a position where we can finally define what a “categorical equivalence” between arbitrary simplicial sets is.

VI.2.9. Definition

A map $f: X \rightarrow Y$ of simplicial sets is a *Joyal equivalence* (or *categorical equivalence*) if the restriction map

$$\pi f^*: \pi \mathrm{Fun}(Y, \mathcal{C}) \rightarrow \pi \mathrm{Fun}(X, \mathcal{C})$$

is an equivalence of categories for every quasicategory $\mathcal{C}$.

VI.2.10. Corollary

A map of quasicategories is a Joyal equivalence in the sense of [VI.1.6](#) if and only if it is a Joyal equivalence in the sense of [VI.2.9](#).

Proof. If $F: \mathcal{C} \rightarrow \mathcal{D}$ admits an inverse $G: \mathcal{D} \rightarrow \mathcal{C}$ witnessed by natural equivalences $GF \sim \mathrm{id}_{\mathcal{C}}$ and $FG \sim \mathrm{id}_{\mathcal{D}}$, then $F^*: \pi \mathrm{Fun}(\mathcal{D}, \mathcal{E}) \rightarrow \pi \mathrm{Fun}(\mathcal{C}, \mathcal{E})$ has an evident inverse, namely G^* .

Conversely, the essential surjectivity of $\pi F^*: \pi \mathrm{Fun}(\mathcal{D}, \mathcal{C}) \rightarrow \pi \mathrm{Fun}(\mathcal{C}, \mathcal{C})$ yields a functor $G: \mathcal{D} \rightarrow \mathcal{C}$ such that $GF \sim \mathrm{id}_{\mathcal{C}}$. Then $\pi G^*: \pi \mathrm{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \pi \mathrm{Fun}(\mathcal{D}, \mathcal{D})$ is also an equivalence, so there exists $F': \mathcal{C} \rightarrow \mathcal{D}$ such that $F'G \sim \mathrm{id}_{\mathcal{D}}$. It follows that $FG \sim F'GFG \sim F'G \sim \mathrm{id}_{\mathcal{D}}$.

VI.2.11. Example

A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ of categories is an equivalence if and only if NF is a Joyal equivalence.

VI.2.12. Lemma

- i) A map of Kan complexes is a Joyal equivalence if and only if it is a homotopy equivalence.
- ii) Every trivial Kan fibration is a Joyal equivalence.
- iii) every inner anodyne map is a Joyal equivalence.

iv) Every Joyal equivalence is a weak equivalence.

Proof. Any Joyal equivalence of quasicategories is a particular kind of simplicial homotopy equivalence. Conversely, if $f: X \rightarrow Y$ is a homotopy equivalence, we find $g: Y \rightarrow X$ and homotopies $h: gf \simeq \text{id}_X$ and $k: gf \simeq \text{id}_Y$. Since $F(X, X)$ and $F(Y, Y)$ are Kan complexes, there are also homotopies $h': \text{id}_X \simeq gf$ and $k': \text{id}_Y \simeq gf$ such that h and h' compose to the constant homotopy on gf , and k and k' compose to the constant homotopy on id_Y (compositions are taken in $F(X, X)$ and $F(Y, Y)$, respectively). The diagram

$$\begin{array}{ccc}
 & gf & \xrightarrow{s(gf)} gf \\
 h' \nearrow & & \searrow h \\
 \text{id}_X & \xrightarrow{s(\text{id}_X)} \text{id}_X & \nearrow h'
 \end{array}$$

in $F(X, X)$ defines a right horn: the bottom face and left face are degenerate with the dotted arrow unlabeled, the right face is the one arising from the definition of h' , and the front face is missing. Filling this horn shows that h' and h also compose to the constant homotopy on id_X . The argument for k and k' is analogous.

Let $p: X \rightarrow Y$ be a trivial Kan fibration. Choose a section $s: Y \rightarrow X$. Then the lifting problem

$$\begin{array}{ccccc}
 X \times \Delta^{\{0,2\}} \sqcup X \times \Delta^{\{1,3\}} & \xrightarrow{\text{pr} \sqcup \text{pr}} & X \sqcup X & \xrightarrow{\text{id}_X \sqcup sp} & X \\
 \downarrow & & & & \downarrow p \\
 X \times \Delta^3 & \xrightarrow{\text{pr}} & X & \xrightarrow{p} & Y
 \end{array}$$

has a solution. Restricting to along the spine inclusion $X \times I_3 \rightarrow X \times \Delta^3$ yields three natural transformations $\tau_1: \text{id}_X \Rightarrow sp$, $\tau_2: sp \Rightarrow \text{id}_X$, $\tau_3: \text{id}_X \Rightarrow sp$ such that id_X is a composition of both τ_2 and τ_1 as well as τ_3 and τ_2 (in $\text{Fun}(X, X)$). Since equivalences in a quasicategory satisfy two-out-of-six (by III.7), it follows that sp and id_X are naturally equivalent.

Let $i: A \rightarrow B$ be inner anodyne. By VI.2.4, the restriction map

$$i^*: \text{Fun}(B, C) \rightarrow \text{Fun}(A, C)$$

is a trivial Kan fibration. The first part of the lemma shows that i^* is a Joyal equivalence, so it follows that πi^* is an equivalence of categories.

Let $f: X \rightarrow Y$ be a Joyal equivalence. Since Kan complexes are quasicategories, f induces for every Kan complex K an equivalence of categories

$$\pi f^*: \pi F(Y, K) \rightarrow \pi F(X, K).$$

Equivalences of categories induce bijections on the sets of connected components, so $\pi_0 f^*$ is a bijection. $\square$

VI.2.13. Remark

It may seem pedantic that we gave proofs for VI.2.12 ii) and i), but here is an example to illustrate that this is necessary: Suppose that $\mathcal{C}$ and $\mathcal{D}$ are Kan-enriched categories, and that we have functors $F: \mathcal{C} \rightarrow \mathcal{D}$ as well as $G: \mathcal{D} \rightarrow \mathcal{C}$ together with natural transformations $u: \text{id}_{\mathcal{C}} \rightarrow GF$ and $c: FG \rightarrow \text{id}_{\mathcal{D}}$ such that u_x is a weak equivalence for every $x \in \mathcal{C}$ and

c_y is a weak equivalence for every $y \in \mathcal{D}$; the adjunction between $|-| : \text{Kan} \rightarrow \text{CW}$ and $\text{Sing} : \text{CW} \rightarrow \text{Kan}$ is an example of this situation.

Applying the coherent nerve, we obtain functors $N^c F, N^c G$ together with natural transformations $\text{id} \rightarrow N^c G N^c F$ and $N^c F N^c G \rightarrow \text{id}$ through equivalences. But this is not yet enough to conclude that $N^c F$ is a Joyal equivalence because we cannot say anything at the moment about the invertibility of the natural transformations. This would either rely on [Theorem E](#), which we have not proved yet, or we would have to provide inverse transformation on the Kan-enriched level.

VI.3. Joins and Slices

The next result from the list announced in Chapter III that we are going to prove is [Theorem C](#), but the proof requires some preparations. It relies on the formalism of *slices*. To motivate their definition, consider the following situation: let $\mathcal{C}$ be a category and $f : x \rightarrow y$ a morphism in $\mathcal{C}$. Then one observes that f is an isomorphism in $\mathcal{C}$ if and only if the forgetful functor

$$\mathcal{C}_{f/} := \{\text{cones under } f\} \rightarrow \{\text{cones under } x\} = x/\mathcal{C}$$

$$\begin{array}{ccc} x & & \\ f \downarrow & \searrow & \\ y & & \end{array} \quad c \mapsto (x \rightarrow x)$$

is an isomorphism of categories. The “only if” part is clear. If $\mathcal{C}_{f/} \rightarrow x/\mathcal{C}$ is an isomorphism, then $x \xrightarrow{\text{id}_x} x$ has a preimage, which is a morphism $g : y \rightarrow x$ such that $gf = \text{id}_x$. Since

$$\begin{array}{ccc} & y & \\ f \nearrow & \uparrow f & \\ x & & x \\ \text{id}_x \searrow & & \end{array}$$

is a morphism in $x/\mathcal{C}$, the pair (id_x, g) is a preimage of the domain and (f, id_y) is a preimage of the target, it follows that $fg = \text{id}_y$.

It will turn out that a similar statement holds for quasicategories, namely that a morphism f in a quasicategory is an equivalence if and only if a certain forgetful functor $\mathcal{C}_{f/} \rightarrow \mathcal{C}_{x/}$ is a trivial Kan fibration. It will then only be a matter of decoding to see that certain outer horn fillers exist. Note that we have not yet made sense of the symbols, but they are supposed to indicate categories or appropriate cones.

In the 1-categorical situation, the following construction formalises the concept of cones:

VI.3.1. Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be ordinary categories. Define their *join* $\mathcal{C} \star \mathcal{D}$ as the category whose objects are $\text{obj } \mathcal{C} \sqcup \text{obj } \mathcal{D}$ and whose morphisms are given by

$$\text{Hom}_{\mathcal{C} \star \mathcal{D}}(x, y) := \begin{cases} \text{Hom}_{\mathcal{C}}(x, y) & x, y \in \mathcal{C} \\ \text{Hom}_{\mathcal{D}}(x, y) & x, y \in \mathcal{D} \\ \{*\} & x \in \mathcal{C}, y \in \mathcal{D} \\ \emptyset & x \in \mathcal{D}, y \in \mathcal{C} \end{cases}$$

together with the obvious composition rules.

So the join connects each object in $\mathcal{C}$ with each object in $\mathcal{D}$ via unique morphism. Now note that objects in $\mathcal{C}_{f/}$ are given by certain diagrams $[1] \star \{c\} \rightarrow \mathcal{C}$, and morphisms in $\mathcal{C}_{f/}$ correspond to certain functors $[1] \star \{c \leq c'\} \rightarrow \mathcal{C}$. This suggests to define an n -simplex in the quasicategory $\mathcal{C}_{f/}$ to be a certain diagram $\Delta^1 \star \Delta^n \rightarrow \mathcal{C}$. Of course, this is in no way specific to the situation of cones under a morphism, and Δ^1 can be replaced by any diagram shape. This leads us to consider joins of arbitrary simplicial sets.

VI.3.2. Definition

Define the functor cut as the representable functor

$$\text{cut} := \text{Hom}_{\text{LinOrd}}(-, [1]) : \text{LinOrd}^{\text{op}} \rightarrow \text{Set}$$

Note that for $L \in \text{LinOrd}$, there is a natural bijection

$$\text{Hom}_{\text{LinOrd}}(L, [1]) \cong \{(L_0, L_1) \mid L_0 \cup L_1 = L, L_0 \cap L_1 = \emptyset, \forall l_0 \in L_0 \forall l_1 \in L_1 : l_0 < l_1\},$$

which explains the name of the functor.

VI.3.3. Definition

Let $c: [n] \rightarrow [1]$ be a cut in $[n]$. Define for $i \in \{0, 1\}$ the simplicial set $\Delta_{c,i}$ as the pullback

$$\begin{array}{ccc} \Delta_{c,i} & \longrightarrow & \Delta^n \\ \downarrow & & \downarrow c \\ \{i\} & \longrightarrow & \Delta^1 \end{array}$$

Observe that $\Delta_{c,i}$ is isomorphic to a standard simplex.

VI.3.4. Definition

Let X and Y be simplicial sets. Define their *join* as follows: for $n \geq 0$, set

$$(X \star Y)_n := \coprod_{c \in \text{cut}[n]} \text{Hom}_{\text{sSet}}(\Delta_{c,0}, X) \times \text{Hom}_{\text{sSet}}(\Delta_{c,1}, Y).$$

Given $\alpha: [m] \rightarrow [n]$, the universal property of the pullback construction induces maps $\Delta_{c\alpha,i} \rightarrow \Delta_{c,i}$, and precomposition with these maps turns $X \star Y$ into a simplicial set.

VI.3.5. Remark

The join of two simplicial sets is slightly awkward to write down since we chose to define Δ as the category with object set $\{[n] \mid n \geq 0\}$. Note that Δ is a skeleton of the category of all non-empty, finite linear orders $\text{LinOrd}_{\text{fin}}$, so

$$\text{sSet} \simeq \text{Fun}(\text{LinOrd}_{\text{fin}}^{\text{op}}, \text{Set}).$$

In the latter category, we could define the join of X and Y by

$$(X \star Y)_n := \coprod_{c \in \text{cut}[n]} X(c^{-1}(0)) \times Y(c^{-1}(1)).$$

This is not the only construction that is easier to define. With the enlarged indexing category, the functor $(-)^{\text{op}}$ is for example simply induced by the endofunctor of $\text{LinOrd}_{\text{fin}}$ which reverses the ordering on each object.

VI.3.6. Example

By inspection, one observes that

- 1) $\Delta^0 \star \Delta^0 \cong \Delta^1$;
- 2) $X \star \emptyset \cong X$ and $\emptyset \star Y \cong Y$.

VI.3.7. Lemma

The join construction defines a functor $- \star -: \text{sSet} \times \text{sSet} \rightarrow \text{sSet}$.

Proof. Given maps $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$, composition with f and g induces maps

$$(X \star Y)_n \rightarrow (X' \star Y')_n$$

for all n , and it is straightforward to check that these define maps of simplicial sets. $\square$

VI.3.8. Corollary

Let X and Y be simplicial sets. Then $X \star -$ and $- \star Y$ factor as

$$X \star -: \text{sSet} \rightarrow X/\text{sSet} \rightarrow \text{sSet} \quad \text{and} \quad - \star Y: \text{sSet} \rightarrow Y/\text{sSet} \rightarrow \text{sSet}.$$

Proof. By VI.3.7, the unique map $\emptyset \rightarrow Y$ induces a map $X \cong X \star \emptyset \rightarrow X \star Y$, which defines an object in X/sSet . Functoriality is obvious. The other case is analogous. $\square$

VI.3.9. Lemma

For every map of simplicial sets $p: X \rightarrow \Delta^1$, there exists a factorization

$$X \xrightarrow{q} X_0 \star X_1 \xrightarrow{c} \Delta^1,$$

where $X_i = p^{-1}(\{i\})$ and $c: X_0 \star X_1 \rightarrow \Delta^0 \star \Delta^0 \cong \Delta^1$ is induced by the projection maps. This factorization is initial among maps

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y \star Z \\ & \searrow & \swarrow \\ & \Delta^1 & \end{array}$$

such that $Y \star Z \rightarrow \Delta^1$ is induced by the projection maps to Δ^0 .

Proof. We need to define q . Let $x: \Delta^n \rightarrow X$ be an n -simplex. Then the composition px determines a cut in $[n]$. Define

$$q(x) := (\Delta_{px,0} \rightarrow \Delta^n \xrightarrow{x} X, \Delta_{px,1} \rightarrow \Delta^n \xrightarrow{x} X).$$

It is straightforward to check that this defines a map of simplicial sets and that it is initial in the claimed sense. $\square$

VI.3.10. Example

Let $\mathcal{C}$ and $\mathcal{D}$ be ordinary categories. The canonical projection $\mathcal{C} \star \mathcal{D} \rightarrow [1]$ induces by VI.3.9 a factorization

$$N(\mathcal{C} \star \mathcal{D}) \rightarrow N\mathcal{C} \star N\mathcal{D} \rightarrow \Delta^1.$$

Since there exists a unique morphism from each object in $\mathcal{C}$ to each object in $\mathcal{D}$, it is easy to see that the first map is an isomorphism. So there is a natural isomorphism

$$N(\mathcal{C} \star \mathcal{D}) \cong N\mathcal{C} \star N\mathcal{D}.$$

In particular, there are canonical isomorphisms

$$\Delta^{n+m+1} \cong \Delta^n \star \Delta^m.$$

VI.3.11. Lemma

If $\mathcal{C}$ and $\mathcal{D}$ are quasicategories, then $\mathcal{C} \star \mathcal{D}$ is also a quasicategory.

Proof. Let $f: \Lambda_k^n \rightarrow \mathcal{C} \star \mathcal{D}$ be an inner horn. If f factors via one of the maps $\mathcal{C} \rightarrow \mathcal{C} \star \mathcal{D}$ or $\mathcal{D} \rightarrow \mathcal{C} \star \mathcal{D}$, the map f obviously extends. Note that this is the case if and only if the composite map $\Lambda_k^n \xrightarrow{f} \mathcal{C} \star \mathcal{D} \xrightarrow{p} \Delta^1$ is constant.

So let us assume that pf is not constant. Since Δ^1 is the nerve of a category, pf extends uniquely to a map $q: \Delta^n \rightarrow \Delta^1$ which we identify with a map $[n] \rightarrow [1]$. By VI.3.9, there is a factorization

$$f: \Lambda_k^n \rightarrow L_0 \star L_1 \xrightarrow{c} \mathcal{C} \star \mathcal{D}.$$

A map $\alpha: [r] \rightarrow [n]$ defines a simplex in L_0 if and only if there exists some $i \neq k$ such that $i \notin \text{im}(\alpha)$ and if $q\alpha$ takes the constant value 0. Since q is not constant, the latter condition implies that $n \notin \text{im}(\alpha)$. As $n \neq k$, this automatically entails $\alpha \in \Lambda_k^n$. Consequently, $L_0 \cong \Delta^{|q^{-1}(0)|}$.

Similarly, if $q\alpha$ takes the constant value 1, then $0 \notin \text{im}(\alpha)$ and $\alpha \in \Lambda_k^n$ since $0 \neq k$. Therefore, $L_1 \cong \Delta^{|q^{-1}(1)|-1}$.

by VI.3.10, $L_0 \star L_1 \cong \Delta^n$, so c is the desired extension of f . $\square$

VI.3.12. Lemma

Let $\mathcal{C}$ be a category and $x \in \mathcal{C}$.

- i) If $\mathcal{C}$ is cocomplete, then $x/\mathcal{C}$ and $\mathcal{C}/x$ are cocomplete. The forgetful functor $x/\mathcal{C} \rightarrow \mathcal{C}$ preserves connected colimits, the forgetful functor $\mathcal{C}/x \rightarrow \mathcal{C}$ preserves all colimits.
- ii) If $\mathcal{C}$ is complete, then $x/\mathcal{C}$ and $\mathcal{C}/x$ are complete. The forgetful functor $x/\mathcal{C} \rightarrow \mathcal{C}$ preserves all limits, the forgetful functor $\mathcal{C}/x \rightarrow \mathcal{C}$ preserves all connected limits.

Proof. We prove only that part of the lemma we will actually need. Suppose $\mathcal{C}$ is cocomplete and let $F: I \rightarrow x/\mathcal{C}$ be a diagram. Then F defines a diagram $G: \{-\infty\} \star I \rightarrow \mathcal{C}$, which admits a colimit by assumption. The structure map $x = G(-\infty) \rightarrow \operatorname{colim} G$ defines an object in $x/\mathcal{C}$.

Let $F^+: I \star \{\infty\} \rightarrow x/\mathcal{C}$ be an arbitrary cone under F . Then F^+ corresponds to a diagram $G^+: \{-\infty\} \star I \star \{\infty\} \rightarrow \mathcal{C}$, which is a cone under G . So there is an unique induced map $\operatorname{colim} G \rightarrow G^+(\infty)$, which is compatible with the structure maps $x \rightarrow \operatorname{colim} G$ and $x \rightarrow G^+(\infty)$ by definition. Therefore, $x \rightarrow \operatorname{colim} G$ is a colimit of F .

Note that there is always a canonical map $\operatorname{colim} G|_I \rightarrow \operatorname{colim} G$. Let us try to turn $\operatorname{colim} G|_I$ into a cone under G . For $i \in I$, choose as $G(i) \rightarrow \operatorname{colim} G|_I$ the respective structure map of the colimit. Let $\sim$ be the equivalence relation on I given by $i \sim j$ if the maps $x \rightarrow G(i) \rightarrow \operatorname{colim} G|_I$ and $x \rightarrow G(j) \rightarrow \operatorname{colim} G|_I$ are the same. By definition of the colimit, $i \sim j$ whenever there exists a morphism $i \rightarrow j$, so there exists only a single equivalence class if I is connected. Consequently, we may pick any $i \in I$ and consider the composition $x \rightarrow G(i) \rightarrow \operatorname{colim} G|_I$ to exhibit $\operatorname{colim} G|_I$ as a cone under G . It follows that there exists an induced map $\operatorname{colim} G \rightarrow \operatorname{colim} G|_I$.

Tracing through the definitions shows that both $\operatorname{colim} G|_I \rightarrow \operatorname{colim} G \rightarrow \operatorname{colim} G|_I$ and $\operatorname{colim} G \rightarrow \operatorname{colim} G|_I \rightarrow \operatorname{colim} G$ are the identity.

For the other claims, see Land's "Introduction to infinity-categories", Lemma 4.13. $\square$

VI.3.13. Corollary

Let $X \in \mathbf{sSet}$. Then $X \star -: \mathbf{sSet} \rightarrow X/\mathbf{sSet}$ and $- \star X: \mathbf{sSet} \rightarrow X/\mathbf{sSet}$ preserve all colimits.

Proof. For $F: I \rightarrow \mathcal{C}$ an arbitrary diagram in a cocomplete category $\mathcal{C}$, note that

$$\operatorname{colim} F \cong \coprod_{J \in \pi_0(I)} \operatorname{colim} F|_J.$$

Hence, it suffices to show that $X \star -$ preserves coproducts and connected colimits.

In $\mathbf{Set}$, observe that

$$\operatorname{Hom}_{\mathbf{Set}}(\emptyset, \operatorname{colim}_I Z) \cong \operatorname{colim}_I \operatorname{Hom}_{\mathbf{Set}}(\emptyset, Z)$$

if and only if I is connected.

By VI.3.12, connected colimits may be computed in $\mathbf{sSet}$, and this can be done degreewise. Let $Y: I \rightarrow \mathbf{sSet}$ be a connected diagram. Using that simplices are tiny, the remark about colimits in $\mathbf{Set}$ (since $\Delta_{c,1} = \emptyset$ for c the constant map $[n] \rightarrow \{0\} \rightarrow [1]$). and that all colimits in $\mathbf{Set}$ commute with products with a fixed set, we have

$$\begin{aligned} (X \star \operatorname{colim}_I Y)_n &= \coprod_{c \in \operatorname{cut}[n]} \operatorname{Hom}(\Delta_{c,0}, X) \times \operatorname{Hom}(\Delta_{c,1}, \operatorname{colim}_I Y) \\ &\cong \coprod_{c \in \operatorname{cut}[n]} \operatorname{colim}_{i \in I} \operatorname{Hom}(\Delta_{c,0}, X) \times \operatorname{Hom}(\Delta_{c,1}, Y(i)) \\ &\cong \operatorname{colim}_I \coprod_{c \in \operatorname{cut}[n]} \operatorname{Hom}(\Delta_{c,0}, X) \times \operatorname{Hom}(\Delta_{c,1}, Y(i)) \\ &\cong \operatorname{colim}_I (X \star Y(i))_n. \end{aligned}$$

So $X \star -$ preserves all connected colimits.

For coproducts, we saw in the proof of VI.3.12 that $\coprod_{i \in I} (X \rightarrow X \star Y(i))$ may be computed as the colimit of the associated diagram $\{-\infty\} \star I \rightarrow \mathbf{sSet}$. It is straightforward to verify that this colimit is precisely $X \star (\coprod_{i \in I} Y(i))$. $\square$

Since $X \star -: \mathbf{sSet} \rightarrow X/\mathbf{sSet}$ and $- \star X: \mathbf{sSet} \rightarrow X/\mathbf{sSet}$ are colimit-preserving, it follows that they admit right adjoints.

VI.3.14. Definition

Let $f: X \rightarrow Y$. Define the *slices* $Y_{f/}$ and $Y_{/f}$ of Y under and over f as the images of f under the right adjoints $X/\mathbf{sSet} \rightarrow \mathbf{sSet}$ of $X \star -$ and $- \star X$, respectively.

VI.3.15. Remark

Note that slices encode cones under and over a given diagram f . So to $f: I \rightarrow \mathcal{C}$ a functor of categories, one must be careful to distinguish the categories $N(f/\mathcal{C})$ and $N(\mathcal{C})_{N(f)/}$; they only agree when $I = *$.

The reformulation of Theorem C that we are planning to prove next is the following (note that we are cheating a little bit in the formulation since we have not yet discussed the functoriality of the slice construction):

VI.3.16. Theorem

Let $\mathcal{C}$ be a quasicategory and let $f: x \rightarrow y$ be a morphism in $\mathcal{C}$. Then f is an equivalence if and only if

$$\mathcal{C}_{f/} \rightarrow \mathcal{C}_{x/}$$

is a trivial Kan fibration.

This will be done in January. On the way to proving this theorem, we will consider a pushout-joint-construction $\boxtimes$ which is analogous to the pushout-product-construction $\boxtimes$.

VI.3.17. Notation

For $i: A \rightarrow B$ and $f: X \rightarrow Y$, denote by $i \boxtimes f$ the map

$$(A \star Y) \bigcup_{A \star X} (B \star X) \rightarrow B \star Y$$

induced by i and f .

The key technical statement about this construction is the following:

VI.3.18. Lemma

Let $i: A \rightarrow B$ and $j: K \rightarrow L$ be injections of simplicial sets.

- i) $i \boxtimes j$ is an injection.
- ii) If i is right anodyne, then $i \boxtimes j$ is inner anodyne.

- iii) If j is left anodyne, then $i \boxtimes j$ is inner anodyne.
- iv) If i is left anodyne, then $i \boxtimes j$ is left anodyne.
- v) If j is right anodyne, then $i \boxtimes j$ is right anodyne.

Just like VI.3.16, the proof has to wait until next year. But we can use it to close the remaining gap in Chapter V:

Recall that $\square^n$ denotes the n -cube $(\Delta^1)^n$, and that

$$\partial \square^n := \{\alpha: [n] \rightarrow [1]^n \mid \exists 1 \leq i \leq n: \text{pr}_i \alpha \text{ constant}\}$$

is the boundary of the n -cube. In order to identify our cubical model $\pi_*(X, x)$ of simplicial homotopy groups with the simplicial model $\pi_*^\Delta(X, x)$, we used the following statement:

VI.3.19. Lemma

For $n \geq 2$, let $C^n \subseteq \square^n$ denote the simplicial subset obtained by removing the simplex

$$\sigma^n: [n] \rightarrow [1]^n, \quad i \mapsto \underbrace{1 \dots 1}_{i \text{ times}} \underbrace{0 \dots 0}_{n-i \text{ times}}.$$

Then the inclusion $\partial \square^n \rightarrow C^n$ is inner anodyne.

Proof. The case $n = 2$ holds by inspection.

For $k \geq 1$, set

$$L_k := \{\alpha: [p] \rightarrow [1]^k \mid \underbrace{1 \dots 1}_{k \text{ times}} \notin \text{im}(\alpha)\} \subseteq \square^k$$

and

$$L'_k := \{\alpha: [p] \rightarrow [1]^k \mid \underbrace{0 \dots 0}_{k \text{ times}} \notin \text{im}(\alpha)\} \subseteq \square^k.$$

Note that L_k already appeared in V.4.14 and the proof of V.4.15. Set $S_k := L_k \cap L'_k$. Observe that there is a canonical isomorphism

$$(\{1\} \times L_k) \cup_{\{1\} \times S_k} (\Delta^1 \times S_k) \cup_{\{0\} \times S_k} (\{0\} \times L'_k) \xrightarrow{\cong} S_{k+1}(\tau, \alpha) \mapsto (i \mapsto \tau(i) \smallfrown \alpha(i)).$$

Then

$$(\{1\} \rightarrow \Delta^1) \boxtimes (S_k \rightarrow L_k)$$

is right anodyne. Consequently, the lower horizontal map in the pushout square

$$\begin{array}{ccc} (\Delta^1 \times S_k) \cup_{\{1\} \times S_k} (\{1\} \times L_k) & \longrightarrow & L_k \times \Delta^1 \\ \downarrow & & \downarrow \\ S_{k+1} & \longrightarrow & X_{k+1} \end{array}$$

is also right anodyne. Note that the induced map $X \rightarrow \square^{k+1}$ is injective, so we can regard X_{k+1} as a simplicial subset of $\square^{k+1}$.

Let now $n > 2$. The simplicial subset $\{0\} \times \partial\Box^{n-1}$ is contained in X_n , so we can form the pushout

$$\begin{array}{ccc} \{0\} \times \partial\Box^{n-1} & \longrightarrow & \{0\} \times C^{n-1} \\ \downarrow & & \downarrow \\ X_n & \longrightarrow & Y_n \end{array}$$

By induction, the bottom horizontal map is inner anodyne. Note again that Y_n is a simplicial subset of $\Box^n$. It follows that $S_n \rightarrow X_n \rightarrow Y_n$ is right anodyne.

By VI.3.18, the map

$$(S_n \rightarrow Y_n) \boxtimes (\emptyset \rightarrow \{1 \dots 1\})$$

is inner anodyne. Finally, the square

$$\begin{array}{ccc} (S_n \star \{1 \dots 1\}) \cup_{S_n \star \emptyset} & \longrightarrow & Y_n \star \{1 \dots 1\} \\ \downarrow & & \downarrow \\ \partial\Box^n & \longrightarrow & C^n \end{array}$$

is a pushout, so we see that $\partial\Box^n \rightarrow C^n$ is inner anodyne as claimed. $\square$

This closes the remaining gap in the proof of V.3.14.

The goal of the present section is to prove:

VI.3.20. Theorem (Joyal's lifting theorem)

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be an inner fibration between quasicategories. Then all outer horns

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{f} & \mathcal{C} \\ \downarrow & \nearrow & \downarrow F \\ \Delta^n & \longrightarrow & \mathcal{D} \end{array} \quad \begin{array}{ccc} \Lambda_n^n & \xrightarrow{g} & \mathcal{C} \\ \downarrow & \nearrow & \downarrow \\ \Delta^n & \longrightarrow & \mathcal{D} \end{array}$$

with

$$\Delta^1 \xrightarrow{\text{incl}} \Lambda_0^n \xrightarrow{f} \mathcal{C} \quad \text{or} \quad \Delta^1 \xrightarrow{(n-1, n)} \Lambda_n^n \xrightarrow{g} \mathcal{C}$$

representing equivalences in $\mathcal{C}$ admit a contractible space of lifts.

The proof will be by reduction to the ? (also to be proven VI.3.16 and VI.3.17). But before that, let's get back into the groove with some corollaries to see the importance of this result.

VI.3.21. Corollary (Theorem C)

Every anima is a Kan complex. In particular, the core of any quasicategory is a Kan complex. We already recorded the converse in III.7.

Proof. Apply VI.3.20 with $\mathcal{D} = \text{pt}$ to obtain fillers for the outer horns. $\square$

VI.3.22. Corollary (Second half of Theorem D)

For quasicategories $\mathcal{C}$ and $\mathcal{D}$,

$$\mathrm{Fun}(\mathcal{C}, \mathrm{core} \mathcal{D}) \subseteq \mathrm{core}(\mathrm{Fun}(\mathcal{C}, \mathcal{D}))$$

is a collection of path components, with equality if either $\mathcal{C}$ or $\mathcal{D}$ is an anima.

Proof. Since $\mathrm{core} \mathcal{D}$ is an anima, it is a Kan complex by the previous corollary, so $\mathrm{Fun}(\mathcal{C}, \mathrm{core} \mathcal{D})$ is a Kan complex by V.2.24, thus an anima by (the trivial) III.7. This also gives the containment and furthermore equality if $\mathcal{D}$ is an anima.

For the remainder, observe first that a functor naturally equivalent to one taking values in the core of $\mathcal{D}$ also does so, as this can (by definition of the core and III.7) be tested in the homotopy category, where it is obvious. To see that it is an inclusion of path components, take an n -simplex σ on the right, whose vertices lie in $\mathrm{Fun}(\mathcal{C}, \mathrm{core} \mathcal{D})$.

We have to show that already $\sigma \in \mathrm{Fun}(\mathcal{C}, \mathrm{core} \mathcal{D})$. This means that regarded as a map $\Delta^n \times \mathcal{C} \rightarrow \mathcal{D}$ all edges of $\Delta^n \times \mathcal{C}$ map to equivalences in $\mathcal{D}$. Again this can be tested in the homotopy category, where for an arbitrary edge (f, g) we find $[(f, g)] = [(f, \mathrm{id})] \circ [(\mathrm{id}, g)]$. But by assumption the edges of σ are natural equivalences which in particular means that edges of the form (f, id) are mapped to equivalences. And also by assumption the vertices of σ are functors into the core, so edges of the form (id, g) are mapped to equivalences. Thus we generally have an inclusion of path components with equality if $\mathcal{C}$ is anima, as then any functor $\mathcal{C} \rightarrow \mathcal{D}$ factors through $\mathrm{core} \mathcal{D}$. $\square$

Of course the inclusion

$$\mathrm{Fun}(\mathcal{C}, \mathrm{core} \mathcal{D}) \subseteq \mathrm{core}(\mathrm{Fun}(\mathcal{C}, \mathcal{D}))$$

is generally strict, and we do not yet have a handle on the target. However, already the fact that transformations between functors to an anima are invertible is far from obvious. Decoding we obtain that a natural transformation $\tau: F \Rightarrow G$ of functors $\mathcal{C} \rightarrow \mathcal{D}$ is invertible if for every $f: x \rightarrow y$ in $\mathcal{C}$ the diagram

$$\begin{array}{ccc} F(x) & \xrightarrow{\tau_x} & G(x) \\ F(f) \downarrow & & \downarrow G(f) \\ F(y) & \xrightarrow{\tau_y} & G(y) \end{array}$$

consists entirely of equivalences, then τ is invertible.

Clearly, the invertibility of $G(f)$ and $F(f)$ should not play a role. That is the content of Theorem E and will have to wait until the next section.

However, we now know that $\mathrm{Core} F(-, -)$ gives a Kan-enrichment on the category $\mathrm{qCat} \subseteq \mathrm{sSet}$.

VI.3.23. Definition

Define the *quasicategory of quasicategories*, usually denoted by Cat_∞ , as $\mathcal{N}^c(\mathrm{qCat})$ using the above enrichment.

Since $F(\mathcal{C}, K)$ is Kan if K is, we also obtain a simplicially enriched functor

$$\mathrm{Kan} \rightarrow \mathrm{qCat}$$

and so a functor $\mathbf{An} \rightarrow \mathbf{Cat}_\infty$. It is degree-wise injective and gives bijections $\mathrm{Hom}_{\mathbf{An}}(K, K') \rightarrow \mathrm{Hom}_{\mathbf{Cat}_\infty}(K, K')$ for every two anima K and K' .

Let me warn you about two things though: We do not yet know the homotopy type of either of these spaces (that is the content of [Theorem H](#)), but even assuming it we find

$$\mathrm{Hom}_{\mathbf{Cat}_\infty}(\mathcal{C}, \mathcal{C}') \simeq \mathrm{CoreFun}(\mathcal{C}, \mathcal{C}').$$

So $\mathbf{Cat}_\infty$ is neither analogous to $\mathbf{Cat}$, the 1-category of small 1-categories, nor $\mathbf{Cat}_2$, the 2-category of small 1-categories, but rather the $(2, 1)$ -category obtained from $\mathbf{Cat}_2$ by discarding non-invertible natural transformations. Indeed, $\mathbf{qCat}$ really is $\mathbf{qCat}$ enriched and thus wants to give a (non-strict) $(\infty, 2)$ -category. Alas, we shall not define such objects (they are not nearly as developed as quasicategories), apart from mentioning that they can be defined using either enriched $(\infty, 1)$ -category theory, or by iterating Rezk's Segal space approach to higher categories.

Let us, however, record that equivalences in $\mathbf{Cat}_\infty$ are exactly the equivalences of quasicategories.

Here's another application of anima being Kan complexes: Let $\mathcal{J} \in \mathbf{Kan}$ denote the nerve of the groupoid having objects 0, 1 and 4 morphisms as indicated on the picture:

$$\mathrm{id} \circlearrowleft 0 \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{array} 1 \circlearrowright \mathrm{id}$$

There is then a canonical inclusion $\Delta^1 \subseteq \mathcal{J}$ and we have:

VI.3.24. Proposition

A morphism f in a quasicategory $\mathcal{C}$ is an equivalence if and only if its classifier $\Delta^1 \rightarrow \mathcal{C}$ lies in the image of the restriction

$$p : \mathrm{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \mathrm{Fun}(\Delta^1, \mathcal{C}).$$

In this case the core of the fibre over f is contractible.

This proposition says that inverses of an equivalence in a quasicategory are unique up to contractible choice!

The statement follows from [VI.3.21](#) and the following Lemma:

VI.3.25. Lemma

The map $\Delta^1 \rightarrow \mathcal{J}$ is both left and right anodyne.

The complex $\mathcal{J}$ has exactly 2 non-degenerate simplices in each dimension (corresponding to the sequences $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \dots$ and $1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \dots$). Its realization is easily checked to be S^∞ with the “hemispherical” cell structure, in particular, contractible, whence [V.4.4](#) shows that $\Delta^1 \subseteq \mathcal{J}$ is at least anodyne, which would suffice for our purposes.

But it is clearly not inner anodyne (it is not even an equivalence of categories), so gives a weak example that

$$\text{inner anodynes} \subsetneq \text{left anodynes} \cap \text{right anodynes}.$$

Proof. To see that the inclusion is left anodyne, let $F_k \subseteq \mathcal{J}$ be the simplicial subset generated by the k -simplex $v_k = 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow \dots$. Then $\Delta^1 = F_1$ and there is a pushout diagram

$$\begin{array}{ccc} \Lambda_0^k & \longrightarrow & F_{k-1} \\ \downarrow & & \downarrow \cap \\ \Delta^k & \xrightarrow{v_k} & F_k \end{array}$$

since $v_k: \Delta^k \rightarrow F_k$ is surjective and $v_k^{-1}(F_{k-1}) = \Lambda_0^k$, because $d_i v_k$ is degenerate for $0 < i < k$ (and $\text{sk}_{k-2} F_{k-1} = \text{sk}_{k-2} \mathcal{J}$, which contains all degenerate $k-1$ simplices), $d_k v_k = v_{k-1} \in F_{k-1}$ and $d_0 v_k = 1 \rightarrow 0 \rightarrow 1 \rightarrow \dots \notin F_{k-1}$. The claim follows. To see that $\Delta^1 \subseteq \mathcal{J}$ is also right anodyne use the filtration spanned by the simplices $\dots 0 \rightarrow 1 \rightarrow 0 \rightarrow 1$ instead. $\square$

Proof of VI.3.24. Since $\mathcal{J}$ is an anima any map $\mathcal{J} \rightarrow \mathcal{C}$ factors through $\text{core } \mathcal{C}$, so both its edges go to equivalences.

Conversely, if f is an equivalence it is represented by a map $\Delta^1 \rightarrow \text{core } \mathcal{C}$, which is a Kan complex by VI.3.21. But then the map

$$\bar{p}: \text{Fun}(\mathcal{J}, \text{core } \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \text{core } \mathcal{C})$$

is a trivial Kan fibration by VI.3.25 and V.2.26, in particular, surjective on 0-simplices. To see the contractibility statement, note that

$$\text{Fun}(\mathcal{J}, \text{core } \mathcal{C}) = \text{core } \text{Fun}(\mathcal{J}, \mathcal{C})$$

by VI.3.22 and that

$$\text{core } p^{-1}(f) = p^{-1}(f) \cap \text{core } \text{Fun}(\mathcal{J}, \mathcal{C}) = \bar{p}^{-1}(f);$$

to see the first equality let $h: \Delta^1 \rightarrow \text{Fun}(\mathcal{J}, \mathcal{C})$ be inverse to $g: \Delta^1 \rightarrow p^{-1}(f)$. Then $p(h)$ is inverse to $p(g) = \text{id}_f$ so $p(h) \sim \text{id}_f$. But p is an inner fibration by VI.2.4 so then we can fill

$$\begin{array}{ccc} \Lambda_1^2 & \xrightarrow{(h, \text{id})} & \text{Fun}(\mathcal{J}, \mathcal{C}) \\ \downarrow & \nearrow G & \downarrow p \\ \Delta^2 & \xrightarrow{H} & \text{Fun}(\Delta^1, \mathcal{C}) \end{array} \quad H = \begin{array}{ccc} & 1 & \\ p(h) \nearrow & & \searrow \text{id}_f \\ 0 & \xrightarrow{\text{id}_f} & 2 \end{array}$$

and $d_1 G \sim h$ is still inverse to g , but also $d_1 G: \Delta^1 \rightarrow p^{-1}(f)$, so $g \in \text{core } p^{-1}(f)$. Finally, $\bar{p}^{-1}(f)$ is contractible by V.3.17. $\square$

VI.3.26. Remark

In fact, something stronger is true: The map $p: \text{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \mathcal{C})$ is a trivial Kan fibration, and thus an equivalence of categories, onto the subcategory of the target spanned by the equivalences. This will be an exercise later. In particular, $p^{-1}(f)$ is itself a contractible Kan complex! The latter statement also follows from VI.3.24 and the pointwise criterion for natural equivalences, which implies that p is conservative.

Continuing with applications, we have:

VI.3.27. Corollary

If $p: \mathcal{C} \rightarrow \mathcal{D}$ is left or right fibration and $\mathcal{D}$ is Kan, then p is a Kan fibration as well. In particular, $\mathcal{C}$ is also a Kan complex.

Specializing to $\mathcal{D} = *$ we see that left and right Kan complexes are automatically Kan complexes, as announced in V.1.6. In particular, the fibres of left/right fibrations are always Kan complexes, removing an assumption from Exercise 4, sheet 8, that constructed a functor

$$\mathrm{st}_p : h\mathcal{D} \rightarrow \mathrm{hoAn}$$

from a left fibration $p: \mathcal{C} \rightarrow \mathcal{D}$. From this point of view VI.3.22 is also very natural. If $\mathcal{D}$ is Kan then $h\mathcal{D}$ is a groupoid, so st_p factors through $\mathrm{core} \, \mathrm{hoAn}$, as does the straightening of any Kan fibration. In fact, it is an exercise on sheet 10 to show that this characterizes Kan fibrations among left fibrations.

VI.3.28. Lemma / Definition

Left and right fibrations are *conservative*, i.e. if $p: \mathcal{C} \rightarrow \mathcal{D}$ is a left/right fibration between quasicategories and $f \in \mathcal{C}_1$ is a morphism such that $p(f)$ is an equivalence, then so is f itself. In particular, if $\mathcal{D}$ is an anima then so is $\mathcal{C}$.

The proof is another exercise on sheet 10; crucially it does not use Joyal's lifting theorem, as it will also enter into its proof!

Proof of VI.3.27. Given a lifting diagram

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow p \\ \Delta^n & \longrightarrow & \mathcal{D} \end{array}$$

with $i \in \{0, n\}$, then we know from VI.3.26 that $\mathcal{C}$ is an anima as $\mathcal{D}$ is one. Thus VI.3.20 applies to give a lift. $\square$

Ley me spell out explicitly another corollary on the technical side:

VI.3.29. Corollary

If $p: \mathcal{C} \rightarrow \mathcal{D}$ is a left (or right) fibration with contractible fibres, then it is a trivial Kan fibration.

This generalizes V.3.18, where we showed the analogous assertion for Kan fibrations. Let me warn you, though: A left fibration, that is a weak homotopy equivalence need not be a Joyal equivalence, let alone a trivial fibration. Counterexamples follow below.

Proof. We follow the proof of V.3.18: Given a diagram

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow p \\ \Delta^n & \longrightarrow & \mathcal{D} \end{array}$$

consider a simplicial contraction $H: \Delta^n \times \Delta^1 \rightarrow \Delta^n$ onto $\{n\}$. Then in the diagram

$$\begin{array}{ccc} \partial\Delta^n \times \{0\} & \xrightarrow{\quad} & \mathcal{C} \\ \downarrow \cap & \nearrow \text{dashed} & \downarrow \\ \partial\Delta^n \times \Delta^1 & \xrightarrow{H} \Delta^n \longrightarrow & \mathcal{D} \end{array}$$

the left vertical map is left anodyne by V.2.9, so there exists a filler G . Now $G|_{\partial\Delta^n \times \{1\}}$ factors through a single fibre, so by assumption G can be extended to a diagram

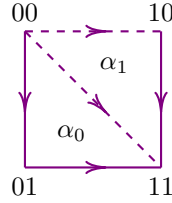
$$\begin{array}{ccc} \partial\Delta^n \times \Delta^1 \cup \Delta^n \times \{1\} & \xrightarrow{G} & \mathcal{C} \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \Delta^n \times \Delta^1 & \xrightarrow{H} \Delta^n \longrightarrow & \mathcal{D} \end{array}$$

which in V.2.9 we could fill since the left hand map is right anodyne (by definition). The restriction of a filler to $\Delta^n \times \{1\}$ in any case is a solution to the original lifting problem. This time we have to use Joyal's theorem: First, as part of the proof of V.2.9, we have seen that the inclusion

$$\partial\Delta^n \times \Delta^1 \cup \Delta^n \times \{1\} \subseteq \Delta^n \times \Delta^1 - \{\alpha_n \cup d_{n+1}\alpha_n\}$$

is actually inner anodyne, where $\alpha_i: [n+1] \rightarrow [n] \times [1]$ is given by

$$k \mapsto \begin{cases} (k, 0) & k \leq i \\ (k-1, 1) & k > i \end{cases}$$



Only α_0 is attached as a right horn. This a filler K certainly exists on $\Delta^n \times \Delta^1 - \{\alpha_n\}$. But then consider

$$\begin{array}{ccccc} & & K \circ \alpha_n & & \\ & \curvearrowright & & \curvearrowright & \\ \Lambda_{n+1}^{n+1} & \xrightarrow{\exists!} & \mathcal{C}' & \longrightarrow & \mathcal{C} \\ \downarrow \cap & & \downarrow p' & & \downarrow p \\ \Delta^{n+1} & \xrightarrow{\alpha_n} \Delta^n \times \Delta^1 \xrightarrow{H} & \Delta^n & \longrightarrow & \mathcal{D} \end{array}$$

Now p' is a left fibration, so $\mathcal{C}'$ is a quasicategory (since Δ^n is one). Also

$$\begin{aligned} \Delta^1 & \xrightarrow{(n, n+1)} \Lambda_{n+1}^{n+1} \rightarrow \mathcal{C}' \xrightarrow{p'} \Delta^n \\ = \Delta^1 & \xrightarrow{(n, n+1)} \Lambda_{n+1}^{n+1} \xrightarrow{\alpha_n} \Delta^n \times \Delta^1 \xrightarrow{H} \Delta^n \\ (0, 1) & \mapsto (n, n+1) \mapsto ((n, 0), (n, 1)) \mapsto (n, n) \end{aligned}$$

is constant, so

$$\Delta^1 \xrightarrow{(n,n+1)} \Lambda_{n+1}^{n+1} \rightarrow \mathcal{C}'$$

factors through a fibre, But the fibres of p' are Kan complexes by VI.3.27, so this map represents an equivalence in $\mathcal{C}'$. Theorem VI.3.20 then supplies the filler. $\square$

A similar argument shows:

VI.3.30. Corollary (Theorem A)

For a quasicategory $\mathcal{C}$, and two objects $x, y \in \mathcal{C}$ the simplicial set $\text{Hom}_{\mathcal{C}}(x, y)$ is Kan.

By VI.3.27 this is immediate from the following:

VI.3.31. Lemma

For a quasicategory $\mathcal{C}$ the outer horns

$$\begin{array}{ccc} \Lambda_0^n & \longrightarrow & \text{Fun}(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow (s,t) \\ \Delta^n & \xrightarrow{f} & \mathcal{C} \times \mathcal{C} \end{array} \quad \begin{array}{ccc} \Lambda_n^n & \longrightarrow & \text{Fun}(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow (s,t) \\ \Delta^n & \xrightarrow{g} & \mathcal{C} \times \mathcal{C} \end{array}$$

admit fillers whenever the composites

$$\begin{array}{c} \Delta^1 \xrightarrow{(0,1)} \Delta^n \xrightarrow{f} \mathcal{C} \times \mathcal{C} \xrightarrow{\text{pr}_n} \mathcal{C} \\ \Delta^1 \xrightarrow{(n-1,n)} \Delta^n \xrightarrow{g} \mathcal{C} \times \mathcal{C} \xrightarrow{\text{pr}_2} \mathcal{C} \end{array}$$

respectively, represent equivalences in $\mathcal{C}$. In particular, the left vertical projections in the diagrams

$$\begin{array}{ccc} X/\mathcal{C} & \longrightarrow & \text{Fun}(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow (s,t) \\ \{X\} \times \mathcal{C} & \longrightarrow & \mathcal{C} \times \mathcal{C} \end{array} \quad \begin{array}{ccc} \mathcal{C}/X & \longrightarrow & \text{Fun}(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow (s,t) \\ \mathcal{C} \times \{X\} & \longrightarrow & \mathcal{C} \times \mathcal{C} \end{array}$$

are a left and right fibration, respectively, for all $X \in \mathcal{C}$.

Proof: Let us treat the left case only. The diagram adjoints to the extension problem

$$\begin{array}{ccc} \Lambda_0^n \times \Delta^1 \cup \Delta^n \times \partial\Delta^1 & \longrightarrow & \mathcal{C} \\ \downarrow \sqcup & \nearrow & \\ \Delta^n \times \Delta^1 & & \end{array}$$

Set $K(0) = \Lambda_0^n \times \Delta^1 \cup \Delta^n \times \partial\Delta^1$ and

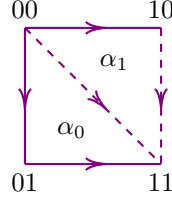
$$K(i+1) = K(i) \cup \{\alpha_i\}$$

where the $\alpha_i: \Delta^{n+1} \rightarrow \Delta^n \times \Delta^1$ are the nondegenerate $n+1$ -simplices as described in ?? and VI.3.29.

Now one readily checks again that

$$K(0) \subseteq K(1) \subseteq \dots \subseteq K(n) = \Delta^n \times \Delta^1 - \{\alpha_n \cup d_0\alpha_n\}$$

are given by filling an inner horn each, where as α_n defines an outer left horn in $K(n)$.



But for $n > 0$ (which is all we need!) the composite

$$\Delta^1 \xrightarrow{(0,1)} \Delta^{n+1} \xrightarrow{\alpha_n} \Delta^n \times \Delta^1$$

$(0,1) \mapsto (0,1) \mapsto ((0,0), (1,0))$

factors as

$$\Delta^1 \xrightarrow{(0,1)} \Delta^n \times \{0\} \subseteq \Delta^n \times \Delta^1$$

and thus

$$\begin{aligned} & \Delta^1 \xrightarrow{(0,1)} \Lambda_0^{n+1} \xrightarrow{\alpha_n} K(n) \rightarrow \mathcal{C} \\ = & \Delta^1 \xrightarrow{(0,1)} \Delta^n \times \{0\} \rightarrow \mathcal{C} \\ = & \Delta^1 \xrightarrow{(0,1)} \Delta^n \rightarrow \text{Fun}(\Delta^1, \mathcal{C}) \xrightarrow{s} \mathcal{C} \\ = & \Delta^1 \xrightarrow{(0,1)} \Delta^n \rightarrow \mathcal{C} \times \mathcal{C} \xrightarrow{\text{pr}_1} \mathcal{C} \end{aligned}$$

represents an equivalence by assumption. By VI.3.20 then there exists an extension

$$\begin{array}{ccc} \Lambda_0^{n+1} & \xrightarrow{\alpha_n} & K(n) \longrightarrow \mathcal{C} \\ \downarrow & & \nearrow \text{dashed} \\ \Delta^{n+1} & & \end{array}$$

which completes the extension to $\Delta^n \times \Delta^1 \rightarrow \mathcal{C}$.

The final statements follow since

$$\text{Fun}(\Delta^1, \mathcal{C}) \rightarrow \text{Fun}(\partial\Delta^1, \mathcal{C}) = \mathcal{C} \times \mathcal{C}$$

is an inner fibration by VI.2.4. □

VI.3.32. Remark

i) An inner fibration $A \rightarrow S \times T$ such that all outer horns

$$\begin{array}{ccc} \Lambda_0^n & \longrightarrow & A \\ \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{f} & S \times T \end{array} \quad \begin{array}{ccc} \Lambda_n^n & \longrightarrow & A \\ \downarrow & & \downarrow \\ \Delta^n & \xrightarrow{g} & S \times T \end{array}$$

in which the composites

$$\begin{aligned} \Delta^1 &\xrightarrow{(0,1)} \Delta^n \xrightarrow{f} S \times T \xrightarrow{\text{pr}_1} S \\ \Delta^1 &\xrightarrow{(n-1,n)} \Delta^n \xrightarrow{g} S \times T \xrightarrow{\text{pr}_2} T \end{aligned}$$

are degenerate and admit fillers is called a *bifibrant*.

Bifibrations are always conservative (with essentially the same proof as VI.3.28), which gives us another instance of the pointwise criterion for the invertibility for natural transformations (which, together with Theorem B clearly also implies Theorem A!)

- ii) The categories $X/\mathcal{C}$, $\mathcal{C}/X$ in VI.3.39 are models for over- and under-categories, see exercise 4, sheet 12. Note, however, that these are different (!) from $\mathcal{C}_{x/}$ and $\mathcal{C}_{/x}$ unless $\mathcal{C}$ is the nerve of an ordinary category. It is a theorem of Joyal that the natural map $\mathcal{C}/X \rightarrow \mathcal{C}_{/x}$ is always a Joyal equivalence. We may discuss this later, or next term.
- iii) VI.3.39 gives us examples of left fibrations that are weak homotopy equivalences, but not Joyal equivalences:
Let $\mathcal{C}$ be an ordinary category. Then for $X \in \mathcal{C}$

$$X/\mathcal{N}(\mathcal{C}) \xrightarrow{t} \mathcal{N}(\mathcal{C})$$

is a left fibration by VI.3.30, and $X/\mathcal{N}(\mathcal{C}) \cong \mathcal{N}(X/\mathcal{C})$ by exercise 4, sheet 12. If for example $\mathcal{C}$ has an initial object then both $\mathcal{N}(\mathcal{C})$ and $X/\mathcal{N}(\mathcal{C})$ are simplicially contractible as

$$\Delta^0 \xrightarrow{*} \mathcal{C} \quad \text{and} \quad \Delta^0 \xrightarrow{\text{id}_X} X/\mathcal{N}(\mathcal{C})$$

are then left adjoint functors by ??, so simplicial homotopy equivalences by exercise 2, sheet 4. But already for $\mathcal{C} = \mathbb{N}[1]$ and $X = 1$, we find

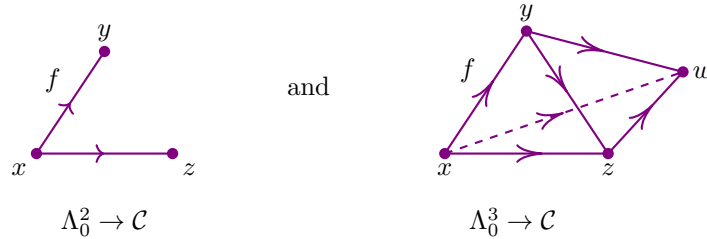
$$1/\Delta^1 = \{\text{id}_1\}$$

not Joyal equivalent to Δ^1 .

Finally, let us record the converse to Joyal's lifting theorem:

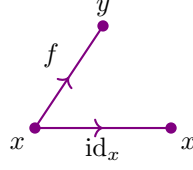
VI.3.33. Observation

If $f: x \rightarrow y$ is a morphism in a quasicategory $\mathcal{C}$ and the lifting problems

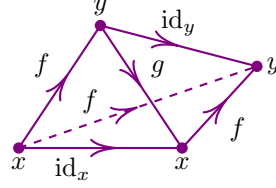


admit solutions, then f is an equivalence.

Proof: A solution to the first problem as below



gives a left inverse g to f . But then a solution to the second one as below



shows that g is also right inverse. $\square$

VI.3.34. Remark

- i) Another way of phrasing this is the argument that the left of these two lifting diagrams admits a filler iff $[f]$ admits a left inverse and the right if $[f]$ is right cancelable, clearly equivalent to $[f] \in \text{mor}(\pi\mathcal{C})$ being an isomorphism.

We also already saw that for the nerve of a category *any* higher dimensional horn admits (unique) fillers in [II.28](#), and the same argument (namely nerves of categories are 2-coskeletal and Λ_0^n and Δ^n have the same 2-skeleton once $n \geq 4$) also gives Joyal's lifting theorem for ordinary categories. In a sense then, it encodes that higher simplices of a quasicategory are automatically invertible, as the fillability of higher outer horns cannot fail once the 1-skeleton consists of equivalences in the necessary spots.

- ii) Finally, let me comment on the role of the assumption that F is an inner fibration. Without it, the theorem becomes false, but only barely: Every functor $\mathcal{C} \rightarrow \mathcal{D}$ can be factored into an inner anodyne map $\mathcal{C} \rightarrow \mathcal{C}'$ followed by an inner fibration using the small object argument [V.2.30](#). Since $\mathcal{D}$ is a quasicategory it follows that $\mathcal{C}'$ is one as well and by [VI.2.12](#) and [VI.2.10](#) an inner anodyne map is an equivalence of quasicategories. So Joyal's theorem applies to $\mathcal{C}' \rightarrow \mathcal{D}$ and the resulting lift $\Delta^n \rightarrow \mathcal{C}'$ can be composed with the inverse equivalence $\mathcal{C}' \rightarrow \mathcal{C}$ to yield a map $\Delta^n \rightarrow \mathcal{C}$ whose restriction to $\partial\Delta^n$ is homotopic to the given one and whose projection to $\mathcal{D}$ is naturally isomorphic to the given diagram. The homotopy extension property finally allows us to alter $\Delta^n \rightarrow \mathcal{C}$ to be an extension of the original lifting problem, and a lift up to natural isomorphism.

Now, to move toward the proof of [VI.3.20](#) we observe, that the map $\Lambda_0^n \rightarrow \Delta^n$ is isomorphic to

$$(\Delta^0 \xrightarrow{d_1} \Delta^1) \boxtimes (\partial\Delta^{n-2} \rightarrow \Delta^{n-2}) \rightarrow \Delta^1 \star \Delta^{n-2};$$

more generally we have:

VI.3.35. Lemma

There are canonical isomorphisms

$$(\Lambda_j^n \rightarrow \Delta^n) \boxtimes (\partial\Delta^m \rightarrow \Delta^m) \cong \Lambda_j^{n+1+m} \rightarrow \Delta^{n+1+m}$$

and

$$(\partial\Delta^m \rightarrow \Delta^m) \boxtimes (\Lambda_j^n \rightarrow \Delta^n) \cong \Lambda_{m+1+j}^{m+1+n} \rightarrow \Delta^{m+1+n}$$

Proof: Since joining preserves (by definition) injections, it also preserves unions of simplicial subsets, i.e. for $A, B \subseteq X$ we have

$$\begin{aligned} (A \cup B) \star Y &= (A \star Y) \cup (B \star Y) \subseteq X \star Y \\ Y \star (A \cup B) &= (Y \star A) \cup (Y \star B) \subseteq Y \star X \end{aligned}$$

as the image of either term in $X \star Y$ can be determined using the pushout of $A \leftarrow \emptyset \rightarrow B$, which $- \star Y$ and $Y \star -$ preserve by VI.3.12 and VI.3.13.

Thus we have:

$$\begin{aligned} \Lambda_i^n \star \Delta^m &= \left(\bigcup_{\substack{0 \leq j \leq n \\ i \neq j}} d_j(\Delta^{n-1}) \right) \star \Delta^m = \bigcup_{\substack{0 \leq j \leq n \\ i \neq j}} (d_j(\Delta^{n-1}) \star \Delta^m) \\ &= \bigcup_{\substack{0 \leq j \leq n \\ i \neq j}} d_j(\Delta^{n+m}) \subseteq \Delta^{n+1+m} \end{aligned}$$

Similarly, we have

$$\begin{aligned} \Delta^n \star \partial\Delta^m &= \Delta^n \star \bigcup_{0 \leq j \leq m} d_j(\Delta^{m-1}) \\ &= \bigcup_{0 \leq j \leq m} \Delta^n \star d_j(\Delta^{m-1}) \\ &= \bigcup_{0 \leq j \leq m} d_{n+1+j}(\Delta^{n+1+m-1}) = \bigcup_{n+1 \leq j \leq n+1+m} d_j(\Delta^{n+m}) \subseteq \Delta^{n+1+m} \end{aligned}$$

Their union is clearly Λ_i^{n+1+m} and since $\boxtimes$ preserves injections (again straight from the definition) the first claim follows. The second is analogous. $\square$

But by adjunction a lifting problem

$$\begin{array}{ccc} (A_0 \rightarrow A_1) \boxtimes (B_0 \rightarrow B_1) & \longrightarrow & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ A_1 \star B_1 & \longrightarrow & Y \end{array}$$

is equivalent to either of

$$\begin{array}{ccc} A_0 & \longrightarrow & X_{/B_1 \rightarrow X} \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ A_1 & \longrightarrow & (X_{/B_0 \rightarrow X})_{Y_{/B_0 \rightarrow Y}} \times (Y_{/B_1 \rightarrow Y}) \end{array} \quad \begin{array}{ccc} B_0 & \longrightarrow & X_{A_1 \rightarrow X /} \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ B_1 & \longrightarrow & (X_{A_0 \rightarrow X /})_{Y_{A_0 \rightarrow Y /}} \times (Y_{A_1 \rightarrow Y /}) \end{array}$$

In our case $A_0 \rightarrow A_1 = \Delta^0 \xrightarrow{d_1} \Delta^1 = \Lambda_0^1 \rightarrow \Delta^1$ is left anodyne, so it would be enough to see that the right hand map in the upper diagram is a left fibration. It is *not*! We will see that it is instead a right fibration. This will turn out to be good enough! Since $B_0 \rightarrow B_1 = \partial\Delta^{n-2} \rightarrow \Delta^{n-2}$, setting $Y = \star$ in the right diagram (together with VI.3.30) recovers the formulation VI.3.16, that we gave earlier.

Let us first deal with the last argument:

VI.3.36. Lemma

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be an inner fibration of quasicategories. Then if $f: x \rightarrow y$ and $g: y \rightarrow x$ are inverse equivalences there exists a solution to one of the lifting problems

$$\begin{array}{ccc} \Delta^0 & \xrightarrow{\quad} & \mathcal{C} \\ \downarrow d_1 & \nearrow & \downarrow F \\ \Delta^1 & \xrightarrow{f} & \mathcal{D} \end{array} \quad \begin{array}{ccc} \Delta^0 & \xrightarrow{\quad} & \mathcal{C} \\ \downarrow d_0 & \nearrow & \downarrow F \\ \Delta^1 & \xrightarrow{g} & \mathcal{D} \end{array}$$

that represents an equivalence in $\mathcal{C}$, if and only if the same is true for the other.

VI.3.37. Definition

An inner fibration admitting lifts as above for all equivalences f and g is called an *isofibration*.

Proof of VI.3.36: Assume the left lifting problem has a solution $h: a \rightarrow b$. Since h represents an equivalence in $\mathcal{C}$ we can choose an inverse k to it. Since $F(h) = f$ we find $F(k)$ inverse to f , so there exists an $H: \Delta^2 \rightarrow \mathcal{D}$ with $d_0H = 0_{\text{sec}}F(k)$, $d_2H = \text{id}_Y$ and $d_1H = g$. But then the lifting problem

$$\begin{array}{ccc} \Lambda_1^2 & \xrightarrow{(\text{id}_Y, k)} & \mathcal{C} \\ \downarrow & \nearrow & \downarrow \\ \Delta^2 & \xrightarrow{H} & \mathcal{D} \end{array}$$

admits a solution G and d_1G is then a lift of g and an equivalence since it is homotopic to the equivalence k . $\square$

VI.3.38. Observation

Both left and right fibrations between quasicategories are isofibrations.

This statement will give the existence part of Joyal's theorem. We will use this to prove a refinement of this lemma below, from which we will deduce uniqueness.

Proof: For a left fibration the left of the two lifting problems admits a solution by definition even without assuming f an equivalence. But if it is an equivalence so is any lift by VI.3.28. $\square$

Next we discuss the slice construction a little further, as its functoriality is a bit funny: Recall that we have functors ls_K, rs_K

$$K/\text{sSet} \rightarrow \text{sSet} \quad (X, f) \mapsto X_{f/} \text{ and } X_{/f}$$

right adjoint to

$$K \star - \text{ and } - \star K: \mathbf{sSet} \rightarrow K/\mathbf{sSet} .$$

Given a map $\varphi: K \rightarrow L$, we obtain the functors

$$\varphi^*: L/\mathbf{sSet} \rightarrow K/\mathbf{sSet} \xrightarrow{\text{ls}_K, \text{rs}_K} \mathbf{sSet} .$$

These composites admit a natural transformation from ls_L, rs_L : For a map $f: L \rightarrow X$ a map

$$X_{/f} \rightarrow X_{/f\varphi}$$

for example, is the same as a map

$$\begin{array}{ccc} & K & \\ \swarrow & & \searrow f\varphi \\ X_{/f} \star K & \dashrightarrow & X \end{array}$$

by adjunction. But the adjunction

$$X_{/f} \star L \rightarrow X$$

provides such a map after precomposition with φ .

The same argument gives the transformation for the left slice functor and I leave it to the reader to check naturality. In summary, the associations

$$(X \xrightarrow{f} Y) \mapsto Y_{/f} \text{ and } Y_{f/}$$

are covariant in the target of f but contravariant in the source. To encode this we have

VI.3.39. Definition

If $\mathcal{C}$ is a category define $\text{TwAr}(\mathcal{C})$, the *twisted arrow category* of $\mathcal{C}$, to have objects the arrows of $\mathcal{C}$ and morphisms $f \rightarrow g$ pairs of arrows $(p, q) \in \text{mor } \mathcal{C}^2$ such that

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ p \uparrow & & \downarrow q \\ X' & \xrightarrow{g} & Y' \end{array}$$

commutes. Composition is just component-wise composition of arrows.

Note that in distinction with the source and target evaluation in the arrow category

$$(s, t): \text{Ar}(\mathcal{C})^1 \rightarrow \mathcal{C} \times \mathcal{C}$$

these combine in to a functor

$$(s, t): \text{TwAr}(\mathcal{C}) \rightarrow \mathcal{C}^{\text{op}} \times \mathcal{C} .$$

It is also readily checked that $\text{TwAr}(\mathcal{C})$ is nothing but the category of elements $Y/\text{Hom}_{\mathcal{C}}$ of the functor

$$\text{Hom}_{\mathcal{C}}: \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$$

We find:

¹ $\text{Ar}(\mathcal{C}) = \text{Fun}(\mathbb{N}[1], \mathcal{C})$

VI.3.40. Observation

The slice constructions assemble into functors

$$\text{ls, rs: TwAr}(\text{sSet}) \rightarrow \text{sSet} .$$

In particular, for every sequence of map

$$A \xrightarrow{i} X \xrightarrow{f} Y \xrightarrow{p} B$$

we obtain maps

$$\begin{aligned} Y_{f/} &\rightarrow Y_{fi/} \times_{B_{pfi/}} B_{pf/} \\ Y_{/f} &\rightarrow Y_{/fi} \times_{B_{/pfi}} B_{pf} . \end{aligned}$$

It is these maps that appear in the proof of Joyal's theorem.

Proof: The first part is immediate from the discussion preceding ???. For the second one we need to show that

$$\begin{array}{ccc} Y_{f/} & \xrightarrow{i^*} & Y_{fi/} \\ \downarrow p_* & & \downarrow p_* \\ B_{pf/} & \xrightarrow{i^*} & B_{pfi/} \end{array}$$

commutes. But this is immediate from the compositions

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ i \uparrow & & \parallel \\ A & \xrightarrow{fi} & Y \\ \parallel & & \downarrow p \\ A & \xrightarrow{pfi} & B \end{array} \quad \text{and} \quad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ \parallel & & \downarrow p \\ X & \xrightarrow{pf} & B \\ \uparrow & & \parallel \\ A & \xrightarrow{pfi} & B \end{array}$$

representing the same morphism $f \rightarrow pfi$ in $\text{TwAr}(\text{sSet})$. □

Let us now address the fibrancy properties of slices. Just as those of function spaces rested in the cofibrancy properties of the pushout-product operation, so do they rest on the cofibrancy properties of the pushout-join, which we recorded in VI.3.18, but did not yet prove. Here's the statement again:

VI.3.18 Lemma: Let $i: A \rightarrow B$ and $j: K \rightarrow L$ be cofibrations. Then $i \boxtimes j$ is again a cofibration and

- ii)-iii) inner anodyne if either i is right anodyne or j is left anodyne
- iv) left anodyne if i is left anodyne
- v) right anodyne if j is right anodyne

We need one more lemma:

VI.3.41. Lemma

If $\Sigma \subseteq \text{mor}(\text{sSet})$ is a saturated class of morphisms then

$$\{i \in \text{mor}(\text{sSet}) \mid i \boxtimes j \in \Sigma\}$$

and

$$\{i \in \text{mor}(\text{sSet}) \mid j \boxtimes i \in \Sigma\}$$

whenever Σ is of the form

$$\Sigma = \{i \in \text{mor}(\text{sSet}) \mid i \text{ admits left lifts against } J\}$$

for some class $J \subseteq \text{mor}(\text{sSet})$.

Note that [V.2.31](#) says that $\Sigma = \text{sat}(S)$ for some set $S \subseteq \text{mor}(\text{sSet})$ with compact sources satisfies the assumptions; in particular this is the case for Σ the left/right/inner/ $\emptyset$ anodyne maps.

VI.3.42. Remark

The analogue of this result for the pushout-product was exercise 3 on sheet 7. It was easier in that for every saturated class $\Sigma \in \text{mor}(\text{sSet})$ the set

$$\{i \in \text{mor}(\text{sSet}) \mid i \boxtimes j \in \Sigma\}$$

is again saturated, without the assumption that Σ was defined by left lifting against some class J . The proof of that statement transfers to the present situation to show closure under retracts (trivially), pushouts and countable compositions, but not coproducts, since the join functor, when regarded as a functor $\text{sSet} \rightarrow \text{sSet}$ only preserves connected colimits.

There are other sufficient conditions on Σ such as right cancelability against cofibrations, i.e. if $i \in \Sigma$, j cofibration and $i \circ j \in \Sigma$ implies $j \in \Sigma$. This is also true in all relevant examples, but nontrivial for all three cases of interest (though we know it for anodynes: By [V.4.4](#) and [V.2.27](#) we have

$$\{\text{anodyne extensions}\} = \{\text{cofibrations}\} \cap \{\text{weak homotopy equivalences}\}$$

and the latter two are clearly right cancelable against all maps).

Proof of [VI.3.41](#): By assumption, $i \boxtimes j \in \Sigma$ is equivalent to admitting left lifts against J , i.e. every diagram

$$\begin{array}{ccc} A \star Y \cup_{A \star X} B \star X & \longrightarrow & K \\ i \boxtimes j \downarrow & \nearrow & \downarrow f \\ B \star Y & \longrightarrow & L \end{array}$$

admitting a filler for all $f \in J$. But by adjunction this is the same as

$$\begin{array}{ccc} A & \longrightarrow & K/B \rightarrow K \\ i \downarrow & \nearrow & \downarrow \\ B & \longrightarrow & (K/A \rightarrow K) \times_{L/A \rightarrow L} (L/B \rightarrow L) \end{array}$$

admitting fillers, which defines a saturated class by V.2.2.

The other case is analogous using right instead of left slices. $\square$

Proof of VI.3.18: Part i) is obvious (and we used it in VI.3.30 already). For part ii) consider the class

$$C = \{k \in \text{mor}(\text{sSet}) \mid k \boxtimes l \text{ is inner anodyne for all cofibrations } l\}$$

By VI.3.34 it is an intersection of saturated classes and thus itself saturated. We need to show that it contains the right anodynes, so it suffices to show

$$\Lambda_i^n \rightarrow \Delta^n \in C$$

for all $0 < i \leq n$. But consider

$$D = \{l \in \text{mor}(\text{sSet}) \mid (\Lambda_i^n \rightarrow \Delta^n) \boxtimes l \text{ is inner anodyne}\}$$

Again, this is saturated by VI.3.34, and we have to show that all cofibrations are contained in D , for which it suffices to treat the boundary inclusions $\partial\Delta^k \rightarrow \Delta^k$ by exercise 2, sheet 7. But lemma VI.3.30 identifies $(\Lambda_i^n \rightarrow \Delta^n) \boxtimes (\partial\Delta^k \rightarrow \Delta^k)$ as an inner horn, which is certainly inner anodyne.

The argument in the other cases is the same. $\square$

VI.3.43. Proposition

Let $A \xrightarrow{i} X \xrightarrow{f} Y \xrightarrow{p}$ be composable morphisms maps of simplicial sets, such that i is a cofibration and p as inner fibration, Then consider the maps

$$s: Y_{/f} \rightarrow Y_{/fi} \times_{B_{/pfi}} B_{/pf}$$

and

$$t: Y_{f/} \rightarrow Y_{fi/} \times_{B_{pfi/}} B_{pf/}.$$

- i) t is a left fibration and s a right fibration
- ii) if p is a right fibration, then t is a Kan fibration, and if p is a left fibration, then s is a Kan fibration
- iii) if p is a trivial fibration, then so are t and s
- iv) if i is right anodyne, then t is a trivial fibration, and if i is left anodyne then s is a trivial fibration.

Proof: We prove statements about t : As discussed, a lifting problem

$$\begin{array}{ccc} S & \xrightarrow{\quad} & Y_{f/} \\ a \downarrow & \nearrow \text{dashed} & \downarrow t \\ T & \xrightarrow{\quad} & Y_{fi/} \times_{B_{pfi/}} B_{pf/} \end{array}$$

is equivalent to

$$\begin{array}{ccc}
 A \star T \cup_{A \star S} X \star S & \longrightarrow & Y \\
 i \boxtimes a \downarrow & \nearrow t & \downarrow p \\
 X \star T & \longrightarrow & B
 \end{array}$$

- i) To show that t is a left fibration we consider the inclusions $\Lambda_i^n \rightarrow \Delta^n$ for $0 \leq i < n$ for a , which are left anodyne by definition. Thus VI.3.18 shows that $i \boxtimes a$ is then inner anodyne, so since p is an inner fibration, the lower diagram admits a lift.
- ii) If we take $a = \Lambda_n^n \rightarrow \Delta^n$ then VI.3.18 gives that $i \boxtimes a$ is still right anodyne so a lift exists whenever p is a right fibration.
- iii) If we take $a = \partial\Delta^n \rightarrow \Delta^n$ then $i \boxtimes a$ is still a cofibration, so a lift exists if p is a trivial fibration.
- iv) If again $a = \partial\Delta^n \rightarrow \Delta^n$, but i is right anodyne, then $i \boxtimes a$ is inner anodyne by VI.3.18, so a lift always exists. $\square$

VI.3.44. Corollary

If $f: \mathcal{I} \rightarrow \mathcal{C}$ is a map to a quasicategory then

$$s: \mathcal{C}_{/f} \rightarrow \mathcal{C} \quad t: \mathcal{C}_{f/} \rightarrow \mathcal{C}$$

are a right and left fibration, respectively.

In particular, $\mathcal{C}_{/f}$ and $\mathcal{C}_{f/}$ are again quasicategories.

Before we finish the proof of Joyal's theorem, a remark:

VI.3.45. Remark

We now have two competing definitions for over- and under-categories of quasicategories: On the one hand for $X: \Delta^0 \rightarrow \mathcal{C}$ there are the categories

$$\mathcal{C}_{/X} \quad \text{and} \quad \mathcal{C}_{X/}$$

defined using joins and on the other hand there are

$$\mathcal{C}/X \quad \text{and} \quad X/\mathcal{C}$$

defined using the arrow category. Worse yet, the latter construction can be extended as follows: Given $F: \mathcal{I} \rightarrow \mathcal{C}$ and $G\mathcal{J} \rightarrow \mathcal{C}$ put

$$\begin{array}{ccc}
 F/G & \longrightarrow & \text{Ar}(\mathcal{C}) \\
 \downarrow & \lrcorner & \downarrow (s,t) \\
 \mathcal{I} \times \mathcal{J} & \longrightarrow & \mathcal{C} \times \mathcal{C}
 \end{array}$$

As (s, t) is an inner fibration f/G is a quasicategory whenever $\mathcal{I}$ and $\mathcal{J}$ are such, and it follows from VI.3.39 that the map $f/G \rightarrow \mathcal{I} \times \mathcal{J}$ is a left fibration if $\mathcal{I} \simeq *$ and a right fibration if $\mathcal{J} \simeq *$. But then

$$\mathcal{C}_{//f} := \text{const}: \mathcal{C} \rightarrow \text{Fun}(\mathcal{I}, \mathcal{C}) \Big/ f: \Delta^0 \rightarrow \text{Fun}(\mathcal{I}, \mathcal{C})$$

is another perfectly reasonable attempt at defining a category of cones over a functor $f: \mathcal{I} \rightarrow \mathcal{C}$.

It will be an exercise to show, that $\mathcal{C}_{//f}$ and $\mathcal{C}_{/f}$ agree for nerves of ordinary categories, but this is not true in general. Furthermore, since by definition both

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(Y, X) & \longrightarrow & \mathcal{C}/X \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{Y} & \mathcal{C} \end{array} \quad \text{and} \quad \begin{array}{ccc} \text{Hom}_{\mathcal{C}}(Y, X) & \longrightarrow & Y/\mathcal{C} \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{X} & \mathcal{C} \end{array}$$

are pullbacks, the diagrams

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}^R(Y, X) & \longrightarrow & \mathcal{C}_{/X} \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{Y} & \mathcal{C} \end{array} \quad \text{and} \quad \begin{array}{ccc} \text{Hom}_{\mathcal{C}}^L(Y, X) & \longrightarrow & \mathcal{C}_{Y/} \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{X} & \mathcal{C} \end{array}$$

give reasonable definitions of mapping spaces (they are Kan by VI.3.44 and VI.3.27). As the notation suggests these are different from $\text{Hom}_{\mathcal{C}}(Y, X)$ and inn fact from one another! There are natural injections

$$\text{Hom}_{\mathcal{C}}^R(Y, X) \hookrightarrow \text{Hom}_{\mathcal{C}}(Y, X) \hookleftarrow \text{Hom}_{\mathcal{C}}^L(Y, X)$$

but it is a non-trivial result that these are homotopy equivalences, and similarly, that the two categories of cones are in fact equivalent.

The formation of fat slice categories also admits a left adjoint, the *fat join*, so one may wonder, why bother with the usual join and slices at all... The answer is that necessarily the fat join of Δ^n and Δ^m is *not* Δ^{n+m+1} , so the one thing they cannot obviously be used for is a proof of Joyal's lifting theorem! Quite the pickle...

These issues will come facts to haunt us when establishing Theorem H, as $\text{Hom}_{\mathcal{C}}(X, Y)$ is more easily compared to $\text{Hom}_{\mathcal{N}^c(\mathcal{C})}^L(X, Y)$ than $\text{Hom}_{\mathcal{N}^c(\mathcal{C})}(X, Y)$.

We are finally ready to prove Joyal's theorem:

VI.3.46. Proof of ?? (existence)

By VI.3.35 the lifting problem

$$\begin{array}{ccc} \Lambda_0^n & \xrightarrow{f} & \mathcal{C} \\ \downarrow & \nearrow \text{dashed} & \downarrow F \\ \Delta^n & \longrightarrow & \mathcal{D} \end{array}$$

is equivalent to

$$\begin{array}{ccc}
 \Delta^0 & \longrightarrow & \mathcal{C}/\Delta^{n-2} \\
 d_1 \downarrow & \nearrow & \downarrow \\
 \Delta^1 & \longrightarrow & \mathcal{C}/\partial\Delta^{n-2} \times_{\mathcal{D}/\partial\Delta^{n-2}} \mathcal{D}/\Delta^{n-2}
 \end{array}$$

As F is an inner fibration and $\partial\Delta^{n-2} \rightarrow \Delta^{n-2}$ a cofibration the right hand vertical map is a right fibration by VI.3.43. Furthermore, by construction there is a commutative diagram

$$\begin{array}{ccccc}
 \Delta^1 & \longrightarrow & \mathcal{C}/\partial\Delta^{n-2} \times_{\mathcal{D}/\partial\Delta^{n-2}} \mathcal{D}/\Delta^{n-2} & \longrightarrow & \mathcal{D}/\Delta^{n-2} \\
 \downarrow (01) & & \downarrow & \nearrow & \downarrow \text{ (magenta)} \\
 \Lambda_0^n & \xrightarrow{f} & \mathcal{C} & \longrightarrow & \mathcal{D}/\partial\Delta^{n-2}
 \end{array}$$

By VI.3.43 these two magenta vertical maps are right fibrations, thus so is the middle vertical composite. This implies first that $\mathcal{C}/\partial\Delta^{n-2} \times_{\mathcal{D}/\partial\Delta^{n-2}} \mathcal{D}/\Delta^{n-2}$ is a quasicategory and secondly ?? implies that

$$\Delta^1 \rightarrow \mathcal{C}/\partial\Delta^{n-2} \times_{\mathcal{D}/\partial\Delta^{n-2}} \mathcal{D}/\Delta^{n-2}$$

represents an equivalence as the composite

$$\Delta^1 \rightarrow \Lambda_0^n \xrightarrow{f} \mathcal{C}$$

does so by assumption. But then VI.3.36 and VI.3.38 give the desired lift. $\square$

To establish uniqueness of the lift in Joyal's theorem it suffices to show:

VI.3.47. Lemma

If $p: \mathcal{C} \rightarrow \mathcal{D}$ is a right fibration between quasicategories, then the fibre of the map

$$r: \text{Fun}(\Delta^1, \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \mathcal{D})_s \times_{\mathcal{D}} \mathcal{C}$$

over a pair (f, x) is a contractible Kan complex, whenever $f: \Delta^1 \rightarrow \mathcal{D}$ is an equivalence.

The proof will use VI.3.27, which in turn relies only on the existence part of VI.3.20, so we avoid circularity.

Proof: Since p is conservative by VI.3.28, the diagram

$$\begin{array}{ccc}
 \text{core}(\mathcal{C}) & \longrightarrow & \mathcal{C} \\
 \text{core}(p) \downarrow & & \downarrow p \\
 \text{core}(\mathcal{D}) & \longrightarrow & \mathcal{D}
 \end{array}$$

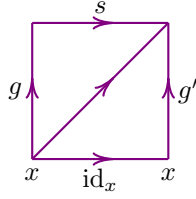
is a pullback. In particular, $\text{core}(p)$ is again a right fibration, and thus a Kan fibration by VI.3.27. But then by V.2.26 the map

$$\bar{r} : \text{Fun}(\Delta^1, \text{core}(\mathcal{C})) \rightarrow \text{Fun}(\Delta^1, \text{core}(\mathcal{D})) \times_{\text{Fun}(\{0\}, \text{core}(\mathcal{D}))} \text{Fun}(\{0\}, \text{core}(\mathcal{C}))$$

is a trivial fibration, so in particular has contractible fibres. It remains to check that

$$\bar{r}^{-1}(f, x) = r^{-1}(f, x)$$

whenever f is an equivalence. But these simplicial sets have the same vertices and edges as p is conservative. This is immediate for the vertices and for edges one decodes that an edge of $r^{-1}(f, x)$ is a diagram $\Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$:



where g, g' are lifts of f and s lifts the identity of $d_0(f)$. In particular all these edges are equivalences by conservativity of p , so the entire diagram factors through $\text{core}(\mathcal{C})$. For higher simplices $\Delta^n \rightarrow r^{-1}(f, x)$, note that the condition that

$$\Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$$

factors through $\text{core}(\mathcal{C})$ depends only on the images of edges, which we just checked. $\square$

VI.3.48. Proof of VI.3.20 (uniqueness)

Apply VI.3.47 to the reformulation of the lifting problem in the proof of existence. $\square$

Finally, we deduce the formulation of Joyal's theorem given in VI.3.16. Given a morphism $f : x \rightarrow y$ in a quasicategory $\mathcal{C}$ consider the functors

$$\mathcal{C}_{/x} \leftarrow \mathcal{C}_{/f} \rightarrow \mathcal{C}_{/y}$$

arising from the sequences $\Delta^0 \xrightarrow{0 \text{ or } 1} \Delta^1 \xrightarrow{f} \mathcal{C} \rightarrow *$. Then by VI.3.43 the left functor is a trivial Kan fibration, so in particular an equivalence of categories by VI.2.12. Choosing an inverse (of which there is a contractible choice by VI.3.24) we obtain a functor

$$f_* : \mathcal{C}_{/x} \rightarrow \mathcal{C}_{/y}$$

that on vertices is given by postcomposition with f . Analogously, there is a functor

$$f^* : \mathcal{C}_{y/} \rightarrow \mathcal{C}_{x/}$$

given by precomposition with f . We then have the following refinement of VI.3.20:

VI.3.49. Proposition

For a morphism $f: x \rightarrow y$ in a quasicategory $\mathcal{C}$ the statements

- i) f is an equivalence
- ii) $\mathcal{C}_{/f} \rightarrow \mathcal{C}_{/y}$ is a trivial Kan fibration
- iii) $\mathcal{C}_{/f} \rightarrow \mathcal{C}_{/y}$ is an equivalence of categories
- iv) 'the' functor $f_*: \mathcal{C}_{/x} \rightarrow \mathcal{C}_{/y}$ is an equivalence of categories

are equivalent.

The analogous statements using under-categories hold as well.

Proof: $i) \rightarrow ii)$: A lifting problem

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & \mathcal{C}_{/f} \\ \downarrow & & \downarrow \\ \Delta^n & \longrightarrow & \mathcal{C}_{/x} \end{array}$$

by using VI.3.35 translates to

$$\begin{array}{ccccc} & & f & & \\ & \nearrow & & \searrow & \\ \Delta^1 & \longrightarrow & \Lambda_0^{n+2} & \longrightarrow & \mathcal{C} \\ & & \downarrow & & \downarrow \\ & & \Delta^{n+2} & \longrightarrow & * \end{array}$$

which has a solution by VI.3.20 if f is an equivalence.

$ii) \rightarrow iii)$: VI.2.12.

$iii) \rightarrow iv)$: Immediate from the homotopy commutative diagram

$$\begin{array}{ccc} \mathcal{C}_{/x} & \xleftarrow{\quad} & \mathcal{C}_{/f} \\ & \searrow f_* & \downarrow \\ & & \mathcal{C}_{/y} \end{array}$$

since the horizontal map is an equivalence. $iv) \rightarrow i)$: If $f_*: \mathcal{C}_{/x} \rightarrow \mathcal{C}_{/y}$ is an equivalence of categories we can in particular pick a $g \in \mathcal{C}_{/x}$ such that f_*g is equivalent to id_y in $\mathcal{C}_{/y}$. But decoding this means that there is a 2-simplex in $\mathcal{C}$

$$\begin{array}{ccc} & & \\ s \nearrow & & \searrow h \\ y & \xrightarrow{\text{id}_y} & y \end{array}$$

where s is an equivalence and h a composite of f and g . But this means that id_y is a composite of f and gs . So f admits a right inverse. But then the images of id_x and gsf in $\mathcal{C}_{/y}$ are equivalent since

$$f_* \text{id}_x \sim f \sim \text{id}_y f \sim fgsf \sim f_*gsf.$$

But as f_* is an equivalence this implies that id_y and gsf are equivalent in $\mathcal{C}_{/x}$. Decoding again, we find this implies that gsf is an equivalence, so f also admits a left inverse, hence is an equivalence. $\square$

VI.3.50. Remark

We have *not* constructed a functor

$$\text{Cat}_{\infty\mathcal{C}_/} \rightarrow \text{Cat}_{\infty}, \quad f: \mathcal{C} \rightarrow \mathcal{D} \mapsto \mathcal{D}_{f/} \text{ or } \mathcal{D}_{/f}$$

or anything similar so far! This is quite difficult, in fact already showing that an equivalence of categories $g: \mathcal{D} \rightarrow \mathcal{D}'$ gives equivalences

$$\mathcal{D}_{f/} \rightarrow \mathcal{D}'_{gf/} \quad \text{and} \quad \mathcal{D}_{/f} \rightarrow \mathcal{D}'_{/gf}$$

is a highly non-trivial endeavor. Here again the fat slices are better behaved; the latter statement is almost a triviality here. To obtain a functor one 'only' needs to understand the interaction of the coherent nerve functor with over- and under-categories to build a simplicial model for $\text{Cat}_{\infty\mathcal{C}_/}$; alternatively the statement will follow from general machinery about limits in quasicategories.

In fact the statement about thin slices is usually proved by reduction to the corresponding one for fat slices!

Chapter VII.

The homotopy hypothesis